User's Manual

Multisizer 4e

Particle Analyzer

PN B23937AA
December 2013

Beckman Coulter, Inc.
250 S. Kraemer Blvd.
Brea, CA 92821 U.S.A.

Multisizer 4e Particle Analyzer User's Manual
PN B23937AA, 12/2013

Find us on the World Wide Web at:
www.beckmancoulter.com

Made in USA.

Revision History

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Software Version 4.03

This document applies to the latest software listed and higher versions. When a subsequent software version affects the information in this document, a new issue will be released to the Beckman Coulter Web site. For labeling updates, go to www.beckmancoulter.com and download the latest version of the manual or system help for your instrument.

Safety Notice

Read all product manuals and consult with Beckman Coulter-trained personnel before attempting to operate instrument. Do not attempt to perform any procedure before carefully reading all instructions. Always follow product labeling and manufacturer's recommendations. If in doubt as to how to proceed in any situation, contact your Beckman Coulter Representative.

Beckman Coulter, Inc. urges its customers to comply with all national health and safety standards such as the use of barrier protection. This may include, but is not limited to, protective eyewear, gloves, and suitable laboratory attire when operating or maintaining this or any other laboratory analyzer.

Alerts for Warning and Caution

Throughout this manual, you will see the appearance of these alerts for Warning and Caution conditions:

WARNING

WARNING indicates a potentially hazardous situation, which, if not avoided, could result in death or serious injury.

CAUTION

CAUTION indicates a potentially hazardous situation, which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.

Safety Precautions

⚠ WARNING

Risk of operator injury if:

- **All doors covers and panels are not closed and secured in place prior to and during instrument operation.**
- **The integrity of safety interlocks and sensors is compromised.**
- **Instrument alarms and error messages are not acknowledged and acted upon.**
- **You contact moving parts.**
- **You mishandle broken parts.**
- **Doors, covers and panels are not opened, closed, removed and/or replaced with care.**
- **Improper tools are used for troubleshooting.**

To avoid injury:

- **Keep doors, covers and panels closed and secured in place while the instrument is in use.**
- **Take full advantage of the safety features of the instrument.**
- **Acknowledge and act upon instrument alarms and error messages.**
- **Keep away from moving parts.**
- **Report any broken parts to your Beckman Coulter Representative.**
- **Open/remove and close/replace doors, covers and panels with care.**
- **Use the proper tools when troubleshooting.**

⚠ CAUTION

System integrity could be compromised and operational failures could occur if:

- **This equipment is used in a manner other than specified. Operate the instrument as instructed in the product manuals.**
- **You introduce software that is not authorized by Beckman Coulter into your computer. Only operate your system's software with software authorized by Beckman Coulter.**
- **You install software that is not an original copyrighted version. Only use software that is an original copyrighted version to prevent virus contamination.**

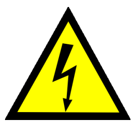
⚠ CAUTION

If you purchased this product from anyone other than Beckman Coulter or an authorized Beckman Coulter distributor, and, it is not presently under a Beckman Coulter service maintenance agreement, Beckman Coulter cannot guarantee that the product is fitted with the most current mandatory engineering revisions or that you will receive the most current information bulletins concerning the product. If you purchased this product from a third party and would like further information concerning this topic, contact your Beckman Coulter Representative.

Electrical Safety

To prevent electrically related injuries and property damage, properly inspect all electrical equipment prior to use and immediately report any electrical deficiencies. Contact a Beckman Coulter Representative for any servicing of equipment requiring the removal of covers or panels.

High Voltage



This symbol indicates the potential of an electrical shock hazard existing from a high-voltage source and that all safety instructions should be read and understood before proceeding with the installation, maintenance, and servicing of all modules.

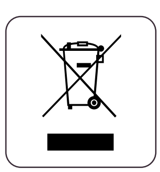
Do not remove system covers. To avoid electrical shock, use supplied power cords only and connect to properly grounded (three-holed) outlets.

International Warning Symbol

This symbol indicates that the user should refer to the user documentation for further instructions, cautions, or warnings that may be associated with the area of the instrument, or specific components, where this symbol is present on the instrument.

Disposal of Electronic Equipment

It is very important that customers understand and follow all laws regarding the safe and proper disposal of electrical instrumentation.



The symbol of a crossed-out wheeled bin on the product is required in accordance with the Waste Electrical and Electronic Equipment (WEEE) Directive of the European Union. The presence of this marking on the product indicates:

- That the device was put on the European Market after August 13, 2005 and

- That the device is not to be disposed via the municipal waste collection system of any member state of the European Union.

For products under the requirement of WEEE directive, please contact your dealer or local Beckman Coulter office for the proper decontamination information and take back program which will facilitate the proper collection, treatment, recovery, recycling, and safe disposal of device.

Moving Parts

WARNING

Risk of personal injury. To avoid injury due to moving parts, observe the following:

- **Never attempt to exchange labware, reagents, or tools while the instrument is operating.**
- **Never attempt to physically restrict any of the moving components of the instrument.**
- **Keep the instrument work area clear to prevent obstruction of the movement.**

Cleaning

Observe the cleaning procedures outlined in this user's manual for the instrument. Prior to cleaning equipment that has been exposed to hazardous material:

- Contact the appropriate Chemical and Biological Safety personnel.
- Review the Chemical and Biological Safety information in the user's manual.

Maintenance

Perform only the maintenance described in this manual. Maintenance other than that specified in this manual should be performed only by service engineers.

IMPORTANT It is your responsibility to decontaminate components of the instrument before requesting service by a Beckman Coulter Representative or returning parts to Beckman Coulter for repair. Beckman Coulter will NOT accept any items which have not been decontaminated where it is appropriate to do so. If any parts are returned, they must be enclosed in a sealed plastic bag stating that the contents are safe to handle and are not contaminated.

RoHS Notice

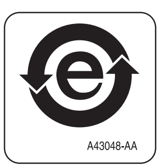
These labels and materials declaration table (the Table of Hazardous Substance's Name and Concentration) are to meet People's Republic of China Electronic Industry Standard SJ/T11364-2006 "Marking for Control of Pollution Caused by Electronic Information Products" requirements.

China RoHS Caution Label

This label indicates that the electronic information product contains certain toxic or hazardous substances. The center number is the Environmentally Friendly Useful Period (EFUP) date, and indicates the number of calendar years the product can be in operation. Upon the expiration of the EFUP, the product must be immediately recycled. The circling arrows indicate the product is recyclable. The date code on the label or product indicates the date of manufacture.

China RoHS Environmental Label

This label indicates that the electronic information product does not contain any toxic or hazardous substances. The center 'e' indicates the product is environmentally safe and does not have an Environmentally Friendly Useful Period (EFUP) date. Therefore, it can safely be used indefinitely. The circling arrows indicate the product is recyclable. The date code on the label or product indicates the date of manufacture.



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Introduction

About the Multisizer 4e

The Beckman Coulter Multisizer 4e is a flexible, multi-channel analyzer employing the Coulter Principle, together with state-of-the-art Digital Pulse Processing (DPP) technology to provide both particle sizing and counting within an overall size range of 0.2 µm to 1600 µm. Data is acquired and processed using proprietary Digital Signal Processing (DSP) circuitry and may be archived, displayed, and re-analyzed on any PC equipped with a CORE or similar microprocessor running Beckman Coulter's Multisizer 4e software on Microsoft Windows 7 operating system.

How the Multisizer 4e Works

The Multisizer 4e uses a unique, mercury-free "isolated piston" system to draw suspensions through the aperture at a steady rate and to meter these liquid volumes precisely enough to measure accurate particle concentrations. The instrument is also equipped with a continuous flow alternative for applications where accuracy of liquid metering is less important or less critical, as well as options that allow the user to define analysis parameters based on elapsed time or particle count.

By using the very latest in digital pulse measurement techniques, an entire analysis (which may consist of many thousands of particle pulses) can be stored either in memory or on disk for recall and reprocessing without the need to re-run the sample. Each pulse is stored at maximum resolution so that, for the first time, the genuine resolution of an analysis can be increased up to a hundred-fold in real-time by narrowing the size range of interest, eliminating the need to re-run the sample. The Convert Pulses to Size feature in the software allows measurements from the lowest to the highest possible resolution to be made on just one pass of sample through the aperture.

Storing the individual values of pulse heights means that it is not necessary to predetermine the number of size classes (also called "channels" or "size bins") into which any size distribution can be divided. This choice is at the discretion of the operator and can be chosen either before or after the analysis or even changed during analysis. Any number of channels from 4 up to 400 may represent a partial or full range of an aperture. A smaller number may improve the appearance of a size distribution by reducing statistical scatter in the results, while details in the results may be enhanced by increasing the number of size classes in regions of particular interest. A higher number of channels provides more detail about the size of the particles. The advanced digital electronics used in the Multisizer 4e allow the number of channels to be changed in real time without losing any of the information acquired with other channel spacing.

Results can also be stored in a fully processed (abridged) form that conserves disk space. Although in this form the raw pulse data is lost and cannot be re-analyzed at a different resolution, the processed results still can be enlarged, or a region of interest selected, using the Zoom facility in the software.

Both number data and size distributions are corrected by proprietary algorithms for the occurrence of multiple particle passage (coincidence) in the aperture. An indicator provides real time particle concentration in the analysis suspension. The Multisizer 4e software includes a pulse monitor window that offers a guide to the electronic behavior of the aperture. The time base of this window can be adjusted to show either pulse concentration or the shapes of individual pulses. Electronic pulse editing facilities ensure that very narrow size distributions can be highly resolved by minimizing the effect of line width broadening in the sensing aperture.

Particle Size Determination

In industrial applications, historically it has been common practice to report particle "size" in linear terms, and so the volume of a particle is converted into an equivalent spherical diameter (ESD). In this case, the "particle size" is the diameter of a sphere whose volume is equal to that of the particle. The biological and biomedical sciences traditionally (when reporting the results of cellular analyses) are more accustomed to using actual volumetric metric units, and in these applications "particle size" is usually presented in terms such as femtoliters or cubic microns (fL, or μm^3).

Particle surface areas can be calculated, with the assumption that the surface area is that derived from (and equates to) the surface area of a perfect sphere whose ESD is the same as that of the volume of the particle.

The International Standard ISO 13319 Determination of Particle Size Distributions—the Coulter Principle describes and gives guidance on the measurement of the size distribution of particles using the Coulter Principle.

Displaying Multisizer 4e Data

The Multisizer 4e can display data graphically as number, volume, mass, and surface area size distribution plots, or as tables of values versus size. Both display types can show data as differential or cumulative frequencies. Full data reduction facilities are included for statistical and trend analysis, with export and import capabilities to text and spreadsheet programs via disk files or the Windows Clipboard. The Multisizer 4e can print results using any printer compatible with the Windows platform and drivers installed on the system.

A report utility allows results to be combined with sample information in a user-selectable way to produce a systematic reporting configuration. Report formats can be individually named for ease of archiving and retrieval. Number and date formats are supported in both conventional as well as in ISO 9276-1 forms.

Multisizer 4e Components

The Multisizer 4e is an integrated fluid handling and pulse processing unit (the Analyzer Module) capable of supporting a personal computer and monitor with a minimum resolution of 1024 by 768, which can be purchased separately. A keyboard, mouse and an ink-jet printer are also optional accessories to this system.

Multisizer 4e Specifications

Operational Parameters

- *Particle Size Range:* 0.2 µm to 1600 µm in diameter
- *Volume:* 0.004 μm^3 to 2.14×10^9 μm^3

NOTE The upper size limit is dictated by solution viscosity and particle density, and the lower limit by the aperture size, solution conductivity, and environmental conditions.

- *Displayable size units:* µm, μm^2, μm^3, fL
- *Number of channels:* Pulse data is stored in unchannelized form and can displayed in up to 400 size classes for any selected size range. The number of channels and their range can be reprocessed as necessary.
- *Digital pulse processor:* High-speed digitalization of the signal allows the definition of pulses by a number of characteristics. This information may be used at a later time to recalculate the size distribution using different pulse parameters.
- *Pulse data:* Up to 525,000 pulses per analysis
- *Linearity for 2% to 60% of aperture diameter:* ± 1% for diameter, ± 3% for volume
- *Aperture tube sizes:* 10 µm to 2000 µm apertures (nominal sizes). Each size allows the measurement of particles in a range of 2% to 80% of aperture diameter. See Table 4.1 in CHAPTER 4, Installing and Calibrating an Aperture Tube.
- *Aperture current range:* 30 µA to 6000 µA in 1 µA steps

- *Aperture current accuracy:* ± 0.4% of setting
- *Polarity error:* Less than 0.5%
- *Time mode range:* 0.01 to 1000 seconds, selectable in 10-ms increments
- *Metering system:* Mercury-free wide range metering pump
- *Metered sample volumes (volumetric mode only):* Continuously selectable from 10 µL to 2000 µL
- *Volumetric pump accuracy:* Better than 99.5%

Regulatory Compliance

This software is FDA 21 CFR Part 11 enabled.

Before You Start: Warnings and Cautions

Warnings

In this document, an orange header box alerts you to a warning message.

WARNING

A warning message describes either a potentially hazardous situation which, if not avoided, could result in death or serious injury to the operator.

Electrical Warnings

- High voltages are present inside the instrument. Always disconnect the instrument from the power supply before removing the cover.
- The instrument must be grounded properly.

Chemical Warnings

- Do not use any electrolyte solutions that are not compatible with the materials of construction. Consult Beckman Coulter, Inc. or its local representative before using any concentrated acids or alkalis in the instrument.
- Toxicity, safety, and proper handling procedures for electrolyte and reagents used in particle and cell analysis should be adhered to at all times. Consult appropriate safety manuals and Material Safety Data Sheets for the items.
- Care should be taken when mixing or exchanging solutions. Reactions can occur between incompatible solvents and electrolyte salts, and may be violent.
- Do not use azide and cyanide salts in acid solutions.
- Flammable electrolyte solutions should be prepared for use in an appropriate environment and brought to the instrument only when required for an analysis.

Mechanical Warnings

- Do not interfere with or attempt to disable the interlocks incorporated in the door and platform assembly.
- Do not use force to remove or replace any glass item. In the event of difficulty, consult Beckman Coulter, Inc. or its local representative.
- Replace an aperture tube empty of electrolyte. Do not attempt to pre-fill a tube before placing it in the aperture tube connector.

Fire Warnings

- Many non-aqueous electrolyte solutions are flammable and contain organic salts that can support or enhance fire. Where possible choose less flammable alternatives.
- Prepare and filter flammable solvents and electrolyte solutions away from the Multisizer 4e, preferably under a suitable fume hood. Extinguish all open flames in the vicinity and observe normal precautions against ignition by hot surfaces or electrical sparks.

Cautions

In this document, a yellow header box alerts you to a caution message:

⚠ CAUTION

A caution message describes a potentially hazardous situation, which if not avoided, may result in minor or moderate injury. It can also be used to alert against unsafe practices.

Electrical Cautions

- Power must be largely free from interference. The Multisizer 4e is CE, UL and CSA compliant with regards to electrical interference but some circumstances may benefit from a constant voltage transformer or regulator.

Chemical Cautions

- If Beckman Coulter ISOTON II is used, the Multisizer 4e must be drained and flushed through with distilled water before carrying out a sodium hypochlorite solution (bleach) disinfection procedure.

⚠ WARNING

Risk of chemical injury from bleach. To avoid contact with bleach, use barrier protection, including protective eyeware, gloves, and suitable laboratory attire. Refer to the Safety Data Sheet for details about chemical exposure before using the chemical.

- If the Multisizer 4e has been subjected to a bleach disinfection procedure, the instrument must be drained and flushed through with distilled water before refilling with ISOTON II.

- Never stand containers of liquids on top of the Multisizer 4e. Repair of instruments damaged or affected by spilled liquids will not be covered by any warranty or guarantee.

  NOTE The Multisizer 4e performs best when it has reached a steady working temperature. This typically takes 15 minutes from powering on the Analyzer.

CHAPTER 1

Installation

General Laboratory Information

IMPORTANT Your Beckman Coulter Representative is responsible for uncrating, installing, and initial setup of the Multisizer 4e instrument. Contact your Beckman Coulter Representative before relocating your Multisizer 4e instrument.

Instrument Specifications

IMPORTANT Heating and air conditioning vents or fans are not recommended directly above the Multisizer 4e instrument because of resulting temperature fluctuation, vibration, and possible dust.

The General Systems Specification and Environmental Requirements are shown in the table below.

Table 1.1 General System Specification and Environmental Requirements

Specification	Requirements
Service Access	28 cm (11 in.) on left side, 28 cm (11 in.) on right side, 31 cm (12 in.) on back side, 31 cm (12 in.) on top side.
Installation Category	II
Pollution Degree	2
Instrument Dimensions	Height: 51 cm (20 in.) Width: 64 cm (25 in.) Depth: 61 cm (24 in.) Weight: 45 kg (99 lbs)
Operating Temperatures	5˚C to 40˚C
Humidity range for instrument storage and operation	Relative humidity:10–80% without condensation
Altitude	Up to 2000 m (6500 ft). Above 7300 m (24,000 ft) unpressurized transport may damage the monitor.
Electrolyte Solutions	All aqueous and non-aqueous electrolyte solutions devised for use with aperture technology are suitable for use with the Multisizer 4e. Electrolytes should be compatible with glass, fluoropolymers, fluoroelastomers, and stainless steel.

The Electrical Requirements are shown in the table below.

Table 1.2 Electrical Requirements

Specification	Requirements
Input Voltage within Set Ranges	100–120 VAC; 230–240 VAC ± 10%; single phase
Supply Frequency	47 to 63 Hz inclusive
Power	Less than 55 VA (watts)
Fuse Types	(100-120 VAC) 2A, Slo-Blo, 250 V, Time Delay (TD) (230-240 VAC) (5x20 mm) 1 A, Slo-Blo, 250 V, Time Delay (TD)

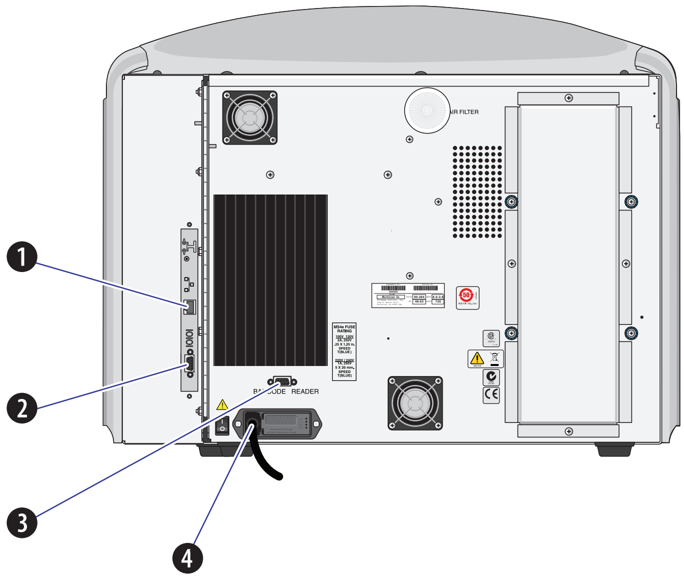
NOTE The use of other fuse types may result in damage to the instrument.

NOTE Power cord replacement must be of equivalent type and electrical ratings as the original, including safety agency recognitions.

System Connections

The Multisizer 4e Connections are shown in Figure 1.1.

Figure 1.1 Multisizer 4e Connections



The Multisizer 4e Connections and Descriptions are listed in the Table 1.3.

Table 1.3 Multisizer 4e Conntactions and Descriptions

	Connection	Description
1	Ethernet	The Ethernet connection point for the Ethernet cable that is attached to the instrument. This cable communicates with and controls the Multisizer.
2	Serial	The Serial connection is used to connect the 9-pin serial cable. This cable is used by service for communication.
3	Bar code Reader	The Bar code Reader connectionis used to connect the bar code reader that is attaced to the USB ports on the PC. This cable allows the use of the bar code reader in several functions.
4	Power Cord	The Power Cord connection is for the main power input cable. **NOTE** The instrument must be configured for the specific voltage being applied prior to connecting the main power cord to the instrument.

Installing Multisizer 4e Software

Multisizer 4e software will be installed by Beckman Coulter personnel upon instrument installation. To install Multisizer 4e software on additional computers, insert the CD into the CD/DVD drive and follow the on-screen prompts.

CHAPTER 2

Analyzer Overview

Overview

Multisizer 4e Features

Multisizer 4e features allow the operator to ensure consistent and accurate results.

Bar Code Reader

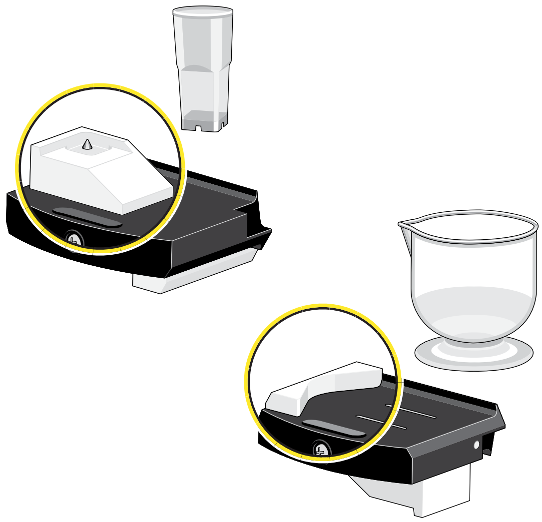
The Bar Code Reader allows the operator to:

- Read the Beckman Coulter ISOTON II label and automatically record the lot number, expiration date, and the current date as the opened date
- Read the Beckman Coulter ST aperture tube so the software can enter and/or load that specific aperture tube's size, serial number, and Kd value after the tube has been calibrated
- Enter the Multisizer 4e Smart Technology Beaker size, which prompts the Analyzer to move the stirrer into the ideal position for uniform particle distribution.
- Automatically input sample information by scanning the sample bar code
- Read the Beckman Coulter Control Material bar-coded assay sheet and will automatically load the size value, lot number, expiration date, and open date into the software's required fields.

Sample Platform Beaker and Accuvette Guides

Beaker guides on the sample platform align Multisizer 4e Smart Technology Beakers and Accuvette STs relative to the aperture tube for optimal and repeatable results. See Figure 2.1.

Figure 2.1 Sample Platform Beaker and Accuvette guide location

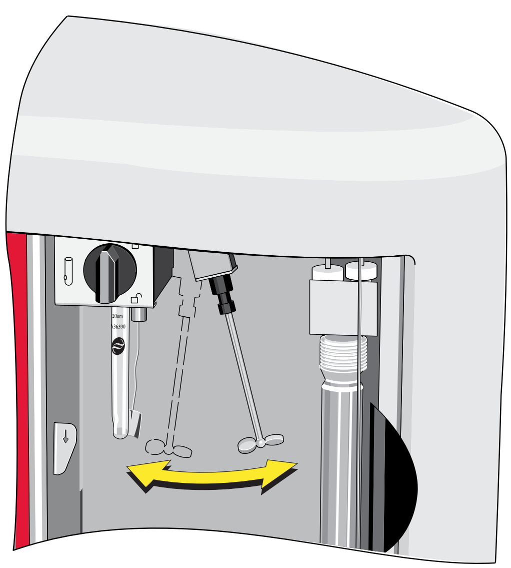


For more information on the Beaker and Accuvette guides, see Sample Platform.

Automatic Stirrer Positioning

Software controls automatically position the stirrer (see Figure 2.2) based on beaker and aperture tube size for consistent results (see Standard Operating Method: Description in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).

Figure 2.2 Automatic Stirrer



In addition, the software allows the operator to select stirrer speed and direction for uniform particle suspension. The Multisizer 4e software records stirrer position, speed, and direction, including adjustments made during an analysis run, so that you can easily reproduce analysis results (see Viewing Control Panel Actions in Run Information).

Automatic Blockage Detection

The Multisizer 4e automatic blockage detection feature monitors analysis runs for aperture blockage and can automatically unblock the aperture and continue a run without operator intervention. The Multisizer 4e allows the operator to select blockage detection settings based on current run data or reference runs, view a blockage monitoring graph during analysis runs, and review blockage monitoring data after a run is complete.

Extended Dynamic Range

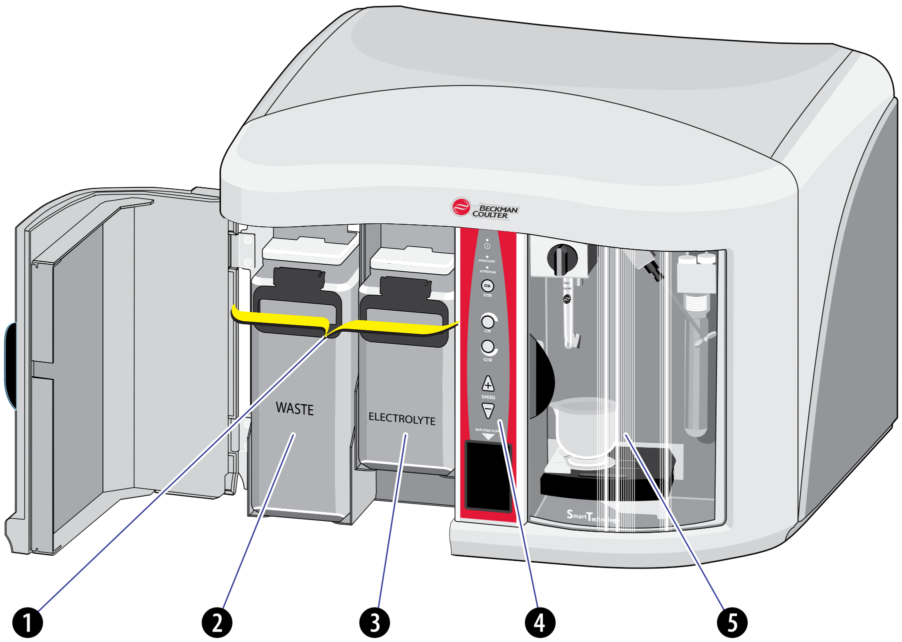
The Multisizer 4e is capable of measuring particles in the range of 2% to 80% of the aperture diameter for all aperture sizes. The standard range is 2% to 60%; an extended dynamic range of 60% to 80% is available to examine broad distributions that extend above the 60% threshold. This feature is intended for examining particles in the small "tail" of a distribution that extends above 60%. For best results and to minimize blockage, it is recommended to use a larger aperture size for samples that contain a significant concentration of particles above 60% of the aperture diameter.

Multisizer 4e Components

The convenient design of the Beckman Coulter Multisizer 4e places the Sample Compartment, Control Panel, and Electrolyte and Waste Jars at the front of the instrument. See Figure 2.3.

Analyzer components are labeled in Figure 2.3.

Figure 2.3 Multisizer 4e Analyzer Components

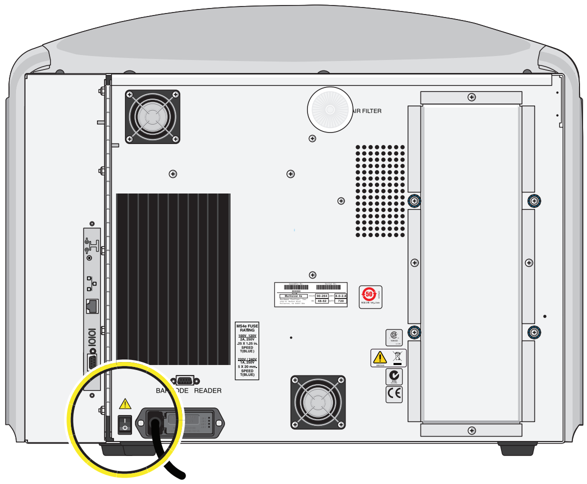


1 Waste and Electrolyte Jar Compartment
2 Waste Jar
3 Electrolyte Jar
4 Control Panel
5 Sample Compartment

Power ON/OFF Switch

The ON/OFF switch is located next to the power cable on the rear panel of the instrument. See Figure 2.4.

Figure 2.4 Power ON/OFF switch location



⚠ CAUTION

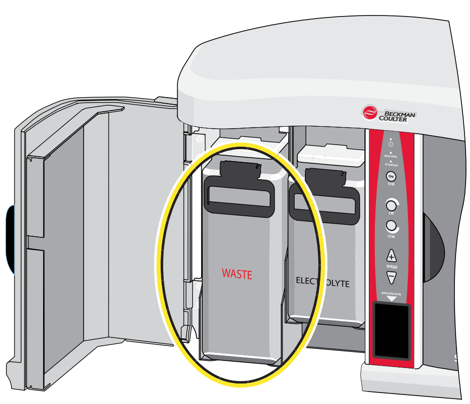
Make sure that the Analyzer has been powered down for at least 20 seconds before restarting. This ensures that the power supply has been depleted of voltage and allows the processor to reset.

NOTE The Multisizer 4e performs best when it has reached a steady working temperature. This typically takes 15 minutes from powering on the Analyzer.

Waste Jar

The Waste Jar is the larger of the two jars, located to the left of the Electrolyte Jar. See Figure 2.5.

Figure 2.5 Waste Jar location

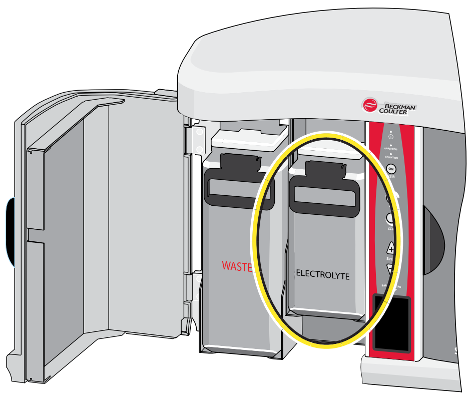


During an analysis, the Analyzer fills the Waste Tank with electrolyte solution drawn through the aperture tube. You can empty the Waste Tank fluid into the Waste Jar using settings in the Multisizer 4e software (see Waste Tank Settings in CHAPTER 10, Troubleshooting).

Electrolyte Jar

The Electrolyte Jar is located next to the Control Panel on the front of the Analyzer. See Figure 2.6.

Figure 2.6 Electrolyte Jar location



Electrolyte is used to fill the system, flush particles out of the system, rinse the aperture tube, and unblock the aperture.

It is preferable that you use the same electrolyte solution in both the Electrolyte Jar and the sample beaker or Accuvette ST. When running an analysis, however, the Analyzer draws only sample beaker electrolyte solution through the aperture orifice.

Electrolyte solution passes through the orifice only during the unblock procedure, when the Analyzer pushes fluid through the aperture in reverse to remove blockages, such as large groups of particles.

Electrolyte and Waste Jar Fluid Levels

The Multisizer 4e software displays the fluid levels in the Electrolyte Jar and Waste Jars on the Status Panel (see Status Panel in CHAPTER 3, Software Overview).

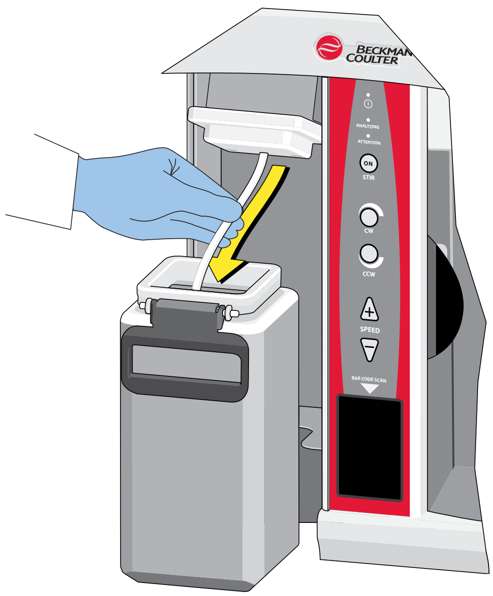
The fluid levels in the Electrolyte and Waste Jars are calculated by weight. It is usually necessary to calibrate the weight sensors only once, when you set up the Analyzer. If you switch to an electrolyte with significantly different density, however, you may need to recalibrate the Electrolyte and Waste Jar sensors (see Calibrating the Electrolyte and Waste Jars in CHAPTER 10, Troubleshooting).

Installing and Removing the Electrolyte and Waste Jars

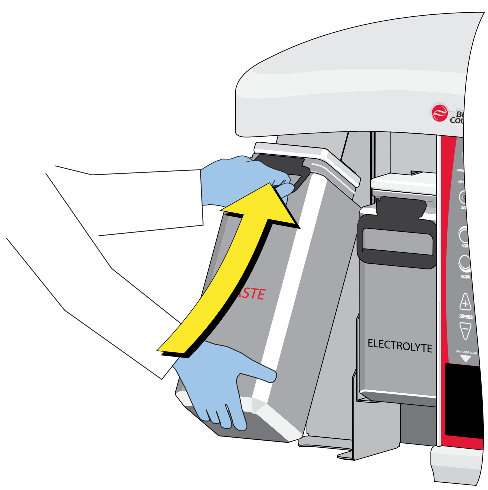
The lids of the Electrolyte and Waste Jars remain attached to the Analyzer.

To Install an Electrolyte or Waste Jar

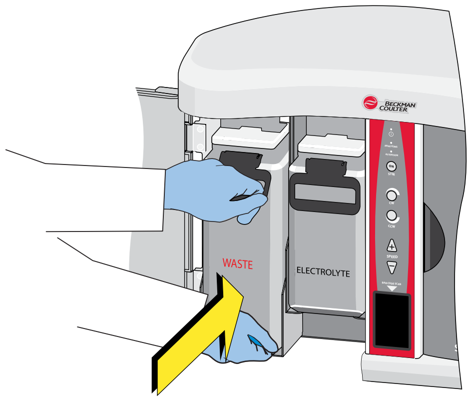
1 Place the Electrolyte or Waste Jar tube inside the jar.



2 Lift the jar until the lid is in place.



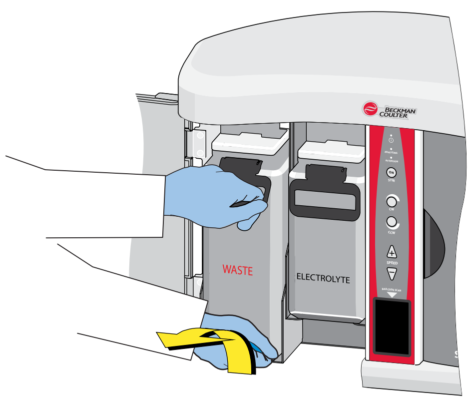
3 Hold the jar from the bottom and slide the jar into the appropriate compartment.



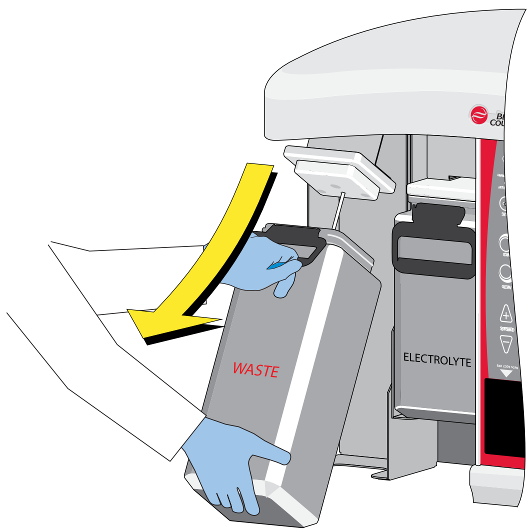
To Remove an Electrolyte or Waste Jar

1 Hold the bottom of the jar.

2 Lift the jar slightly and pull the jar away from the Analyzer.



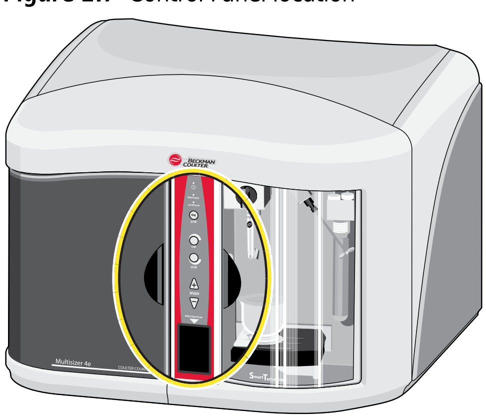
3 Tilt the jar and pull the jar away from the lid and tube.



Control Panel

The Control Panel is located in the center of the front panel. See Figure 2.7.

Figure 2.7 Control Panel location

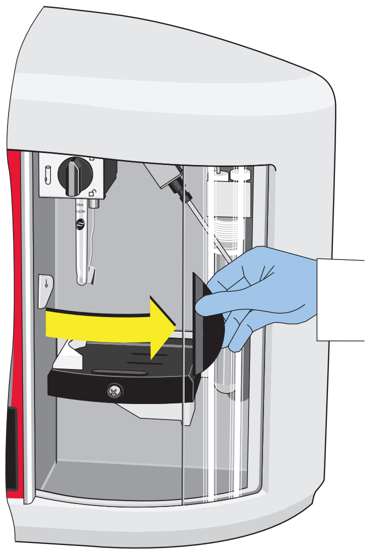


For information on Control Panel buttons, status lights, and the Bar Code Reader, see Control Panel.

Sample Compartment

The Sample Compartment is located at the front of the Analyzer on the right. See Figure 2.8.

Figure 2.8 Sample Compartment location



The door of the Sample Compartment consists of two layers of glass laminated on either side of a conductive material. When closed, the conductive material in the door reduces electrical noise created by electromagnetic fields (EMFs) in the vicinity. For information on other causes of noise in sample analyses, see Reducing Noise in CHAPTER 10, Troubleshooting.

NOTE The Multisizer 4e software will display an error message if you attempt to start a sample run while the door is open.

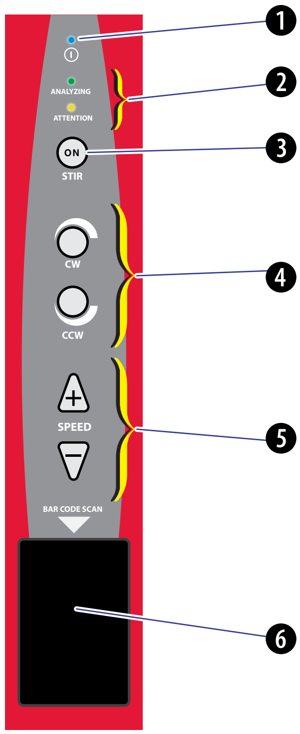
NOTE For information on components inside the Sample Compartment, see Sample Compartment.

Control Panel

The Control Panel includes an ON light, status lights, stirrer control buttons, and the Bar Code Reader.

Control Panel functions are labeled in Figure 2.9.

Figure 2.9 Control Panel functions



1 Analyzer ON Light
2 **Analyzing** (Green) and Attention (Amber) Lights
3 **Stirrer ON** button
4 Stirrer Direction buttons (**CW** and **CCW**)
5 **Stirrer Speed** buttons
6 Bar Code Reader

Disabling the Control Panel Buttons

You can use the Control Panel buttons during a sample run to turn the stirrer on or off, reverse the stirrer direction, and increase or decrease the stirrer speed.

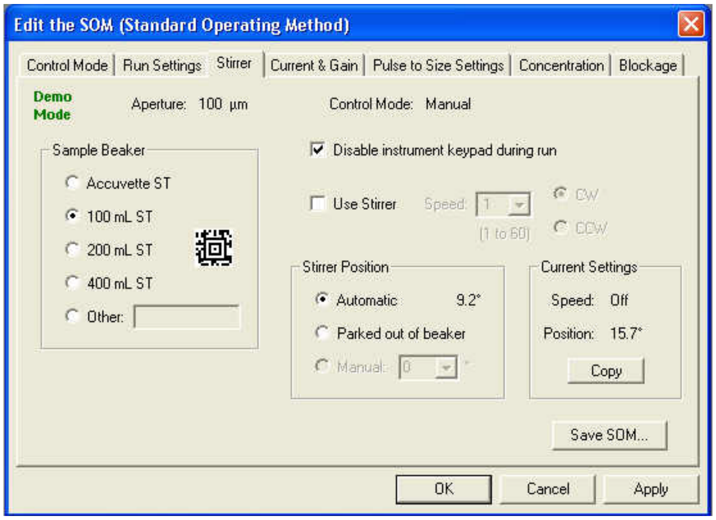
To disable the stirrer control buttons during sample runs, use the Edit the SOM (Standard Operating Method) window in the Multisizer 4e software.

To Disable the Control Panel Stirrer Buttons

1 In the Multisizer 4e software Main Menu, select **SOP > Edit the SOM.**

2 In the Edit the SOM (Standard Operating Method) window, select the Stirrer tab.

3 On the Stirrer tab, check the box next to Disable instrument keypad during run, and click **OK**.



NOTE If Accuvette ST or Parked out of beaker is selected, Disable will not be available.

Viewing Control Panel Actions in Run Information

If you use Control Panel buttons to adjust stirrer operation during an analysis run, the Multisizer 4e software stores all changes in stirrer operation, direction, and speed with sample run information.

To View Sample and Run Information

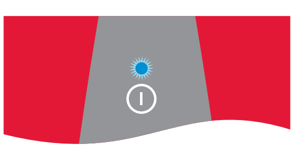
1 Open an analysis file.

2 Select **RunFile** > **Get Info** on the Run Menu bar.

ON Light

The blue **ON** light is located at the top of the Control Panel (see Figure 2.10).

Figure 2.10 ON light location



Analyzing and Attention Lights

There are two status lights (see Figure 2.11): Analyzing and Attention, below the **ON** light.

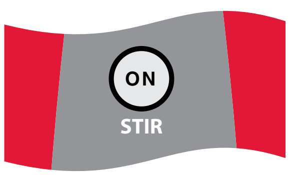
Figure 2.11 Analyzing and Attention lights location

- The green Analyzing light is on when the Analyzer is running an analysis. It duplicates the function of the green LED light in the Sample Compartment.
- The amber Attention light flashes when an error prevents the Analyzer from starting a run (for example, when the door is open or the sample platform is not in place). The amber Attention light duplicates the function of the amber LED light in the Sample Compartment. When the amber light flashes, the Multisizer 4e software displays an error message.

Stir ON Button

The **Stir ON** button is located below the Analyzing and Attention lights on the Control Panel (see Figure 2.12).

Figure 2.12 Stir ON button location

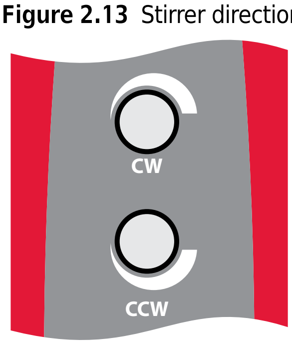


The toggle button turns the stirrer on or off. It is lit when the stirrer is **ON**. The Stirrer operates only when the platform is in the raised position for sample analysis. It will not spin when the platform is in the loading position.

Stirrer Direction Controls

You can select or change stirrer direction before or during an analysis run using the clockwise (CW) or counter-clockwise (CCW) button (see Figure 2.13).

Figure 2.13 Stirrer direction controls location



The direction of the stirrer influences particle suspension. The clockwise (CW) direction circulates fluid and sample particles downwards and inwards toward the base of an ST Beaker. The counter-clockwise (CCW) direction circulates fluid and particles upwards and outwards toward the edges of the beaker.

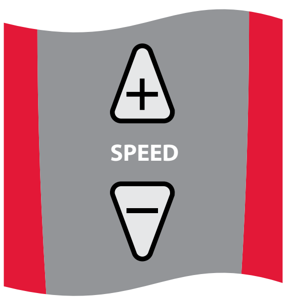
NOTE Beckman Coulter recommends the clockwise (CW) direction when using the stirrer. In general, the counter-clockwise (CCW) direction is less effective in maintaining uniform particle distribution and suspension.

For information on changing the stirrer direction using Standard Operating Method settings, see Standard Operating Method: Stirrer Settings in CHAPTER 5, Selecting Analysis Settings: SOM and SOP.

Stirrer Speed Controls

You can increase or decrease the stirrer speed using the Control Panel Speed buttons, **+** and **-** (see Figure 2.14).

Figure 2.14 Stirrer speed controls location



Bar Code Reader

You can use the Bar Code Reader (see Figure 2.9, (6)) to scan information about the following directly into the Multisizer 4e software:

- Beckman Coulter ISOTON II labels
- Beckman Coulter Smart Technology Aperture Tubes (size and serial number)
- Beckman Coulter material assay sheets (assayed value, lot number, expiration date, and open date)
- Beckman Coulter Multisizer 4e Smart Technology Beakers
- Automatically input the sample information by scanning the customer's sample bar code.

To configure the bar code reader, see Bar Code Reader Configuration Options in CHAPTER 9, Configuring Software and Security.

To Use the Bar Code Reader

In the Multisizer 4e software, a Bar Code Reader symbol indicates that you can use the scanner to enter information automatically into appropriate text fields.

1 When you see the Bar Code Reader symbol in a window, hold the ISOTON II label, ST Aperture Tube, beaker, or sample bar code in front of the Bar Code Reader.

NOTE The bar code reader will illuminate the bar code with red light. Hold the bar code in the vertical green illumination bar to be read.

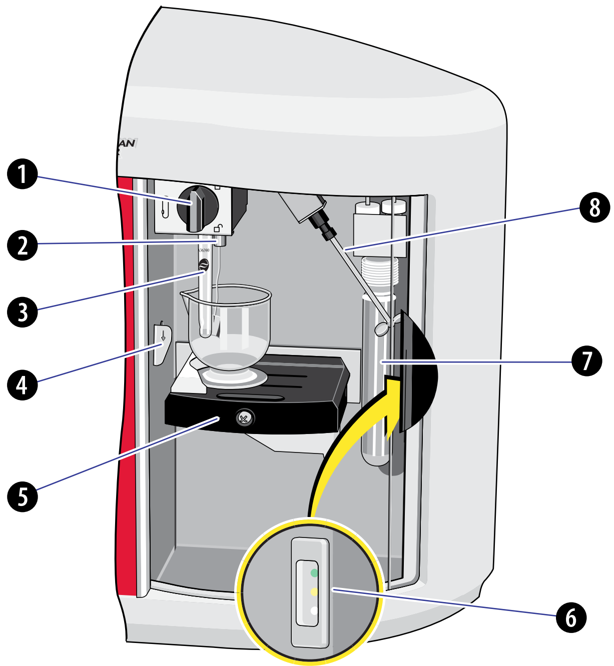
2 The reader will flash a green light when it has successfully read the presented bar code.

3 The bar code information appears in the correct fields in the Multisizer 4e software window.

Sample Compartment

Sample Compartment components are labeled in Figure 2.15.

Figure 2.15 Sample Compartment components

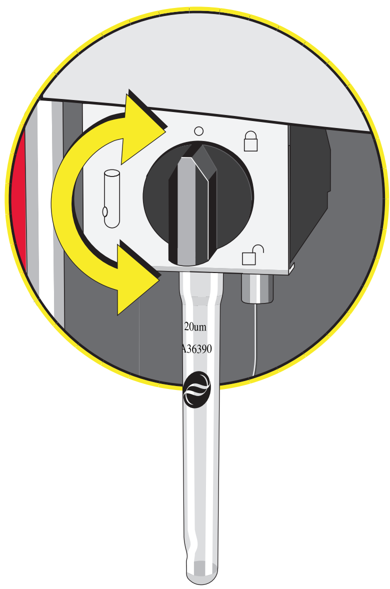


1 Aperture tube knob
2 External Electrode
3 Aperture tube
4 Platform Release
5 Sample Platform
6 LED green, amber, and white status lights (illuminate entire chamber)
7 Particle Trap
8 Stirrer

Aperture Tube Knob

The aperture tube knob (see Figure 2.16) secures the aperture tube in place. The knob is in the locked position when pointing up (12 o'clock position). To remove an aperture tube, first unlock the aperture tube knob by turning the knob in either direction until it points down (6 o'clock position).

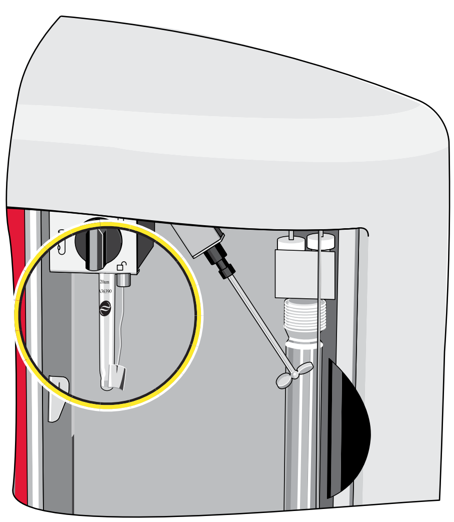
Figure 2.16 Aperture tube knob



Aperture Tube

The aperture tube, when in place, is located in the upper left corner of the Sample Compartment (see Figure 2.17).

Figure 2.17 Aperture tube location



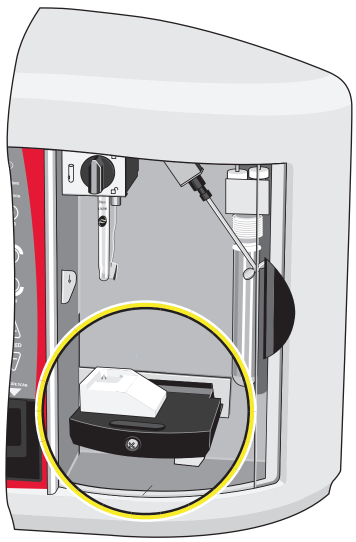
For information on installing or removing an aperture tube, see Installing an Aperture Tube in CHAPTER 4, Installing and Calibrating an Aperture Tube.

For information on aperture tube maintenance, see Aperture Maintenance in CHAPTER 4, Installing and Calibrating an Aperture Tube.

Sample Platform

The sample platform (see Figure 2.18) holds a Multisizer 4e Smart Technology Beaker (ST Beaker) or Accuvette ST. To ensure consistent results, the sample platform locks into position for analysis runs. ST Beakers and Accuvette STs are designed to ensure optimal fluid use and circulation with the locked platform height.

Figure 2.18 Sample platform location



To lower the platform, use the platform release (see Platform Release).

The platform rotates and has two usable sides. One side of the platform has a guide for Multisizer 4e Smart Technology Beakers (see Figure 2.19). The other side of the platform has a built-in guide for Accuvette STs (see Figure 2.20).

The guides ensure that a beaker or Accuvette ST will be in the correct position relative to the aperture tube, external electrode, and stirrer when you raise the platform. The guides ensure consistent results by maintaining an optimal distance for fluid circulation between the aperture tube and the side and base of the beaker or Accuvette ST.

For information on how to turn the platform, see Reversing the Sample Platform.

Figure 2.19 Platform in Loading Position: Beaker Side

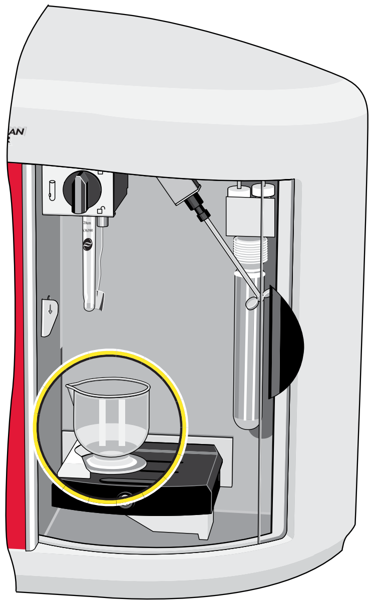
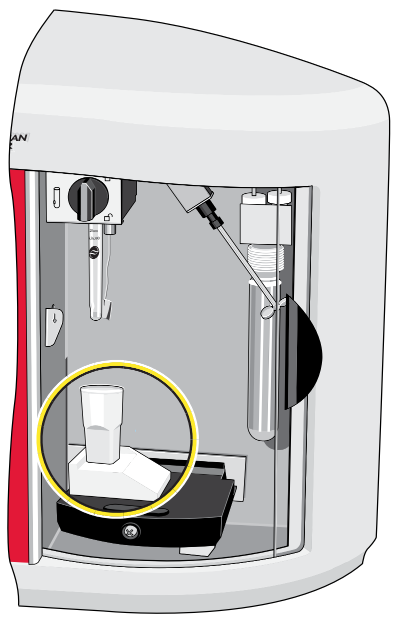


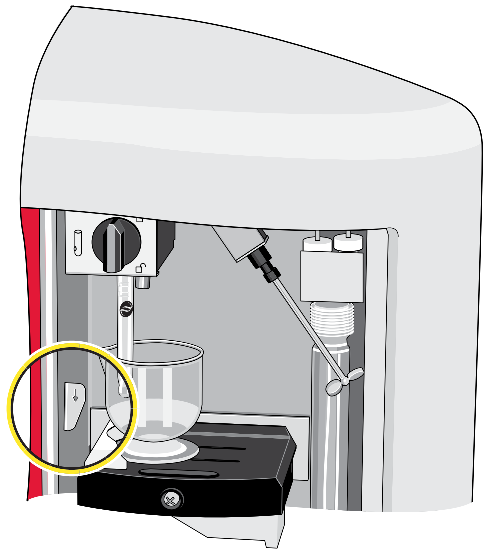
Figure 2.20 Platform in Loading Position: Accuvette ST Side



Platform Release

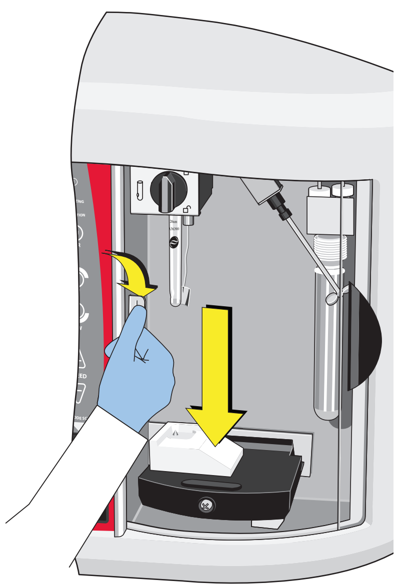
The platform release is located on the left edge of the Sample Compartment, above the sample platform in its lowered position. See Figure 2.21.

Figure 2.21 Platform release location



To lower the platform from its locked analysis position, push the top of the release towards the rear of the compartment. See Figure 2.22.

Figure 2.22 Lowering the platform from its locked analysis position



NOTE It is not necessary to use the platform release when raising the platform to the analysis run position.

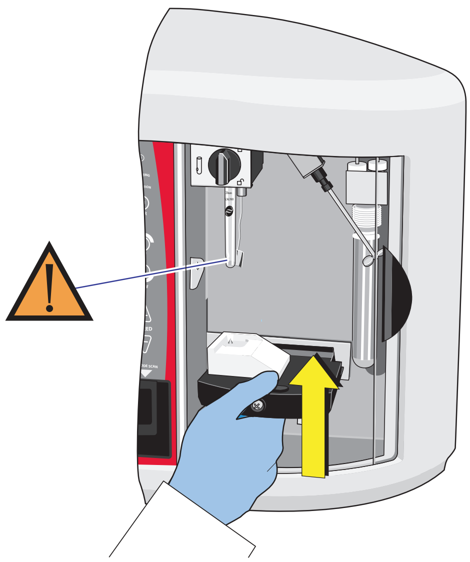
Reversing the Sample Platform

To Turn the Sample Platform from the Beaker to Accuvetter ST Side, or Vice Versa

1 Lower the platform to the loading position, if necessary. See Figure 2.22.

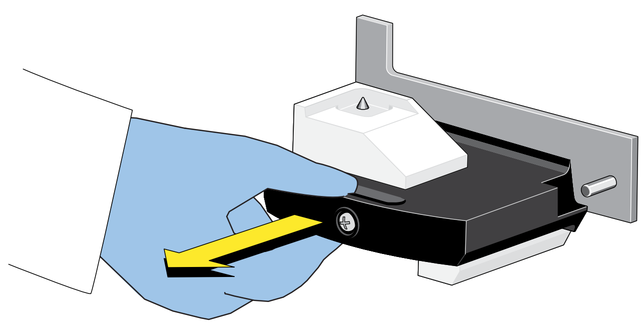
NOTE Rotate the platform only when it is located at or near the loading position. This ensures that the platform does not bump the aperture tube or stirrer while rotating.

2 Grasp the platform and lift the platform slightly.

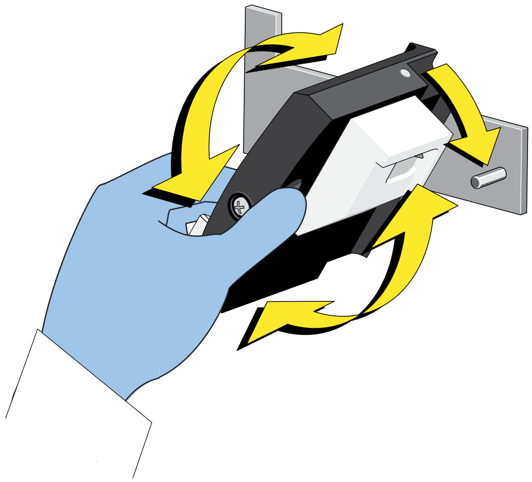


NOTE It is not necessary to use the platform release.

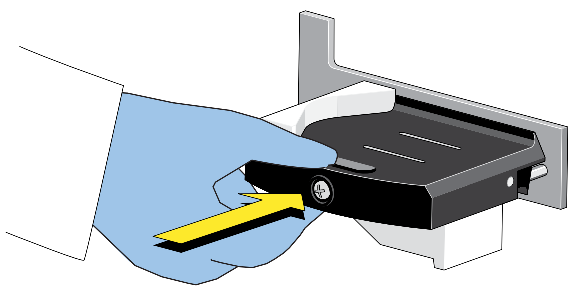
3 Pull the platform toward you. You will feel the resistance of a spring mechanism.



4 Rotate the platform until it contacts the stop pin.



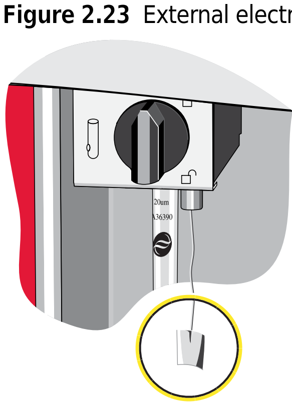
5 Slowly release the platform.



External Electrode

The external electrode is located in the upper left corner of the Sample Compartment, just behind or to the side of the aperture tube. See Figure 2.23.

Figure 2.23 External electrode location

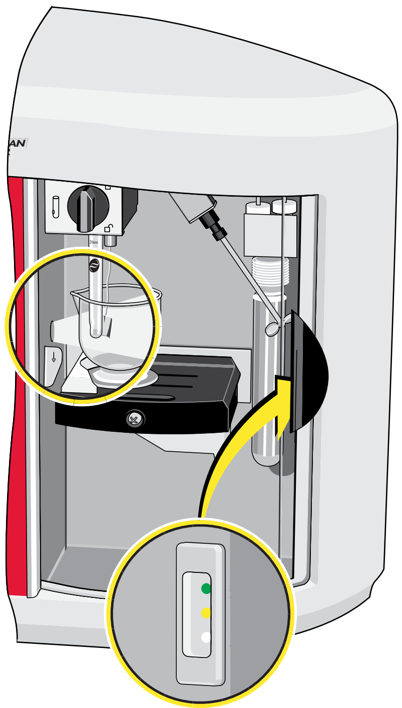


The external electrode creates an electrical current that passes through the aperture orifice. As particles suspended in the electrolyte solution flow through the orifice during an analysis run, the Analyzer determines particle size by measuring the electrical pulse generated by each particle.

Status and Illumination Lights

There are four status and illumination lights located throughout the Sample Compartment.

Figure 2.24 Status and Illumination Lights location



There are three LED status lights located at the right of the Sample Compartment, next to the Particle Trap (see Figure 2.24):

- The green (Analyzing) light is on when the Analyzer is running an analysis.
- The amber (Attention) light flashes when the Analyzer cannot start a run (for example, because the door is open or the sample platform is not in place). When the amber light flashes, the Multisizer 4e software displays an error message.
- The white light illuminates the Sample Compartment, when the door is open. When you close the door, the light turns off after a 10-second delay.

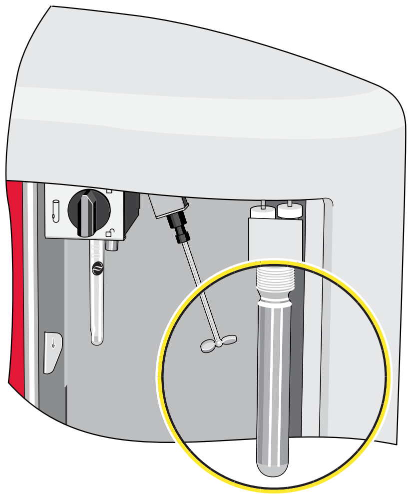
There is one light that illuminates the aperture tube located at the left of the Sample Compartment (see Figure 2.24):

- The light on the left side of the Sample Compartment illuminates the aperture tube. This light is on when the Sample Compartment door is open; when a sample run is in progress; and when the system is filling, flushing, or draining.

Particle Trap

The Particle Trap is located at the far right of the Sample Compartment. See Figure 2.25.

Figure 2.25 Particle Trap location



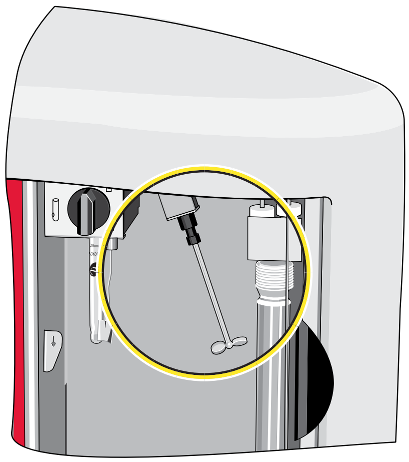
The Particle Trap is designed to keep large particles from entering the system, where they could block the system or abrade valves. Electrolyte and sample fluid from the aperture enter the Particle Trap, where large particles drift to the bottom.

NOTE Beckman Coulter recommends that you clean the Particle Trap when particles have settled and filled one-quarter of the Trap. If you allow particles to collect above the one-quarter mark, large particles may recirculate and damage the system.

Stirrer

The stirrer is located at the top of the Sample Compartment, to the right of the aperture tube and electrode. See Figure 2.26.

Figure 2.26 Stirrer location



NOTE The Multisizer 4e software automatically determines stirrer position for sample runs based on Standard Operating Method (SOM) settings. To change the stirrer position, or park the stirrer outside the beaker, select the appropriate checkboxes in the **Edit the SOM** > **Stirrer** tab. For information on the Edit the SOM (Standard Operating Method) window, see Standard Operating Method: Stirrer Settings in CHAPTER 5, Selecting Analysis Settings: SOM and SOP. It is possible to manually reposition the stirrer if you are not using Multisizer 4e ST Beakers and Accuvette STs.

Accessories

Beakers

Multisizer 4e Smart Technology Beakers (ST Beakers) are available in 100 mL, 200 mL, and 400 mL sizes.

Figure 2.27 Beakers

The unique design of ST Beakers optimizes particle suspension. A spherical interior interrupted by a raised cone creates an unstable area below the stirrer that provides uniform particle distribution.

The base of each ST Beaker is designed to fit a built-in guide on the sample platform (see Sample Platform). When the platform is raised to the analysis position, the guide positions the beaker for optimum fluid circulation, ensuring that the aperture tube is approximately 1/8 inch from the beaker wall. The raised base of the beaker maximizes the volume of fluid the Analyzer is able to extract before drawing air through the aperture orifice.

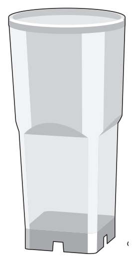
The combination of ST Beakers and the built-in guide allows the Multisizer 4e to deliver consistent, optimal results not possible to obtain with beakers designed by other manufacturers.

NOTE If you are using a beaker produced by another manufacturer, it may be necessary to manually adjust the position of the beaker on the platform. Variations in the position of the beaker relative to the aperture tube, electrode, and stirrer during analysis runs can produce inconsistent results.

Accuvette STs

Beckman Coulter Accuvette STs are disposable and include a secure lid to prevent spills and contamination. Rounded corners in the interior encourage re-suspension of settled particles.

Figure 2.28 Accuvette ST



Use a 20 mL Accuvette ST in combination with an aperture tube 100 µm or smaller.

The base of each Beckman Coulter Accuvette ST is designed to fit a built-in guide on the sample platform (see Sample Platform). When the platform is raised to the analysis position, the guide positions the Accuvette ST for optimum results and maximizes the volume of fluid the Analyzer is able to extract before drawing air through the aperture orifice.

The combination of Accuvette STs and the built-in guide allows the Multisizer 4e to deliver consistent, optimal results not possible to obtain with sample vials designed by other manufacturers.

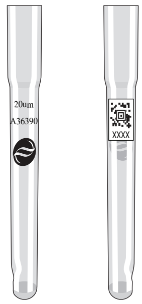
Calibrators

Beckman Coulter provides calibrator beads in sizes ranging from 2 µm to 90 µm. For information on calibrating the aperture tube and selecting the appropriate calibrator size, see Calibrating the Aperture in CHAPTER 4, Installing and Calibrating an Aperture Tube. For more information on scanning Beckman Coulter Control material assay sheets, see Bar Code Reader.

Aperture Tubes

Beckman Coulter manufactures Smart Technology Aperture Tubes designed specifically for the Multisizer 4 or Multisizer 4e Analyzer. Aperture tubes designed for use with the Multisizer 3 will not fit the Multisizer 4 or the Multisizer 4e.

Figure 2.29 Smart Technology Aperture Tubes



Multisizer 4e Smart Technology Aperture Tubes:

- Include size and serial number information in a bar code on the tube for automatic data entry (for more information, see Bar Code Reader).
- Are compatible with ST Beakers and Accuvette STs, and work in concert with the enhanced Multisizer 4e sample platform design (see Sample Platform).
- Are designed for convenience and to ensure consistent and accurate results.

For information on selecting the appropriate aperture tube size for your sample range, see Selecting an Aperture Size in CHAPTER 4, Installing and Calibrating an Aperture Tube.

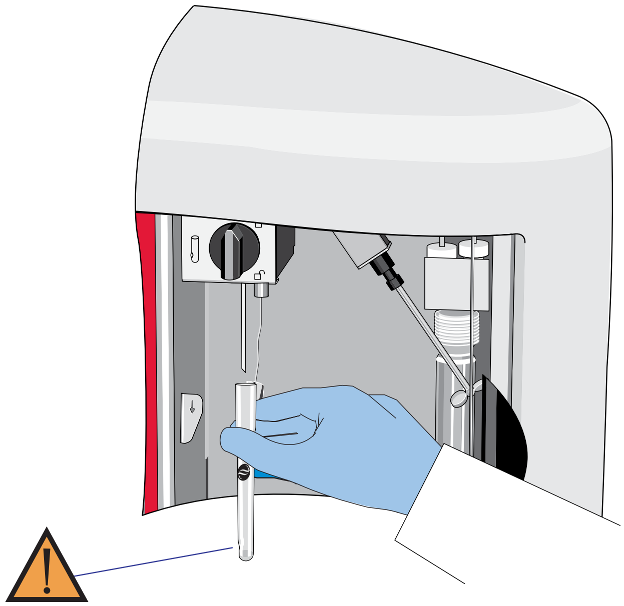
Aperture tubes are fragile and require proper handling. For information on small Aperture Tube handling, cleaning, and operating procedures, see Cleaning the Aperture Tube, Beakers, and Accuvette STs and Using Small Aperture Tubes (10 µm, 20 µm and 30 µm) in CHAPTER 4, Installing and Calibrating an Aperture Tube.

Handling Aperture Tubes

It is highly recommended that you wear powder-free latex or nitrile gloves when handling aperture tubes and other system components and accessories, such as the Electrolyte Jar, ST Beakers, and Accuvette STs.

Hold the aperture tube near the top during installation and removal (see Figure 2.30), and avoid touching the aperture tube at the bottom, near the orifice.

Figure 2.30 Proper Handling of an Aperture Tube



Cleaning the Aperture Tube, Beakers, and Accuvette STs

Before operating the Analyzer, rinse all system components and accessories in pre-filtered ISOTON II (see Pre-Filtering the Electrolyte and Cleaning Fluid).

CAUTION

- **If Beckman Coulter ISOTON II is used, the Multisizer 4e must be drained and flushed through with distilled water before carrying out a sodium hypochlorite solution (bleach) disinfection procedure.**
- **If the Multisizer 4e has been subjected to a bleach disinfection procedure, the instrument must be drained and flushed through with distilled water before refilling with ISOTON II.**

WARNING

Risk of chemical injury from bleach. To avoid contact with bleach, use barrier protection, including protective eyeware, gloves, and suitable laboratory attire. Refer to the Safety Data Sheet for details about chemical exposure before using the chemical.

System components and accessories include:

- Electrolyte Jar
- ST Aperture Tube
- External Electrode
- Particle Trap
- ST Beaker or Accuvette ST

When handling Aperture Tubes and other system components, it is highly recommend that you wear powder-free latex or nitrile gloves.

NOTE Cleaning the Aperture Tube an Electrode may help to lower background counts. See To Clean the External Aperture Tube and Electrode and To Clean the 10 µm and 20 µm Aperture tubes.

NOTE Any unused apertures should be cleaned and dried for long term storage. For short term storage, refer to Storing Small Aperture Tubes in CHAPTER 4, Installing and Calibrating an Aperture Tube.

To Clean the 10 µm and 20 µm Aperture tubes

1 Soak the aperture in hot water for at least 5 minutes in order to wash/dissolve any particle contaminants.

CAUTION

Risk of tube breakage. Avoid contact with hard surfaces as it may result in damage to the aperture.

2 Gently scrub the aperture inside and out using the aperture cleaning brushes included in the cleaning kit, hot water and concentrated glassware soap Micro 90.

3 Rinse thoroughly using filtered DI water or filtered electrolyte to remove soap and possible particle contaminants.

NOTE The following figures are for illustrative purposes only. The aperture can be visually inspected under a microscope to determine if more cleaning is required. See Figure 2.31 for an example of what a completly clogged aperture looks like. See Figure 2.32 for an example of an aperture with a dirty surface that may need extra rinsing. See Figure 2.33 for an example of a clean aperture.

Figure 2.31 Clogged Aperture

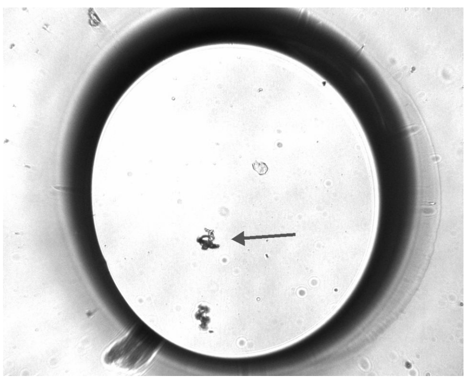


Figure 2.32 Dirty Aperture

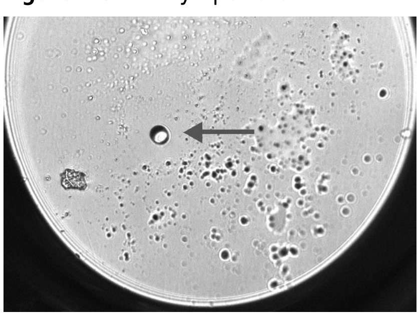
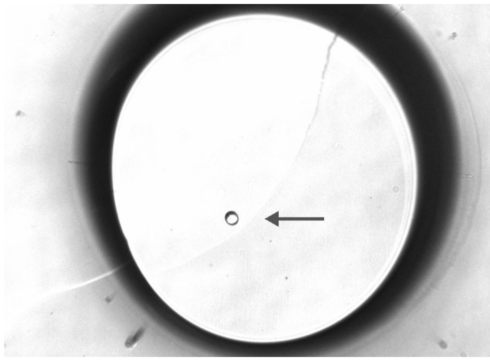


Figure 2.33 Clean Aperture



4 Install the 10 µm or 20 µm aperture in the instrument using the Change Aperture Tube Wizard. See The Change Aperture Tube Wizard in CHAPTER 4, Installing and Calibrating an Aperture Tube.

5 Set the aperture current to +200 µA for the 10 µm aperture or +600 µA for the 20 µm aperture.

6 Set the Preamp Gain to 4 for the 10 µm or 20 µm aperture.

7 Set the Control Mode to Volumetric Mode, 50 µL.

IMPORTANT Accuvettes should be rinsed 2 to 3 times with clean, filtered Isoton II before use with a sample. This is done to eliminate any possible particle contaminants that may result from the manufacturing process.

8 Use a clean Accuvette filled above the 20 mL mark with filtered Isoton II.

9 Rinse the aperture and electrode with fresh filtered Isoton II using a new Accuvette in a spiral motion.

NOTE This can be done 2 or 3 times always using a new Accuvette and fresh filtered Isoton II.

10 Replace the old Accuvette with a new Accuvette using fresh filtered Isoton II.

NOTE The Accuvette should be filled to the regular level (20 mL).

11 Fill the system and perform two consecutive flushes.

NOTE The electrolyte in the electrolyte jar should also be clean and filtered.

12 Run the small aperture maintenance cycle. Refer to Aperture Blockage in CHAPTER 10, Troubleshooting.

IMPORTANT Use a new Accuvette with fresh filtered electrolyte for each background.

13 Run backgrounds when the small aperture maintenance cycle is complete.

NOTE Determine background counts 2 or 3 times until background counts are consistent.

⚠ CAUTION

Risk of damage to the aperture. Sonication is not recommended as it may result in damage to the aperture.

To Clean the External Aperture Tube and Electrode

1 Press the platform release (see Figure 2.21) towards the back of the Sample Compartment and lower the platform to the loading position.

2 Select an ST Beaker that is at least one size larger than the size you used for the previous run (or a 400-mL ST Beaker if you used that size for the previous run).

3 Fill the clean ST Beaker with clean or pre-filtered ISOTON II or de-ionized water (see Pre-Filtering the Electrolyte and Cleaning Fluid). Make sure that the fluid level is higher than the fluid level during the previous run.

4 Lift the ST Beaker to immerse the aperture tube and electrode in the ST Beaker’s filtered liquid. Gently swirl to remove remaining particles.

NOTE An Accuvette ST can be used for cleaning when very low background counts are required. Ensure the level in the Accuvette ST is higher than the fluid level for the previous run sample.

5 Place the ST Beaker on the platform below the aperture tube and allow remaining liquid to drip into the ST Beaker.

NOTE An alternate method may be to use a squeeze bottle filled with clean or pre-filtered ISOTON II to rinse the aperture tube, external electrode, and the stirrer.

Cleaning the Analyzer

Clean the system thoroughly before running analyses. An effective cleaning procedure includes filling and draining the system multiple times and in multiple cycles, alternating between de-ionized water and 98% Isopropyl Alcohol (IPA).

NOTE Cleaning the Analyzer may help to lower background counts.

1. On the Multisizer 4e software Main Menu bar, select **Run** > **Drain System**.

2. Remove and empty the Electrolyte and Waste Jars. See Installing and Removing the Electrolyte and Waste Jars.

3. Replace the Waste Jar. See Installing and Removing the Electrolyte and Waste Jars.

4. Rinse and fill the Electrolyte Jar with clean or pre-filtered de-ionized water.

5. Replace the Electrolyte Jar. See Installing and Removing the Electrolyte and Waste Jars.

6. On the Multisizer 4e software Instrument Toolbar, click **Empty** to empty the Waste Tank.

7. On the Multisizer 4e Main Menu bar:
   - **a.** Select **Run** > **Fill System**, or click the **Fill** button on the Instrument Toolbar.
   - **b.** When the system has filled, select **Run** > **Drain System**.
   - **c.** Fill and drain the system two more times (for a total of three times).

8. Remove and empty the Electrolyte Jar. See Installing and Removing the Electrolyte and Waste Jars.

9. Fill the Electrolyte Jar with 98% Isopropyl Alcohol.

10. Repeat Steps 5 through 8 above.

11. Rinse and fill the Electrolyte Jar with de-ionized water.

12. Repeat Steps 5 through 8 above.

13 Fill the Electrolyte Jar with filtered ISOTON II.

14 Install the aperture tube.

15 On the Multisizer 4e software Instrument Toolbar, click the **Flush** button to flush the aperture tube.

16 Flush the aperture tube three times.

Pre-Filtering the Electrolyte and Cleaning Fluid

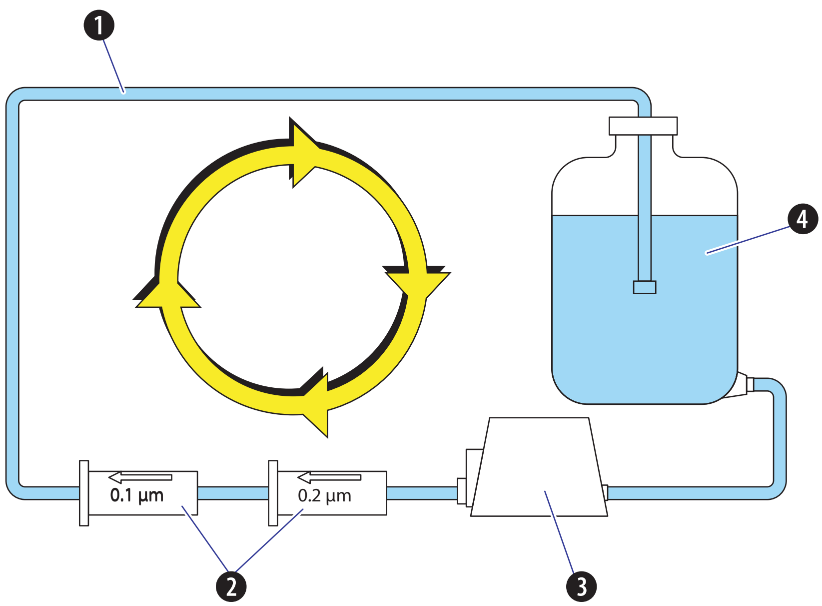
When working with small apertures, it is necessary to pre-filter de-ionized water and/or ISOTON II used to fill, flush, or clean the system and accessories.

Filter the Isoton II or diluent solution with a compatible 0.2 µm filter. A 0.1 µm filter can also be used in series with a 0.2 µm filter for additional filtering. Contact a filtration supplier to determine the appropriate filtration components for your application.

NOTE Different filter materials can have varying degrees of extractables and can shed particles into the electrolyte.

One possible configuration for filtering large amounts of diluent is shown in the diagram below (see Figure 2.34).

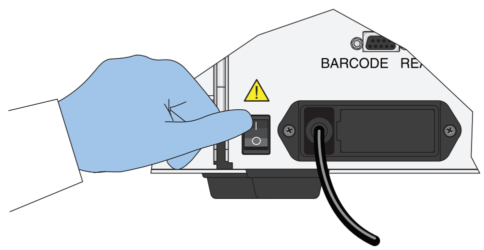
Figure 2.34 Closed Circuit Filtration System



1 Tubing
2 Filters
3 Pump
4 Carboy

Starting the Analyzer

1 Power ON the Analyzer.



2 Empty the Waste Jar, if necessary. See Waste Jar.

3 Fill the Electrolyte Jar with electrolyte, if necessary. See Electrolyte Jar.

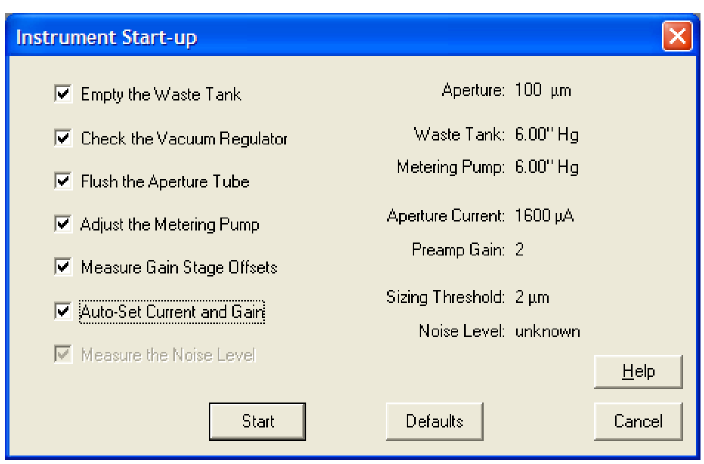
4 Open the Multisizer 4e software and connect the software to the Analyzer.

5 On the Multisizer 4e software Main Menu bar, select **Run** > **Instrument Start-Up**.

6 In the Instrument Start-Up dialog box:

a. Select checkboxes for each function you would like the Analyzer to perform on startup.

b. Click **Start**.



NOTE The Multisizer 4e performs best when it has reached a steady working temperature. This typically takes 15 minutes from powering on the Analyzer.

Preparing the Analyzer for Sample Runs

To Prepare the Analyzer for a Calibration, Control, or Sample Analysis Run

1 Start the Analyzer and connect the software to the Analyzer (see Starting the Analyzer).

2 Wash the ST Aperture Tube and ST Beaker or Accuvette ST (see Cleaning the Aperture Tube, Beakers, and Accuvette STs).

3 Fill an ST Beaker or Accuvette ST with clean electrolyte solution.

NOTE For aperture tubes up to 100 µm, you can use an Accuvette ST (20 mL). Use an ST Beaker for any aperture tube size larger than 100 µm.

4 Add a few drops of the calibrator, control sample, or sample to the electrolyte solution.

- For more information on calibration, see Calibrating the Aperture in CHAPTER 4, Installing and Calibrating an Aperture Tube.
- For more information on control sample runs, see Running a Concentration Control Sample in CHAPTER 4, Installing and Calibrating an Aperture Tube.
- For more information on running a sample analysis, see Analyzing a Sample in CHAPTER 7, Analyzing a Sample.
- For more information on scanning Beckman Coulter Control material assay sheets, see Bar Code Reader.

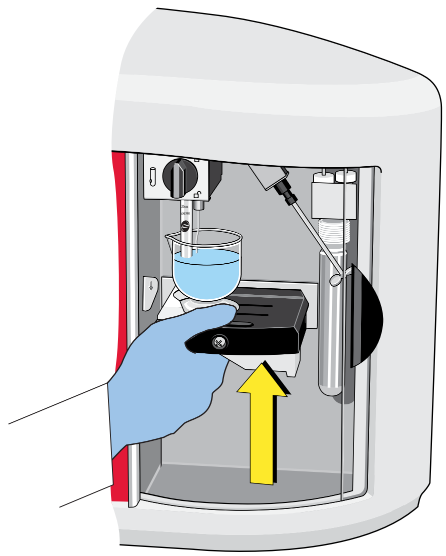
5 If necessary, use the platform release and lower the platform to the loading position. See Platform Release.

6 If necessary, turn the platform so that the correct side (ST Beaker or Accuvette ST) is on top (see Reversing the Sample Platform).

NOTE The sample platform has two usable sides. The ST Beaker side has a built-in guide to ensure that a Multisizer 4e Smart Technology Beaker of any size is in the correct position when you raise the platform. The Accuvette ST side has a smaller guide for Accuvette STs. The guides ensure that the ST Beaker or Accuvette ST will be in the correct position relative to the aperture tube, electrode, and stirrer when you raise the platform.

7 Place the ST Beaker or Accuvette ST on the sample platform.

8 Raise the platform.



NOTE It is not necessary to use the platform release to lift the platform into position.

Software Overview

Opening the Multisizer 4e Software

1 On your desktop, double click the Multisizer 4e software shortcut, or select the program from your program list.

2 The Beckman Coulter Multisizer 4e startup window opens. The window displays the software version, the current date and time, and your security configuration.

Connecting the Software to the Analyzer

Connecting Analyzer and Software on Startup

To connect the software to the analyzer on startup, click **OK** when the Multisizer 4e dialog box appears.

If you do not want to connect on startup, you can select **Run > Connect to Multisizer 4e** in the Main Menu after you have opened the software.

NOTE If the Administrator has selected a security mode that requires a username and password, a Login window will appear when the software opens. If the user does not have a valid username or password, it will be necessary to exit the system. For more information, see Logging In in CHAPTER 9, Configuring Software and Security.

Connecting to the Analyzer Using the Software Main Menu

To connect the software to the Analyzer, select **Run > Connect to Multisizer 4e** on the Main Menu bar.

Disconnecting from the Analyzer Using the Software Main Menu

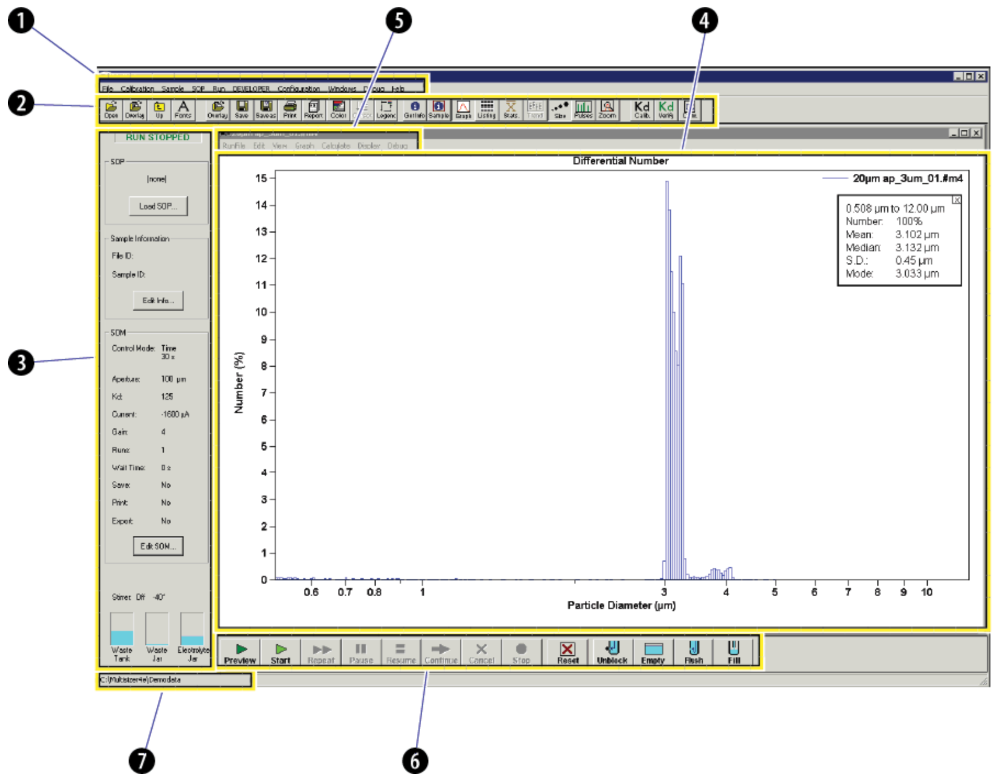
To disconnect the software from the Analyzer, select **Run > Disconnect from Multisizer 4e** on the Main Menu bar.

Multisizer 4e Software Window

The Multisizer 4e software window is illustrated in Figure 3.1.

All Multisizer 4e software functions are available from drop-down menu items in the Main Menu (**1**) and Run Menu (**5**). Many menu item functions are duplicated by shortcut buttons on the Main Toolbar (**2**) or the Instrument Toolbar (**6**).

Figure 3.1 Multisizer 4e software window



1 Main Menu
2 Main Toolbar
3 Status Panel
4 Viewing Area
5 Run Menu
6 Instrument Toolbar
7 Status Bar

Main Menu

The Main Menu bar appears in the upper left corner of the Multisizer 4e software window. For information on Main Menu bar drop-down menus, see Table 3.1.

The Main Menu bar and its drop-down menu items change based on security settings. For example, if you are running the Multisizer 4e software in no security mode, a Configuration drop-down menu appears. If you are running the Multisizer 4e software in security mode with Administrator or Supervisor privileges, an Administrator or Supervisor drop-down menu appears that includes configuration and other security-related items.

Menu items in the Configuration, Administrator, or Supervisor drop-down menu allow the user to remove items from the drop-down menus (see Customizing Main Menus in CHAPTER 9, Configuring Software and Security).

NOTE If you are running the Multisizer 4e software in security mode, certain menu items will appear only to user types with sufficient privileges. The system administrator sets privileges for each user type.

In the table below, Main Menu items that may not be available based on security settings are italicized.

Table 3.1 Main Menu: Drop-Down Menus

The menu...	Allows you to...
File	• Open a file • Open a file for overlay • Use **File Tools** to merge runs, create a size trend, overlay analyses, or to access other file functions (see Working with Multiple Analysis Files in CHAPTER 8, Working with Analysis Files). • Select a recently used file or folder, or create a new folder • Print a report, the open window, or the current screen • Select print settings • Exit the software
Calibration	• *Calibrate the aperture* • Verify aperture calibration • Run a concentration control sample • *Change aperture calibration options* • *Change control sample options*
Sample	• Enter, load and save sample information • Create sample specifications • Load or remove a blank run
SOP	• *Create*, remove, or load a Standard Operating Procedure • Create, edit, or load a Standard Operating Method • Create, edit, or load Preferences • Review or print SOM information • Review or print Preference information

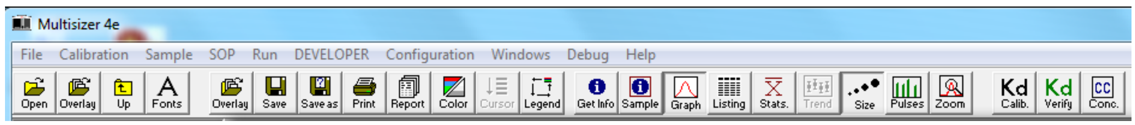
Table 3.1 Main Menu: Drop-Down Menus

The menu...	Allows you to...
Run	• Start, stop, cancel, pause, resume, reset, preview, repeat, or start a 60% to 80% run • Initiate system functions (empty the Waste Tank, fill the system, unblock the aperture tube, etc.) • Turn the stirrer and lamp on and off • Select Analyzer start-up functions • Enter electrolyte information • Use the Change Aperture Tube Wizard • Troubleshoot system functions • Disconnect from the Analyzer Many **Run** menu items duplicate functions available as shortcuts in the Instrument Toolbar.
Configuration	• *Customize menu items* • *Customize sample information terms* • *Customize the default Preferences save setting* • *Set the security level* • *Choose a language* • *Configure file and User's Manual locations* • *Change Multisizer 4e connection parameters* • *Configure Bar Code Reader*
Windows	• Copy or save the current window as a bitmap • Select view options for open files • Minimize or restore all open files to the Viewing Area • Customize the Main Toolbar and Instrument Toolbar • Customize the Status Panel • Select the file that will appear in the Viewing Area (if more than one file is open)
Help	• Open the Multisizer 4e User's Manual • Convert particle sizes (using diameter, surface area, or volume) • View certification information, if available • View log-in information, if running in security mode • Contact Beckman Coulter Field Service • View software version information

Main Toolbar Buttons

The Main Toolbar is located below the Main Menu at the top of the Multisizer 4e software window. The Main Toolbar buttons are shortcuts that duplicate functions available in the Main Menu.

Figure 3.2 Main Toolbar



You can remove buttons from the Main Toolbar, re-order buttons, change button appearance, or remove the Main Toolbar as a whole by selecting **Windows > Toolbars** on the Main Menu bar (see Customizing Toolbars in CHAPTER 9, Configuring Software and Security).

Default Main Toolbar Configuration

In the default Main Toolbar configuration, buttons are organized in four groups:

- The first group duplicates features available in the File drop-down menu and **SOP > Edit Preferences** window.
- The second group duplicates features available in the File drop-down menu and **SOP > Edit Preferences (Graph Options)** window.
- The third group duplicates features available in the Main Menu and Run Menu.
- The fourth group duplicates features available in the Calibration drop-down menu.

Main Toolbar Buttons: First Group

The first group, at the far left (see Figure 3.3), duplicates features available in the File drop-down menu and the **SOP > Edit Preferences** window.

Figure 3.3 Main Toolbar Buttons: First Group

Table 3.2 Main Toolbar: Buttons (First Group)

Click...	To...
Open	Open a file. To change the folder that automatically opens when you click this button, go to **File** > **Change Folder** in the Main Menu, or use the **Up** button. The current folder is displayed in the Status Bar in the bottom left corner of the Multisizer 4e software window.
Overlay	Open files for overlay.

Table 3.2 Main Toolbar: Buttons (First Group)

Click...	**To...**
Up	Change the current folder for files to the folder one level above. The current folder is displayed in the Status Bar in the bottom left corner of the Multisizer 4e software window.
Fonts	Open the Fonts & Colors tab of the Edit Preferences window. This button allows you to change fonts for the file currently open in the Viewing Area, and save these settings for future use.

Main Toolbar Buttons: Second Group

The second group (see Figure 3.4) duplicates features available in the File drop-down menu and the **SOP > Edit Preferences (Graph Options)** window.

Figure 3.4 Main Toolbar Buttons: Second Group

NOTE This group of buttons is active only when one or more analysis files are open in the Viewing Area.

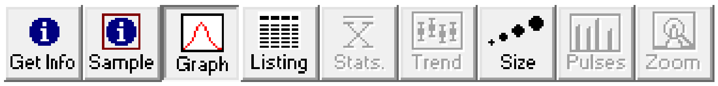
Table 3.3 Main Toolbar Buttons (Second Group)

Click...	**To...**
Overlay	Open files for overlay.
Save	Save the open file. If the file has not been saved previously, the Save As window opens.
Save As	Save the open file with a new name.
Print	Print the open file. The Multisizer 4e software will print the report using current Preferences settings.
Report	Open the Print Report window. The Print Report window allows you to change print settings or update Preferences before printing.
Color	Change the appearance of the analysis file in the Viewing Area. Toggle to display graph lines in color or black and white. This button is only available when viewing a graph.
Cursor	Bring a different graph line to the forefront of an overlay file. This button is a shortcut for the Cursor menu on the Run Menu bar. The button is only available if you have opened multiple files for an average, overlay, or trend. For more information, see Moving Between Multiple Files (Cursor Menu and Shortcut) in CHAPTER 8, Working with Analysis Files.
Legend	Move the legend from corner to corner in the open analysis, or remove the legend. This button is only available when viewing a graph.

Main Toolbar Buttons: Third Group

The third Main Toolbar group (see Figure 3.5) duplicates functions available in the Main Menu (**Sample** drop-down menu items) and the Run Menu (**RunFile** and **View** drop-down menu items). The Run Menu appears at the top of an open analysis file.

Figure 3.5 Main Toolbar Buttons: Third Group



NOTE This group of buttons is active only when an analysis file is open in the Viewing Area.

NOTE If you are running the Multisizer 4e software in security mode, certain Main Toolbar buttons will appear only to user types with sufficient privileges (for example, the **Sample** button). The system administrator sets privileges for each user type.

In Table 3.4, Main Toobar buttons that may not be available based on security settings are italicized.

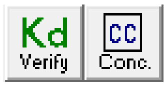
Table 3.4 Main Toolbar: Buttons (Third Group)

Click...	To...
Get Info	Display sample and run information associated with the open analysis file.
Sample	*Open the Edit Sample Info window to update sample information for the open analysis file. The Edit Sample Info window is available only in no security mode and to Administrator and Supervisor user types.*
Graph	Display the open analysis file as a graph.
Listing	Display the open analysis file as a listing.
Stats	Display statistics associated with the open analysis file.
Trend	View a size trend graph or listing. The **Trend** button is greyed out when the open file does not have trend data. Only trend and average files, which end in .#tr or .#av, have trend data.
Size	Display the particle size distribution as a graph or listing.
Pulses	Display electrical pulses as a graph or listing.
Zoom	Display only the area of the graph between two cursors. Before clicking the **Zoom** button, first insert two cursors on a graph as follows: a single click within the graph area creates a single cursor. Click, drag, and un-click to create two cursors: one at the click point, the other at the un-click point. After displaying a zoomed image, you can click the **Zoom** button again to view the full size distribution, or insert new cursors and click the **Zoom** button to see a smaller portion of the size distribution.

Main Toolbar Buttons: Fourth Group

The fourth group (see Figure 3.6) duplicates functions available in the Calibration drop-down menu.

Figure 3.6 Main Toolbar Buttons: Fourth Group



NOTE If you are running the Multisizer 4e software in security mode, certain Main Toolbar buttons will appear only to user types with sufficient privileges (for example, the **Kd Calib.** button). The system administrator sets privileges for each user type.

In the table below, Main Toobar buttons that may not be available based on security settings are italicized.

Table 3.5 Main Toolbar: Buttons (Fourth Group)

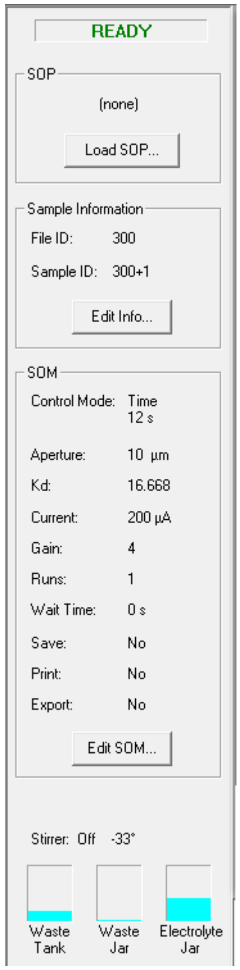
Click...	To...
Kd Calib.	*Open the Calibrator Size window and calibrate the aperture.*
Kd Verify	Open the Calibrator Size window and verify aperture calibration.
CC Conc.	Open the Concentration Control window and run a concentration control sample.

Status Panel

The Status Panel, located on the left side of the Multisizer 4e software window (see Figure 3.7), displays elapsed run time, particle count, and concentration information during an analysis run.

Before an analysis run begins, the Status Panel displays READY or RUN COMPLETE if the Analyzer is not busy.

Figure 3.7 Status Panel



The Status Panel displays three panes with basic information: the name of the SOP (if loaded), Sample Information (if entered), and SOM information (if entered).

- In the SOP pane, click **Load SOP** to load a Standard Operating Procedure.
- In the Sample Information pane, click **Edit Info** to enter or update sample information.
- In the SOM pane, click **Edit SOM** to make changes to the Standard Operating Method.

Below the SOP, Sample Information, and SOM panes, the Status Panel displays the status of the stirrer and fluid levels in the Waste Tank, Waste Jar, and Electrolyte Jar.

Customizing the Status Panel

You can customize colors and other aspects of the Status Panel display. For more information, see Customizing the Status Panel in CHAPTER 9, Configuring Software and Security.

Viewing Area

The Viewing Area, located in the middle of the Multisizer 4e software window, displays analysis files as graphs, listings, or interpolations.

During an analysis run, the Viewing Area displays a graph or listing showing the particle size distribution in progress.

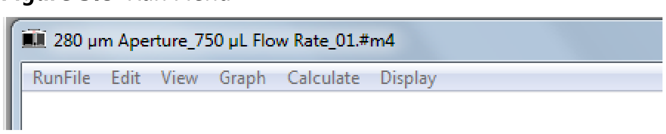
You can work with multiple analysis files. To change how the Viewing Area displays multiple files, open the Windows drop-down menu in the Main Menu bar and select one of the following options:

- **Cascade**
- **Tile**
- **Tile - Horizontal**
- **Tile - Vertical**
- **Minimize All**
- **Restore All**

Run Menu

The Run Menu bar is located in the Viewing Area, at the top of each open analysis file. See Figure 3.8.

Figure 3.8 Run Menu



Items displayed on the Run Menu bar vary based on selections made in the View drop-down menu. For more information on Run Menu behaviors, see Viewing Analysis Files in CHAPTER 8, Working with Analysis Files.

Instrument Toolbar Buttons

The Instrument Toolbar is located at the bottom of the Multisizer 4e software window, below the Viewing Area.

Figure 3.9 Instrument Toolbar

Instrument Toolbar buttons duplicate functions also available in the Run drop-down menu. The buttons are divided into three sections:

- **Run** buttons
- **Reset** button
- **Fluid control** buttons

Instrument Toolbar: Run Buttons

The Instrument Toolbar Run buttons allow you to preview, start, repeat, stop, pause, resume, continue, or cancel an analysis run.

Figure 3.10 Instrument Toolbar: Run Buttons

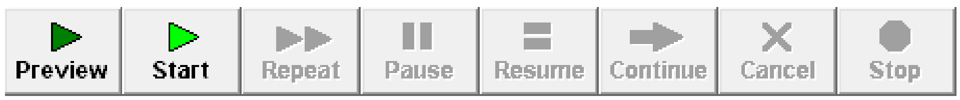


Table 3.6 Instrument Toolbar: Run Buttons

Click...	To...
Preview	Preview an analysis run to check concentration. If you adjust the concentration, the Analyzer automatically continues the preview run when you shut the Sample Compartment door.
Start	Start an analysis run.
Repeat	Repeat an analysis run with the same settings.
Pause	Pause an analysis run.
Resume	Resume an analysis run after pausing. If you pause a 30-second timed analysis after 10 seconds, the **Resume** button will run only the 20-second remainder of the analysis time. If you pause a run and then press **Continue**, the software will re-run the 30-second analysis with your original settings, and add the data to the data already gathered during the abbreviated 10-second analysis.
Continue	Continue a run from the beginning, adding the data to data previously gathered from a run with the same settings.
Cancel	Cancel an analysis run. If you click the **Cancel** button, the software will cancel the run and display Run Canceled above the analysis in the Viewing Area. You will not be able to save data from the analysis.
Stop	Stop an analysis run. If you click the **Stop** button, the software will stop the analysis currently in progress and display Run Complete above the analysis data in the Viewing Area. You can save the analysis file.

Instrument Toolbar: Reset Button

Click 

to clear the Viewing Area of analysis data after running a sample.

If you have not canceled the run, the software will prompt you to save the analysis data before clearing the Viewing Area.

Instrument Toolbar: Fluid Control Buttons

The Instrument Toolbar fluid control shortcuts allow you to empty the Waste Tank into the Waste Jar and use electrolyte solution to unblock the aperture, flush the system of particles, or fill the system.

Figure 3.11 Fluid Control Buttons

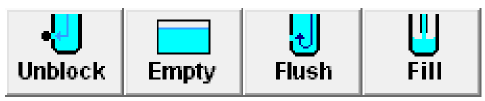


Table 3.7 Instrument Toolbar: Function Buttons

Click...	To...
Unblock	Reverse the flow of electrolyte solution through the aperture orifice to unblock the aperture tube.
Empty	Empty the Waste Tank fluid into the Waste Jar.
Flush	Flush the system with electrolyte solution to remove particles.
Fill	Fill the system with electrolyte solution.

Status Bar

The Status Bar is located in the lower left corner of the Multisizer 4e software window. See Figure 3.12.

Figure 3.12 Status Bar

The Status Bar text displays the current path and folder (if you are not mousing over a drop-down menu item) or a brief description of a highlighted menu item.

CHAPTER 4

Installing and Calibrating an Aperture Tube

Overview

When to Install a New Aperture Tube

Install a new aperture tube if:

- You are using the Analyzer for the first time
- You are using different aperture tube sizes to obtain different particle size distributions
- Your aperture tube has broken

To Install a New Aperture Tube

1. Select an aperture tube size.

2. Use the Change Aperture Tube Wizard to install the new aperture tube.

Selecting an Aperture Size

Select an aperture tube size based on the size of the particles you will be measuring. The Analyzer can accurately measure particles between 2% and 80% of the aperture size. See Table 4.1.

CAUTION

If you select a 10 µm, 20 µm or 30 µm aperture, Beckman Coulter recommends that you follow special cleaning, handling, and operating procedures to accommodate the sensitivity of small apertures. For more information, see Using Small Aperture Tubes (10 µm, 20 µm and 30 µm).

Table 4.1 Aperture Sizes

Aperture (Nominal Diameter, µm)	Diameter Range (µm)			Volume Range μm^3 or fL		
	Total	Standard	Extended	Total	Standard	Extended
10[a]	0.2–8	0.2–6	6–8	0.004–268	0.004–113	113–268
20[a]	0.4–16	0.4–12	12–16	$0.034–2.14\times10^3$	0.034–905	$905–2.14\times10^3$
30[a]	0.6–24	0.6–18	18–24	$0.113–7.24\times10^3$	$0.113–3.05\times10^3$	$3.05\times10^3–7.24\times10^3$
50	1.0–40	1.0–30	30–40	$0.524–33.5\times10^3$	$0.524–14.1\times10^3$	$14.1\times10^3–33.5\times10^3$
70	1.4–56	1.4–42	42–56	$1.44–92.0\times10^3$	$1.44–38.8\times10^3$	$38.8\times10^3–92.0\times10^3$
100	2.0–80	2.0–60	60–80	$4.19–268\times10^3$	$4.19–113\times10^3$	$113\times10^3–268\times10^3$
140	2.8–112	2.8–84	84–112	$11.5–736\times10^3$	$11.5–310\times10^3$	$310\times10^3–736\times10^3$
200	4.0–160	4.0–120	120–160	$33.5–2.14\times10^6$	$33.5–905\times10^3$	$905\times10^3–2.14\times10^6$
280	5.6–224	5.6–168	168–224	$92.0–5.88\times10^6$	$92.0–2.48\times10^6$	$2.48\times10^3–5.88\times10^6$
400	8.0–320	8.0–240	240–320	$268–17.2\times10^6$	$268–7.24\times10^6$	$7.24\times10^6–17.2\times10^6$
560	11.2–448	11.2–336	336–448	$736–47.1\times10^6$	$736–19.9\times10^6$	$19.9\times10^6–47.1\times10^6$
800[b]	16–640	16–480	480–640	$2145–137\times10^6$	$2145–57.9\times10^6$	$57.9\times10^6–137\times10^6$
1000[b]	20–800	20–600	600–800	$4189–268\times10^6$	$4189–113\times10^6$	$113.1\times10^6–268\times10^6$
2000[b]	40–1600	40–1200	1200–1600	$33.5\times10^3–2.14\times10^9$	$33.5\times10^3–905\times10^6$	$905\times10^6–2.14\times10^9$

a. Range depends on system cleanliness and environmental electrical noise.

b. Range depends on sample density.

Installing an Aperture Tube

This section describes how to correctly place an aperture tube in the Analyzer. After installing the aperture tube, you will need to:

- Calibrate the aperture
- Follow the step-by-step guide provided in the Change Aperture Tube Wizard. The Change Aperture Tube Wizard allows you to verify calibration, test aperture settings, and enter the new aperture tube size and serial number in the Multisizer 4e software.

To Install an Aperture Tube

1 Clean the new ST Aperture Tube. For instructions on the proper aperture cleaning procedure, see Cleaning the Aperture Tube, Beakers, and Accuvette STs in CHAPTER 2, Analyzer Overview.

CAUTION

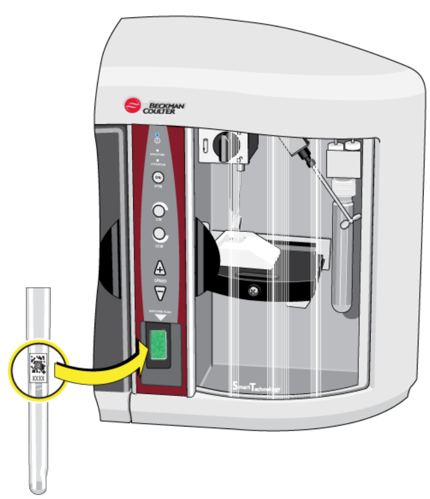
Small apertures (10 µm, 20 µm and 30 µm) are fragile. Beckman Coulter recommends that you handle them with great care. For more information on small aperture handling, cleaning, and operating procedures, see Cleaning the Aperture Tube, Beakers, and Accuvette STs in Analyzer Overview and Using Small Aperture Tubes (10 µm, 20 µm and 30 µm).

2 If you have not already done so, power on the Analyzer, open the Multisizer 4e software, and connect the software to the Analyzer. See Connecting the Software to the Analyzer in CHAPTER 3, Software Overview.

3 In the Multisizer 4e software, select **Run > Change Aperture Tube Wizard** on the Main Menu bar.

4 In the Change Aperture Tube Wizard, click **Aperture Tube**. The Select Aperture Tube window opens. For more information, see Step 1: Select the new aperture tube.

5 Hold the ST Aperture Tube bar code in front of the Bar Code Reader on the Control Panel. See Bar Code Reader in CHAPTER 2, Analyzer Overview for detailed instructions on using the bar code reader.

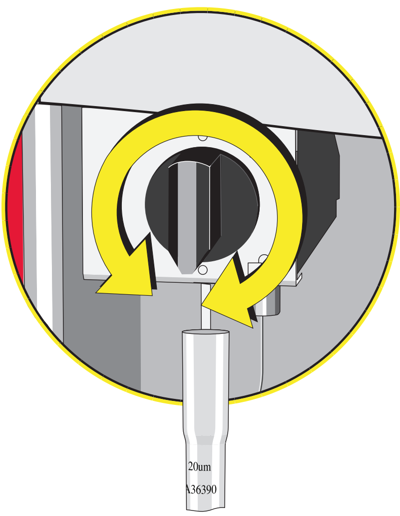


- If an operator has already calibrated the aperture and entered ST Aperture Tube information in the software, the software highlights the aperture tube size, serial number, and Kd. Click **OK**.
- If the ST Aperture Tube has not been entered into the system, the software displays the aperture tube size and serial number. To enter this information, use the Add New Aperture Tube window.

6 Open the Sample Compartment door. See Sample Compartment in CHAPTER 2, Analyzer Overview.

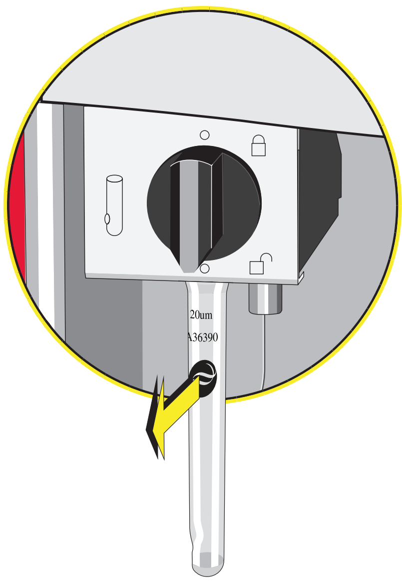
7 Lower the platform to the loading position using the platform release (see Platform Release in CHAPTER 2, Analyzer Overview).

8 Turn the aperture tube knob in either direction until it points down. The aperture knob is locked only when the knob points up (12 o'clock position).



9 If necessary, remove the currently installed ST Aperture Tube, and click **Next**.

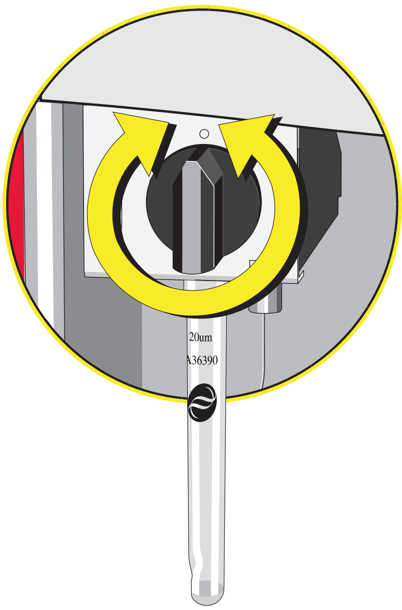
10 Insert the new ST Aperture Tube in the connection head with the logo facing forward. Insert the tube as far as possible until you feel the edge of the aperture tube pass by the O-ring and snap into the set position at the top of the O-ring.



CAUTION

Risk of possible sample contamination, sample leakage, and/or increased noise levels. The Multisizer 4e aperture tube connection forms a greaseless seal with O-rings. Do not grease the aperture tube.

11 Turn the aperture tube knob until it points up in the vertical position. Turning the knob until it is precisely vertical (in the 12 o'clock position) locks the aperture tube in place.

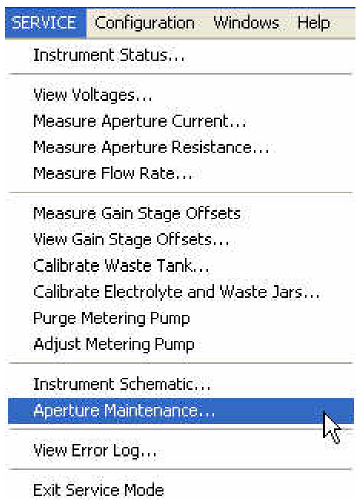


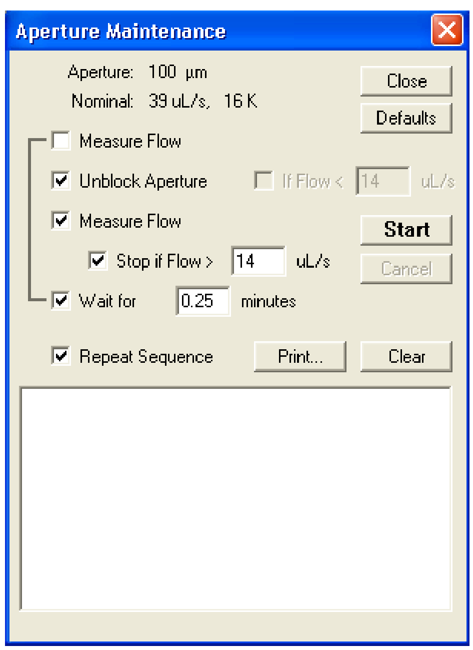
12 Return to the Change Aperture Tube Wizard and proceed through the remaining steps.

Aperture Maintenance

In order to perform aperture maintenance, you must be in Service Mode. See Opening Service Mode in CHAPTER 10, Troubleshooting.

1 Once in Service Mode, select **SERVICE > Aperture Maintenance** from the main menu bar. The Aperture Maintenance window will appear.





2 Select **Start** to begin the maintenance cycle.

NOTE You can control the sequence by selecting the checkboxes of each task desired. If you make any changes, the system will remember these changes until the instrument is shutdown. If you would like to reset to the default cycle, select the **Defaults** button.

NOTE Aperture Maintenance performs automatic, unattended unblocking of the aperture tube. A series of unblocking and subsequent measurement of aperture tube flow is performed. The series stops when aperture tube flow reaches the minimum for the aperture tube installed.

NOTE If the **Stop if Flow >** is deslected and the **Repeat Sequence** is selected, the sequence will continue to cycle. You may cancel the cycle at any point by selecting **Cancel**.

NOTE Aperture maintenance can be performed on all aperture tube sizes. With tube sizes greater than 200 µm, aperture maintenance can still be performed, as long as the measure flow feature is not selected in the cycle.

Using Small Aperture Tubes (10 µm, 20 µm and 30 µm)

Small apertures (10 µm, 20 µm and 30 µm) are fragile and require special handling, cleaning, and operating procedures. See Handling Aperture Tubes in CHAPTER 2, Analyzer Overview and Cleaning the Aperture Tube, Beakers, and Accuvette STs in CHAPTER 2, Analyzer Overview.

Because small apertures measure extremely small particles, it is necessary to take steps to safeguard the aperture tube and reduce electrical noise.

- Handle the aperture tube properly
- Remove environmental factors that can contribute to electrical noise
- Pre-filter cleaning fluids and electrolyte
- Clean the system thoroughly prior to using a small aperture tube
- Clean ST Beakers, Accuvette STs, jars, ST Aperture Tubes, and other accessories
- Run a noise profile to verify a low background count
- Use appropriate current and gain settings
- Store the aperture tube properly
- Adhere to nano-particle calibration standards, if applicable

Unblocking Small Aperture Tubes

The orifices of 10 µm, 20 µm and 30 µm aperture tubes block easily. To unblock the aperture, use the Unblock button in the Instrument Toolbar, or use Standard Operating Method settings to automatically unblock the aperture (see Standard Operating Method: Blockage Detection in CHAPTER 5, Selecting Analysis Settings: SOM and SOP.

NOTE The Aperture Maintenance function is the preferred method to be used to automatically unblock a small tube or an aperture tube with marginal flow issues. For more information of the Aperture Maintenance function, see Aperture Blockage in CHAPTER 10, Troubleshooting.

⚠ CAUTION

Risk of damage to the aperture tube. Never use an object or your finger to unblock the aperture.

Reducing Environmental Factors That Create Noise

When measuring extremely small particles with a 10 µm, 20 µm or 30 µm aperture tube, take steps to ensure that the background count produced by electrical noise remains as low as possible.

Follow the instructions below to reduce environmental causes of electrical noise. For further information on possible sources of electrical noise, see Reducing Noise in CHAPTER 10, Troubleshooting.

Create a dust-free lab environment

It will be almost impossible to obtain accurate particle counts if the instrument is located in a dust-laden atmosphere. Minimize Analyzer exposure to dust, debris, and airborne contaminants.

Reduce acoustic noise levels

Small apertures are extremely sensitive to ambient noise (banging, fan motors, even loud voices). Power off other machines and equipment in the vicinity, if possible.

Reduce vibration

Do not place the Analyzer on a table top that transmits vibrations from another machine.

Block electrical interference

Use an AC line conditioner on the Analyzer power cord to block electrical interference as required by local power conditions.

Cleaning the Analyzer Prior to Using a Small Aperture

Before running analyses with a small aperture tube, clean the system. An effective cleaning procedure includes filling and draining the system multiple times and in multiple cycles, alternating between pre-filtered, de-ionized water and 98% isopropyl alcohol. For step-by-step instructions, see Cleaning the Analyzer in CHAPTER 2, Analyzer Overview. For step-by-step instructions on cleaning system components and accessories, see Cleaning the Aperture Tube, Beakers, and Accuvette STs in CHAPTER 2, Analyzer Overview.

NOTE When working with small apertures, it is necessary to pre-filter de-ionized water and/or ISOTON II used to fill, flush, or clean the system and accessories. See Pre-Filtering the Electrolyte and Cleaning Fluid in CHAPTER 2, Analyzer Overview.

Small Aperture Noise Profiles

All Multisizer instruments will exhibit some noise counts at the smallest size on the particle diameter size scale for an individual aperture tube and electrolyte. These noise or debris counts can mask data for sizes of interest.

Care should be exercised to keep instrument elements clean to minimize any addition contribution of contamination to analysis run data.

Instrument elements that should be kept clean and free of contamination include the:

- Electroyte jar
- Electrolyte pickup tube
- Aperture tube
- External electrode
- Stirrer (if used)

Refer to Cleaning the Aperture Tube, Beakers, and Accuvette STs in CHAPTER 2, Analyzer Overview and Cleaning the Analyzer in CHAPTER 2, Analyzer Overview for step-by-step instructions on cleaning the required elements.

The images below illustrate a typical noise profile obtained using a clean system, and a noise profile obtained using a system that requires additional cleaning for both the 10 µm and 20 µm aperture tubes.

NOTE The 10 µm aperture tube is more sensitive to contamination. This is because the flow rates and volumes are smaller in the 10 µm aperture tube than the flow rates and volumes in aperture tubes 20 µm and larger.

The noise profiles obtained for both the 10 µm and 20 µm aperture tubes using a clean Analyzer system shows a background count primarily in the regions described in Table 4.2.

Table 4.2 Clean Analyzer background count regions

Aperture size (µm)	Region of background count (µm)
10 µm	0.2 µm and 0.3 µm (see Figure 4.1)
20 µm	0.4 µm and 0.5 µm (see Figure 4.4)

NOTE Some background noise remains when using small apertures.

The noise profile obtained using an Analyzer system that has not been properly cleaned (see Figure 4.2 and Figure 4.5) and shows a high background count due to system contamination . See

Cleaning the Aperture Tube, Beakers, and Accuvette STs and Cleaning the Analyzer in CHAPTER 2, Analyzer Overview for step-by-step instructions on cleaning the system, aperture tubes, beakers, and accuvette STs.

Figure 4.1, Figure 4.2, and Figure 4.3 were obtained using a 10 µm ST Aperture Tube and filtered ISOTON II. Runs were completed in volumetric mode (10 µL for the 10 µm aperture tube) with current set to 200 µA and gain set to 4. For information on current and gain settings appropriate for small apertures, see Step 8: Set Current and Gain.

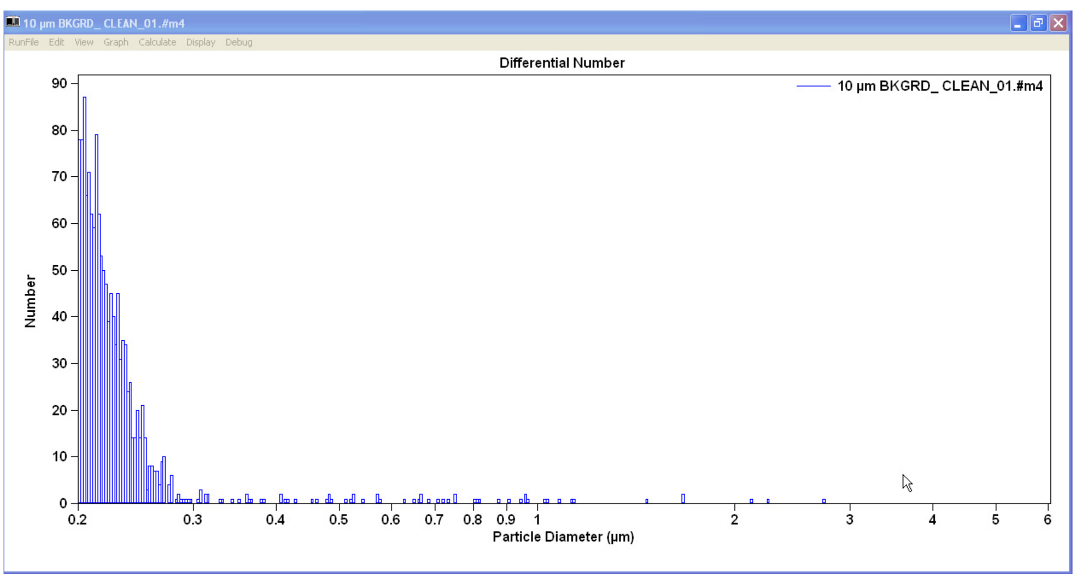
Figure 4.4 and Figure 4.5 were obtained using a 20 µm ST Aperture Tube and filtered ISOTON II. Runs were completed in volumetric mode (50 µL for the 20 µm aperture tube) with current set to 600 µA and gain set to 4. For information on current and gain settings appropriate for small apertures, see Step 8: Set Current and Gain.

If you are unable to obtain results similar to those in Figure 4.1 for the 10 µm aperture tube or Figure 4.4 for aperture tubes larger than 20 µm after thoroughly cleaning the system (see Cleaning the Analyzer in CHAPTER 2, Analyzer Overview), your aperture tube may be damaged or blocked.

NOTE Figure 4.1 is an example of a clean noise profile for the 10 µm aperture tube. Figure 4.4 is an example of a clean noise profile for the 20 µm aperture tube.Factors such as aperture size, current and gain settings, and electrolyte can affect results. The noise profile graphs below show the relative difference between a system that is performing well (see Figure 4.1 and Figure 4.4) and one that may need cleaning (see Figure 4.2 and Figure 4.5).

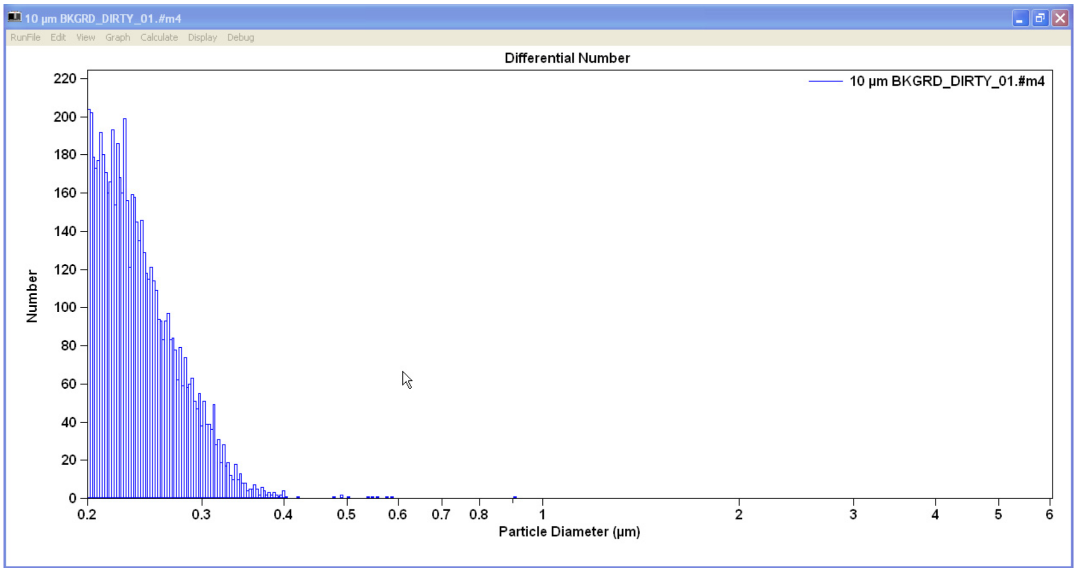
IMPORTANT These distributions are for illustrative purposes only. Actual values may change due to the instrument, electrolyte used, and the external environment.

Figure 4.1 Noise Profile, Clean System (10 µm aperture)



NOTE The noise level or debris starts at a much lower particle diameter size scale. The amplitude of the noise is also reduced. A further indication of acceptable level of cleaning is the ability to observe individual particle counts along the particle diameter size scale.

Figure 4.2 Noise Profile, Dirty System (10 µm aperture)



NOTE The noise level or debris starts at a larger particle diameter size scale. This can mask sample data in an analysis run. The amplitude also gives an indication of the magnitude of contamination that is required to be removed.

Figure 4.3 Noise Profile, Clean/Dirty Overlay (10 µm aperture)

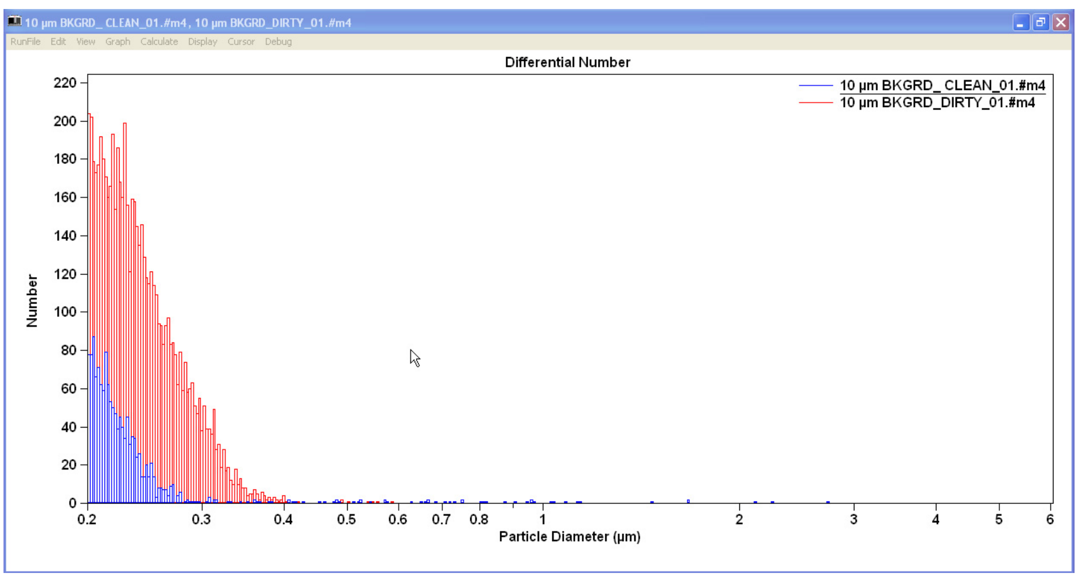


Figure 4.4 Noise Profile, Clean System (20 µm aperture)

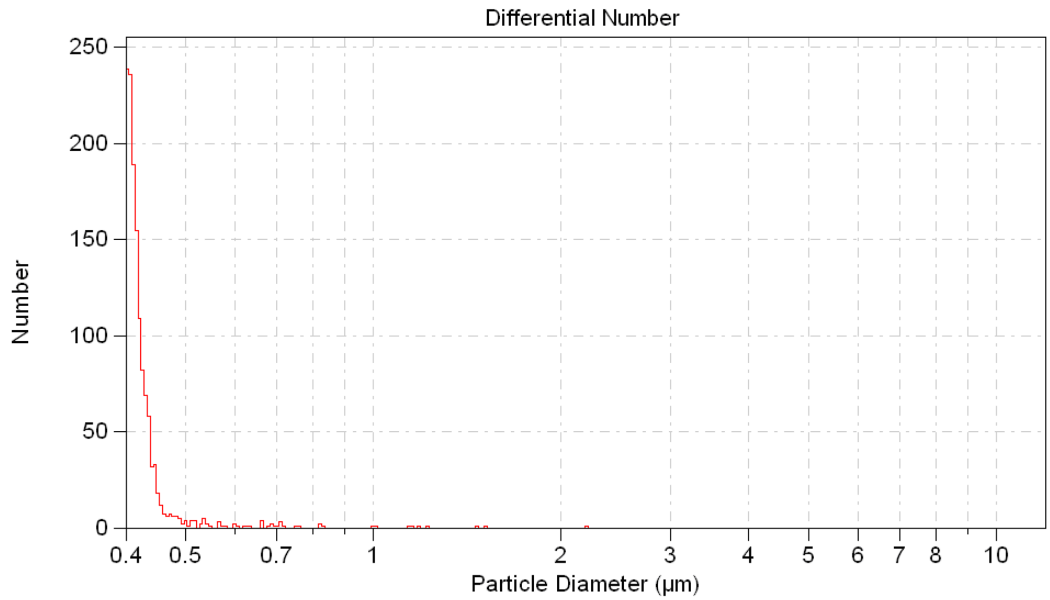
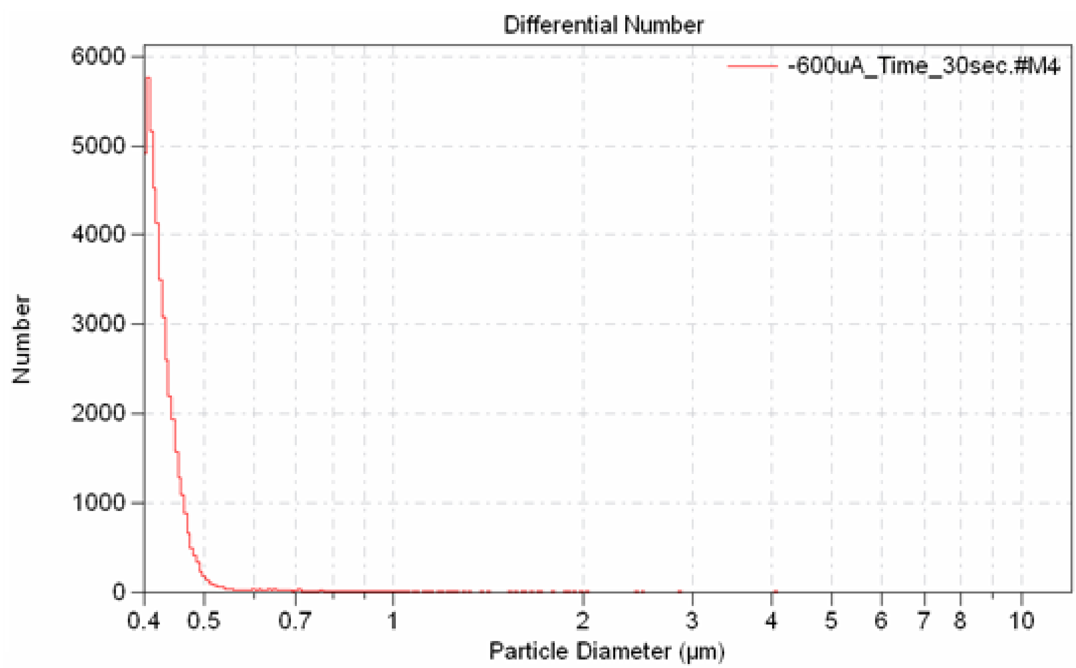


Figure 4.5 Noise Profile, Contaminated System (20 µm aperture)



Using Appropriate Current and Gain Settings

The recommended current setting for the 10 µm aperture tube is 200 µA. Higher currents can damage the aperture orifice and create noise as a result of the electrolyte overheating or boiling.

NOTE Beckman Coulter recommends that you set the 10 µm aperture gain to 4 and the current to 200 µA.

The recommended current setting for 20 µm and 30 µm aperture tubes is 600 µA. Higher currents can damage the aperture orifice and create noise as a result of the electrolyte overheating or boiling.

NOTE Beckman Coulter recommends that you set the 20 µm and 30 µm aperture gain to 4 and the current to 600 µA.

You can set current and gain using the Change Aperture Tube Wizard (see Step 8: Set Current and Gain) or the Edit the SOM window (see Standard Operating Method: Threshold, Current and Gain in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).

Storing Small Aperture Tubes

IMPORTANT Any unused apertures should be cleaned and dried for long term storage. For short term storage, follow these steps:

1. Remove the aperture tube from the Sample Compartment.
2. Rinse the tube (inside and outside) several times with pre-filtered, de-ionized water (see Cleaning the Aperture Tube, Beakers, and Accuvette STs in CHAPTER 2, Analyzer Overview).
3. Fill the ST Aperture Tube with pre-filtered, de-ionized water.
4. Fill a beaker with pre-filtered, de-ionized water.
5. Place the Aperture Tube in a clean beaker filled with pre-filtered, de-ionized water.
6. Seal the top of the ST Aperture Tube with parafilm or plastic film.
7. Store in a dust-free environment.

Using Large Apertures (≥400 µm)

Turbulence due to fast flow of low viscosity electrolyte solutions through the 2000, 1000, 800, 560, and 400 µm orifices can create an audible noise that can actually be heard as a 'whistle' and can be picked up as 'noise'. To overcome this, electrolyte solutions can be used with the addition of, for example, 30-50% glycerol for the 2000 and 1000 µm orifices and 10% glycerol for the 560 and 400 µm orifices when analyzing large, low density particles, for example PVC. If the addition of glycerol is not an acceptable solution, then vacuum can be lowered to 3 to 2 inches of Hg. Large sample beakers are helpful. Viscosity should not generally be higher than the equivalent of 90% glycerol. In fact, the addition of glycerol has a two fold effect; by increasing viscosity the turbulent flow is reduced and at the same time, suspension of large, dense, particles is facilitated.

The size levels obtainable with 1000 µm and 2000 µm aperture tubes are very dependent upon the density of the material being analyzed and the practical upper limit is in the order of 1200 µm with particles of density approximately 1 g/mL.

Calibrating the Aperture

When you calibrate the aperture, the Multisizer 4e runs an analysis to measure calibrator beads of a predetermined size. Calibrate the aperture with a particle standard that is 10% to 20% of the aperture size (20% is preferred).

Variations in the Analyzer's measurement of the calibrator beads determine the calibration constant (Kd) required for the Analyzer to measure the beads at their actual size.

The Kd takes into account many factors associated with the instrument and the aperture tube, including fluid flow characteristics, electrical resistance with the electrolyte, and imperfections of the orifice such as chip damage or wear.

When you verify aperture calibration (see Verifying the Calibration), the Multisizer 4e Analyzer uses the Kd (or the mean Kd, if you have obtained 10 Kd values and calculated the mean) to verify that the instrument is measuring calibrator bead size accurately. Unlike the aperture calibration procedure, which returns a calibration constant (Kd) for the aperture orifice, the verification procedure returns the measured calibrator bead size.

NOTE Calibration options are only available to user types with sufficient security privileges. If the Multisizer 4e software is running with no security, every operator can calibrate the aperture, create an SOP, and configure the software.

Calibrate the aperture after you:

1. Select a new aperture tube size (see Selecting an Aperture Size).

2. Install the new aperture tube (see Installing an Aperture Tube).

Choosing the Calibrator Size

Before you calibrate the aperture, use the table below to choose the appropriate calibrator size. Beckman Coulter recommends that you calibrate the aperture with a particle standard that is 10% to 20% of the aperture size (20% is preferred).

The calibrator beads supplied with the Multisizer 4e Analyzer are suitable for use with a 100 µm Aperture Tube.

Table 4.3 Choosing Calibrator Size

Aperture Size (µm)	Calibrator Mean or Modal Size (µm)
10	1 – 2
20	2 – 4
30	3 – 6
50	5 – 10
70	7 – 14
100	10 – 20
140	14 – 28
200	20 – 40
280	28 – 56
400	40 – 80
560	56 – 112
800	80 – 160
1000	100 – 200
2000	200 – 400

⚠ **CAUTION**

Sizing and statistical errors will occur if you do not use the proper procedure and/ or calibrator size.

Nano-Size Particle Calibration Standards

If you use polystyrene nano-particle calibrators, Beckman Coulter recommends that you keep the concentration level between 2% and 4%.

Polystyrene nano-particles can bind to system components. After the Multisizer 4e has analyzed a high concentration of nano-particles, it is necessary to clean the aperture tube and external electrode thoroughly.

Systems with high levels of nano-particle contamination where the background count cannot be returned to previous normal levels may require the aperture tube and external electrode to be removed from the Multisizer 4e and cleaned seperatly. For information on the recommended cleaning procedure, see Cleaning the Aperture Tube, Beakers, and Accuvette STs in CHAPTER 2, Analyzer Overview.

Selecting Calibration Options

1 Select **Calibration > Aperture Calibration Options** on the Main Menu bar. The Aperture Calibration Options window opens.

NOTE The **Calibration** > **Aperture Calibration Options** menu item is only available to user types with sufficient security privileges. If the Multisizer 4e software is running with no security, every operator can calibrate the aperture.

2 In the Aperture Options window:

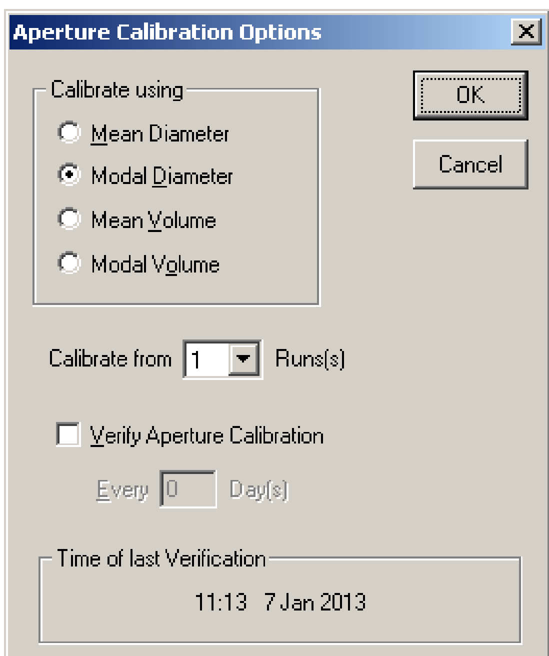
a. In the Calibrate using pane, select the basis for calibration (for example, the Mean Diameter or Modal Diameter). Beckman Coulter recommends that you calibrate the system using the standards on the assay sheet supplied with the calibrator.

b. If you would like more than one run, the operator can select the number of desired runs from the drop down list.

c. If you would like the Multisizer 4e software to prompt the operator to verify calibration at regular intervals:

- Select Verify Aperture Calibration.
- In the Day(s) field, enter the number of days between reminders.

d. Click **OK**.



Preparing the Analyzer for Calibration

1 Fill a clean ST Beaker or Accuvette ST with electrolyte solution.

NOTE For aperture tubes up to 100 µm, you can use an Accuvette ST (20 mL) and pre-filtered ISOTON II (see Pre-Filtering the Electrolyte and Cleaning Fluid in CHAPTER 2, Analyzer Overview). Use an ST Beaker for any aperture tube size larger than 100 µm.

2 Add a few drops of the calibrator to the electrolyte solution.

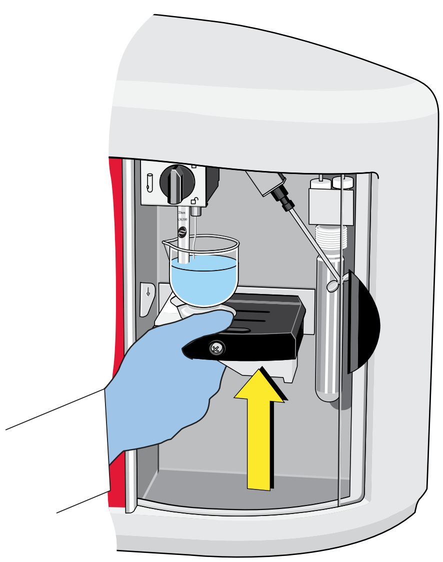
3 If necessary, use the platform release (see Platform Release in CHAPTER 2, Analyzer Overview) and lower the platform to the loading position.

4 If necessary, turn the platform so that the correct side (with the ST Beaker or Accuvette ST guide) is on top (see Reversing the Sample Platform in CHAPTER 2, Analyzer Overview).

NOTE The beaker or Accuvette ST guide on the platform ensures that the aperture tube, electrode, and stirrer are correctly positioned inside the beaker or Accuvette ST when you raise the platform.

5 Place the filled Accuvette ST or beaker on the platform.

6 Raise the platform.



NOTE It is not necessary to use the platform release to lift the platform into position.

7 Select **Run > Preview** on the Main Menu bar. The Status Panel displays Progress.

8 On the Status Panel, verify that the Concentration index bar shows a value as close as possible to 10%.

NOTE Start with a low concentration and build up to 10%. It may be difficult to clean a high concentration of calibrator beads from the system. If necessary, dilute the sample to lower the concentration to an acceptable level.

If the concentration of calibrator beads is greater than 10%, the Concentration index bar will display High. If the concentration of particles is too low, calibration time will increase. In either case, stop the calibration, adjust the calibrator concentration, and recalibrate the aperture.

CAUTION

Risk of possible sample contamination. If you are using a polystyrene nano-particle calibrator, keep the concentration level between 2% and 4%.

Running the Calibration

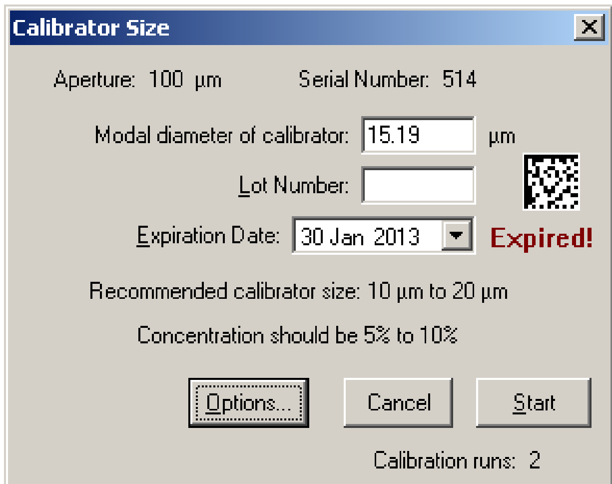
Before you run the calibration, select a calibrator size and prepare the Analyzer for calibration.

To Calibrate the Aperture

1 Select **Calibration > Calibrate Aperture** on the Main Menu bar. The Calibrator Size window opens.

NOTE The **Calibration** > **Calibrate Aperture** menu item is only available to user types whose security settings allow them to create a Standard Operating Procedure (SOP). If the Multisizer 4e software is running with no security, every operator can calibrate the aperture.

2 In the Calibrator Size window, enter the Modal diameter of calibrator. This value appears on the assay sheet supplied with the calibrator. To select the appropriate calibrator for your aperture tube size, see Choosing the Calibrator Size.



NOTE The recommended calibrator size range is 10% to 20% of the aperture diameter; however, the software allows entries from 3% to 50% of the aperture diameter.

3 Use the Date Picker to manually select the expiration date in the Calibrator Size Dialogue Box, or use the bar code reader for automatic entry (see Bar Code Reader in CHAPTER 2, Analyzer Overview).

4 Click **Start**.

5 On the Status Panel, verify that the Concentration index bar shows a value as close as possible to 10%.

NOTE Start with a low concentration and build up to 10%. It may be difficult to clean a high concentration of calibrator beads from the system. If necessary, dilute the sample to lower the concentration to an acceptable level.

If the concentration of calibrator beads is greater than 10%, the Concentration index bar will display High. If the concentration of particles is too low, calibration time will increase. In either case, stop the calibration, adjust the calibrator concentration, and recalibrate the aperture.

CAUTION

Risk of possible sample contamination. If you are using a polystyrene nano-particle calibrator, keep the concentration level between 2% and 4%.

Pausing or Stopping Calibration

During calibration, the Multisizer 4e software displays a graph in the Viewing Area. The Status Panel displays elapsed time, total particle count, and particle concentration.

To Pause or Stop Calibration

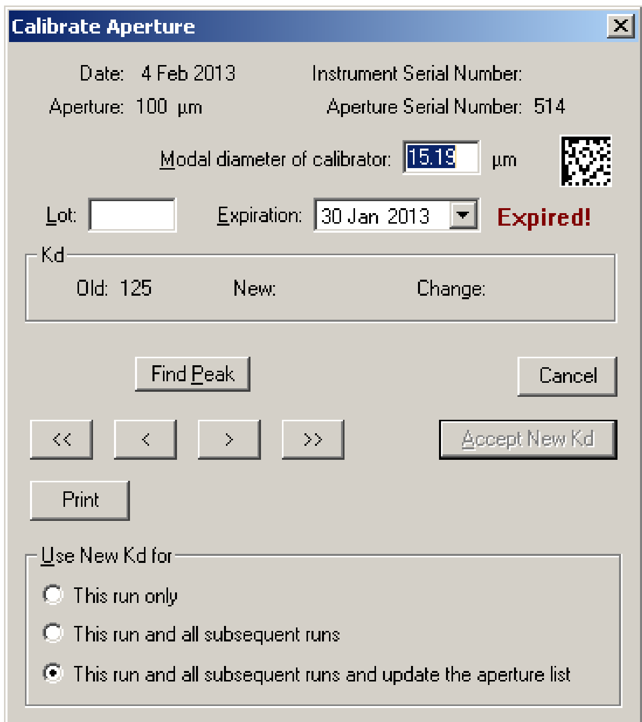
In the Instrument Toolbar, click **Pause**, **Stop**, or **Cancel** to interrupt the calibration process.

Table 4.4 Pausing / Stopping Calibration

Click...	To...
Pause	Pause the calibration.
Stop	Stop the calibration. The Multisizer 4e software displays calibration results from the stopped run in the Calibrate Aperture window.
Cancel	Cancel the calibration. The Multisizer 4e software does not display calibration results. To restart calibration, select **Calibration** > **Calibrate Aperture** on the Main Menu bar.

Completing the Calibration Run (Accepting the Kd)

1 When the calibration run is complete or stopped, a Calibrate Aperture window opens.



2 Use the Date Picker to manually select the expiration date in the Calibrator Size Dialogue Box, or use the bar code reader for automatic entry (see Bar Code Reader in CHAPTER 2, Analyzer Overview).

3 In the Calibrate Aperture window, the Kd pane displays the Old and New Kd values.

NOTE If this is the first calibration run, the Old Kd value is assigned an estimated value based on aperture size.

a. Click **Find Peak** to place cursors at or on either side of the peak of the calibration graph. If you are performing a modal calibration, clicking **Find Peak** automatically places a cursor on the modal channel. If you are performing a mean calibration, **Find Peak** automatically places two cursors on the graph, one on either side of the peak. The cursor(s) define the mode or median used to calculate the new Kd.

b. Click **Print** to print the window. This button does not display a dialog box or allow you to change print settings.

c. In the Use New Kd for pane, select an option to use the new Kd for the next analysis run only, this next analysis run and all subsequent runs and update the aperture list.

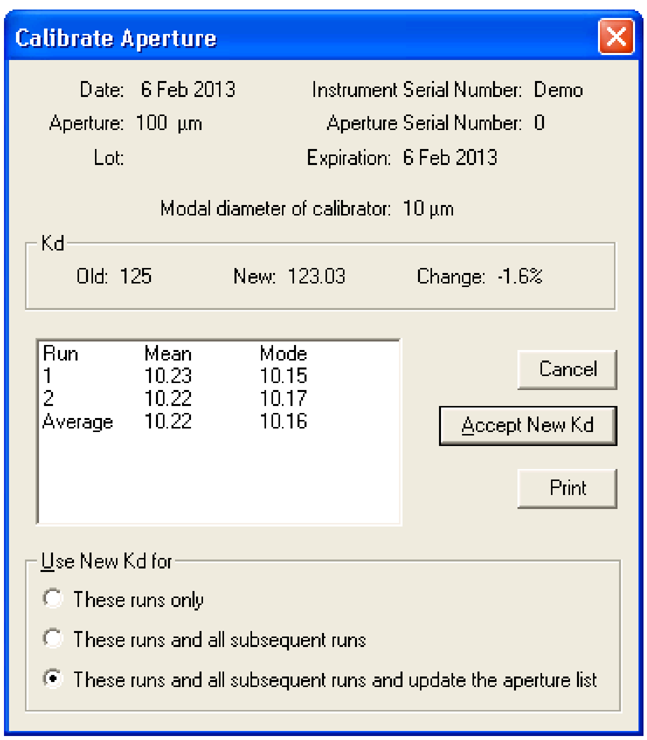
d. Click **Accept New Kd** to enter the new Kd and close the window.

NOTE When you calibrate the aperture, the Multisizer 4e runs an analysis to measure calibrator beads of a predetermined size (ranging from a nominal modal value of 2 µm to 90 µm). Deviations in the Analyzer's measurement of the calibrator beads determine the calibration constant (Kd) required for the Analyzer to measure the beads at their actual size.

The Kd takes into account many factors associated with the instrument and the aperture tube, including fluid flow characteristics, electrical resistance with the electrolyte, and imperfections of the orifice such as chip damage or wear.

4 At the end of a calibration, a new Multi Run Calibrate Kd Dialog will display.

NOTE The Multi Run Calibrate Kd Dialog will display the mean and mode of each run and the average of the means and modes.



Manually Obtaining the Mean Kd from Multiple Runs

For consistent results, perform the aperture calibration procedure at least 10 times and calculate the mean calibration constant (Kd). In future calibration runs, the Kd should be within 4% (+ or -) of the mean Kd value.

1. Calibrate the aperture.
2. Record the Kd value.
3. Repeat the calibration procedure nine times until you have 10 Kd values.
4. Add the 10 values and divide by 10 to obtain the mean value.

   NOTE The mean Kd value is valid only for the aperture tube, electrolyte, and Analyzer combination used for the calibration.

Example

This example uses a 10 µm calibrator and a 100 µm Aperture Tube.

Table 4.5 Obtaining Mean Kd Value

Run	Kd
1	124.15
2	125.82
3	125.13
4	125.23
5	125.43
6	125.13
7	124.25
8	125.62
9	124.25
10	125.03
MEAN	125

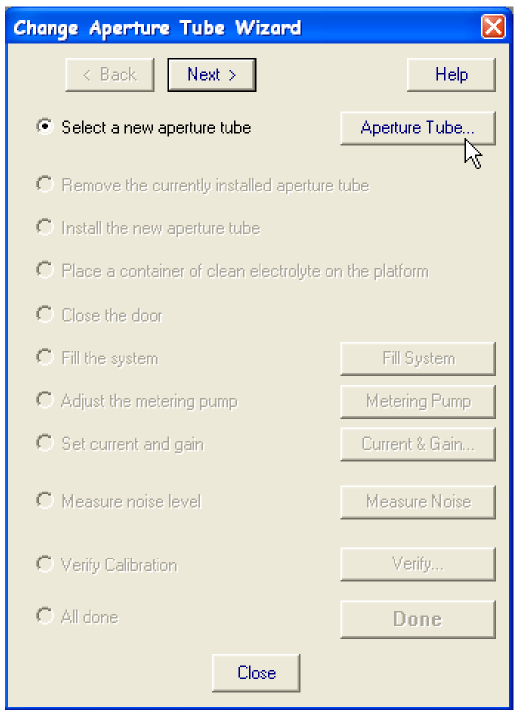
Updating the Mean Kd Value

If you have calculated the mean calibration constant (Kd) value for the aperture (see Manually Obtaining the Mean Kd from Multiple Runs), you can manually enter the Kd value using the Change Aperture Tube Wizard.

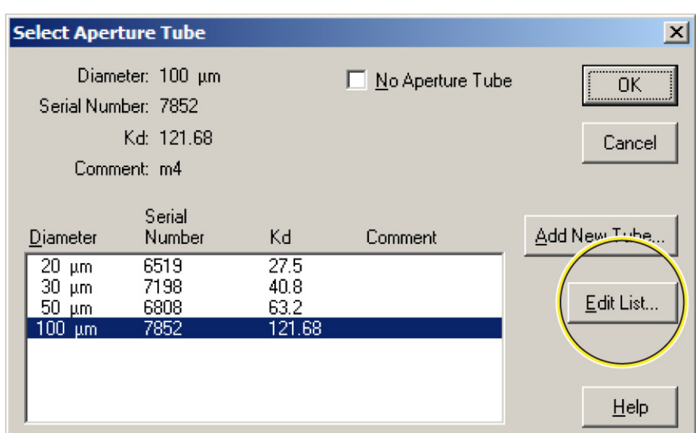
To Enter the Mean Kd Value

1 Select **Run > Change Aperture Tube Wizard** on the Main Menu bar.

2 In the Change Aperture Tube Wizard, click **Aperture Tube** to the right of the Select a new aperture tube step.



3 In the Select Aperture Tube window, click **Edit List**.



4 In the Edit Aperture Tube List window:

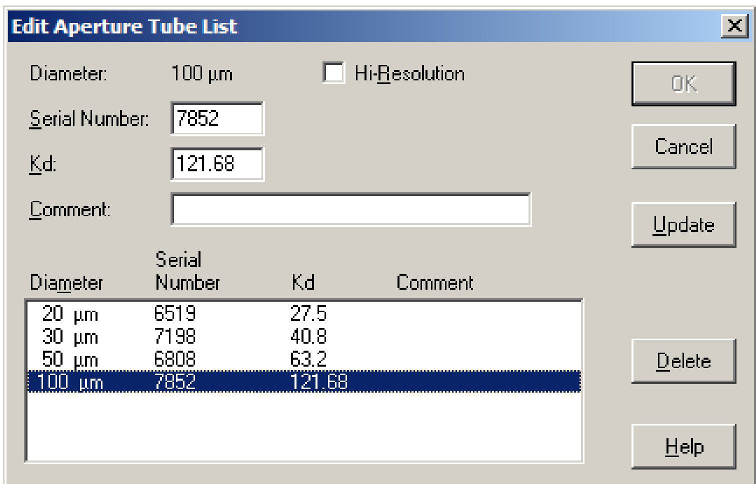
a. In the Diameter/Serial Number/Kd/Comment box, click the row you want to edit.

b. Enter the new mean Kd in the Kd field.

c. Click **Update**.

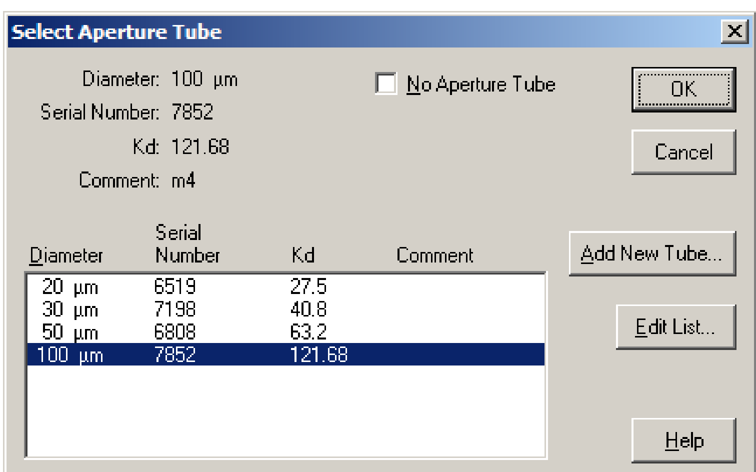
d. Click **OK** to return to the Select Aperture Tube window.

NOTE If you enter a Kd value that is too low or too high for the expected operation, the software will display a warning message.



5 In the Select Aperture Tube window:

a. In the Diameter/Serial Number/Kd box, click to highlight an Aperture Tube.



b. Click **OK** to return to the Wizard.

6 Click **Close** to exit the Wizard.

Verifying the Calibration

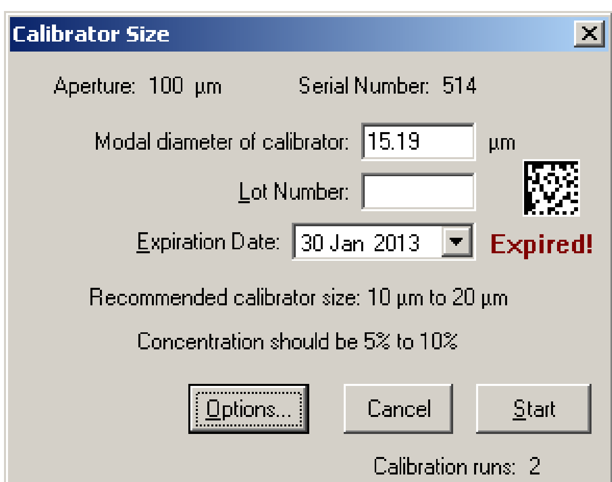
When you verify aperture calibration, the Multisizer uses the Kd (or the mean Kd, if you have obtained 10 Kd values and calculated the mean) to verify that the instrument is measuring calibrator bead size accurately. Unlike the aperture calibration procedure (see Running the Calibration), which returns a calibration constant (Kd) for the aperture orifice, the verification procedure returns the measured calibrator bead size.

Because the Analyzer uses constant current, the instrument holds the calibration for an indefinite time, provided that the working conditions are stable and you are using the same aperture tube and electrolyte combination. If working conditions are not stable, you can select an option to prompt the operator to verify the calibration at regular intervals using the Aperture Calibration Options window (see Selecting Calibration Options).

To Verify Aperture Calibration

1 Select **Calibration > Verify Aperture Calibration** on the Main Menu bar.

2 In the Calibration Size window, enter the Modal diameter of calibrator in the field. This value appears on the assay sheet supplied with the calibrator.



NOTE The recommended calibrator size range is 10% to 20% of the aperture diameter; however, the software allows entries from 3% to 50% of the aperture diameter.

3 Click **Start**.

4 On the Status Panel, verify that the Concentration index bar shows a value as close as possible to 10%.

NOTE Start with a low concentration and build up to 10%. It may be difficult to clean a high concentration of calibrator beads from the system.

If the concentration of calibrator beads is greater than 10%, the Concentration index bar will display High. If the concentration is too low, calibration time will increase. In either case, stop the calibration, adjust the calibrator concentration, and recalibrate the aperture.

⚠ CAUTION

If you are using a polystyrene nano-particle calibrator, keep the concentration level between 2% and 4%.

5 When the run is complete, the Verify Aperture Calibration window opens.

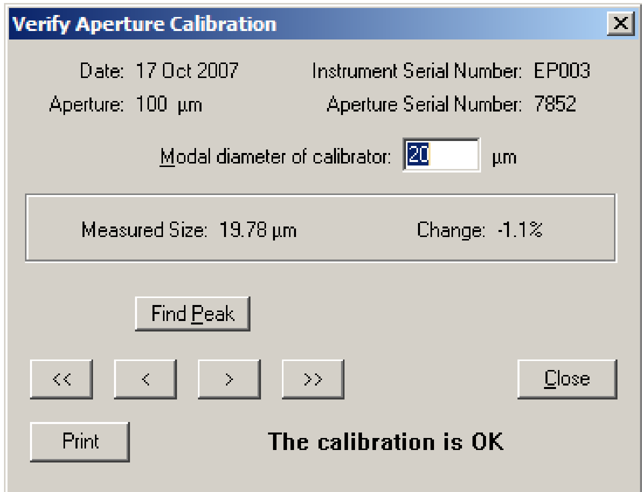
6 In the Verify Aperture Calibration window:

Table 4.6 Verifying Aperture Calibration

If the window displays...	Then...
The Calibration is OK	The measured particle size is within 3% of the actual calibrator size, and the calibration was successful. Click **Close** to exit the window.
The instrument needs to be recalibrated	The measured variation is too great, and the calibration was not successful. **1.** Click **Close** to exit the window. **2.** In the Instrument Toolbar, click **Reset**. **3.** If necessary, adjust the amount of calibrator in the electrolyte solution. **4.** Restart the calibration procedure.

- If desired, click **Find Peak** to place cursors at or on either side of the peak of the calibration graph. If you are performing a modal calibration, clicking **Find Peak** automatically places a cursor on the modal channel. If you are performing a mean calibration, **Find Peak** automatically places two cursors on the graph, one on either side of the peak. The cursor(s) define the mode or median used to calculate the new Kd.

- If desired, click **Print** to print the window. This button does not display a dialog box or allow you to change print settings.



Saving the Calibration File

After you have calibrated the aperture (see Calibrating the Aperture), the calibration results appear in the Viewing Area. Results are displayed as a size graph.

1 Select **RunFile > Save** on the Run Menu bar, or click **Save** in the Main Toolbar.

2 In the Save As window, navigate to the appropriate folder.

3 In the File name field, enter a name for the calibration file.

NOTE The Multisizer 4e software uses the .#m4 extension for all run data files, including calibration runs.

4 Click **Save**.

The Change Aperture Tube Wizard

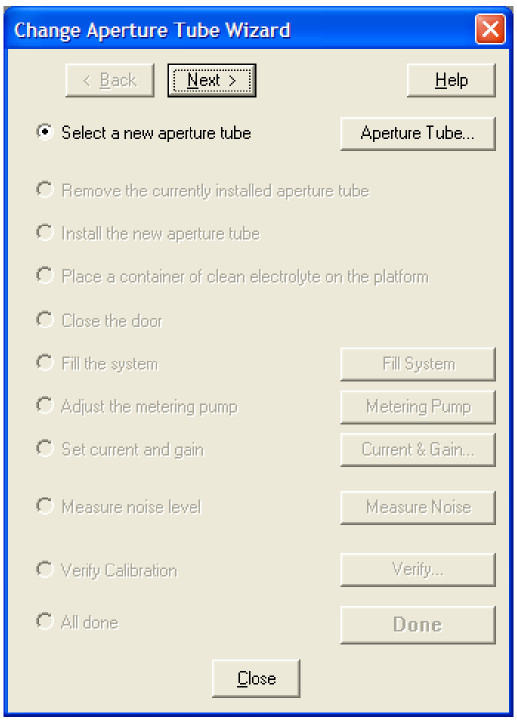
The Change Aperture Tube Wizard in the Multisizer 4e software provides step-by-step instructions to guide you through the steps you need to follow when installing a new aperture tube. The Wizard allows you to verify aperture calibration, test aperture settings, and enter the new aperture tube size and serial number.

Use the Change Aperture Tube Wizard after you have:

- Powered ON the Analyzer and connected the software to the Analyzer
- Selected the aperture tube size
- Calibrated the aperture, and
- Emptied the Waste Tank

To Use the Change Aperture Tube Wizard

Select **Run > Change Aperture Tube Wizard** on the Main Menu bar. The Change Aperture Tube Wizard window opens.



The following sections describe how to use the Change Aperture Tube Wizard:

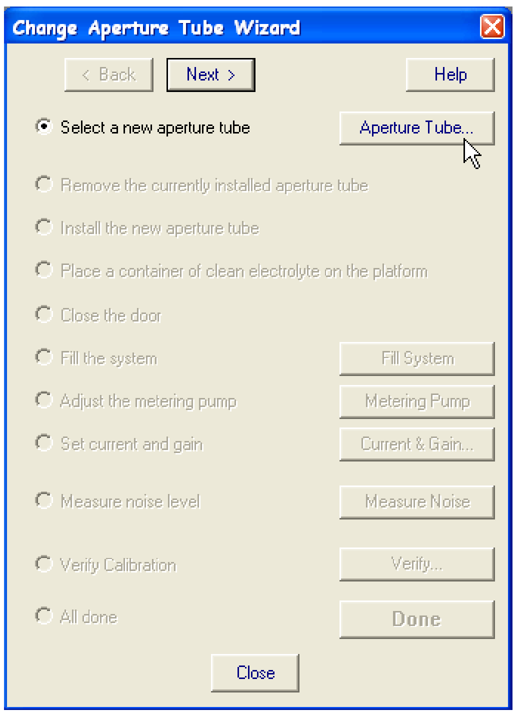
- Step 1: Select the new aperture tube
- Step 2: Remove the Currently Installed Aperture Tube
- Step 3: Install the New Aperture Tube
- Step 4: Place a Beaker of Clean Electrolyte on the Platform
- Step 5: Close the Door
- Step 6: Fill the System
- Step 7: Adjust the Metering Pump
- Step 8: Set Current and Gain
- Step 9: Measure Noise Level
- Step 10: Verify Calibration
- Step 11: All Done

In the Change Aperture Tube Wizard window, click **Next** at the top of the window to proceed to the next step. Click **Back** to return to the previous step.

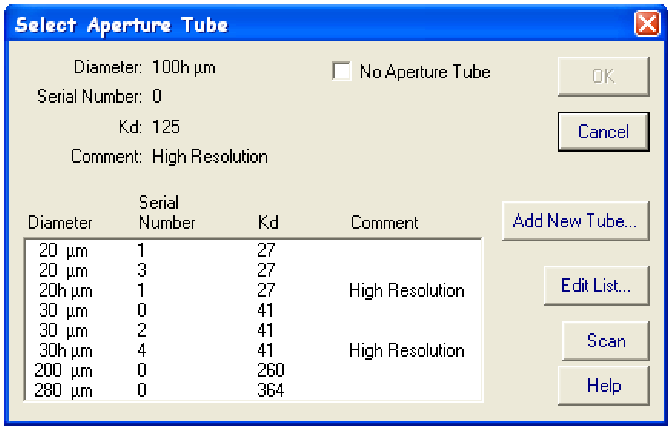
Step 1: Select the new aperture tube

When you install a new aperture tube, you will need to enter the aperture tube size and serial number in the Multisizer 4e software.

1 In the Change Aperture Tube Wizard, click **Aperture Tube** to the right of the Select a new aperture tube radio button.



2 The Select Aperture Tube window opens.



3 If the aperture tube (identified by size, serial number, and Kd) has been entered into the software, click to highlight the correct aperture tube. To ensure that you select the correct aperture tube serial number and Kd, you can scan the aperture tube bar code. For information on using the Bar Code Reader, see Bar Code Reader in CHAPTER 2, Analyzer Overview.

If the aperture tube does not appear in the list, you will need to enter the aperture tube size, serial number, and Kd (after aperture tube calibration). See Entering and Editing Aperture Tube Information.

4 Click **OK** to return to the Wizard. The Wizard automatically highlights the next step, Remove the currently installed aperture tube (see Step 2: Remove the Currently Installed Aperture Tube).

Entering and Editing Aperture Tube Information

To enter or update the size, serial number, and Kd of a new aperture tube in the Multisizer 4e software, use the Change Aperture Tube Wizard.

To Add a New Aperture Tube

1 In the Change Aperture Tube Wizard, click **Aperture Tube**.

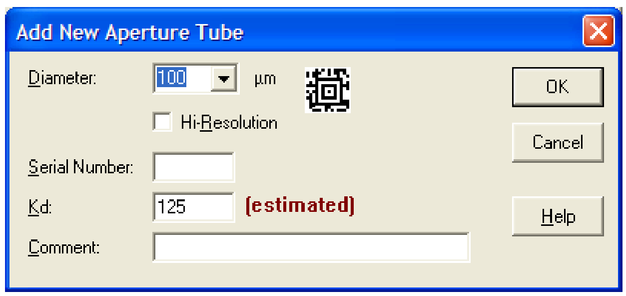
2 In the Select Aperture Tube window, click **Add New Tube**.

3 The Add New Aperture Tube window opens.

4 In the Add New Aperture Tube window, select the aperture size from the Diameter drop-down list and enter the Serial Number and mean Kd value.

To enter the tube size and serial number automatically, hold the aperture tube bar code in front of the Bar Code Reader. The Multisizer 4e software will enter the aperture Diameter, Serial Number, and estimated Kd value in the appropriate text fields. For information on using the Bar Code Reader, see Bar Code Reader in CHAPTER 2, Analyzer Overview.

NOTE For information on aperture tube calibration, see Calibrating the Aperture. For information on obtaining the mean Kd, see Manually Obtaining the Mean Kd from Multiple Runs.



5 Click **OK** to return to the Select Aperture Tube window.

6 In the Select Aperture Tube window:

a. In the Diameter/Serial Number/Kd/Comment box, click to highlight the aperture tube you will be installing.

b. Click **OK** to return to the Wizard.

The Wizard automatically highlights the next step, Remove the currently installed aperture tube (see Step 2: Remove the Currently Installed Aperture Tube).

To Edit the Aperture Tube List

1 In the Change Aperture Tube Wizard, click **Aperture Tube**.

2 In the Select Aperture Tube window, click **Edit List**.

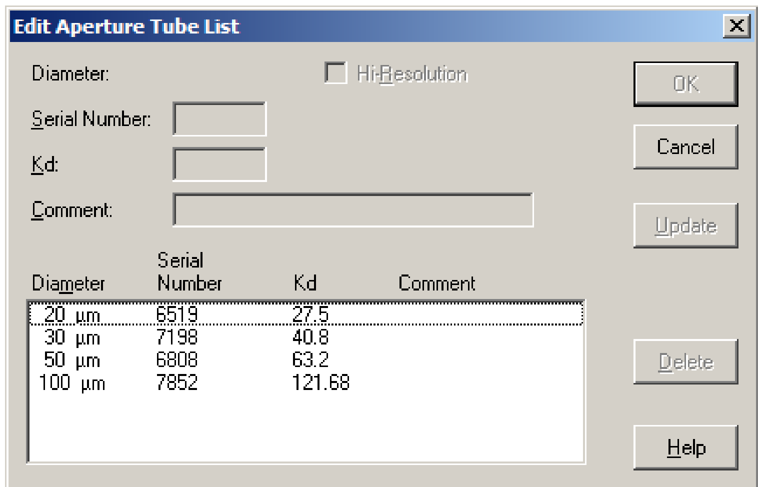
3 The Edit Aperture Tube List window opens.

4 In the Edit Aperture Tube List window:

a. In the Diameter/Serial Number/Kd/Comment box, click the row you would like to edit.

b. Enter a new Serial Number and/or the mean Kd in the text fields (see Manually Obtaining the Mean Kd from Multiple Runs).

c. Click **Update**.



5 Click **OK** to return to the Select Aperture Tube window.

6 In the Select Aperture Tube window:

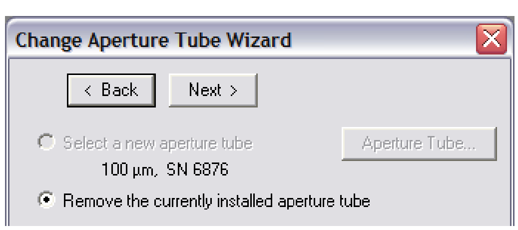
a. In the Diameter/Serial Number/Kd/Comment box, click to highlight the row displaying serial number and Kd of the aperture tube you installed.

b. Click **OK** to return to the Wizard.

The Wizard automatically highlights the next step, Remove the currently installed aperture tube (see Step 2: Remove the Currently Installed Aperture Tube).

Step 2: Remove the Currently Installed Aperture Tube

The Change Aperture Tube Wizard prompts you to remove the currently installed aperture tube from the Sample Compartment, if necessary.



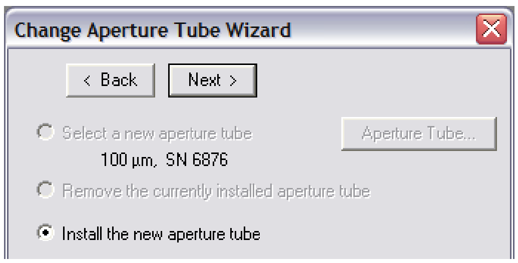
For information on removing an aperture tube, see Installing an Aperture Tube.

In the Change Aperture Tube Wizard, click **Next** to proceed to the next step, Install the new aperture tube (see Step 3: Install the New Aperture Tube).

Step 3: Install the New Aperture Tube

After you remove the previously installed aperture tube from the Sample Compartment, the Change Aperture Tube Wizard prompts you to install the new aperture tube.

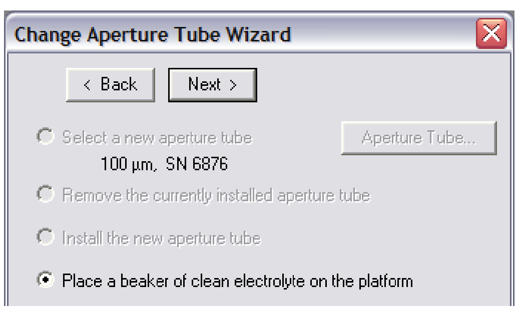
For instructions, see Installing an Aperture Tube.



In the Change Aperture Tube Wizard, click **Next** to proceed to the next step, Place a beaker of clean electrolyte on the platform.

Step 4: Place a Beaker of Clean Electrolyte on the Platform

It is necessary to place a beaker of clean electrolyte on the platform before filling the system. If the aperture is not immersed in electrolyte while the system fills, the Analyzer will draw air through the aperture and into the system during the fill procedure.



To Place a Beaker of Clean Electrolyte on the Platform

1. Wash the beaker. For instructions on the proper cleaning method, see Cleaning the Aperture Tube, Beakers, and Accuvette STs in CHAPTER 2, Analyzer Overview.

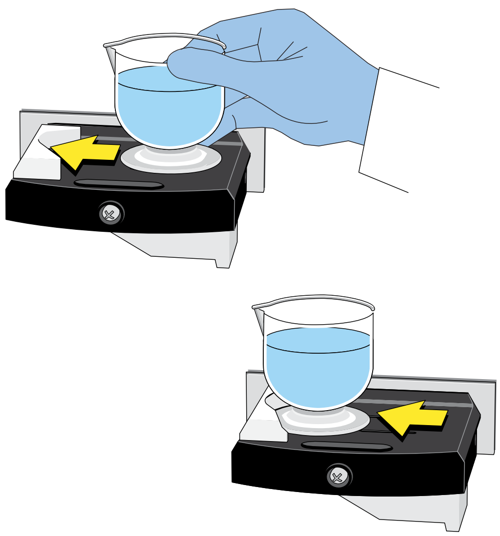
2. Fill the beaker with electrolyte solution.

3. Open the Sample Compartment door. See Sample Compartment in CHAPTER 2, Analyzer Overview.

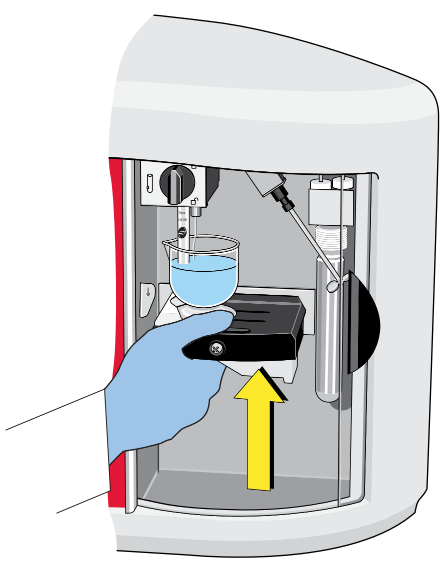
4 If necessary, press the platform release (see Platform Release in CHAPTER 2, Analyzer Overview) and lower the platform to the loading position.

5 Place the beaker on the platform (beaker side up). The beaker guides ensure that a Multisizer 4e Smart Technology Beaker of any size will be in the correct position in relation to the aperture tube, electrode, and stirrer when you raise the platform.

If you are using an Accuvette ST, reverse the platform to use the Accuvette ST guide (see Sample Platform in CHAPTER 2, Analyzer Overview).



6 Raise the platform until it locks into position for an analysis run. It is not necessary to use the platform release when you raise the platform.



7 Fill the Electrolyte Jar with the same electrolyte solution used in the ST Beaker.

In the Change Aperture Tube Wizard, click **Next** to proceed to the next step, Close the door (see Step 5: Close the Door).

Step 5: Close the Door

If the software detects that the Sample Compartment door is closed, the Wizard will skip this step.

If the Sample Compartment door remains open when you attempt to start an analysis run, the amber light will flash, and the software will display an error message.

In the Change Aperture Tube Wizard, click **Next** to proceed to the next step, Fill the System.

Step 6: Fill the System

When you install a new aperture tube, it is necessary to remove air from the system. The system fills from the Electrolyte Jar into the Waste Tank. (Electrolyte does not pass through the aperture orifice unless you are using the Unblock function to clear the orifice.) This procedure is similar to flushing the system to remove particles.

To Fill the System

1 In the Change Aperture Tube Wizard, click **Next** until you reach the step Fill the system if it is not already highlighted.

2 In the Status Panel, verify that the Waste Tank is empty. If the Waste Tank is not empty, exit the Change Aperture Tube Wizard and click **Empty** in the Instrument Toolbar. Re-open the Wizard and click **Next** until the **Fill System** button is selected.

3 To the right of the Fill the system step, click **Fill System**.

4 The Status Panel displays progress while the system is filling.

5 When the process is complete, the Status Panel displays READY and the Wizard automatically highlights the next step, Adjust the metering pump (see Step 7: Adjust the Metering Pump).

Step 7: Adjust the Metering Pump

This step appears in the Change Aperture Tube Wizard only if you are using an aperture tube smaller than 400 µm. The metering pump is required only when using volumetric mode. Aperture tubes 400 µm and larger can be used only in time and count modes.

For information on control modes (volumetric, time, and count) see Standard Operating Method: Control Mode in CHAPTER 5, Selecting Analysis Settings: SOM and SOP.

To Adjust the Metering Pump

1. In the Change Aperture Tube Wizard, click **Next** until you reach the Adjust the metering pump step if it is not already highlighted.

2. To the right of the Adjust the metering pump step, click **Metering Pump**.

3. The Status Panel displays Adjusting Metering Pump.

4. When the process is complete, the Status Panel displays READY and the Wizard automatically highlights the next step, Set current and gain (see Step 8: Set Current and Gain).

Step 8: Set Current and Gain

Setting current and gain determines the electrical pulse size (current) and amplification (gain) that the Analyzer will use as particles pass through the aperture. Unless you are running Service Mode (see Opening Service Mode in CHAPTER 10, Troubleshooting), the Multisizer 4e software sets current and gain automatically.

CAUTION

If you are using a 10 µm aperture tube, the recommended current setting is 200 µA. Higher currents can erode the aperture and create electrical noise. Beckman Coulter recommends that you set the gain to 4, and the current to 200 µA. For information on how to change current and gain settings, see Standard Operating Method: Threshold, Current and Gain in CHAPTER 5, Selecting Analysis Settings: SOM and SOP.

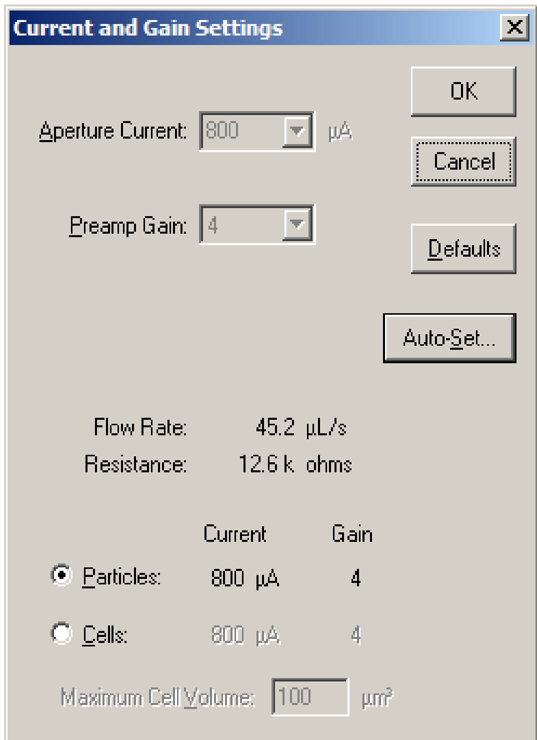
CAUTION

If you are using a 20 µm or 30 µm aperture tube, the recommended current setting is 600 µA. Higher currents can erode the aperture and create electrical noise. Beckman Coulter recommends that you set the gain to 4, and the current to 600 µA. For information on how to change current and gain settings, see Standard Operating Method: Threshold, Current and Gain in CHAPTER 5, Selecting Analysis Settings: SOM and SOP.

To Set Current and Gain

1 In the Change Aperture Tube Wizard, click **Next** until you reach the step Set Current and Gain if it is not already highlighted.

2 To the right of the Set Current and Gain step, click **Current & Gain**. The Current and Gain Settings window opens.



3 In the Current and Gain Settings window, click **Auto-Set**. If you are using 10 µm, 20 µm or 30 µm aperture tubes, Auto-Set is not available. Use the system defaults by clicking the **Defaults** button.

a. The window expands to display automatic settings for particles and cells.

b. The Analyzer measures electrical resistance through the aperture. MEASURING RESISTANCE appears in the Status Panel.

During the Measuring Resistance procedure, the software selects a current setting based on the size of the aperture tube and the concentration of the electrolyte solution. Using an electrolyte solution at a lower concentration increases the electrical resistance and requires a higher current setting. If you are using a more concentrated electrolyte solution, the software will select a lower current setting.

A low current with high gain will increase the noise level. A current set too high could saturate the amplifier, damage the aperture orifice, distort your results, or damage cells. The Auto-Set feature selects the optimum conditions based on the aperture size and electrolyte solution.

NOTE A blocked aperture or air in the aperture tube will raise resistance and cause an error message to display. Click **Cancel** to close the wizard, then, in the Instrument Toolbar, click **Unblock** to clear the aperture orifice, or click **Flush** to flush particles out of the system before proceeding.

When the system finishes the unblocking process, re-open the wizard by selecting **Run** > **Calibrate Aperture Tube Wizard** and clicking **Next** until Set Current and Gain is selected. Click the **Current & Gain** button, then click **Auto-Set**.

4 In the expanded Current and Gain Settings window, select Particles or Cells. If you select Cells, enter the Maximum Cell Volume.

CAUTION

Selecting Particles when you are analyzing Cells can cause cell damage due to a higher current setting.

5 To return to system default settings, click **Defaults**.

6 Click **OK** to return to the Change Aperture Tube Wizard.

7 The Wizard automatically highlights the next step, Measure noise level (see Step 9: Measure Noise Level).

Step 9: Measure Noise Level

When the system sets current and gain (see Step 8: Set Current and Gain), the software selects an electrical current setting based on the size of the aperture tube and the concentration of the electrolyte solution. If you are using a small aperture tube or a concentrated electrolyte solution, the software will select a low current/high gain setting.

For information on reducing noise levels when using small apertures, see Reducing Environmental Factors That Create Noise.

For information on current and gain settings appropriate for small apertures, see Using Appropriate Current and Gain Settings.

To Measure the Noise Level

1 In the Change Aperture Tube Wizard, click **Next** until you reach the step Measure noise level if it is not already highlighted.

2 To the right of the Measure noise level step, click **Measure Noise**.

3 The Status Panel displays MEASURING NOISE LEVEL.

4 When the process is complete, the Status Panel displays READY. The noise level is displayed as a percentage of aperture size below the Measure noise level step (in grey text). The Wizard automatically highlights the next step, Verify Calibration (see Step 10: Verify Calibration).

Step 10: Verify Calibration

This step in the Change Aperture Tube Wizard assumes that you have already calibrated the currently installed aperture tube (see Calibrating the Aperture). The Multisizer 4e software will automatically verify the calibration of the aperture tube using the serial number you entered in Step 1: Select the new aperture tube.

If you have not calibrated the aperture tube, exit the Wizard and calibrate the aperture (see Calibrating the Aperture).

To Verify Aperture Tube Calibration

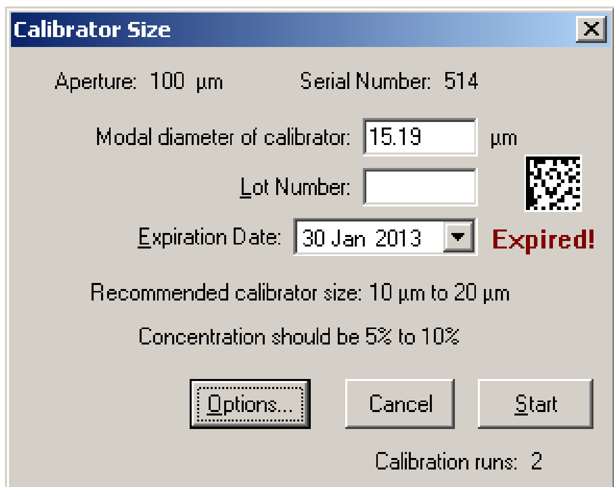
1 In the Change Aperture Tube Wizard, click **Next** until you reach the step Verify Calibration.

2 To the right of the Verify Calibration step, click **Verify**. The Calibrator Size window opens.

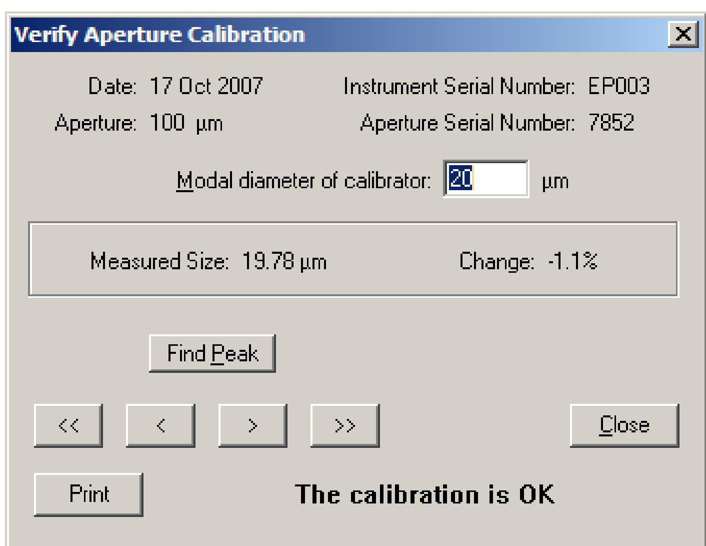
3 Enter the appropriate calibrator size.

4 Prepare the Analyzer for calibration.

5 In the Calibrator Size window, click **Start**. For more information on the verification procedure, see Verifying the Calibration.



6 When the verification is complete, the Verify Aperture Calibration window opens.



7 In the Verify Aperture Calibration window:

Table 4.7 Verify Aperture Calibration Window

If the window displays...	Then...
The calibration is OK	The measured particle size is within 3% of the actual calibrator size, and the calibration was successful.
The instrument needs to be recalibrated	The measured variation is too great, and/or the calibration was not successful. **1.** Click **Close** to exit the window. **2.** In the Instrument Toolbar, click **Reset**. **3.** If necessary, adjust the amount of calibrator in the electrolyte. **4.** Restart the calibration procedure.

- If desired, click **Find Peak** to place cursors at or on either side of the peak of the calibration graph. If you are performing a modal calibration, clicking **Find Peak** automatically places a cursor on the modal channel. If you are performing a mean calibration, **Find Peak** automatically places two cursors on the graph, one on either side of the peak. The cursor(s) define the mode or median used to calculate the new Kd.
- If desired, click **Print** to print the window. This button does not display a dialog box or allow you to change print settings.

8 Click **Close** to exit the Verify Aperture Calibration window and the Change Aperture Tube Wizard.

Step 11: All Done

Click **Close** or **Done** to exit the Wizard. If you have verified calibration, the Wizard closes automatically when verification is complete.

Running a Concentration Control Sample

Analyzing a concentration control sample at regular intervals allows you to verify count accuracy.

The software automatically uses the following settings when running a control sample:

- **Electrolyte solution volume:** 20 mL (Accuvette ST only)
- **Calibrator volume:** 0.2 mL (200 µL)
- **Dilution factors:** none
- **Mode:** Volumetric (500 µL)
- **Flush before run:** Yes

- **Number of runs**: 1
- **File save, print, or export**: Off
- **Size bins**: 400 log diameter size bins from 8 µm to 12 µm
- **Coincidence correction**: On
- **Edit**: Off
- **View**: Differential number size graph, linear diameter X axis; graph displays number per mL when run is complete

Setting Control Sample Options

To Specify a Control Sample Run and Set Recurring Reminders

1 Select **Calibration > Control Sample Options** on the Main Menu bar.

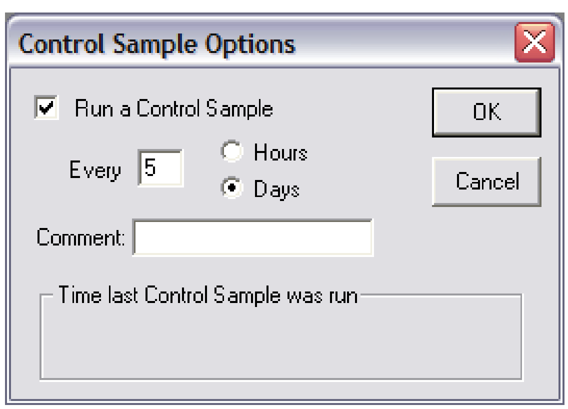
NOTE The Control Sample Options menu item is only visible to user types with sufficient security privileges. The system administrator configures security settings. If the Multisizer 4e software is running with no security, every operator can set control sample options, create an SOP, and configure the software.

2 In the Control Sample Options window:

a. Select Run a Control Sample.

b. If you would like the Multisizer 4e software to prompt you to run a control sample at regular intervals:

- Select Hours or Days for the intervals between reminders.
- In the Every field, enter the number of hours or days between control sample runs.



c. Click **OK**.

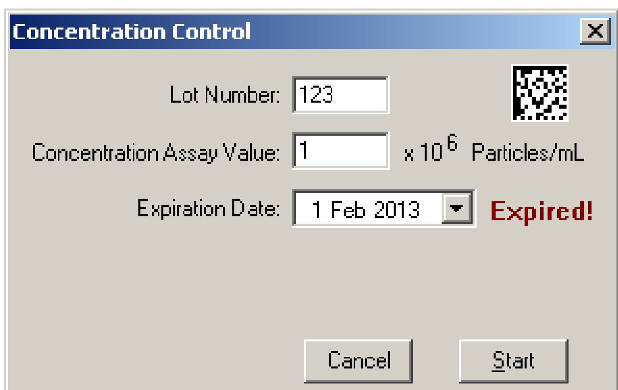
Running a Concentration Control Sample

1. Fill a clean Accuvette ST with 20 mL electrolyte solution and 0.2 mL (200 µL) calibrator.

2. Place the Accuvette ST on the Analyzer platform and raise the platform. For more information on preparing the Analyzer, see Preparing the Analyzer for Sample Runs in CHAPTER 2, Analyzer Overview.

3. Select **Calibration > Run Concentration Control** on the Main Menu bar. The Concentration Control window opens.

4. In the Concentration Control window, enter the Lot Number, the Concentration Assay Value, and the Expiration Date listed on the assay sheet.



5. Click **Start** to begin the concentration control run.

6. The Multisizer 4e software flushes the system and begins the run.

7. When the run is complete, a second Concentration Control window opens and displays results. The window displays a message that the concentration is low or high if the Analyzer measurement deviates from the concentration control by 10% or more. If this occurs, contact Beckman Coulter Service. For information on how to do this, see Working with Field Service in CHAPTER 10, Troubleshooting.

8. Click **Close**.

CHAPTER 5

Selecting Analysis Settings: SOM and SOP

Definitions: SOM, Preferences, and SOP

Standard Operating Method (SOM)
Use a Standard Operating Method (SOM) to specify sample analysis settings such as the control mode, file name, pulse settings (current and gain), the number of size bins, and sample concentration information.

For more information on creating a Standard Operating Method, see Using the Create SOM Wizard.

Preferences
Create a Preferences file to specify view and print settings, including graph, statistics, trend, page setup, and export settings.

You can save your Preferences to a file and load the Preferences file into the Multisizer 4e software when needed. You can also include the Preferences file in a Standard Operating Procedure (SOP).

For more information on setting Preferences, see Creating a Preferences File in CHAPTER 6, Setting View and Print Preferences.

Standard Operating Procedure (SOP)
A Standard Operating Procedure (SOP) combines one Preferences file and one Standard Operating Method (SOM) into an overall settings file for a particular type of analysis. Within the SOP, the SOM file determines sample run settings, while the Preferences file determines view and print specifications.

If you have saved default or custom SOM and Preferences files, you can create one or more Standard Operating Procedures (SOPs). Standard Operating Procedures are helpful if you are running several types of analyses that require different settings.

For more information on creating a Standard Operating Procedure, see Creating an SOP.

Using a Standard Operating Method (SOM)

Using the Create SOM Wizard

If you are creating a Standard Operating Method (SOM) for the first time, use the Create SOM Wizard.

If you are already familiar with using an SOM, or if you are changing only a few settings, bypass the Wizard and select **SOP > Edit the SOM** on the Main Menu bar (see Editing SOM Settings).

NOTE The Multisizer 4e software will not allow you to open the SOM Wizard if a Standard Operating Procedure (SOP) is loaded. To remove an SOP, click **RemoveSOP** in the Status Panel, or select **SOP > Remove the SOP** on the Main Menu bar.

To Use the Standard Operating Method (SOM) Wizard

1. If you have not connected to the Analyzer, select **Run > Connect to Multisizer 4e** on the Main Menu bar.

2. Select **SOP > Create SOM Wizard** on the Main Menu bar. The SOM (Standard Operating Method) Wizard opens.

The SOM Wizard includes the following steps:

- Standard Operating Method: Description
- Standard Operating Method: Control Mode
- Standard Operating Method: Run Settings
- Standard Operating Method: Stirrer Settings
- Standard Operating Method: Threshold, Current and Gain
- Standard Operating Method: Pulse to Size Settings
- Standard Operating Method: Concentration Information
- Standard Operating Method: Blockage Detection
- SOM Wizard: Summary of Settings

NOTE You can exit the SOM Wizard at any time; however, your changes will be lost unless you save them. Go to Step 9 in the SOM Wizard, and click **Save**.

Standard Operating Method: Description

Use Step 1 of the SOM Wizard (see Using the Create SOM Wizard) to enter descriptive information, aperture and electrolyte specifications, and automatic file name defaults. You can also enter file name settings using the Run Settings tab in the Edit the SOM window (see Editing SOM Settings).

Figure 5.1 SOM Window

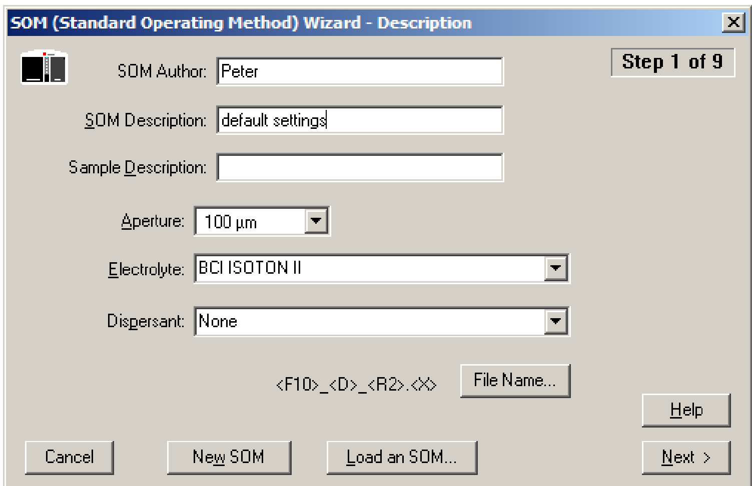


Table 5.1 SOM Wizard (Step 1): Field Descriptions

In the field...	Enter or select...
SOM Author	An author or operator name, if desired. Information entered in this field is displayed when you select **RunFile** > **Get Info** on the Run Menu bar.
SOM Description	A description for the SOM. Information entered in this field is displayed when you select **RunFile** > **Get Info** on the Run Menu bar.
Sample Description	A description of the sample, if desired. Information entered in this field is displayed when you select **RunFile** > **Get Info** on the Run Menu bar.
Aperture	Aperture size. Select the aperture size from the drop-down list. This list includes all aperture sizes that have been entered in the Change Aperture Tube Wizard (see The Change Aperture Tube Wizard in CHAPTER 4, Installing and Calibrating an Aperture Tube).
Electrolyte	The type of electrolyte. Scan or enter the electrolyte name or select the electrolyte from the drop-down list.
Dispersant	The type of dispersant, if applicable. Enter the dispersant name or select the dispersant from the drop-down list.

In the Description Window

- Click **FileName** to set automatic file name generation settings (see Creating Automatically Generated Analysis File Names).
- Click **New SOM** to set all fields to their default values and enter a new Standard Operating Method.
- Click **LoadSOM** to load previously saved SOM settings into the Wizard.

Creating Automatically Generated Analysis File Names

To automatically generate a file name, you will need to enter sample information before each analysis run (or set of analysis runs). To enter sample information, select **Sample** > **Enter Sample Info** on the Main Menu bar. For more information, see Entering Sample Information in CHAPTER 7, Analyzing a Sample.

By selecting file name options when you create a Standard Operating Method (SOM), you can use information entered in the Enter Sample Info for Next Run text fields to create individual or sequential file names that will help you organize your data.

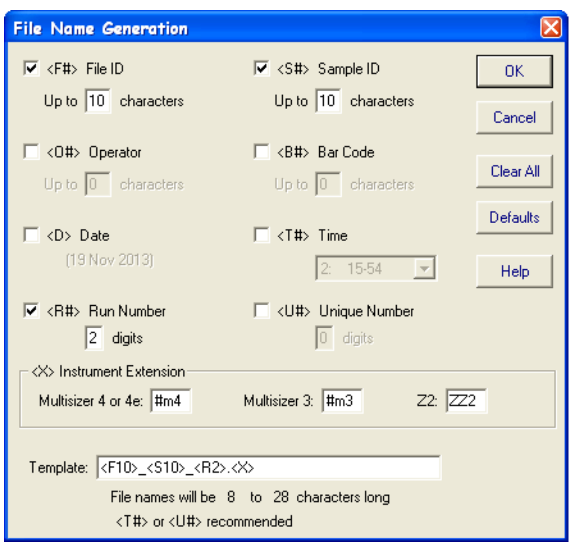
To Automatically Generate a File Name for Each Analysis

1 Select **SOP** > **Create SOM Wizard** on the Main Menu bar.

2 In Step 1 of the SOM Wizard, click **File Name**.

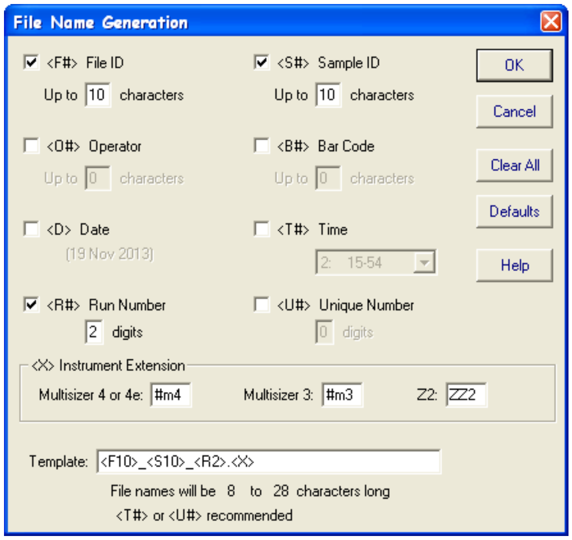
3 The File Name Generation window opens.

NOTE If you are creating or updating an SOM using the Edit the SOM window (see Editing SOM Settings), the **File Name** button is located on the Run Settings tab.



4 In the File Name Generation window, select each sample information field you want to include in file names. As you select text fields to include, associated template codes appear in the Template field at the bottom of the window. For information on these codes, see Using File Name Template Codes.

NOTE If necessary, you can change the Template field by entering new information manually. The character range specified below the Template field will change automatically based on the number of fields and character limits you select.



5 For File ID, Sample ID, Operator, and Bar Code, you can limit the number of characters that will be included in the file name. To set a limit, enter the maximum number of characters for each selected field in the Up to field.

6 In the Instrument Extension pane, accept the defaults or, if desired, enter the extension you would like to use for your analysis files in the Multisizer 4e field.

NOTE File extensions do not affect file format. You can open any Multisizer 4e analysis with Multisizer 3, or 4 software. Features specific to the Multisizer 4e will not be available when opened in earlier software versions.

7 Click **OK** to save your settings.

Using File Name Template Codes

In the File Name Generation window (see Creating Automatically Generated Analysis File Names), each sample information field is preceded by a code that the Multisizer 4e software uses when creating a template for the file name. For example, the template:

<F10>_<S10>_<R1>_<U2>.<X>

will create the following file name:

FileID_Sample_RunNumber_UniqueNumber.Extension

In this example, the File ID and Sample ID (<F10> and <S10>, respectively) will be included up to a maximum of 10 characters. The Run number will have a maximum of one digit, and the Unique Number will have a maximum of two digits.

Table 5.2 File Name Generation Codes

Code	Enter Sample Info Field
<F#>	File ID (Group Name)
<S#>	Sample ID
<O#>	Operator
<B#>	Bar Code
<D#>	Date (when run was saved)
<T#>	Time (when run was saved)
<R#>	Run Number
<U#>	Unique Number (File Name Generation window only). Including a unique number is recommended to avoid accidental file overwrites (see Using Unique Numbers in Analysis Files).
<X>	File Extension. You can select the Multisizer 4e default (#M4) or choose a new extension.

Using Unique Numbers in Analysis Files

Adding a Unique Number to file names ensures that analysis files are not accidentally overwritten.

For example, if you perform more than one run on a sample with the same File ID and Sample ID, or if you accidentally reset the Run Number to one (1) in the Enter Sample Information window before you have completed the total number of sample runs, it is possible that the Multisizer 4e software could overwrite run files.

Write-Protecting Saved Analysis Files

To Ensure That an Analysis File is Not Overwritten

1. Open the file in the Multisizer 4e software.
2. Select **RunFile > Save** or **RunFile > Save As** on the Run Menu bar.
3. In the Save As dialog box, enter a new file name or location, if desired.
4. In the middle of the window, select Save as read-only, and click **Save**.

In the SOM Wizard, click **Next** to proceed to the next step.

Standard Operating Method: Control Mode

Use Step 2 of the SOM Wizard (see Using the Create SOM Wizard) or the Control Mode tab of the Edit the SOM window (see Editing SOM Settings) to enter settings that will determine when an analysis run will end. The Multisizer 4e can end an analysis run based on elapsed time, total count, modal count, or the volume of sample that has passed through the aperture.

Figure 5.2 SOM Wizard - Control Mode

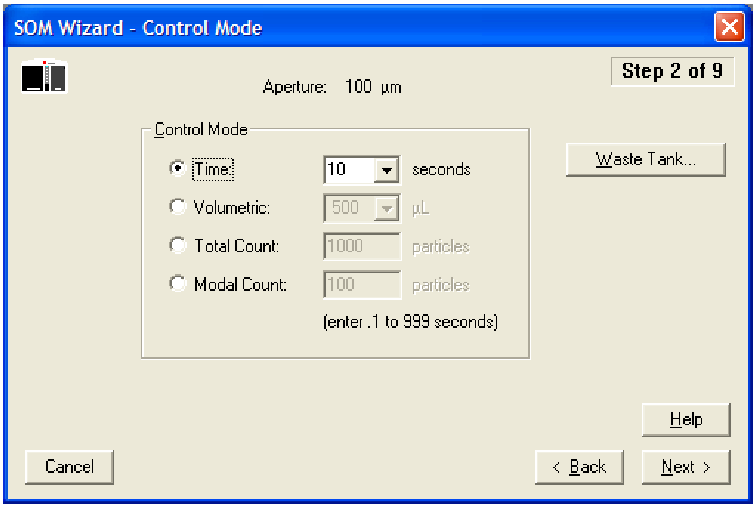


Table 5.3 Control Mode (Step 2): Selections

Select and specify...	To end the run when...
Time	The specified number of seconds have elapsed since the run began. Enter any time from 1 to 999 seconds, or select a number from the drop-down list. When you click **Next**, the valid range is displayed. Time mode uses constant pressure to move sample through the orifice for a pre-determined length of time.
Volumetric	The specified volume of electrolyte has passed through the aperture tube. Enter a volume from 10 to 2000 µL, or select a number from the drop-down list. When you click **Next**, the valid range is displayed. Use Volumetric mode to move a precise volume of sample through the orifice of the aperture tube with a metering pump. **NOTE** Volumetric mode is not allowed on aperture sizes greater than 360 µm. The largest standard aperture size allowed is 280 µm.

Table 5.3 Control Mode (Step 2): Selections

Select and specify...	To end the run when...
Total Count	The specified number of particles or cells (corrected for coincidence) has passed through the aperture. Enter a number from 50 to 500,000. **NOTE** To specify particles or cells, exit the SOM Wizard and select **SOP** > **Edit Preferences** on the Main Menu bar. In the Edit Preferences window, select the View Options tab and under Measuring, select Particles or Cells. To save changes you made in the Wizard, go to step 9 and click **Save** before exiting. Re-open the Wizard and return to Control Mode, step 2.
Modal Count	The specified number of particles or cells has accumulated in the modal channel. Enter a number from 10 to 100,000.

Click **WasteTank** to specify how often the Waste Tank will empty. In the Waste Tank window, select the intervals by run or time. If you do not select an option to automatically empty the Waste Tank, you can do so manually whenever a run is not in progress by clicking **Empty** on the Instrument Toolbar (see Main Toolbar Buttons in Software Overview).

NOTE For particle concentration analyses, use volumetric mode. For size distribution analyses, use any available control mode.

In the SOM Wizard, click **Next** to proceed to the next step.

Standard Operating Method: Run Settings

Use Step 3 of the SOM Wizard (see Using the Create SOM Wizard) or the Run Settings tab of the Edit the SOM window (see Editing SOM Settings) to specify pre-run prompts; determine the number of runs; and enter post-run settings (including automatic file save, export, and print settings).

NOTE While Preferences allows you to specify how you will view and print analysis results, Step 3 in the SOM Wizard allows you to specify whether files will automatically print or export to another file type after a run. Print and export settings will be based on your selections in Preferences (see Creating a Preferences File in CHAPTER 6, Setting View and Print Preferences).

Figure 5.3 SOM Wizard - Run Settings

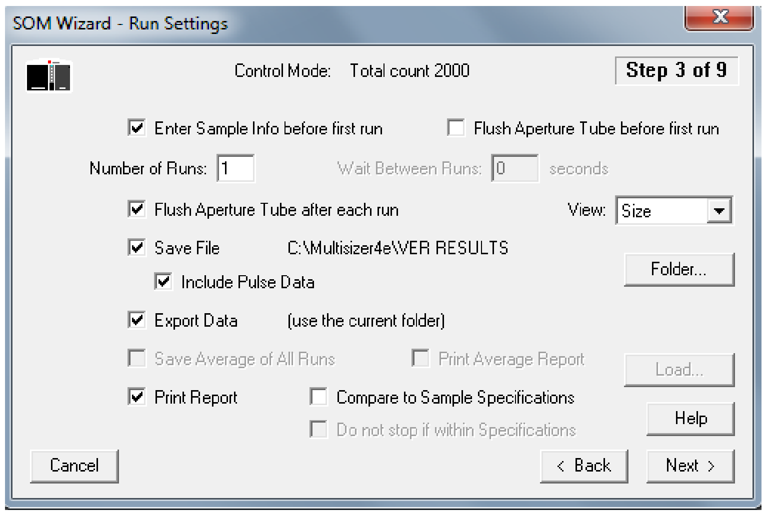


Table 5.4 Run Settings (Step 3): Entries and Selections

Enter or select...	To...
Enter Sample Info before first run	Display the Enter Sample Info window before the first run.
Flush Aperture Tube before first run	Automatically flush the aperture tube before the first run. Flushing the aperture tube removes heavy particles that remain trapped in the aperture tube.
Number of Runs	Determine how many times the sample will run. To activate this option, enter the number of runs.
Wait between Runs	Enter the number of seconds between runs.
Flush Aperture Tube after each run	Automatically flush the aperture tube after each run. Flushing the aperture tube removes any heavy particles that remain trapped in the aperture tube.
Save File	Automatically save analysis results to a file. The results will include pulse data only if Include Pulse Data is selected. Files will use name parameters specified in Step 1 of the SOM Wizard (see Creating Automatically Generated Analysis File Names). The Multisizer 4e software will save the file to the folder displayed in the Status Bar (at the lower left corner of the screen). To change the destination folder, click **Folder**.
Include Pulse Data	Save pulse data in the file with analysis results. Saving pulse data allows you to change range, resolution, and other parameters of the analysis without re-running the sample. **NOTE** Including pulse data increases the size of saved files.
Export Data	Automatically export analysis results to an external file type, based on Export settings in Preferences.

Table 5.4 Run Settings (Step 3): Entries and Selections

Enter or select...	To...
Print Report	Automatically print analysis results in a report. Statistics, graph, and page setup options will be determined by the Print Report settings in the current Preferences file.
Compare to Sample Specifications	Include a comparison to sample specifications in the printed report. Use this option only if you have entered specifications (select **Sample > Create Sample Specifications** on the Main Menu bar). For more information, See Using Sample Specifications in CHAPTER 7, Analyzing a Sample. Click the **Load** button to select the sample specifications to compare to.
Save Average of All Runs	Save an average size distribution or particle concentration, based on the total number of runs.
Print Average Report	Print the average size distribution or particle concentration, based on the total number of runs.
BUTTON: Folder	Change the default folder in which the Multisizer 4e software will automatically save new analysis files. For information on automatic file name generation, See Creating Automatically Generated Analysis File Names.
BUTTON: Load	Load sample specifications. This button is only available if you select Compare to Sample Specifications.
Do Not Stop if within Specifications	If the Number of Runs is greater than one, both **Compare to Sample Specifications** and **Do Not Stop within Specifications** are checked and the statistical results are within the specifications, then at the end of each run, statistical results display briefly before automatically advancing to the next run.

NOTE When using multiple runs in the Run Settings of the SOM, ensure that the waste tank and waste jar are prepared to handle the total volume of fluid for the series of runs. If the waste tank or waste jar becomes full during a run, the series will not be completed, and an error will result. Empty the waste tank and the waste jar prior to starting a series of multiple runs.

In the SOM Wizard, click **Next** to proceed to the next step.

Standard Operating Method: Stirrer Settings

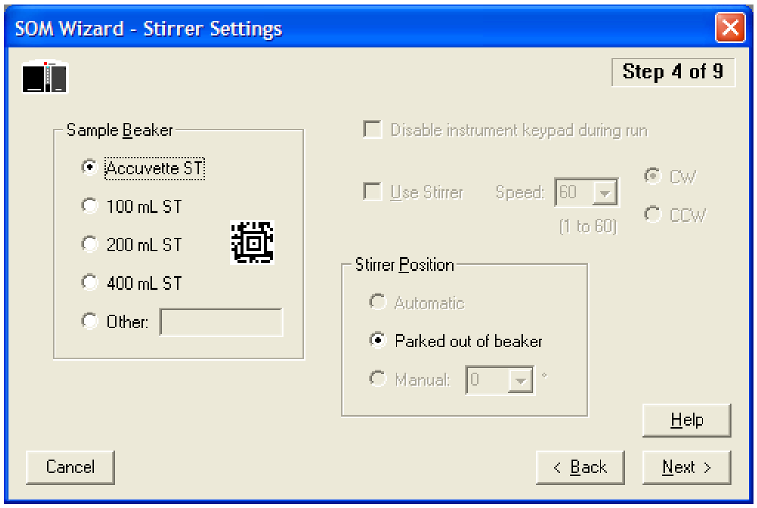
Use Step 4 of the SOM Wizard (see Standard Operating Method: Stirrer Settings) or the Stirrer tab of the Edit the SOM window (see Editing SOM Settings) to enter beaker size and stirrer position.

You can also use the Stirrer window to disable stirrer buttons on the Analyzer Control Panel for the duration of a run. If not disabled, these Control Panel buttons allow the operator to modify stirrer speed or direction and turn the stirrer on or off (see Control Panel in CHAPTER 2, Analyzer Overview).

The Edit the SOM / Stirrer tab includes an additional feature that allows the operator to copy manual settings into the SOM (see Copying Manual Stirrer Settings).

NOTE The Multisizer 4e will automatically position the stirrer based on the beaker size you select in the SOM. Beckman Coulter recommends that you use Multisizer 4e Smart Technology Beakers with automatic stirrer positioning to ensure consistency and accuracy of results.

Figure 5.4 SOM Wizard - Stirrer Settings



To Use the SOM/ Stirrer Settings Window

1 In the Sample Beaker pane, select Accuvette ST or the beaker size you will be using. To enter the beaker size automatically, hold the bar code in front of the Bar Code Reader on the Analyzer Control Panel. For more information on the Bar Code Reader, see Bar Code Reader in CHAPTER 2, Analyzer Overview.

In the Other field, enter a description (maximum length: 40 characters) for a non-standard beaker size.

2 In the Stirrer Position pane:

a. Select Automatic to automatically position the stirrer inside the beaker or Accuvette ST. For consistency and accuracy of results, Beckman Coulter recommends that you use Smart Technology Beakers with automatic stirrer positioning.

You can turn the stirrer off by pressing the stirrer ON/OFF button on the Analyzer Control Panel or by selecting **Run > Stirrer Off** on the Main Menu bar.

b. Select Parked out of beaker to position the stirrer outside the beaker during the run.

c. Select Manual to set the angle of the stirrer when you are using a beaker not manufactured by Beckman Coulter. The Manual radio button is available only if you have selected Other and entered a non-standard beaker size in the Sample Beaker pane.

d. In the Manual field, enter the stirrer position in degrees (from -20 to 10) or select an angle from the drop-down list. Zero (0) degrees is vertical, and the stirrer will touch the aperture tube at approximately 10 degrees. A negative position is farther from the aperture tube and is useful for larger, non-standard beakers.

3 Select Disable instrument keypad during run if you would like to prevent the operator from using buttons on the Analyzer Control Panel to change the speed or direction of the stirrer, or turn the stirrer on or off (see Control Panel in CHAPTER 2, Analyzer Overview). This option is not available if you select Parked out of beaker.

4 Select Use Stirrer to run an analysis with the stirrer positioned inside the beaker and rotating at the speed you select. This checkbox is available only if you select Automatic or Manual in the Stirrer Position pane.

5 In the Speed field, enter the stirrer rotation speed as any integer between 1 and 60, or select the speed from the drop-down list.

To find the appropriate speed for your beaker size and electrolyte combination, position the stirrer and adjust the stirrer speed using the Speed (+ and -) buttons on the Analyzer Control Panel. In the Edit SOM / Stirrer Settings window (see Editing SOM Settings), the Speed field will automatically display the number corresponding to the actual stirrer speed.

6 Select CW (clockwise) or CCW (counterclockwise) to determine stirrer direction. For information on choosing stirrer direction, see Stirrer Direction Controls in CHAPTER 2, Analyzer Overview.

Before you raise the platform, the Analyzer will automatically position the stirrer at the correct angle. If you need to change the stirrer position, lower the platform and enter the correct settings in the Edit SOM / Stirrer Settings window. To set a specific angle for the stirrer position, select Other and enter the angle in the Manual field in the Stirrer Position pane.

NOTE After you save stirrer settings, it is not possible to reposition the stirrer using the SOM Wizard. To reposition the stirrer, or to adjust stirrer speed using buttons on the Analyzer Control Panel, use the Edit SOM / Stirrer Settings window (see Editing SOM Settings).

Copying Manual Stirrer Settings

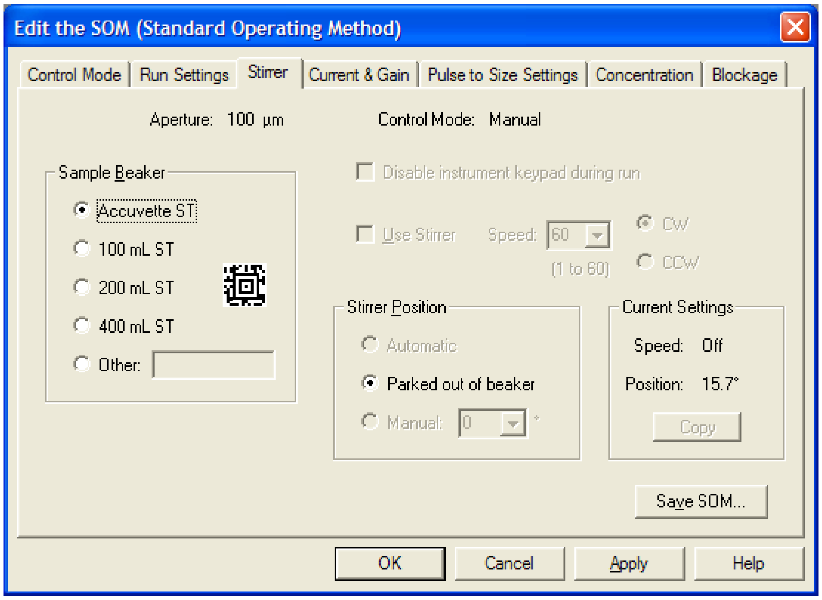
The Edit the SOM / Stirrer tab includes an additional feature that allows you to copy manual stirrer settings into the SOM.

NOTE The Multisizer 4e software allows the operator to position the stirrer manually only when the operator has entered a non-standard beaker size.

To Copy Manual Settings

1 Select **SOP > Edit the SOM** on the Main Menu bar.

2 In the Edit the SOM window, select the Stirrer tab.



3 On the Stirrer tab, select Other in the Sample Beaker pane and enter a non-standard beaker size.

4 In the Stirrer Position pane, select Manual.

5 In the Sample Compartment, manually position the stirrer.

6 If desired, change stirrer speed and direction.

7 In the Current Settings pane, select **Copy**. The current stirrer angle appears in the Manual field.

8 Click **Apply** or **Save SOM** to save your settings. The stirrer will move after you click **Apply**.

Standard Operating Method: Threshold, Current and Gain

Use Step 5 of the SOM Wizard (see Using the Create SOM Wizard) or the Current and Gain tab of the Edit the SOM window to enter the lower limit of a size distribution as well as the electrical pulse size (current) and amplification (gain) that the Analyzer will use as particles pass through the aperture.

Figure 5.5 SOM Wizard - Threshold, Current and Gain

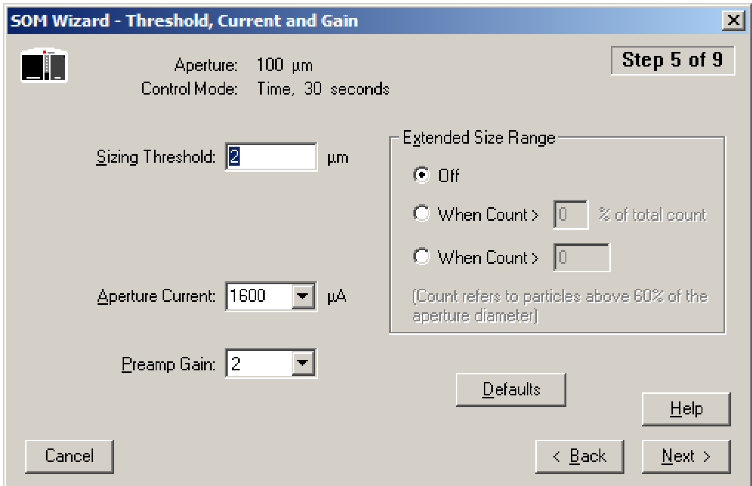


Table 5.5 Threshold, Current and Gain (Step 5): Entries or Selections

In the field...	Enter or select...
Sizing Threshold	The size above which particles are counted (or the lower limit of a size distribution). If the size entry is high or low, a warning message is displayed as follows: "Low" message is displayed for sizes less than 1.9% of aperture diameter "Too Low" message is displayed for sizes less than 1.5% of aperture diameter "High" message is displayed for sizes greater than 20% of aperture diameter "Too High" message is displayed for sizes greater than 50% of aperture diameter **NOTE** This setting cannot be changed during or after an analysis.
Aperture Current	The current that will pass through the aperture. Click **Defaults** to set the current and gain automatically for the aperture size you are using. If the aperture current entry is high or low, a warning message is displayed as follows: "Low" message is displayed for currents less than 100 µA "Too Low" message is displayed for currents less than 10 µA "High" message is displayed for currents greater than 800 µA when measuring cells "High" message is displayed for currents greater than 3200 µA "Too High" message is displayed for currents greater than 6000 µA If you are using a small aperture, current set too high can damage the aperture. For information on small aperture settings, see Using Small Aperture Tubes (10 µm, 20 µm and 30 µm) in CHAPTER 4, Installing and Calibrating an Aperture Tube.
Preamp Gain	The amplification factor. Click **Defaults** to set the current and gain automatically for the aperture size you are using.
Extended Size Range	Controls what happens if particles larger than 60% of the aperture diameter are detected. You can specify the large particle count either as an absolute number, as a percentage of the total count, or turn off this feature.

Click **Defaults** to set current and gain, and sizing threshold, automatically, based on the aperture size.

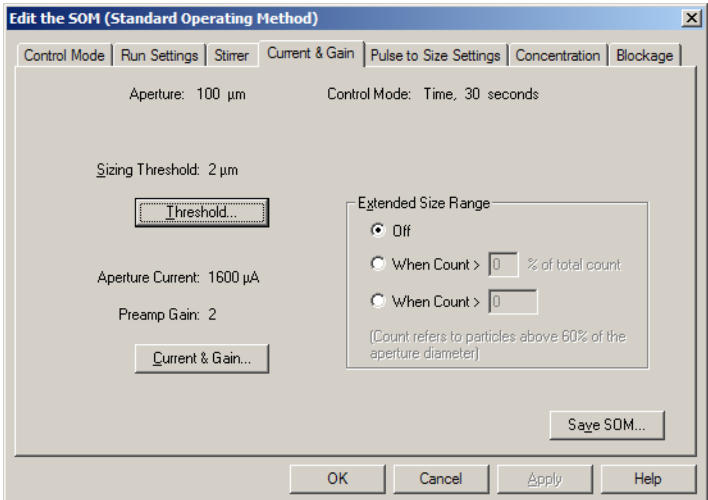
In the SOM Wizard, click **Next** to proceed to the next step.

Using the Current & Gain Tab in the Edit the SOM Window

The Edit the SOM / Current & Gain tab differs from the SOM Wizard / Threshold, Current and Gain window. Unlike the Wizard, the Edit the SOM / Current & Gain tab allows you to interact with the

Analyzer as you edit settings. You can use the **Threshold** and **Current & Gain** buttons on the Current & Gain tab to measure noise level and resistance using the currently installed aperture tube.

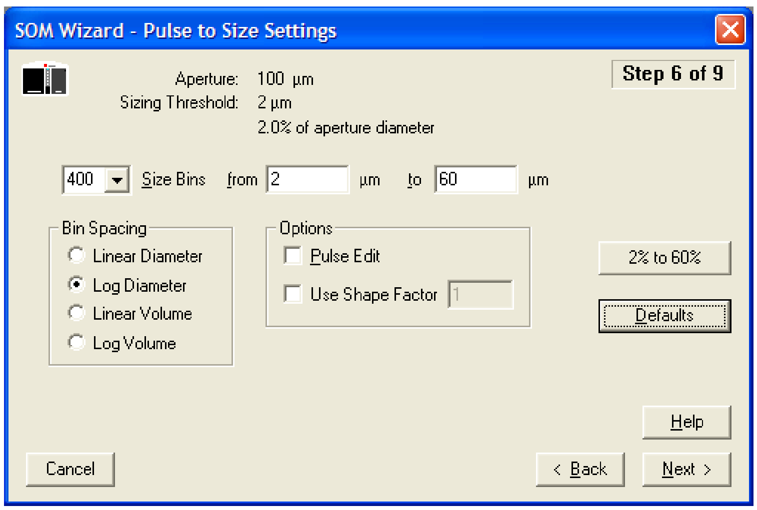
Figure 5.6 Edit the SOM



Standard Operating Method: Pulse to Size Settings

Use Step 6 of the SOM Wizard (see Using the Create SOM Wizard) or the Pulse to Size Settings tab of the Edit the SOM window (see Editing SOM Settings) to enter the number of bins (channels), bin spacing, and other information that will determine the range and resolution of analysis results. After running the analysis, you can change these settings by selecting **Calculate > Convert Pulses to Size** on the Run Menu bar. The aperture information on this screen changes accordingly, depending on the size of the aperture tube you are using.

Figure 5.7 SOM Wizard - Pulse to Size Settings



1 In the Size Bins field, enter the number of bins you want to use, or select the number of bins from the drop-down list (from 4 to 400).

2 In the from and to fields, enter the low and high limits for the range of particle diameter sizes you will be analyzing. The defaults are 2% to 60% of aperture diameter.

NOTE The smallest bin must be greater than or equal to sizing threshold (normally, 2% of aperture diameter).

NOTE The largest bin must be less than or equal to 60% of aperture diameter.

3 In the Bin Spacing pane, select how the Multisizer 4e software will determine bin spacing:

- By an incremental count of particle diameters (Linear Diameter)
- By a geometric progression of particle diameter sizes (Log Diameter)
- By a linear progression of particle volume (Linear Volume)
- By a geometric progression of particle volume size (Log Volume)

4 In the Options pane, select one or both of the following options:

- Pulse Edit. Select this option if you have a narrow distribution and would like to ensure accurate analysis results by eliminating large pulses. Particles that do not pass through the center of the aperture tend to produce wider and larger pulses. Pulse Edit removes the widest pulses from statistical analysis and display results. When the ratio of largest bin diameter to smallest bin diameter is greater than 6, a warning message appears.
- Use Shape Factor. Select this option if you would like to match your results to another particle sizing instrument, or if you need to adjust analysis results based on a known factor. Entering a shape factor of 1.03, for example, increases particle diameter by 3%.

5 Click **2% to 60%** to select a size range from **2% to 60%**.

6 Click **Defaults** to return the settings to Multisizer defaults.

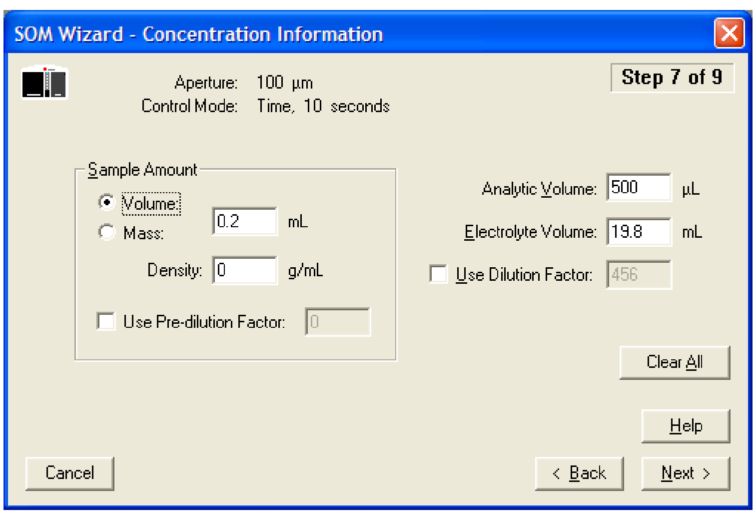
In the SOM Wizard, click **Next** to proceed to the next step.

Standard Operating Method: Concentration Information

Use Step 7 of the SOM Wizard (see Using the Create SOM Wizard) or the Concentration tab of the Edit the SOM window (see Editing SOM Settings) to enter information about sample volume, density, and dilution.

NOTE For particle concentration analyses, use volumetric mode. For size distribution analyses, use any available control mode.

Figure 5.8 SOM Wizard - Concentration Information



1 In the Sample Amount pane:

a. Select Volume if you are entering the volume of a sample suspended in liquid. Enter the volume of liquid sample in the mL field.

b. Select Mass if you are entering the mass of a dry sample of particles. Enter the density of the dry sample in the Density field (available in a chemical handbook).

2 Select Use Pre-dilution Factor if you are diluting your sample (for example, if you are adding electrolyte to a dry sample) before you add the sample to the electrolyte solution in the beaker. Enter the ratio of electrolyte solution to sample (the pre-dilution factor).

If you select Use Pre-dilution Factor, the software will not allow you to enter a dilution factor. The only possible combination is Pre-dilution factor (ratio of electrolyte solution to sample) added to electrolyte solution volume.

3 In the Analytic Volume text field, enter the volume of electrolyte solution and sample that will pass through the aperture during the run.

To enter an accurate value in the Analytic Volume field, it is necessary to run your analysis in volumetric mode. If it is not possible to use volumetric mode (for example, if your aperture is larger than 360 µm) you can estimate the analytic volume. To estimate analytic volume, run a timed analysis several times and average the volume of sample that passes through the orifice.

4 In the Electrolyte Volume field, enter the volume of electrolyte in the beaker. If you enter the volume of electrolyte in this field, the software will not allow you to calculate a dilution factor in the Use Dilution Factor field.

5 Select Use Dilution Factor if you are entering the ratio of electrolyte to sample. Dilution factor is an alternative to entering the electrolyte volume. If you select Use Dilution Factor and enter a dilution factor, the software will not allow you to enter a pre-dilution factor.

6 In the SOM Wizard, click **Next** to proceed to the next step.

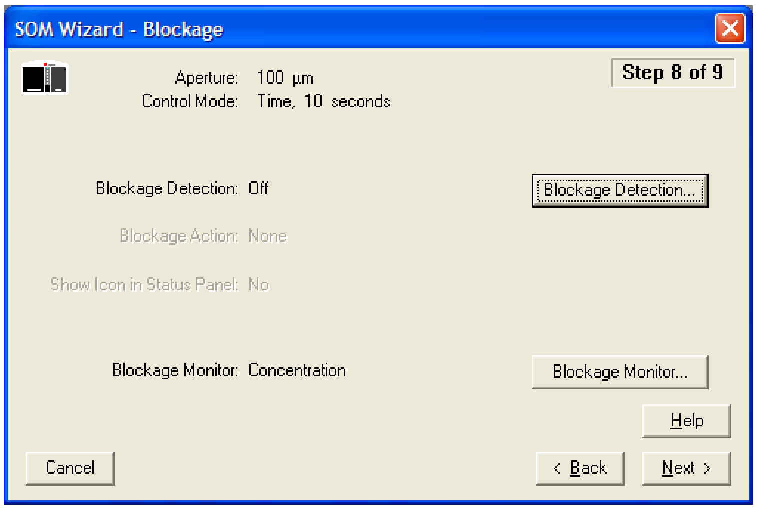
Standard Operating Method: Blockage Detection

Use Step 8 of the SOM Wizard or the Blockage tab of the Edit the SOM window (see Editing SOM Settings) to enter blockage detection settings. Based on your settings, the Multisizer 4e will automatically unblock the aperture or stop the run when the system detects a blockage.

⚠ CAUTION

Do not attempt to unblock the aperture using an object or your finger. The Multisizer 4e unblock procedure uses a reverse flow of electrolyte through the aperture orifice to remove stuck particles. Any other method of removing the trapped particles could damage the aperture. For more information, see **Unblocking Small Aperture Tubes in CHAPTER 4, Installing and Calibrating an Aperture Tube.**

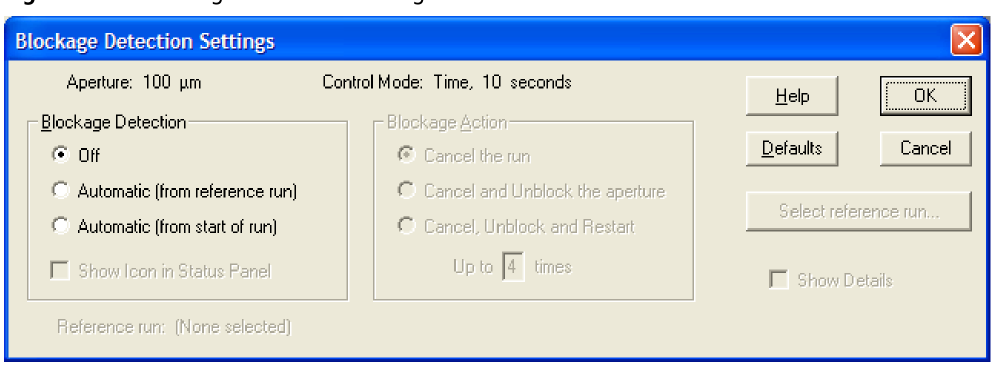
Figure 5.9 SOM Wizard - Blockage



To Enter Blockage Detection Settings

In the SOM Wizard - Blockage window, review the Blockage Detection and Blockage Action settings. To change either of these settings, click **Blockage Detection**. The Blockage Detection Settings window opens.

Figure 5.10 Blockage Detection Settings



In the Blockage Detection Settings window:

---

1 In the Blockage Detection pane:

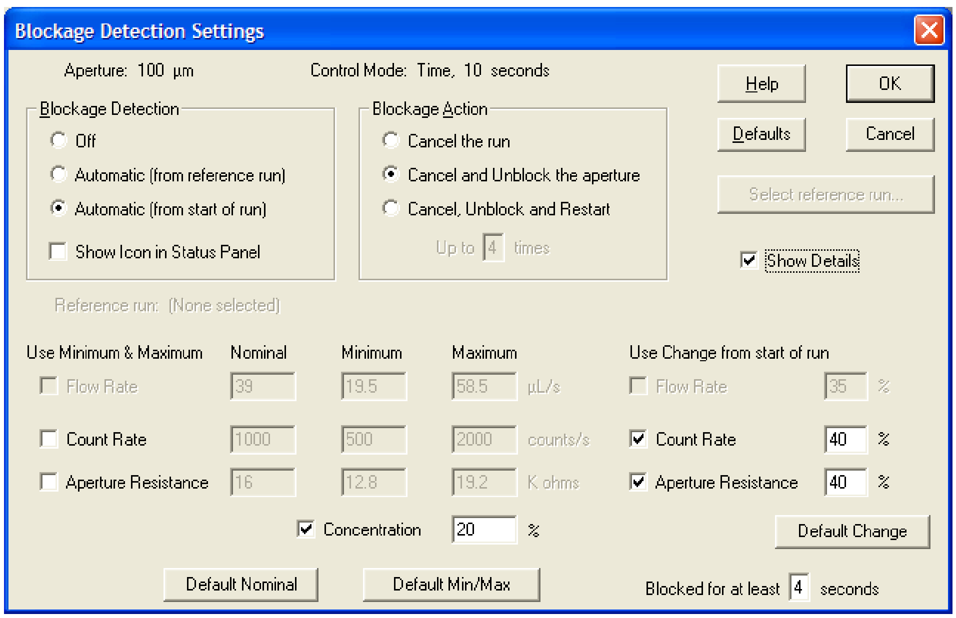
a. Select Off to disable automatic blockage detection.

b. Select Automatic (from reference run) to set blockage detection based on previous run data. To select the previous run, click **Select Reference Run** and browse to select the file. The reference run must use the same aperture tube size.

c. Select Automatic (from start of run) to set blockage detection based on current run data.

2 In the Blockage Action pane, select Cancel the run; Cancel and Unblock the aperture; or Cancel, Unblock and Restart the analysis. If you select the final option, enter the number of times to repeat the unblock procedure in the Up to [X] times field.

3 Select Show Details to see detailed blockage detection settings. The window expands.



The expanded Blockage Detection Settings window displays the Flow Rate, Count Rate, Aperture Resistance, and Concentration settings that will alert the Multisizer 4e to an aperture blockage and trigger the unblock procedure.

- **Flow Rate.** This field is available only in volumetric mode. A low flow rate or a significant change in flow rate can indicate aperture blockage. A high flow rate can result from air passing through the aperture, indicating that the sample fluid level is too low.
- **Count Rate.** In general, the count rate will decrease if the aperture is blocked. With some small aperture and particle combinations, however, the count rate can increase.
- **Aperture Resistance.** Unusually high electrical resistance can indicate blockage (when particles trapped in the aperture act as an electrical insulator) or air in the aperture (when sample fluid level is too low).
- **Concentration.** A brief spike in particle concentration indicates aperture blockage. If the concentration returns to normal, it is possible that the blockage has cleared. You can set the

number of seconds for a momentary blockage If other parameters (flow, aperture resistance, and/or count) indicate a blockage, however, you can assume that the aperture remains blocked.

In the Expanded Blockage Detection Settings Window

1. If you are using 10 µm, 20 µm and 30 µm aperture tubes, click **Defaults** to set the defaults automatically for the aperture size you are using. The default settings are:
   - Automatic (from start of run)
   - Cancel Unblock and Restart, Up to 4 times
   - Use change from start of run, Flow rate: 35%
   - Use change from start of run, Count rate: 40%
   - Use change from start of run, Aperture Resistance: 40%
   - Use Concentration
   - Blocked for at least 4 seconds

2. In the Use Minimum & Maximum column, select each parameter you want to use. In the corresponding fields, enter the Nominal, Minimum, and Maximum values that will trigger an unblock procedure. Flow Rate is only available when you are running volumetric mode.

3. In the Use change from start of run column, select Flow Rate, Count Rate, or Aperture Resistance. In the corresponding fields, enter the percentage of change that will trigger an unblock procedure. To set default values in the selected fields, click **Default Change**. Flow Rate is only available when you are running volumetric mode.

4. You can specify that a momentary blockage occurring for fewer than a preset number of seconds be ignored. Enter the number of seconds in the Blocked for at least [X] seconds field.

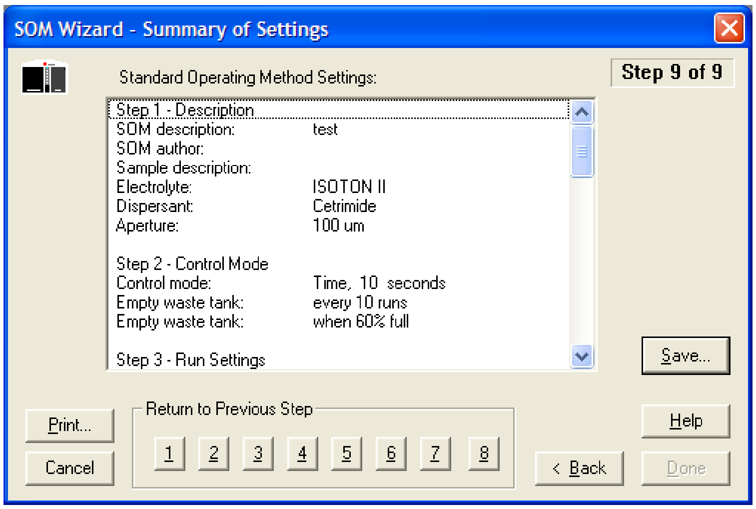
5. Click **OK** to accept the blockage detection settings.

6. In the SOM Wizard, click **Next** to proceed to the next step.

SOM Wizard: Summary of Settings

Step 9 of the SOM Wizard displays a summary of your Standard Operating Method settings. To see a summary of SOM settings when you are not using the SOM Wizard, select **SOP > SOM Info** on the Main Menu bar.

Figure 5.11 SOM Wizard - Summary of Settings



- Return to any previous step by clicking the step number in the Return to Previous Step pane.
- To save the SOM settings to a file, click **Save**.
- To print the SOM settings, click **Print**.

  You can print the SOM summary in its entirety, or specify a range of lines to print. To print selections from the SOM settings, first highlight the lines you would like to print in the Summary of Settings window, click **Print**, and select Print selected lines.

  You can also use the **Print** button to copy the SOM settings, or selected settings, to the clipboard.

  Click **Done** to exit the SOM Wizard.

Saving an SOM to a File

To save a Standard Operating Method to a file that you can load into the Multisizer 4e software, you can use the SOM Wizard or the Edit SOM window.

NOTE The Multisizer 4e software will not allow you to open the SOM Wizard or load, edit, or save SOM settings using the Edit the SOM window if a Standard Operating Procedure (SOP) is loaded. To remove an SOP, click **RemoveSOP** in the Status Panel, or select **SOP > Remove the SOP** on the Main Menu bar.

If You are Using the SOM Wizard

1 Select **SOP > Create SOM Wizard** on the Main Menu bar.

2 In the SOM Wizard, click **Next** until you reach Step 9: Summary of Settings.

3 In the Summary of Settings window, click **Save.**

4 In the Save a Standard Operating Method (SOM) window:

a. Navigate to the appropriate folder (the Multisizer 4e default is the SOP folder).

b. Enter a file name in the File Name field, and click **Save.**

NOTE SOM files are identified by the extension .som.

If You are Using the Edit the SOM Window

1 Select **SOP > Edit the SOM** on the Main Menu bar.

2 In the Edit the SOM window, click **Save SOM.** This button appears on each tab.

3 In the Save a Standard Operating Method (SOM) window:

a. Navigate to the appropriate folder (the Multisizer 4e default is the SOP folder).

b. Enter a file name in the File Name field, and click **Save.**

NOTE SOM files are identified by the extension .som.

Editing SOM Settings

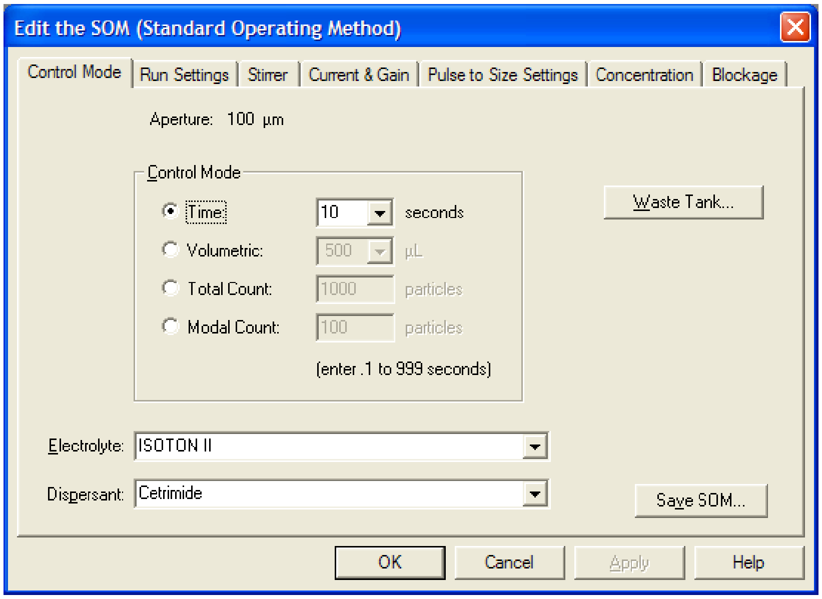
If you are familiar with SOM settings or have already created an SOM using the SOM Wizard (see Using the Create SOM Wizard), you can edit SOM settings.

NOTE The Multisizer 4e software will not allow you to load, edit, or save SOM settings if a Standard Operating Procedure (SOP) is loaded. To remove an SOP, click **Remove SOP** in the Status Panel, or select **SOP** > **Remove the SOP** on the Main Menu bar.

To Edit the Current SOM

Select **SOP > Edit the SOM** on the Main Menu bar. The Edit the SOM window opens.

Figure 5.12 Edit the SOM



The Edit the SOM window displays tabs that duplicate Step 2 through Step 8 of the SOM Wizard (see Using the Create SOM Wizard).

The tabs include:

- **Control Mode.** Select control mode settings to determine when an analysis run will end. The Multisizer 4e can end a run based on elapsed time, total count, modal count, or the volume of sample that has passed through the aperture.

  For more information on the Control Mode window as it appears in the SOM Wizard, see Standard Operating Method: Control Mode.

- **Run Settings.** Select run settings to specify pre-run prompts, determine the number of runs, and enter post-run settings (including automatic file save, export, and print settings).

  For more information on the Run Settings window as it appears in the SOM Wizard, see Standard Operating Method: Run Settings.

- **Stirrer.** Use the Stirrer tab to enter stirrer settings or disable stirrer buttons on the Analyzer Control Panel for the duration of a run.

  For more information on the Stirrer window as it appears in the SOM Wizard, see Standard Operating Method: Stirrer Settings.

  NOTE The Multisizer 4e will automatically position the stirrer based on the Smart Technology Beaker size you select in the SOM.

- **Current & Gain.** Enter the lower limit of a size distribution as well as the electrical field (current) and amplification (gain) that the Analyzer will use as particles pass through the aperture.

  For more information on the Current and Gain window as it appears in the SOM Wizard, see Standard Operating Method: Threshold, Current and Gain. For information on additional features available in the Edit the SOM window, see Using the Current & Gain Tab in the Edit the SOM Window.

- **Pulse to Size Settings.** Enter the number of bins (channels), bin spacing, and other information that will determine the range and resolution of analysis results. After running the analysis, you can change these settings by selecting **Calculate > Convert Pulses to Size** on the Main Menu bar.

  For more information on the Pulse to Size Settings window as it appears in the SOM Wizard, see Standard Operating Method: Pulse to Size Settings.

- **Concentration.** Enter information about sample volume, density, and dilution.

  For more information on the Concentration window as it appears in the SOM Wizard, see Standard Operating Method: Concentration Information.

- **Blockage.** Enter settings that will alert the software to an aperture blockage and select steps the Multisizer 4e will follow when it detects a blockage.

  For more information on the Blockage window as it appears in the SOM Wizard, see Standard Operating Method: Blockage Detection.

CAUTION

Do not attempt to unblock the aperture using an object or your finger. The Multisizer 4e unblock procedure uses a reverse flow of electrolyte through the aperture orifice to remove stuck particles. Any other method of removing the trapped particles could damage the aperture. For more information, see Unblocking Small Aperture Tubes in CHAPTER 4, Installing and Calibrating an Aperture Tube.

Click **Save SOM**, available on every tab, to save the current SOM settings.

The SOM Settings (SOP Loaded) Window

If you select **SOP > Edit the SOM** on the Main Menu bar and the SOM Settings (SOP Loaded, no changes allowed) window opens, remove the SOP before editing current SOM settings.

To remove an SOP, click **Remove SOP** in the Status Panel, or select **SOP > Remove the SOP** on the Main Menu bar.

Differences between the SOM Wizard and Edit the SOM windows

The seven tabs of the Edit the SOM window duplicate Step 2 through Step 8 of the SOM Wizard.

The first and last steps of the SOM Wizard (Step 1 - Description and Step 9 - Summary of Settings) do not appear on the Edit the SOM tabs.

If you are not using the SOM Wizard:

- To enter electrolyte and dispersant information, use the Edit the SOM / Control Mode tab.
- To set automatic file name generation settings (see Creating Automatically Generated Analysis File Names), use the Run Settings tab.
- To view current SOM settings, select **SOP > SOM Info** on the Main Menu bar.

- To manually set the stirrer position and copy the angle into the SOM, use the Edit the SOM / Stirrer tab.

Click **Save SOM**, available on every tab, to save the current SOM settings.

Loading an SOM

If you saved different Standard Operating Methods to one or more file(s), you can load a different SOM depending on the type of analysis you will be running.

NOTE If a Standard Operating Procedure (SOP) is loaded, the Multisizer 4e software will not allow you to edit, save, or load an SOM. To remove an SOP, click **Remove SOP** in the Status Panel, or select **SOP** > **Remove the Sop** on the Main Menu bar.

To Load a Saved SOM File

1. Select **SOP** > **Load SOM** on the Main Menu bar.

2. In the Load Standard Operating Method (SOM) window, navigate to the appropriate folder.

3. Select the desired SOM, and click **Open**.

   NOTE SOM files are identified by the extension .som.

Using a Standard Operating Procedure (SOP)

Creating an SOP

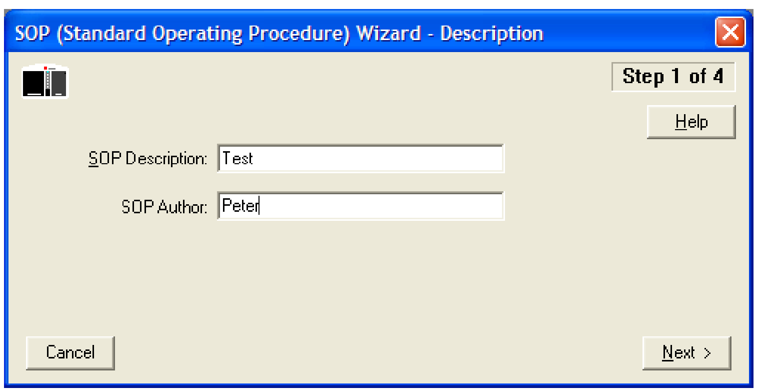
A Standard Operating Procedure (SOP) combines one Preferences file and one Standard Operating Method (SOM) into an overall settings file for a particular type of analysis. Within the SOP, the SOM file determines sample run settings, while the Preferences file determines view and print specifications.

If you have saved default or custom SOM and Preferences files, you can create one or more Standard Operating Procedures (SOPs). Standard Operating Procedures are helpful if you are running several types of analyses that require different settings.

NOTE If you are running the Multisizer 4e software in security mode, the option to create a Standard Operating Procedure will not be available to all user types.

To Create a Standard Operating Procedure

1 Select **SOP > Create SOP Wizard** on the Main Menu bar. The SOP (Standard Operating Procedure) Wizard - Description window opens.



NOTE If you are running the Multisizer 4e software in security mode, the **SOP** > **Create SOP Wizard** menu item will appear only to user types with sufficient privileges. The system administrator sets privileges for each user type.

2 In the SOP Description field, enter a description to identify the SOP. This field is used as the default SOP file name.

3 In the SOP Author field, enter the name of author.

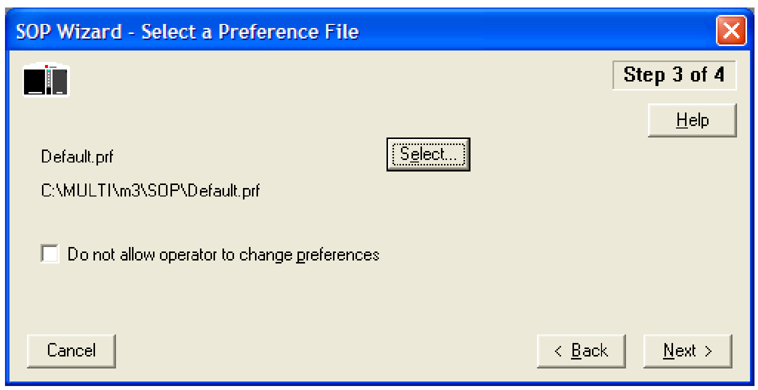
NOTE You can exit the SOP Wizard at any time; however, your changes will be lost unless you save them. Go to Step 4 in the Wizard, and click **Save**.

4 Click **Next** to proceed to the next window in the Wizard and select an SOM File.

5 In the Select an SOM File window, click **Select**.

6 In the Load a Standard Operating Method (SOM) window, navigate to the SOM you want to include, and click **Open**. The name of the selected SOM file appears in the Wizard.

7 Click **Next** to proceed to the next window in the Wizard, Select a Preference File.

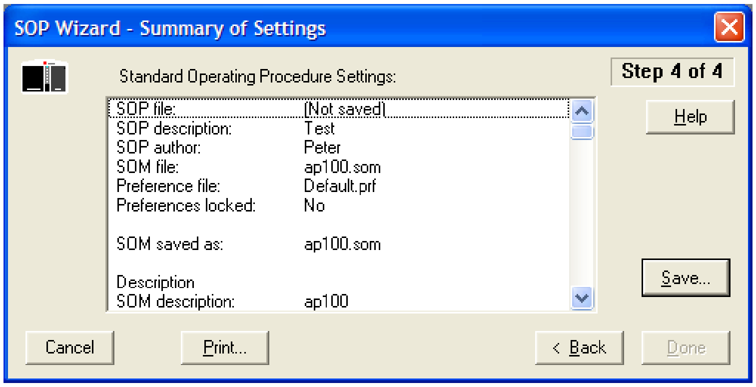


8 In the Select a Preference File window, click **Select**.

9 In the Load Preferences window, navigate to the Preferences file you would like to include, and click **Open**. The name of the selected Preferences file appears in the Wizard.

10 Select Do not allow operator to change preferences to safeguard the format of print and view settings.

11 Click **Next** to proceed to the final step, Summary of Settings.



12 In the Summary of Settings window, click **Save**.

13 In the Save a Standard Operating Procedure window, navigate to the appropriate folder (the Multisizer 4e default is the SOP folder), and click **Save**.

14 To print the SOP summary, click **Print**.

15 To close the SOP, click **Done**.

Loading an SOP

1 Select **SOP > Load an SOP** on the Main Menu bar.

2 In the Load a Standard Operating Procedure (SOP) window, navigate to the appropriate folder.

3 Select the desired SOP, and click **Open**.

NOTE SOP files are identified by the extension .sop.

CHAPTER 6

Setting View and Print Preferences

Creating a Preferences File

To set viewing, exporting, and printing preferences for analysis files, you can use the Create Preferences Wizard or the Edit Preferences window.

You can save your preferences to a file or create a default preferences file. After saving a preferences file, you can include the file in a Standard Operating Procedure. When you load the Standard Operating Procedure (SOP), preference settings in the SOP will determine how analysis graphs and data appear, both on-screen and in printed reports.

If you use the Create Preferences Wizard, the Multisizer 4e software guides you through nine steps. You will only be able to save your preferences in the last step of the Wizard.

If you use the Edit Preferences window, you can move between eight tabs to create or change specific settings. You can save your changes from any tab within the window.

NOTE If you open an analysis file and change its appearance using options on the Run Menu bar, the format of the printed report will not change. To change the format of a printed report, use the Create Preferences Wizard or the Edit Preferences window. Samples of data file formats are included in a subfolder of the Multisizer 4e software called Demodata.

To Create or Change Your Preferences Using the Wizard

1. If the software is not connected to the Analyzer, select **Run > Connect to Multisizer 4e** on the Main Menu bar.

2. Select **SOP > Create Preferences Wizard** on the Main Menu bar. The Create Preferences Wizard opens.

   NOTE If you have not connected the Multisizer 4e software to the Analyzer, the Create Preferences Wizard is available by selecting **Preferences > Create Preferences Wizard** on the Main Menu bar. Different drop-down menus appear in the Main Menu bar when the software is not connected to the Analyzer.

The Create Preferences Wizard guides you through the following steps:

- Step 1: Printed Report

- Step 2: Statistics
- Step 3: Averaging and Trend
- Step 4: Export
- Step 5: Page Setup
- Step 6: Graph Options
- Step 7: Fonts and Colors
- Step 8: View Options
- Step 9: Summary of Settings

To Create or Change your Preferences Using the Edit Preferences Window

1 If the software is not connected to the Analyzer, select **Run > Connect to Multisizer 4e** on the Main Menu bar.

2 On the Main Menu bar, select **SOP > Edit Preferences**. The Edit Preferences window opens.

NOTE If the Multisizer 4e software to the Analyzer is not connected, the Edit Preferences window is available by selecting **Preferences** > **Edit Preferences** on the Main Menu bar. Different drop-down menus appear in the Main Menu bar when the software is not connected to the Analyzer.

The Edit Preferences window displays the following tabs:

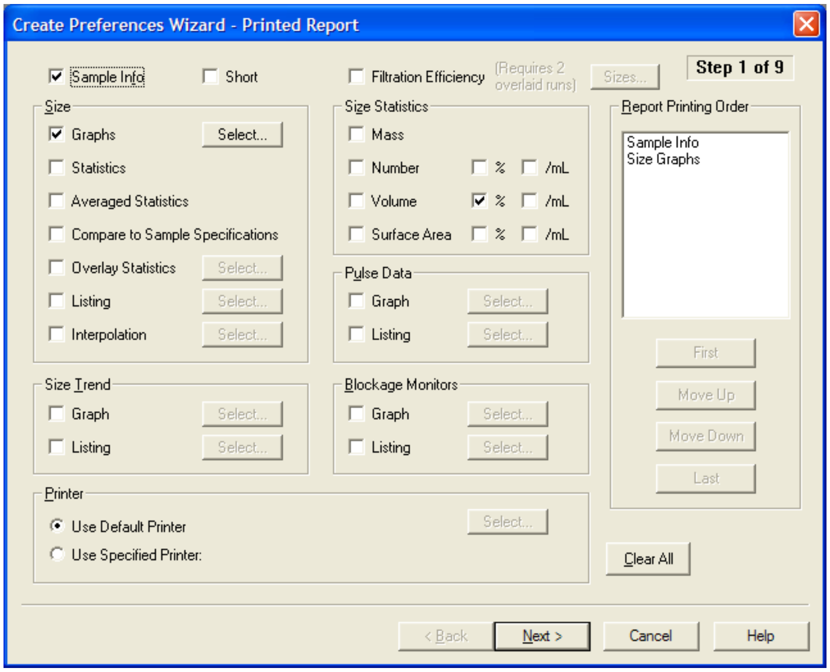
- Printed Report
- Statistics
- Averaging and Trend
- Export
- Page Setup
- Graph Options
- Fonts and Colors
- View Options

3 To see a summary of currently loaded preferences, select **SOP > Preferences Info** on the Main Menu bar.

Preferences: Printed Report

Use Step 1 of the Create Preferences Wizard (Printed Report) or the Printed Report tab of the Edit Preferences window to select analysis data, graphs, or listings that will appear on the screen and in printed reports.

Figure 6.1 Create Preferences Wizard - Printed Report



Printing Sample Information

The Enter Sample Info dialog box allows you to print either an extensive or short version of the sample information in your report.

To Choose All Sample Information

To print all sample and run information in the report, including information previously entered in the Enter Sample Info dialog box (see Entering Sample Information in CHAPTER 7, Analyzing a Sample), check the Sample Info box in the top left corner of the Printed Report window.

To Choose Less Sample Information

To include less information, select both Sample Info and Short at the top of the Printed Report window.

The table below shows the information that will print if you select Sample Info or both Sample Info and Short.

Table 6.1 Printing Sample Information in Reports

	Sample Info	Sample Info and Short
Date	X	X
File name	X	X
Group ID	X	X
Sample ID	X	X
Operator	X	
Run number	X	
Electrolyte	X	
Dispersant	X	
Aperture diameter	X	
Aperture current	X	
Size bins	X	X
Total count	X	X
Count	X	
Control method	X	
Acquired	X	X
Electrolyte volume	X	
Analytic volume	X	
Sample	X	
Background run	X	X

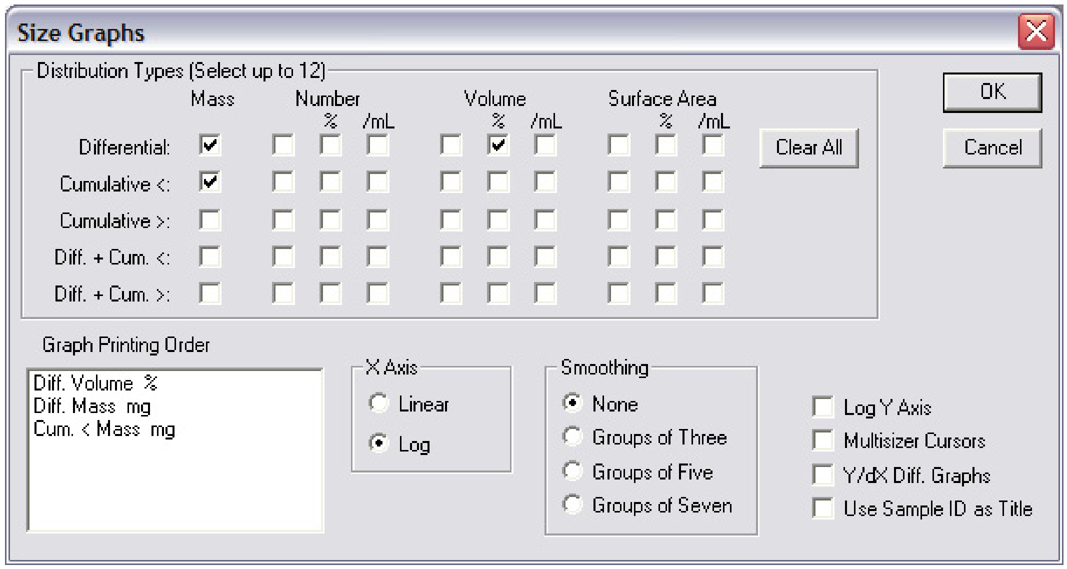
Selecting Graphs, Statistics, and Data for Printing

In the Preferences / Printed Report window, options in the Size pane allow you to select the analysis graphs and data you want to print. When you select an option in the Size pane, the selection appears in Report Printing Order at the right of the window.

Including a Size Distribution Graph in Printed Reports

1 In the Size pane of the Print Report window (see Preferences: Printed Report), select Graphs.

2 Click **Select** to choose different options for viewing the size distribution. The Size Graphs window opens.



3 In the Size Graphs window, select each type of graph you want to print. You can select up to 12 different graph types. Selections appear in the Graph Printing Order field at the bottom of the window.

Table 6.2 Size Graphs Window: Selections

Select...	To plot a graph using...
Differential	The number of particles, total volume, or total surface area in each bin.
Cumulative	The total number of particles, volume, or surface area below or above each bin edge.
Mass	Differential or cumulative results by particle mass (volume times density).
Number	Differential or cumulative results based on the number of particles in each bin.
Volume	Differential or cumulative results by particle volume.
Surface Area	Differential or cumulative results by particle surface area.

4 In the X Axis pane, select whether you want to use a linear or a logarithmic X axis to display bin sizes.

5 In the Smoothing pane, select the amount of smoothing in each graph. Smoothing averages each bin with its neighbors (in groups of three, five, or seven bins) to reduce the resolution and increase clarity.

6 Select Log Y Axis to use a logarithmic Y axis to display the amounts in each bin.

7 Select Multisizer Cursors to print cursors in each graph where they were placed when you saved the analysis.

8 Select Y/Dx Diff. Graphs to normalize the height of each bin based on the bin's width. This option is useful when the bins are not evenly spaced.

9 Select Use Sample ID As Title to include a sample ID title in printed reports. You can select a different analysis title in Preferences / Page Setup window by selecting Include Custom Title.

10 Click **OK** to save your changes.

Including Statistics in Printed Reports

In the Size pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Statistics to include statistical information in printed reports. You can choose which statistics to include in Step 2 of the Create Preferences Wizard or the Statistics tab of the Edit Preferences window.

Including Averaged Statistics in Printed Reports

If you have completed several sample runs and averaged the results, you can include these results in printed reports. In the Size pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Averaged Statistics to include this information.

Including a Comparison to Sample Specifications in Printed Reports

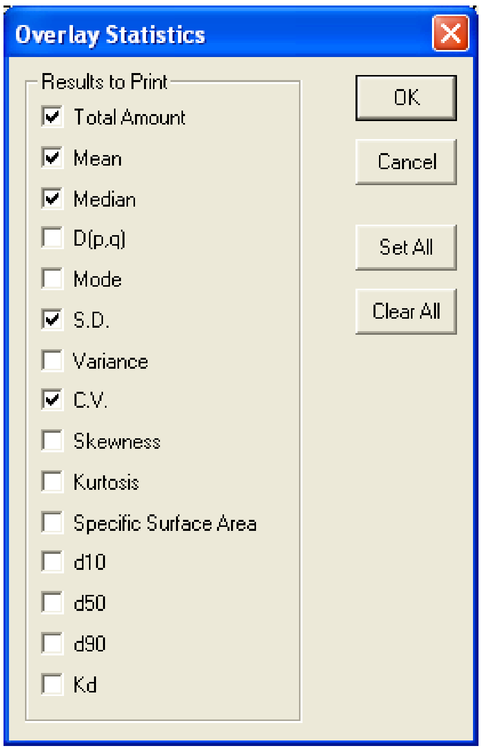
The Multisizer 4e software allows you to compare data files to previously entered sample specifications (see Viewing Sample Specifications in CHAPTER 7, Analyzing a Sample. In the Size pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Compare to Sample Specifications to include this information in printed reports.

Including Overlay Statistics in Printed Reports

If you have created an overlay file (see Working with Multiple Analysis Files in CHAPTER 8, Working with Analysis Files), you can include overlay statistics in printed reports.

1 In the Size pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Overlay Statistics.

2 Click **Select** to choose the overlay statistics that will appear in the report. The Overlay Statistics window opens.



3 In the Overlay Statistics window, select each statistic you want to include.

Table 6.3 Overlay Statistics Options

Select...	To print...
Total Amount	Total number, volume, or surface area in each distribution (the sum of the bins).
Mean	The mean of each distribution.
Median	The median of each distribution.
D(p,q)	Results of a formula used to calculate particle size, surface area, weight, or volume based on particle diameter. Set the D(p,q) formula on the Preferences Statistics tab (see Preferences: Statistics).
Mode	The mode of each distribution.
S.D.	The statistical deviation.
Variance	The variance of each distribution.
C.V.	The co-efficient of variation.
Skewness	The degree of asymmetry in the frequency distribution.
Kurtosis	The steepness of the graphical curve near the mean of the distribution.
Specific Surface Area	The specific surface area.

Table 6.3 Overlay Statistics Options

Select...	To print...
d10, d50, d90	The particle diameter that divides the distribution by percentage (for example, d10 is the diameter at which 10% of the particles are smaller and 90% are larger; d50 is the median).
Kd	The calibration constant.

4 To print all statistics, click **Set All**. To clear the fields, click **ClearAll**.

5 Click **OK** to save your settings.

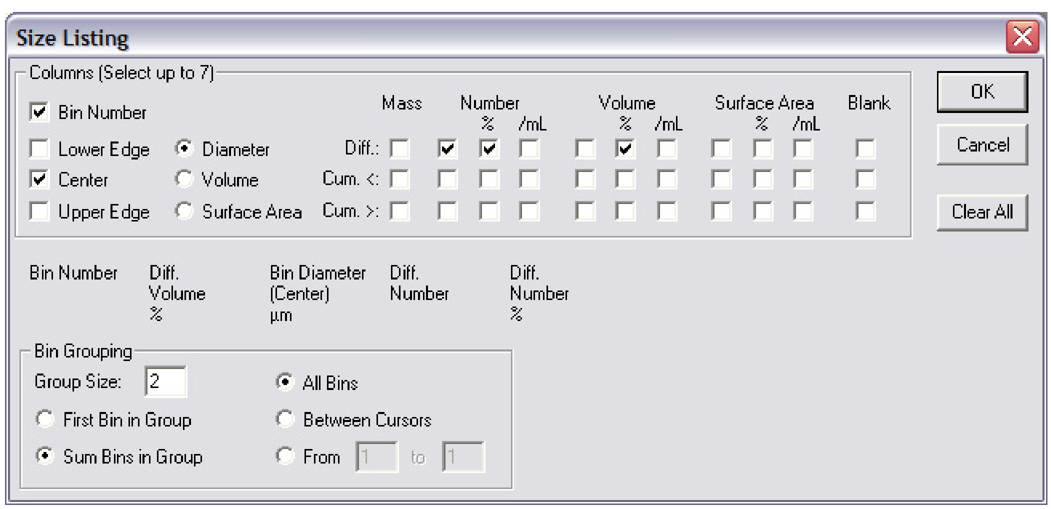
Including a Listing in Printed Reports

You can print analysis results as a listing.

To Include a List of Data for Each Size Distributions Bin

1 In the Size pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Listing.

2 Click **Select** to choose which data will appear in the report. The Size Listing window opens.



3 In the Size Listing window, select each statistic you want to include. You can select up to seven columns of statistics.

a. In the Columns pane:

- Select Bin Number to include the bin number next to each bin value.

- Select Lower Edge, Upper Edge, or Center to print the smallest, largest, or median particle size, volume, or surface area in each bin.
- Select up to seven columns of statistics.

b. In the Bin Grouping pane:

- Enter a number in the Group Size field to group bins, if desired. Grouping bins shortens the listing. Select First Bins in Group to display the value of the first bin in each group, or select Sum Bins in Group to display the value of the sum of the bins in each group.
- Select All Bins to display results from the full range of the analysis.
- Select Between Cursors to display only results between your defined cursors. For information on setting and moving cursors, see Using Cursors in Graphs in CHAPTER 8, Working with Analysis Files.
- Select From and enter specific points in the From and to fields to define statistical limits.

4 To clear all fields, click **ClearAll**.

5 Click **OK**.

Including Interpolation Data in Printed Reports

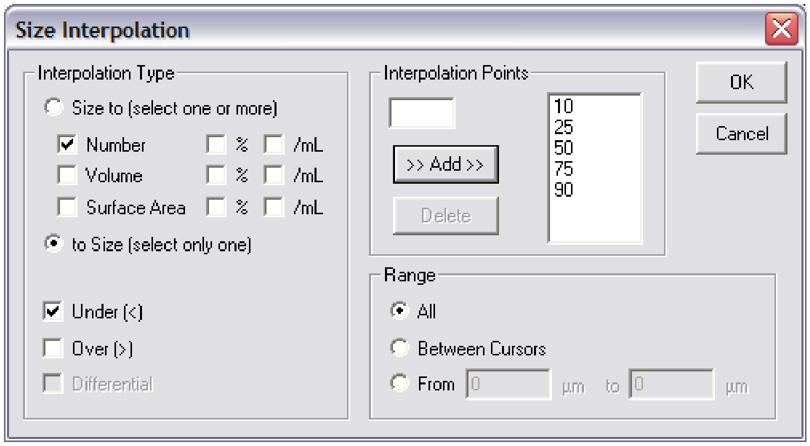
You can create sample data reports that display statistical results within defined boundaries, or interpolation points.

Using interpolation points allows you to define your own bin edges and create any number of bins, of different or equal sizes. Interpolations are similar to percentiles, but are more flexible. Percentiles are limited to eight equally spaced bins.

To Include a List of Interpolation Point Data

1 In the Size pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Interpolation.

2 Click **Select** to set the interpolation points and range. The Size Interpolation window opens.



3 In the Interpolation Type pane of the Size Interpolation window:

a. Select Size to if you want to create interpolation points based on particle diameter size (that is, particle diameters define bin edges). You can select up to nine statistics to display based on particle size interpolation points.

b. Select to Size if you want to create interpolation points based on the statistic you select in the Interpolation Type pane (Number, Volume, Surface Area). You can select only one statistical interpolation point to correlate to particle size.

c. Select Number, Volume, Surface Area,%, and/or /mL to display selected statistics in columns.

d. Select Under (<), Over (>), or (if you selected Size to), Differential.

Table 6.4 Interpolation Display Data

If you select...	Then columns will display...
Under (<)	Statistical results below the interpolation point. The interpolation point becomes the upper edge of the bin.
Over (>)	Statistical results above the interpolation point. The interpolation point becomes the lower edge of the bin.
Differential	The number of particles between interpolation points.

4 In the Interpolation Points pane:

a. To add an interpolation point, enter the new point and click **Add**.

b. To delete an interpolation point, select the point and click **Delete**.

5 In the Range pane:

a. Select All to display results from the full range of the analysis.

b. Select Between Cursors to display only results between your defined cursors. For information on setting and moving cursors, see Using Cursors in Graphs in CHAPTER 8, Working with Analysis Files.

c. Select From and enter specific points to limit the range without inserting cursors in a graph.

6 Click **OK** to return to the Create Preferences Wizard.

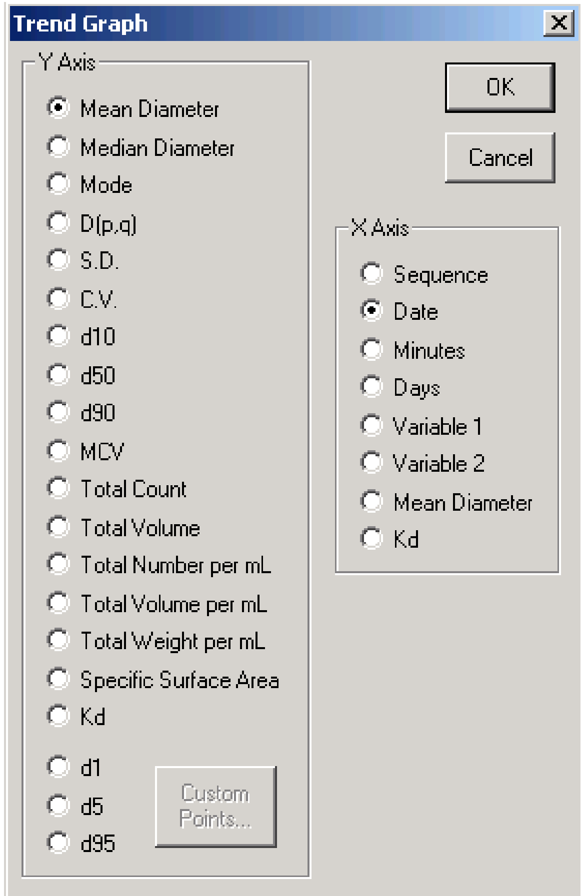
Including Size Trend Information in Printed Reports

In the Size Trend pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Graph, Listing, or both to include size trend information in printed reports. For more information on creating a trend analysis from multiple files, see Creating a Size Trend Analysis in CHAPTER 8, Working with Analysis Files.

To Include a Trend Graph

1 In the Size Trend pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Graph.

2 Click **Select.** The Trend Graph window opens.



3 In the Trend Graph window, select variables for the X and Y axes.

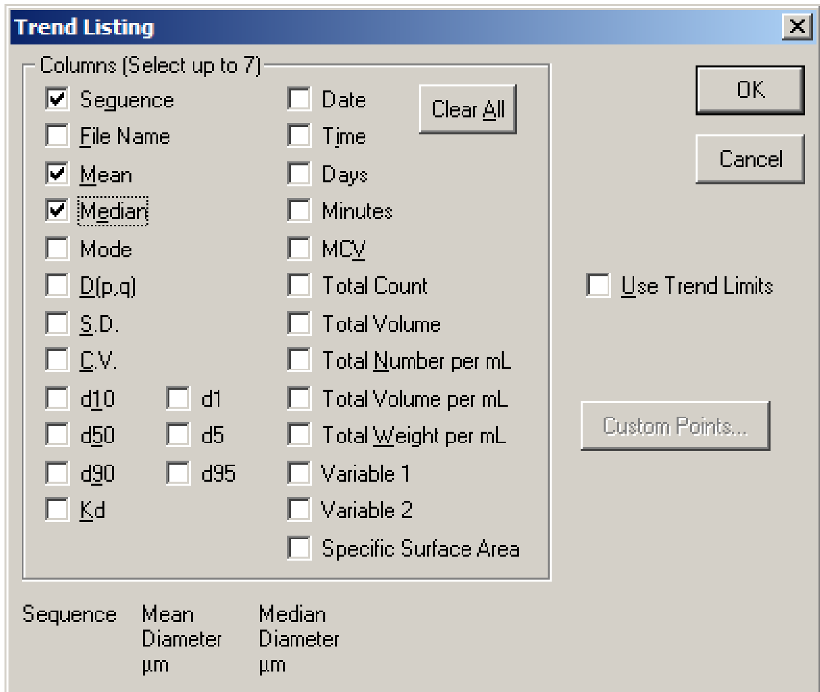
To change the d(x) values to custom percentages, click **Custom Points** and enter new values in the Custom Trend Points window. These values allow you to divide the particle distribution of each run by percentage of particle size above and below a certain diameter. For example, d10 divides the particle distribution by the diameter at which 10% of the particles are smaller and 90% are larger.

4 Click **OK** to save your settings.

To Include a Trend Listing

1 In the Size Trend pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Listing.

2 Click **Select**. The Trend Listing window opens.



3 In the Trend Listing window, select variables to display in columns.

- To change the d(x) values to custom percentages, click **Custom Points** and enter new values in the Custom Trend Points window. These values allow you to divide the particle distribution of each run by percentage of particle size above and below a certain diameter. For example, d10 divides the particle distribution by the diameter at which 10% of the particles are smaller and 90% are larger.
- Select Use Trend Limits to use the trend limits selected in the Preferences / Averaging and Trend window (see Preferences: Averaging and Trend).
  - — Select Use Trend Limits if you want the software to recalculate trend limits based on all included runs when you add one or more analysis runs to your trend file.
  - — Deselect Use Trend Limits if you do not want trend limits to display automatically on the graph when you open a trend file.

4 Click **OK** to save your settings.

Including Size Statistics in Printed Reports

You can select the type of size distribution for printing or on-screen display. In the Size Statistics pane of the Preferences / Printed Report window (see Preferences: Printed Report), select whether you want to include data on Mass, Number, Volume, and Surface Area by %, and/or mL.

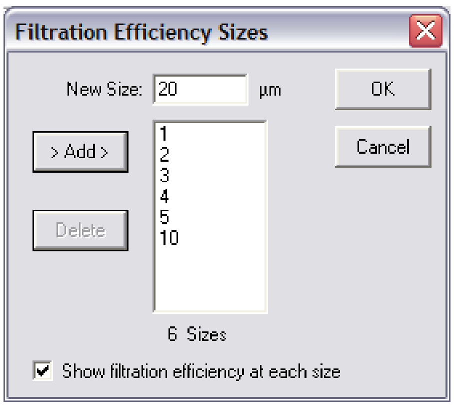
Including Filtration Efficiency in Printed Reports

In order to include filtration efficiency in printed reports, you will need to compare two or more sample analyses so that the Multisizer 4e software can calculate filtration efficiency.

To Include Filtration Efficiency Data

1 In the upper right of the Preferences / Printed Report window (see Preferences: Printed Report), select Filtration Efficiency.

2 Click **Sizes** to the right of Filtration Efficiency. The Filtration Efficiency Sizes window opens.



3 In the Filtration Efficiency Sizes window:

- **a.** In the New Size field, enter a particle size for which you want to show filtration efficiency.
- **b.** Click **Add**.
- **c.** Repeat Step a and Step b for each particle size.
- **d.** Select Show filtration efficiency at each size to include filtration efficiency data for each particle size you entered. If this option remains deselected, the printed report will display overall filtration efficiency.
- **e.** Click **OK** to save your settings.

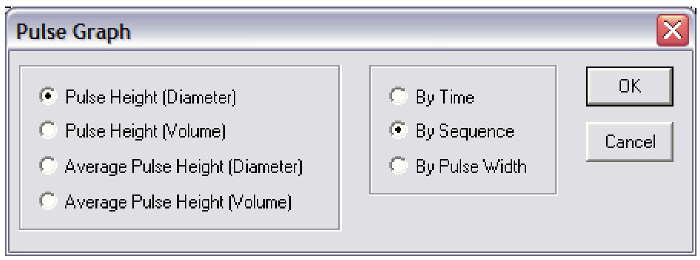
Including Pulse Data in Printed Reports

If you selected the option to save pulse data when you created a Standard Operating Method (see Standard Operating Method: Pulse to Size Settings in CHAPTER 5, Selecting Analysis Settings: SOM and SOP), you can use the Pulse Data pane in the Preferences / Printed Report window (see Preferences: Printed Report) to select how pulse data will appear in the printed report. You can display pulse data as a Graph, a Listing, or both.

To Show Pulse Data in a Graph

1 In the Pulse Data pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Graph.

2 Click **Select**. The Pulse Graph window opens.



3 In the Pulse Graph window, select variables for the X and Y axes.

4 Click **OK** to save your settings.

To Show Pulse Data in a Listing

1 In the Pulse Data pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Listing.

2 Click **Select**. The Pulse Listing window opens.

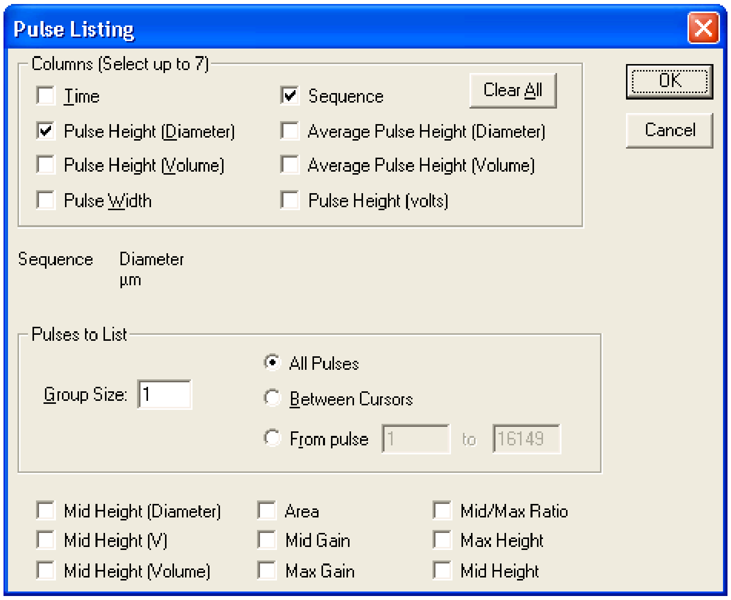
3 In the Pulse Listing window:

a. In the Columns pane, select the values you want to include in the listing.

b. In the Pulses to List pane, enter the Group Size.

- Select All Pulses to display results from the full range of the analysis.
- Select Between Cursors to display only results between your defined cursors. For information on setting and moving cursors, see Using Cursors in Graphs in CHAPTER 8, Working with Analysis Files.

- Select From pulse and enter specific points to limit the range without inserting cursors in a graph.



4 Click **OK** to save your settings.

Including Blockage Monitor Data in Printed Reports

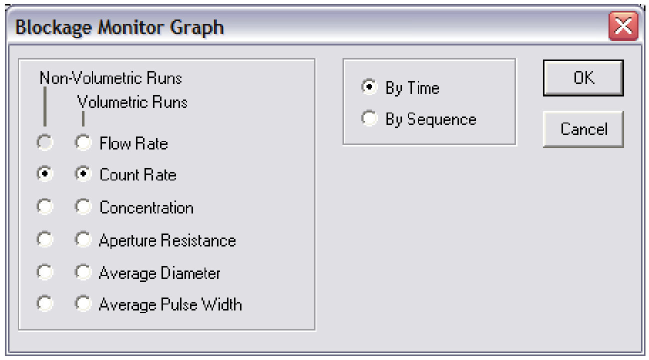
Use the Blockage Monitors pane in the Preferences / Printed Report window (see Preferences: Printed Report) to select how blockage monitor information will appear in the printed report. You can display blockage monitor data as a Graph, a Listing, or both.

NOTE The Multisizer 4e software saves blockage monitor data with every analysis. Printed reports can display blockage monitor data independently of blockage settings in the SOM.

To Show Blockage Monitor Data in a Graph

1 In the Pulse Data pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Graph.

2 Click **Select**. The Blockage Monitor Graph window opens.



3 In the Blockage Monitor Graph window:

a. Select how blockage monitor data will appear for Volumetric Runs and Non-Volumetric Runs.

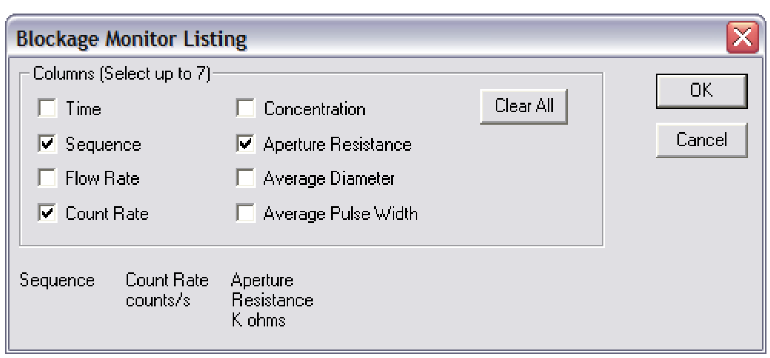
b. Select By Time or By Sequence.

c. Click **OK** to save your settings.

To Show Blockage Monitor Data in a Listing

1 In the Blockage Monitor Data pane of the Preferences / Printed Report window (see Preferences: Printed Report), select Listing.

2 Click **Select**. The Blockage Monitor Listing window opens.



3 In the Blockage Monitor Listing window, select the blockage monitor data you want to include in the listing. The display sequence appears in the lower left corner of the window.

4 To clear all selections, click **Clear All**.

5 Click **OK** to save your settings.

Setting the Print Order for Reports

When you select data and display options in the Preferences / Printed Report window (see Preferences: Printed Report), the options you select appear in the Report Printing Order pane.

1 In the Report Printing Order field, click the row you want to move.

2 Click First, Last, **Move Up**, or **Move Down** to move the row to the desired position.

Preferences: Statistics

Use Step 2 of the Create Preferences Wizard (Statistics) or the Statistics tab of the Edit Preferences window to select the statistics you want to display on-screen and in printed reports.

To proceed from Step 1 to Step 2, click **Next.**

NOTE When you open an analysis file, you can select different statistics and viewing options using the Run Menu bar (see Run Menu in CHAPTER 3, Software Overview). Display changes you make using Run menu items will not change the content of the printed report.

Figure 6.2 Create Preferences Wizard - Statistics

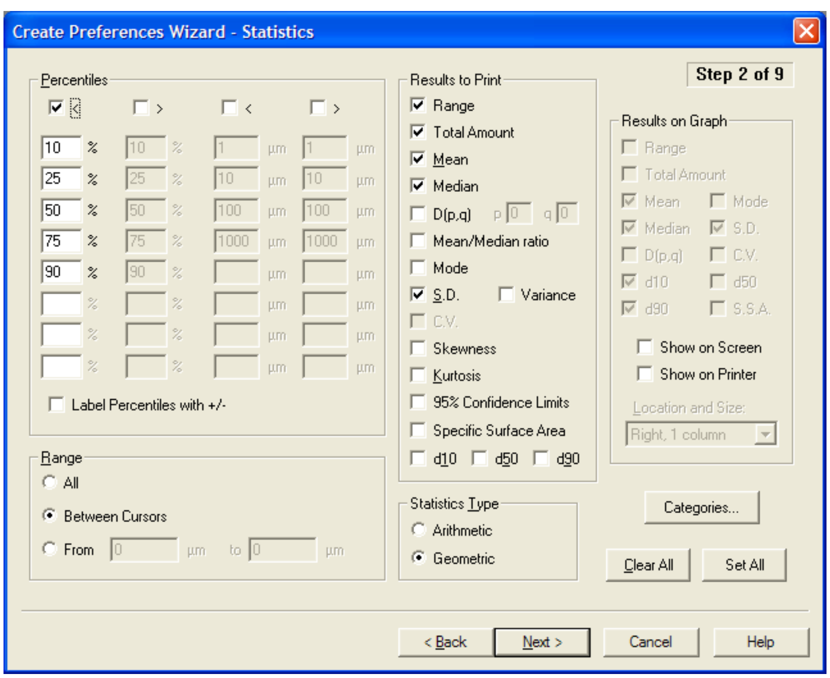


Table 6.5 Statistics Display Preferences

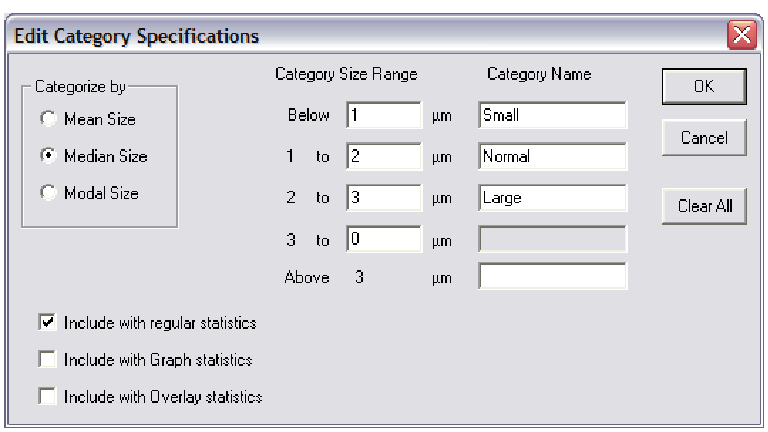
Statistic Type	Description
Percentile	To display percentiles, select < less than and > greater than and enter percentile boundaries. You can define up to eight percentiles.
Range	• Select All to display results from the full range of the analysis • Select Between Cursors to display only results between your defined cursors. For information on setting and moving cursors, see Using Cursors in Graphs in CHAPTER 8, Working with Analysis Files. • Select From and enter specific points to limit the range without inserting cursors in a graph.
Results to Print	Select the variables you want to display. Your selections determine both printed statistics and the statistics that display when you select **Calculate** > **Statistics** in the Analysis menu.
Statistics Type	Select Arithmetic or Geometric. **NOTE** Select Arithmetic if you selected a Linear X axis in the Preferences / Printed Report window. Select Geometric if you chose a Logarithmic X axis.
Results on Graph	If you selected Graphs in Step 1 of the Wizard (Printed Report), select the statistics you want to display with a graph. Your selections will display in a box of statistics overlaid on each graph.
Location and Size	Specifies the placement and size of the text box containing the statistics you selected in the Results on Graph pane.

Using Statistical Category Names

In the Preferences / Statistics window, click **Categories** to define and name size categories. Size category specifications allow you to simplify statistical results for easy interpretation in printed reports.

To Define and Name Statistical Categories

1 In the Preferences / Statistics window, click **Categories.** The Edit Category Specifications window opens.



2 In the Edit Category Specifications window:

a. In the Categorize by pane, select Mean, Median, or Modal Size.

b. In the Category Size Range column, enter the upper limit for each category. The software displays the upper limit you enter as the lower limit for the next category.

c. In the Category Name column, enter the name of each category.

d. Click **OK** to return to the Preferences / Statistics window (see Preferences: Printed Report).

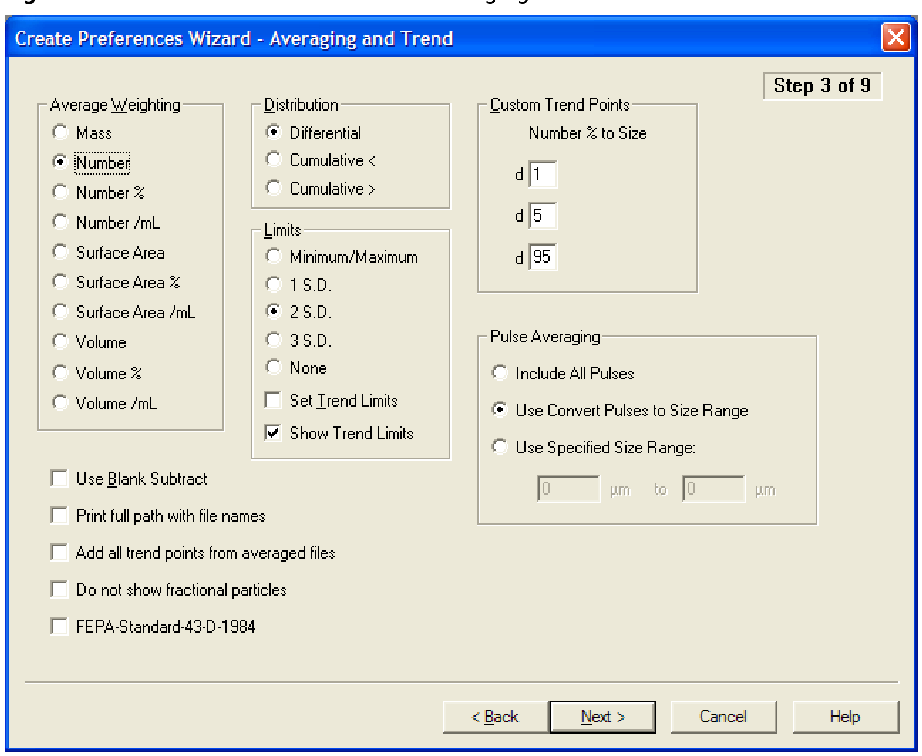
Preferences: Averaging and Trend

Use Step 3 of the Create Preferences Wizard (Averaging and Trend) or the Averaging & Trend tab of the Edit Preferences window to select statistics from averaged or trend analyses that you want to display on-screen and in printed reports.

To proceed from Step 2 to Step 3, click **Next**.

For more information on averaging and trend statistics, see Averaging Analysis Files in CHAPTER 8, Working with Analysis Files and Creating a Size Trend Analysis in CHAPTER 8, Working with Analysis Files.

Figure 6.3 Create Preferences Wizard - Averaging and Trend



In the Preferences /Averaging and Trend window:

- In the Average Weighting pane, select how you will look at analysis results (typically Number/mL) when multiple files are averaged.
- In the Distribution pane:
  - Select Differential to view the number of particles, total mass, total surface area, or total volume in each bin.
  - Select Cumulative < or Cumulative > to view the total number of particles, mass, surface area, or volume below or above each bin edge, respectively.
- In the Limits pane, select trend limit specifications.

Select other options as desired.

NOTE Select Use Blank Subtract to include blank runs previously saved with analysis data when the software averages the runs. After the software has averaged the data (with or without a blank subtract), you cannot change how you view the results.

Use Blank Subtract only applies when you select analysis files for an average or trend file. It functions independently of the **Sample > Load Blank Run** menu item on the Main Menu bar, which applies to new analysis runs.

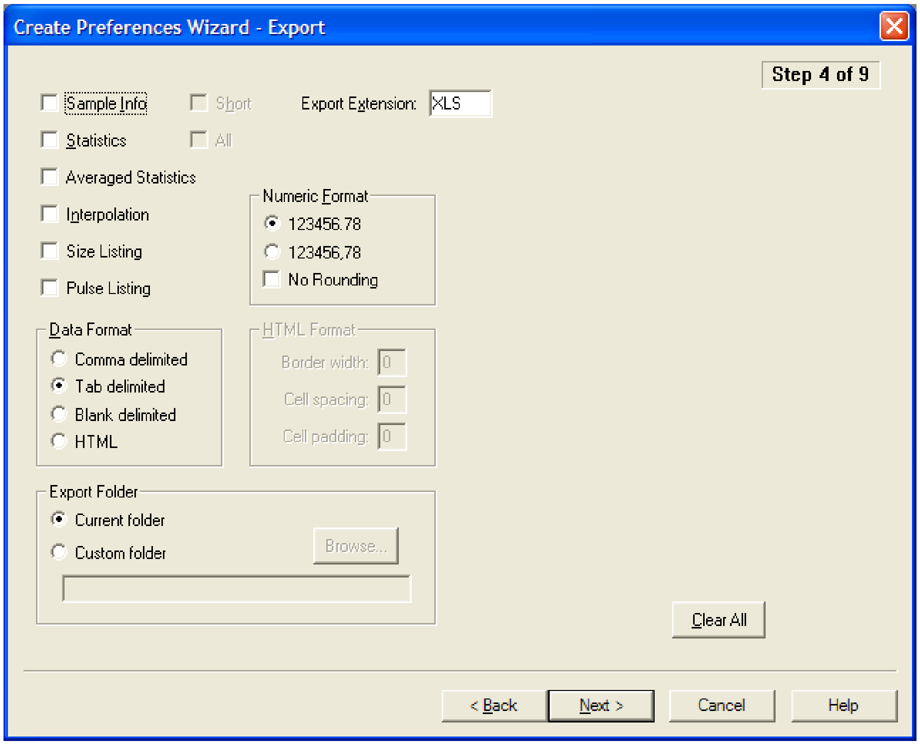
- In the Custom Trend Points pane, set trend percentiles, if desired. d[X] values allow you to divide the particle distribution of each run by percentage of particle size above and below a certain diameter. For example, d10 divides the particle distribution by the diameter at which 10% of the particles are smaller, and 90% larger.
- In the Pulse Averaging pane, select pulse data to be included when viewing pulses as average pulse height or average pulse width.

Preferences: Export

Use Step 4 of the Create Preferences Wizard (Export) or the Export tab of the Edit Preferences window to select statistics you want to export to another file type, such as a Microsoft Excel spreadsheet.

To proceed from Step 3 to Step 4, click **Next**.

Figure 6.4 Create Preferences Wizard - Export



To Export Statistics to Another File Type

1 Select the statistics you want to export.

2 In the Data Format pane, select the type of list formatting you want. Exported list items can be separated by commas, tabs, or blank spaces, or converted into an HTML table.

3 If you select HTML, select table formatting for the exported file in the HTML Format pane.

4 In Export Extension, enter the three-letter extension for the destination file type (for example, XLS for Microsoft Excel). The most common file types are CSV and TXT.

5 In the Numeric Format pane, select the numbers format you want.

6 In the Export Folder pane, allow the Current folder (default) or select the Custom folder and browse to a new file location. The new file path appears in the field.

The exported file name is the analysis file name with the export extension. For information on automatically generated analysis file names, see Creating Automatically Generated Analysis File Names in CHAPTER 5, Selecting Analysis Settings: SOM and SOP.

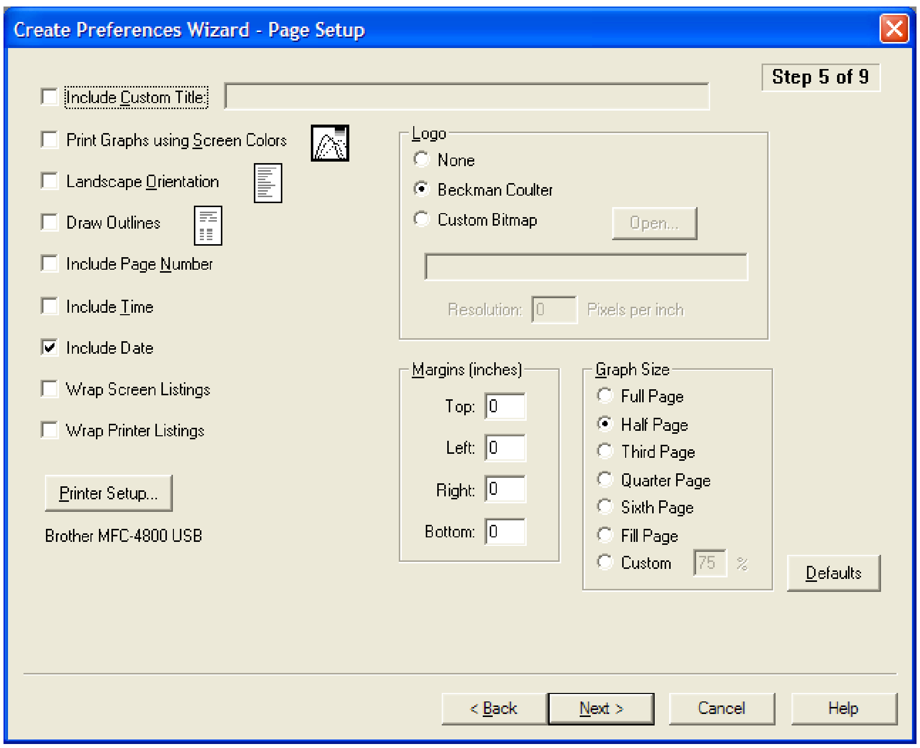
Preferences: Page Setup

Use Step 5 of the Create Preferences Wizard (Page Setup) or the Page Setup tab of the Edit Preferences window to select formatting options and page elements that will appear in the printed report.

To proceed from Step 4 to Step 5, click **Next**.

For information on how to open the Create Preferences Wizard and the Edit Preferences windows, see Creating a Preferences File.

Figure 6.5 Create Preferences Wizard - Page Setup



NOTE In the Graph Size pane, select Full Page to print each graph using an entire page. Click Fill Page to print a graph on a page if the page is empty or less than half full.

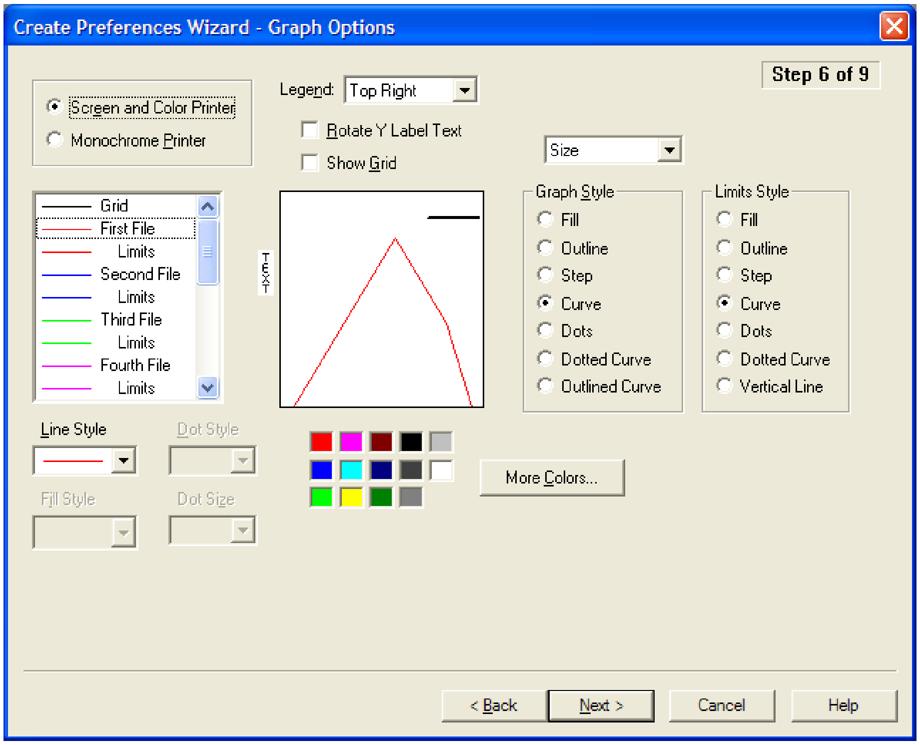
Preferences: Graph Options

Use Step 6 of the Create Preferences Wizard (Graph Options) or the Graph Options tab of the Edit Preferences window to select display, style, and color options for printed graphs.

To proceed from Step 5 to Step 6, click **Next**.

For information on how to open the Create Preferences Wizard and the Edit Preferences windows, see Creating a Preferences File.

Figure 6.6 Create Preferences Wizard - Graph Options



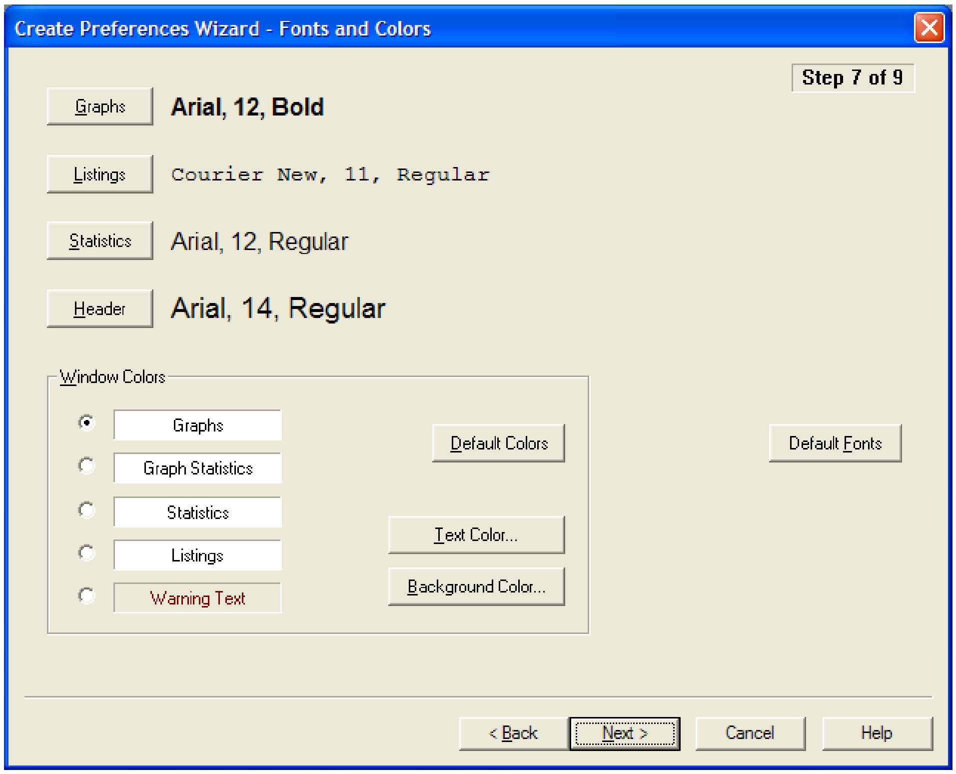
Preferences: Fonts and Colors

Use Step 7 of the Create Preferences Wizard (Fonts and Colors) or the Fonts & Colors tab of the Edit Preferences window to change fonts, text color, and background color for headings, graphs, and statistics in printed reports and on-screen.

To proceed from Step 6 to Step 7, click **Next**.

For information on how to open the Create Preferences Wizard and the Edit Preferences windows, see Creating a Preferences File.

Figure 6.7 Create Preferences Wizard - Fonts and Colors



To Change Font Style and Size in Printed Reports

1 Click **Graphs** > **Listings** > **Statistics** > **Header**.

2 In the Font window, click to select a Font, Font style, and Size.

3 Click **OK** to save your settings.

To Change Text Color on the Screen and in Printed Reports

1 In the Window Colors pane, select Graphs, Graph Statistics, Statistics, Listings, or Warning Text.

2 To change text colors for the selection only (for example, Graphs):

a. Click **Default Colors** to return to Multisizer 4e defaults.

b. Click **Text Color** to open the Color window. Select a color in the palette or define a custom color.

c. Click **OK** to save your settings.

3 To change the background color, click **Background Color** to open the Color window. Select a color in the palette or define a custom color.

Preferences: View Options

Use Step 8 of the Create Preferences Wizard (View Options) or the View Options tab of the Edit Preferences window to select the default graph or listing type that will appear when you complete an analysis, and to select other analysis viewing preferences. These settings will apply to both on-screen graphs or listings and printed reports.

For information on how to open the Create Preferences Wizard and the Edit Preferences windows, see Creating a Preferences File.

Figure 6.8 Create Preferences Wizard - View Options

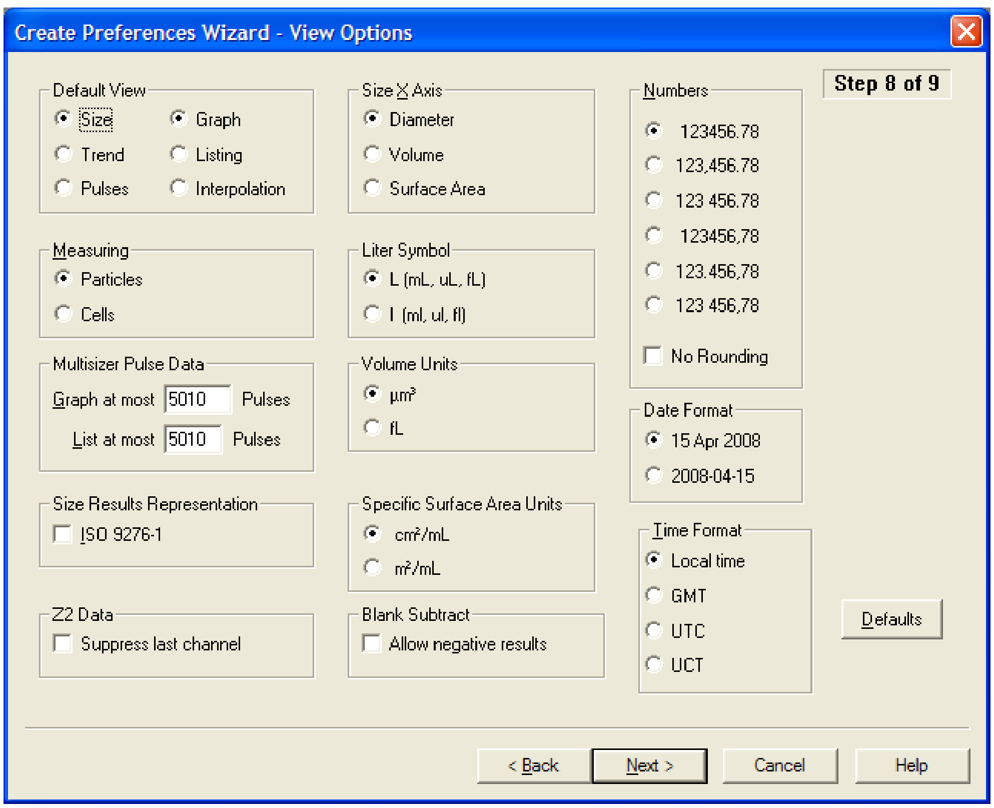


Table 6.6 Preferences Wizard: Set View Options

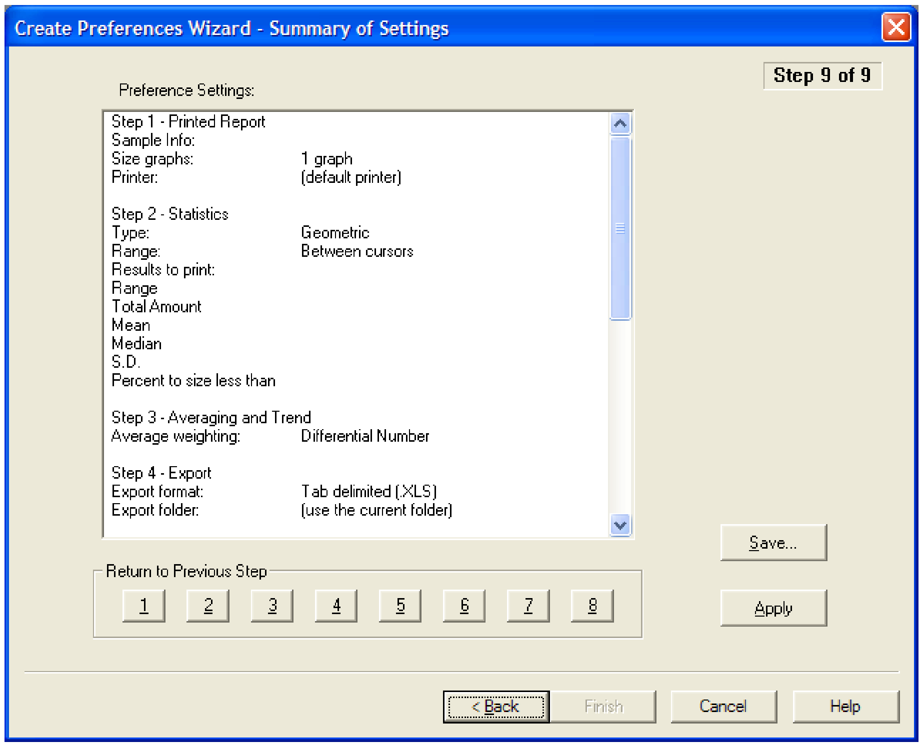
View Preferences	Description
Default View	Select one option in each column to choose how the analysis run is first displayed when the run is complete. You can select Interpolation view only if you select Size in the left column.
Measuring	Select Particles or Cells to change the description that appears on graphs and listings. This setting will not affect analysis specifications.
Multisizer Pulse Data	Select a maximum number of pulses to appear in a graph or listing.
Size Results Representation	Use ISO 9276-1 specifications for Size graph axis labels.
Z2 Data	Select Suppress last channel.
Size X Axis	Select the value that will automatically appear on the X axis of the graph (Diameter, Volume, or Surface Area).
Liter Symbol	Select the symbol you want to use.
Volume Units	Select the volume unit you want to use.
Specific Surface Area Units	Select the surface area units you want to use.
Blank Subtract	Select Allow negative results if you want to display the negative result when a bin in the blank run contains more particles than a bin in the sample run. If you do not select this option, the displayed value will be zero (0).
Numbers	Select the numerical format you want to use.
Date Format	Select the date format you want to use.
Time Format	Select the time format you want to use.

Reviewing Your Preferences Settings

To Review Your Preferences Using the Create Preferences Wizard

1 Select **SOP** > **Create Preferences Wizard** on the Main Menu bar.

2 In the Create Preferences Wizard, click **Next** until you reach the Summary of Settings window (Step 9).



3 In the Summary of Settings window, review your settings. To change a setting, click **Back** or one of the numbered buttons below the text box to return to a previous step.

4 If you clicked **Back** and updated your preferences, click **Apply**, then click **Finish** to exit the Wizard.

NOTE In the Create Preferences Wizard / Summary of Settings window, **Finish** is not available until you click **Apply**.

To View a Summary of Current Preferences from the Main Menu Bar

1 Select **SOP** > **Preferences Info** on the Main Menu bar. The Preference Information window opens.

Saving a Preferences File

If you will use different print and view preferences for different analysis types, you can save different preferences settings to a individual files and load them into the software.

If you are using only one set of preferences, you can configure the Multisizer 4e software to automatically save updates to your preferences to a default file when you exit the software.

To Save Your Preferences Using the Create Preferences Wizard

1. Select **SOP** > **Create Preferences Wizard** on the Main Menu bar.
2. In the Create Preferences Wizard, click **Next** until you reach the Summary of Settings window (Step 9).
3. In the Summary of Settings window, click **Save**.
4. In the Save Preferences dialog box, enter the Author name and Description, if desired.
5. Click **Save**.
6. In the second Save Preferences window, navigate to a new folder (if desired), and enter a file name in the File Name field. Preferences files use the extension .prf.
7. Click **Save**.

Automatically Updating the Default Preferences File

You can automatically save your printing preferences to the default preferences file when you exit the Multisizer 4e application.

To Save Printing Preferences

1. Select **Configuration** > **Configure Preferences** on the Main Menu bar. The Configure Preferences window opens.

NOTE If you are running the Multisizer 4e software in security mode, certain user types may not have sufficient security privileges to view the Configuration menu. In security mode, the Configure Preferences menu item appears in the Administrator or Supervisor drop-down menu if the user has logged in as an administrator or supervisor, respectively. The Configuration menu is available to all users when running the software in no security mode.

2 In the Configure Preferences window, check the box to automatically save your preferences to the default .prf file when you exit the Multisizer 4e software.

⚠ CAUTION

If you check the Automatically save box, the Multisizer 4e software will overwrite previous default preference settings when you exit the application. If you will be working with different preference settings, save your preferences to files with different names (see Saving a Preferences File).

3 Click **OK** to save your settings.

Analyzing a Sample

Analyzing a Sample

Analyzer Checklist

Beckman Coulter recommends that you follow this checklist before each sample run, even if no change in Analyzer settings is required. For information on preparing for a sample run, see Preparing the Analyzer for Sample Runs in CHAPTER 2, Analyzer Overview.

Before Starting a Sample Run

1. Power on the Analyzer, open the Multisizer 4e software, and connect to the Analyzer (see Opening the Multisizer 4e Software in CHAPTER 3, Software Overview).

2. Fill the system with electrolyte solution.

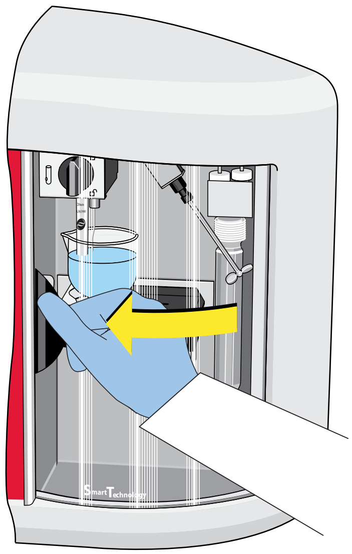
3. Secure the correct aperture tube in place (see Installing an Aperture Tube in CHAPTER 4, Installing and Calibrating an Aperture Tube).

4. Empty the Waste Tank, if necessary.

5. Place a Multisizer 4e Smart Technology Beaker or Accuvette ST with clean electrolyte and sample on the platform. See Step 4: Place a Beaker of Clean Electrolyte on the Platform in CHAPTER 4, Installing and Calibrating an Aperture Tube.

6. Raise the platform and make sure that the aperture is immersed in the electrolyte. See Step 4: Place a Beaker of Clean Electrolyte on the Platform in CHAPTER 4, Installing and Calibrating an Aperture Tube.

7 Close the Sample Compartment door.



Selecting Analysis Settings

Before running an analysis, you can select settings for analyzing a sample and viewing analysis files using the SOM Wizard (see Using the Create SOM Wizard in CHAPTER 5, Selecting Analysis Settings: SOM and SOP) and the Create Preferences Wizard (see Creating a Preferences File in CHAPTER 6, Setting View and Print Preferences).

If you saved default or custom SOM files and Preferences, you can create one or more Standard Operating Procedures (see Using a Standard Operating Procedure (SOP) in CHAPTER 5, Selecting Analysis Settings: SOM and SOP). Standard Operating Procedures (SOPs) are helpful if you are running several types of analyses that require different settings.

A Standard Operating Procedure combines one Preferences file and one Standard Operating Method (SOM) into an overall settings file for a particular type of analysis. Within the SOP, the SOM file determines sample run settings, while the Preferences file determines view and print specifications.

- For information on creating a Standard Operating Method, see Using a Standard Operating Method (SOM) in CHAPTER 5, Selecting Analysis Settings: SOM and SOP.
- For information on setting Preferences, see Creating a Preferences File in CHAPTER 6, Setting View and Print Preferences.
- For information on creating a Standard Operating Procedure, see Using a Standard Operating Procedure (SOP) in CHAPTER 5, Selecting Analysis Settings: SOM and SOP.

Loading an SOM

If you saved one or more Standard Operating Method(s), you can load an SOM specific to the type of analysis you will be running.

NOTE If you loaded a Standard Operating Procedure (SOP), it is not possible to load a Standard Operating Method (SOM), make changes to the SOM, and save the SOM under another name. To edit run settings in the SOM and save it under another name, first remove the SOP (see Removing an SOP).

1 Select **SOP** > **Load an SOM** on the Main Menu bar.

2 In the Load a Standard Operating Method (SOM) window, navigate to the appropriate folder, select the desired SOM, and click **Open**.

NOTE SOM files are identified by the extension .som.

Loading Preference Settings

If you saved different Preferences settings to one or more file(s), you can load a different Preferences file depending on how you want to view and print an analysis.

1 Select **SOP** > **Load Preferences** on the Main Menu bar.

2 In the Load Preferences window, navigate to the appropriate folder, select the desired Preferences file, and click **Open**.

NOTE Preferences files are identified by the extension .prf.

Loading an SOP

1 Select **SOP** > **Load an SOP** on the Main Menu bar.

2 In the Load a Standard Operating Procedure (SOP) window, navigate to the appropriate folder, select the desired SOP, and click **Open**.

NOTE SOP files are identified by the extension .sop.

Removing an SOP

When you load a Standard Operating Procedure, the Multisizer 4e software will not allow you to open an SOM, make changes to it, and then save it under another name.

The Create SOM Wizard menu item is not visible when an SOP is loaded. When you select Edit the SOM, a read-only window opens: SOM Settings (SOP loaded, no changes allowed).

There are two ways to remove an SOP:

- In the Status Panel, click **Remove SOP**.
- Select **SOP** > **Remove the SOP** on the Main Menu bar.

Verifying Preference Settings

To See the View and Print Preferences Currently in Effect:

1 Select **SOP** > **Preference Info** on the Main Menu bar.

2 In the Preference Information window, use the scroll bar to see the complete list of preference settings.

3 Click **Close**.

To Copy and Print Preference Settings

To print your preferences or copy them to the clipboard, click **Print** in the Preference Information window. The following table describes the print and copy options.

Table 7.1 Copy and Print Preference Settings

If you want to...	Then...
Print the Preferences list	**1.** Click **Print**. **2.** In the Print or Copy List Box Contents window, select Print all lines. **3.** Click **OK**.
Print a portion of the Preferences list	**1.** In the Preference Information box, drag to highlight the text you want to print. **2.** Click **Print**. **3.** In the Print or Copy List Box Contents window, select Print selected lines. **4.** Click **OK**.

Table 7.1 Copy and Print Preference Settings

If you want to...	Then...
Copy the Preferences list to the clipboard	**1.** Click **Print**. **2.** In the Print or Copy List Box Contents window, select Copy all lines to clipboard. **3.** Click **OK**.
Copy a portion of the Preferences list to the clipboard	**1.** In the Preference Information box, drag to highlight the text you want to copy. **2.** Click **Print**. **3.** In the Print or Copy List Box Contents window, select the Copy selected lines to clipboard. **4.** Click **OK**.

To change Preference settings, select **SOP > Edit Preferences** on the Main Menu bar (see Creating a Preferences File in CHAPTER 6, Setting View and Print Preferences) or select **SOP > Create Preferences Wizard.**

Verifying SOM Settings

To See the Standard Operating Method Currently in Effect

1 Select **SOP** > **SOM Info** on the Main Menu bar.

2 In the SOM (Standard Operating Procedure) Information window, use the scroll bar to see the complete list of SOM settings.

3 Click **Close.**

To Copy and Print SOM Settings

To print your SOM settings or copy them to the clipboard, click **Print** in the SOM Information window. There are four print and copy options.

Table 7.2 Verify SOM Settings

If you want to...	Then...
Print the SOM settings	**1.** Click **Print**. **2.** In the Print or Copy List Box Contents window, select Print all lines. **3.** Click **OK**.
Print a portion of the SOM settings	**1.** In the SOM Information box, drag to highlight the text you want to print. **2.** Click **Print**. **3.** In the Print or Copy List Box Contents window, select Print selected lines. **4.** Click **OK**.
Copy the SOM settings to the clipboard	**1.** Click **Print**. **2.** In the Print or Copy List Box Contents window, select Copy all lines to clipboard. **3.** Click **OK**.
Copy a portion of the SOM settings to the clipboard	**1.** In the SOM Information box, drag to highlight the text you want to copy. **2.** Click **Print**. **3.** In the Print or Copy List Box Contents window, select the Copy selected lines to clipboard. **4.** Click **OK**.

To change SOM settings and save the SOM under a new name, select **SOP > Edit the SOM** on the Main Menu bar (see Using a Standard Operating Method (SOM) in CHAPTER 5, Selecting Analysis Settings: SOM and SOP), or select **SOP > Create SOP Wizard**.

Entering Sample Information

Before you begin a run, enter sample information after selecting **Sample > Enter Sample Info** on the Main Menu bar.

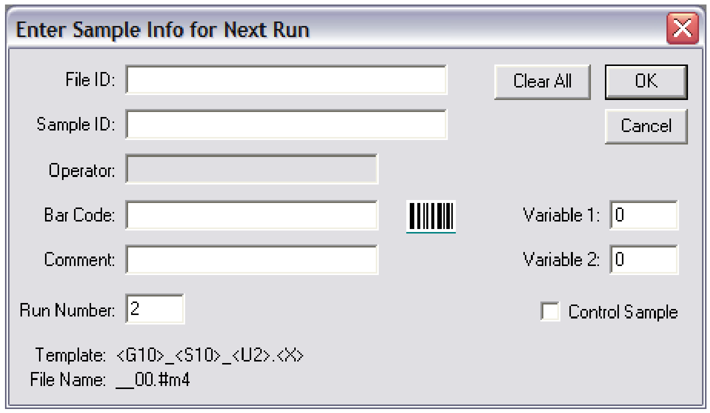
No information that you enter in the Enter Sample Info window will affect analysis results, but it is helpful in organizing your data and preventing accidental file overwrites.

Identifying sample information is particularly important if you are completing more than one analysis run. You can use information entered in the Enter Sample Info for Next Run window to create sequential file names that will help you organize your data for comparing and overlaying files (see Creating Automatically Generated Analysis File Names in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).

When you create a Standard Operating Method (SOM), you can choose to display the Enter Sample Info window automatically before each analysis run to prompt the operator to enter sample information. For more information, see Standard Operating Method: Run Settings in CHAPTER 5, Selecting Analysis Settings: SOM and SOP.

To Enter Sample Information Before a Run

1 Select **Sample > Enter Sample Info** on the Main Menu bar. The Enter Sample Info for Next Run window opens.



2 In the Enter Sample Info for Next Run window, enter information in the text fields according to the table below. The maximum number of characters you can enter in these fields is specified either in Step 1 of the SOM Wizard or by selecting **SOP > Edit the SOM**.

Table 7.3 Entering Sample Info for Next Run

In the text field...	Enter...
File ID	A name or abbreviation that clearly identifies the group or general sample information. Information entered in this and other sample information fields can be used to automatically create part of the file name (see Creating Automatically Generated Analysis File Names in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).
Sample ID	A name or abbreviation that identifies the current sample. Information entered in this and other fields can be used to automatically create part of the file name (see Creating Automatically Generated Analysis File Names in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).
Operator	The operator's name or initials. If you are logged in, the software populates this field automatically.
Bar Code	Bar code information on the sample packet or process sheet, if available. To enter the sample bar code, hold the bar code in front of the Bar Code Reader on the Analyzer Control Panel. See Bar Code Reader in CHAPTER 2, Analyzer Overview for more information on using the Bar Code Reader, To scan bar codes on sample packets or process sheets, it may be necessary to configure the Bar Code Reader (see Bar Code Reader Configuration Options in CHAPTER 9, Configuring Software and Security).
Comment	Detailed information about the sample as desired.

Table 7.3 Entering Sample Info for Next Run

In the text field...	Enter...
Run Number	The number of the run. In general, enter "1" in this field, and reset the run number to "1" if you are re-running the sample. The Multisizer 4e software will automatically increment the run number for multiple runs. You can use this field to create sequential file names for each run by setting file name preferences in the SOM (see Creating Automatically Generated Analysis File Names in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).
Variable 1 and 2	Any variables your analysis requires. These fields will not affect analysis results. For example, you can use a variable field to record a pH value.

Select Control Sample if you are running a sample with known values for comparison.

NOTE The File Name and the file name Template appear at the bottom of the window. The template codes are generated by settings in the Standard Operating Method (see Definitions: SOM, Preferences, and SOP in CHAPTER 5, Selecting Analysis Settings: SOM and SOP). Using SOM settings, you can configure the Multisizer 4e software to automatically create file names based on information in the Enter Sample Info window. For information on automatically generated file names, see Creating Automatically Generated Analysis File Names in CHAPTER 5, Selecting Analysis Settings: SOM and SOP.

Editing Sample Information

After you complete an analysis run, you can enter or update sample information. For more on sample information, see Entering Sample Information.

NOTE The option to edit sample information after a run is available only to user types with sufficient security privileges. If the Multisizer 4e software is running with no security, every operator can edit sample information.

To Update or Enter Sample Information After an Analysis Run

1 Open the analysis file for which you will be entering sample information.

2 Select **Edit** > **Edit Sample Info** on the Run Menu bar. The Edit Sample Info window opens.

NOTE If the user does not have sufficient security privileges to edit sample information, the **Edit Sample Info** menu item will not appear in the Edit drop-down menu.

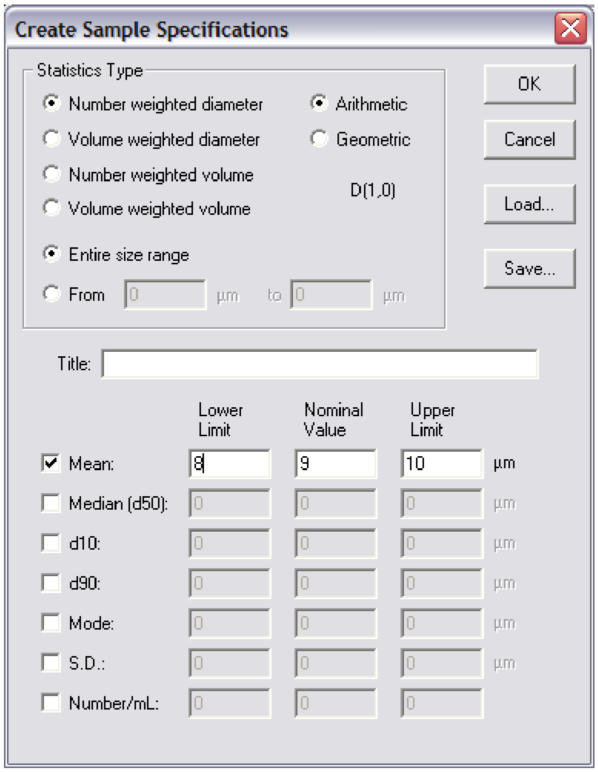
3 Enter information in the text fields and click **OK** to save your settings.

Using Sample Specifications

For applications such as quality control, you can enter sample specifications before running an analysis to verify that the particle size and/or distribution of your sample(s) fall within specified parameters.

To Enter Sample Specifications

1 Select **Sample > Create Sample Specifications** on the Main Menu bar. The Create Sample Specifications window opens.



2 In the Create Sample Specifications window:

a. In the Statistics Type pane, enter the type of bin weighing and the particle size range for comparison.

NOTE Settings that you select in the Create Sample Specifications window will not affect or be affected by selections you make in Preferences. See Creating a Preferences File in CHAPTER 6, Setting View and Print Preferences. Statistics you select in the Preferences Wizard or Edit Preferences window control view and print settings only.

b. Enter a title for the sample specifications, if desired.

c. Select one or more options to specify the sample specification value, and enter a Lower Limit, Upper Limit, and Nominal Value for the specifications you select.

d. To save the sample specifications to a file, click **Save**. Specifications files are identified with the extension .spec.

e. Click **OK**.

Editing Sample Specifications

After completing an analysis run, you can enter or update sample specifications. For more information on sample specifications, see Using Sample Specifications.

NOTE The option to edit sample specifications is available only to user types with sufficient security privileges. If the Multisizer 4e software is running with no security, every operator can edit sample specifications.

To Update or Enter Sample Specifications After an Analysis Run

1. Open the analysis file for which you will be editing sample specifications.

2. Select **Edit > Edit Sample Specifications** on the Run Menu bar. The Edit Sample Specifications window opens.

   NOTE If a user does not have sufficient security privileges to edit sample specifications, an Edit Sample Specifications (Read-Only) window opens.

3. Enter information in the text fields and click **OK** to save your settings.

Including Sample Specifications in a Printed Report

1. Select **SOP > Edit Preferences** on the Main Menu bar. The Edit Preferences window opens.

2. In the Printed Report window (Size pane), select Compare to Sample Specifications.

3. Click **OK**.

Viewing Sample Specifications

To Compare an Analysis Run to Sample Specifications

1. Select **File > Open** on the Main Menu bar.

2. In the Open window, select an analysis file in the appropriate folder, and click **Open**.

3 Select **Calculate > Compare to Sample Specifications** on the Run Menu bar. A new window opens and displays the comparison.

To update sample specifications, see Editing Sample Specifications.

Editing Size Data

If necessary, you can manually edit size data after an analysis run. This feature is included primarily for compatibility with previous versions of the Multisizer software. Editing size data using the Edit Size Data window overrides analysis results.

To import size data from another instrument, use the file import feature (see Importing a File in CHAPTER 8, Working with Analysis Files).

NOTE The option to edit size data is available only to user types with sufficient security privileges. If the Multisizer 4e software is running with no security, every operator can edit size data.

1 Open the analysis file for which you will be editing size data.

2 Select **Edit > Edit Size Data** on the Run Menu bar. The Edit Size Data window opens.

NOTE If the user does not have sufficient security privileges to edit size data, the **Edit Size Data** menu item will not appear in the Edit drop-down menu.

3 Enter information in the text fields and click **Replace** to change measured data.

4 Click **OK** to save your settings.

Running an Analysis

To run an analysis, follow the steps below to prepare the Analyzer and Multisizer 4e software and run the analysis.

1 Prepare the Analyzer (see Preparing the Analyzer for Sample Runs in CHAPTER 2, Analyzer Overview).

2 Set view and print Preferences (see Creating a Preferences File in CHAPTER 6, Setting View and Print Preferences).

3 Create a Standard Operating Method (SOM) and a Standard Operating Procedure (SOP) (see Using a Standard Operating Method (SOM) CHAPTER 5, Selecting Analysis Settings: SOM and SOP).

4 Enter sample information (see Entering Sample Information).

5 Enter sample specifications, if desired (see Using Sample Specifications).

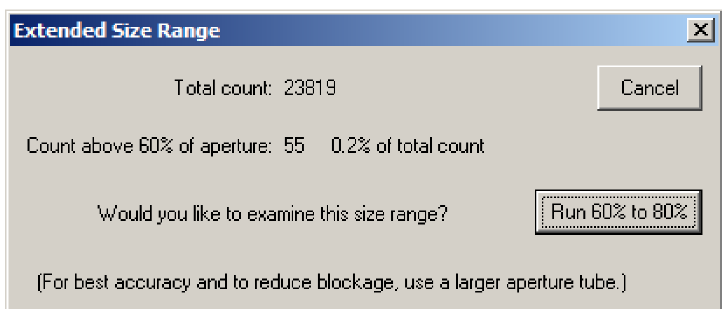
6 Select **Run > Preview** on the Main Menu bar (or click **Preview** in the Instrument Toolbar). Wait for the preview analysis to complete its run.

7 In the Status Panel, verify that the sample concentration is less than 10%.

NOTE If the concentration is greater than 10%, adjust the sample concentration and preview the analysis again.

8 Select **Run > Start** on the Main Menu bar (or click **Start** in the Instrument Toolbar).

- If the *Extended Size Range* feature is enabled in the SOM (see Standard Operating Method: Threshold, Current and Gain in CHAPTER 5, Selecting Analysis Settings: SOM and SOP) and the large particle count exceeds the specified threshold, a notification dialog will appear:



- If you click “Run 60% to 80%,” then an extended dynamic range run will start and display the size distribution from 60% to 80% of the aperture diameter. This feature is intended for examining particles in the small “tail” of a distribution that extends above 60%.

NOTE A 60% to 80% run uses special SOM settings which override some parameters that the user may have set, such as aperture current and sizing threshold. To obtain results for larger particles using customized settings, select a larger aperture tube.

NOTE Aperture blockage is common when sample diameters are larger than 60% of the aperture diameter. The extended dynamic range should only be used for low concentrations of spherical particles larger than 60%. For best results, it is recommended to use a larger aperture size for samples that contain a significant concentration of particles above 60% of the aperture diameter.

Sample Analysis Variations

Running a Multi-Tube Overlap Analysis

A multi-tube overlap analysis allows you to combine up to five particle distributions using apertures of different sizes. After you filter the sample and run it through incrementally smaller aperture tubes, the Multisizer 4e software can merge these runs into one distribution that includes a range of particle sizes greater than can be measured with a single aperture.

NOTE You can run up to five analyses using different aperture sizes.

To Run Multiple Analysis with Different Apertures

1. Select an aperture tube suitable for the largest particles in the sample (for information on selecting the correct aperture tube size, see Selecting an Aperture Size in CHAPTER 4, Installing and Calibrating an Aperture Tube.

2. Follow the steps in the Change Aperture Tube Wizard to install a new tube and verify aperture calibration (see Verifying the Calibration in CHAPTER 4, Installing and Calibrating an Aperture Tube).

3. Run the analysis (see Running an Analysis).

4. When the analysis run is complete, save the file (select **RunFile > Save As** on the Run Menu bar) if you did not configure the Multisizer 4e software to automatically generate analysis file names. To set automatic file name generation, (see Creating Automatically Generated Analysis File Names in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).

5. After the analysis is complete, select and install a smaller aperture tube.

6. Sieve, filter, or use another suitable method to remove the particles from the suspension that will block the smaller aperture.

7. Repeat Steps 2 through 5. The Multisizer 4e can combine a maximum of five analyses in one multi-tube overlap file.

To Create the Multi-Tube Overlap File

1. Select **File > File Tools > Multi-tube Overlap** on the Main Menu bar.

2 In the Multi-tube Overlap window:

- **a.** In the Folders box, double-click a folder name. The folder contents appear in the Files box.
- **b.** In the Files box, hold the Ctrl key down and select each analysis you want to include in the overlap file.
- **c.** Click **Add**. The files appear in the Selected Files box.
- **d.** Click **OK**.

3 In the Viewing Area, a graph showing the offset points appears automatically.

4 Select **OK** in the Run Menu bar if the selected Slope Ratio is acceptable.

NOTE Zoom the Area of the Runs for review with cursors and select **View** in the Run Menu bar.

5 Review the resulting distribution and save the file.

6 To save the file, select **RunFile > Save As** on the Run Menu bar.

7 In the Save As dialog box, navigate to the appropriate folder and enter the new file name in the File Name field.

8 Click **Save**.

Determining Particle Concentration Using Time Mode

Because of stabilizing time before analysis and finishing time after, the volume of electrolyte that passes through the aperture is more than total analysis electrolyte. To calculate the amount of electrolyte used during the analysis run, you could do 30-, 60-, and 90-second runs and calculate the pre- and post-run volume, which will remain the same for every analysis length.

If you are measuring particle concentration using an aperture tube 360 µm or larger, it is necessary to use time mode. Using volumetric mode is not possible because the maximum volume allowed by the metering pump is 2 mL. In larger aperture tubes, this volume of fluid passes through the aperture too quickly to provide accurate statistical results.

NOTE The Multisizer 4e uses small amounts of electrolyte for pre- and post-analysis stabilization. As a result, calculated volume is approximate.

NOTE If you are measuring heavy particles, Beckman Coulter recommends that you thicken the electrolyte to increase particle suspension, reduce the flow rate, and produce less noise. Recommended thickening agents include glycerol and sucrose.

To Determine Particle Concentration Using Time Mode

1. Fill the beaker to approximately 9/10 of its capacity and measure the exact amount of electrolyte you will be using. Place the beaker on the platform.

2. If you have already loaded an SOP (see Loading an SOP in CHAPTER 5, Selecting Analysis Settings: SOM and SOP), select **SOP > Remove the SOP** on the Main Menu bar. Removing the SOP allows you to change Standard Operating Method Settings.

3. Select **SOP > Edit the SOM** on the Main Menu bar.

4. In the Edit the SOM / Control Mode window, select Time, and then select 60 seconds.

5. Click **OK**.

6. Select **Run > Start** on the Main Menu bar (or click **Start** in the Instrument Toolbar).

7. When the run is complete, measure the exact amount of electrolyte remaining in the beaker.

8. Subtract the remaining electrolyte from the original electrolyte volume.

9. Select **Edit > Edit Sample Info** on the Run Menu bar. In the Analytic Volume field, enter the mL multiplied by 1000. The software automatically calculates concentration. To see particle concentration, graph or list the number, volume, or surface area per mL.

Using Blank Runs

A blank run (also called a blank analysis or background analysis), is an analysis of electrolyte without sample material.

When analyzing low concentration samples, a background analysis is required for accurate results.

NOTE Beckman Coulter recommends that you run a background (blank) analysis for every batch of analyses that uses the same electrolyte.

Running a Blank Analysis

1. Load the Standard Operating Procedure (see Loading an SOP in CHAPTER 5, Selecting Analysis Settings: SOM and SOP) or Standard Operating Method (see Loading an SOM in CHAPTER 5, Selecting Analysis Settings: SOM and SOP) you will be using for the sample analysis.

   NOTE If you set automatic file name generation (see Creating Automatically Generated Analysis File Names in CHAPTER 5, Selecting Analysis Settings: SOM and SOP), change the file name to indicate that the run is blank, either before or after the analysis run.

2. Prepare the Analyzer (see Preparing the Analyzer for Sample Runs in CHAPTER 2, Analyzer Overview).

3. Fill a beaker with clean electrolyte and place the beaker on the sample platform.

4. Select **Run** > **Start** on the Main Menu bar (or click **Start** in the Instrument Toolbar).

Saving the Blank Analysis

1. Leave the blank analysis open in the Viewing Area.

2. Select **RunFile** > **Save As** on the Run Menu bar.

3. In the Save As dialog box, enter a file name for the background, and then select Save as read-only.

4. Click **Save**.

Enabling Blank Subtraction for Analysis Runs

IMPORTANT Clean electrolyte and a clean instrument are preferable to mathematical data manipulation described below. See Cleaning the Analyzer in CHAPTER 2, Analyzer Overview and Pre-Filtering the Electrolyte and Cleaning Fluid in CHAPTER 2, Analyzer Overview for detailed instructions on cleaning the analyzer and prefiltering the electrolyte.

After you run a blank analysis (see Running a Blank Analysis), you can set the software to automatically subtract electrolyte background from sample analysis runs.

NOTE Background subtraction should only be employed for difficult electrolyte background issues that cannot be resolved by other means.

To Subtract the Electrolyte Background from Analysis Runs

1. Select **Sample > Load Blank Run** on the Main Menu bar.

2. In the Load a Blank Run window, navigate to the appropriate folder and select the desired blank analysis file.

3. Click **Open**.

The blank analysis will be subtracted from each subsequent analysis until you select **Sample > Remove Blank** on the Main Menu bar.

Disabling Blank Analysis Subtraction

To disable blank analysis subtraction, select **Sample > Remove Blank** on the Main Menu bar.

Loading a Blank Run into an Open Analysis File

To Load a Blank Run for a Single Analysis

1. If the analysis is not open in the Viewing Area, open a saved analysis (see Opening a Saved Analysis File in CHAPTER 8, Working with Analysis Files).

2. Select **Calculate > Load Blank Run** on the Run Menu bar.

3. In the Load a Blank Run window, navigate to the saved blank analysis and click **Open**. The software automatically subtracts the background from analysis data.

Removing a Blank Run from an Open Analysis File

To remove a blank run from the open analysis, select **Calculate > Remove Blank Run** on the Run Menu bar.

The **Remove Blank Run** menu item removes the blank run from the open analysis file only.

If you selected **Sample > Load Blank Run** on the Main Menu bar and want to disable blank analysis subtraction for all future analysis runs, select **Sample > Remove Blank** on the Main Menu bar.

Working with Analysis Files

Setting File Preferences

When an analysis run is complete, there are a number of options for displaying statistical results. You can set viewing, printing, and statistical analysis defaults using the Create Preferences Wizard or the Edit Preferences windows (see Creating a Preferences File in CHAPTER 6, Setting View and Print Preferences).

If you save a set of Preferences, the saved Preferences file can be associated with one or more Standard Operating Procedures (SOPs). Creating SOPs allows you to set specific run, view, and print preferences for different types of analysis runs (see Using a Standard Operating Procedure (SOP) in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).

You can change how the Multisizer 4e software displays an analysis file in the following ways:

- To change default display preferences, or preferences associated with a specific type of analysis, select **SOP > Edit Preferences** on the Main Menu bar (see Creating a Preferences File in CHAPTER 6, Setting View and Print Preferences).
- To temporarily change the display of an open analysis file, select **View** on the Run Menu bar (see Run Menu in CHAPTER 3, Software Overview). You can use items in this menu to select display and analysis settings.
- Sample data files can be found in the *Demodata* file folder on the instrument controller. Use these files to practice the file preference options described in this chapter.
- The Run Menu bar changes based on the display and analysis options you select in the View drop-down menu (see Viewing Analysis Files).

Changing Print Settings

Changes you make using drop-down menus on the Run Menu bar affect on-screen analysis settings only. If you select **RunFile > Print Report**, the Multisizer 4e will use print settings in the current Preferences file, not on-screen analysis view settings.

There are two options for changing print settings:

- If you want to change Printed Report defaults or save print settings and associate them with a Standard Operating Procedure, change settings in the Edit Preferences window. To open the Edit Preferences window:
  — Select **SOP > Edit Preferences** on the Main Menu bar.

— Select **RunFile > Print Report** on the Run Menu bar. In the Print Report window, click **Report** to open the Edit Preferences window.

- If you want to print a graph or listing as it appears on the screen, without changing Print Report settings, select **RunFile > Print** on the Run Menu bar.

Viewing Analysis Files

Changing the Display of an Open File

1 Select **View** on the Run Menu bar (located at the top of the open analysis file).

2 In the View menu, select one display type: **Graph**, **Listing**, or **Interpolation**.

3 In the View menu, select one analysis type: **Size** (particle size), **Pulse** (pulse height and width), or **Blockage Monitors** (aperture resistance over time or by sequence).

Run Menu Behaviors

Graphs

If you select **View > Graph**, three menus will appear to the right of the View menu (**Graph**, **Calculate**, and **Display**). The Graph drop-down menu, and the options available in the Calculate drop-down menu, will differ based on the analysis type you select (Size, Pulses, or Blockage Monitors).

Listings and Interpolation

If you select **View > Listing** or **View > Interpolation**, the drop-down menu to the right of the **View** menu will change to Listing or Interpolation, respectively. Options available in the **Calculate** drop-down menu will vary depending on the analysis type you selected (**Size**, **Pulses**, or **Blockage Monitors**).

Viewing Graphs

1 Select **View > Graph** on the Run Menu bar.

2 Select the **View** drop-down menu a second time and select the type of graph (**Size**, **Pulses**, or **Blockage Monitors**).

Graph Menu

The **Graph** drop-down menu offers different options for X and Y axis values based on the type of graph you select.

NOTE If you select **Differential** + **Cumulative** < or **Differential** + **Cumulative** > in the **Graph** drop-down menu, a new **Cursor** menu appears to the right of the **Display** menu. The **Cursor** menu allows you to select the trace on the graph to which the cursor readout (below the graph) refers. When you are viewing overlaid or multiple runs, menu items also allow you to arrange the runs manually or by start time.

Calculate Menu

For information on available menu items in the **Calculate** drop-down menu, see Viewing Statistics (see Viewing Statistics).

Display Menu

The **Display** drop-down menu items allow you to change the appearance of the graph and select the scale of the X and Y axes.

Using Cursors in Graphs

You can place one or two cursor points in a graph. Placing two cursors in a graph allows you to zoom in on one part of a graph or view statistics between two points.

Table 8.1 Using Cursors in Graphs

If you want to...	Then...
Insert one cursor (or the first cursor)	Click once in the graph to insert a cursor. The placement of the cursor on the X axis appears in numerical form below the graph.
Insert a second cursor	Left-click on the first cursor and drag it to another position on the graph. Un-click to create a second cursor. The first cursor will remain in place as you drag the second cursor line.
Reposition the cursor(s)	Click in the graph at the point you want to place the new cursor. You can use the left and right arrow keys to move the cursor, and use the Tab key to switch between cursors.
Remove the cursor(s)	Click outside the graph.
Change cursor lines from vertical to horizontal	Select **View** > **Horizontal Cursors** (pulse graphs ONLY).
Change cursor lines to box cursors	Select **View** > **Box Cursors** (pulse graphs ONLY).

Table 8.1 Using Cursors in Graphs

If you want to...	Then...
View a portion of the graph between two cursor points	Select **View** > **Zoom In**. To return to the full graph, select **View** > **Zoom Out**.
View statistics between two cursor points	Select **Calculate** > **Statistics**.

Viewing Listings

1 Select **View > Listing** on the Run Menu bar.

2 Open the **View** drop-down menu a second time and select the type of listing (**Size**, **Pulses**, or **Blockage Monitors**).

Listing Menu

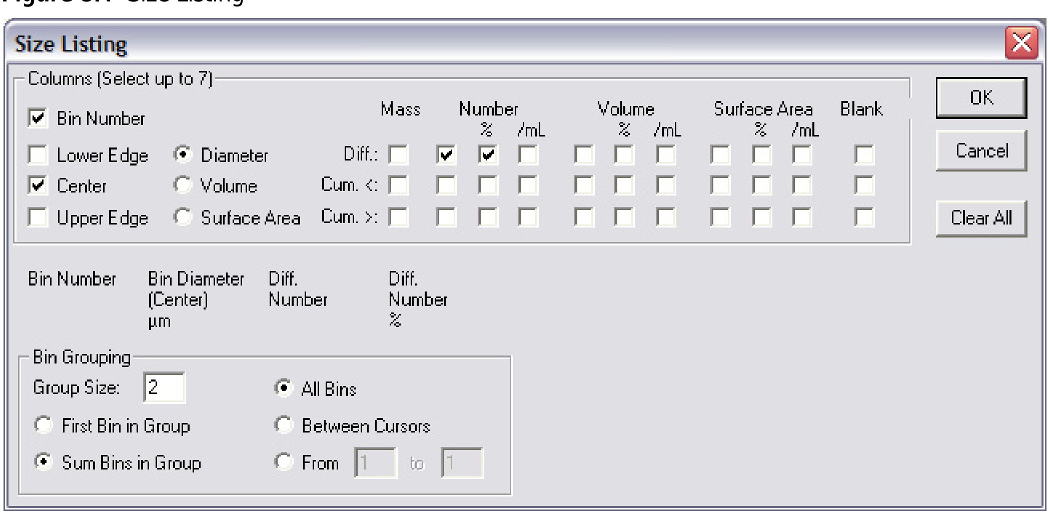
The **Listing** drop-down menu offers the following options: **Column Format**, **Wrap Screen Listings**, and **Wrap Printer Listings**.

When you select **Listing > Column Format**, a different window opens depending on which listing type you selected (**Size**, **Pulses**, or **Blockage Monitors**).

Size Column Format

If you select **View > Listing** then select **Size**, the Size Listing window opens when you select **Listing > Column Format**.

Figure 8.1 Size Listing



To Select the Columns of Data that Will Appear in the Size Listing

1 In the Columns pane:

a. Select Bin Number to include the bin number next to each bin value.

b. Select Lower Edge, Upper Edge, or Center to print the smallest, largest, or median particle size, volume, or surface area in each bin.

c. Select up to seven options to display columns of statistics.

2 In the Bin Grouping pane:

a. Enter a number in the Group Size text field to group bins, if desired. Grouping bins shortens the listing. Select First Bin in Group or Sum Bins in Group.

b. Select All Bins to display results from the full range of the analysis.

c. Select Between Cursors to display only results between your defined cursors. For information on setting and moving cursors, see Using Cursors in Graphs.

d. Select From and enter specific points in the From and to fields to define statistical limits.

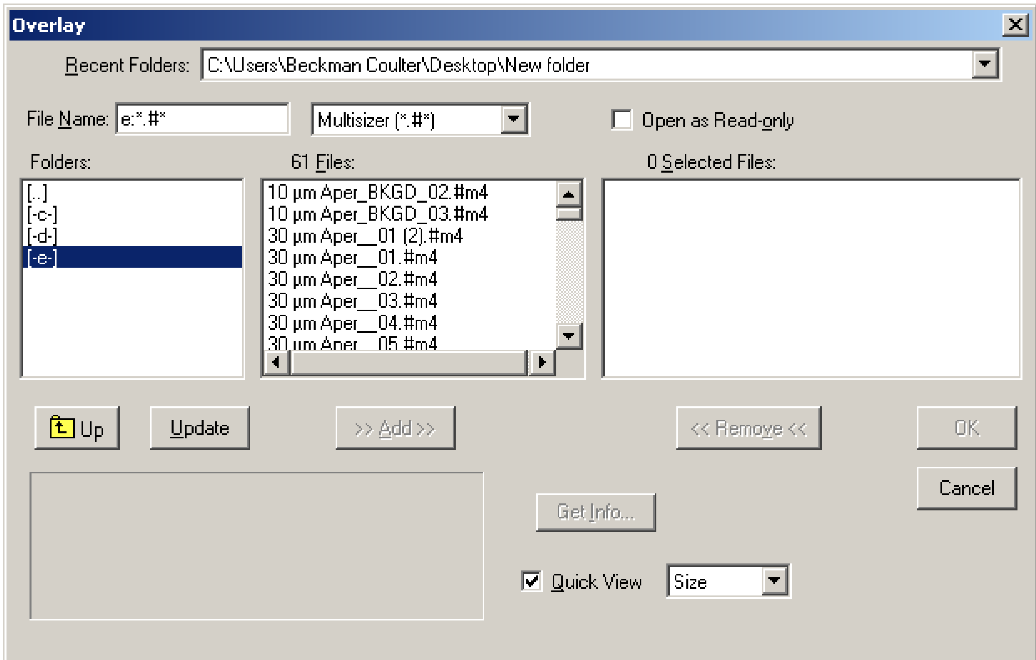
3 To clear all selections, click **Clear All**.

4 Click **OK**.

Pulses Column Format

If you select **View** > **Listing** then select **Pulses**, the Pulses Listing window opens when you select **Listing** > **Column Format.**

Figure 8.2 Overlay



To Select the Columns of Data that will Appear in the Pulse Listing

1. In the Pulse Listing window, select the statistics in the Columns pane that you want to display.

2. In the Pulse Grouping pane, enter the Group Size in the field.

   a. Select All Pulses to display results from the full range of the analysis.

   b. Select Between Cursors to display only results between your defined cursors. For information on setting and moving cursors, see Using Cursors in Graphs.

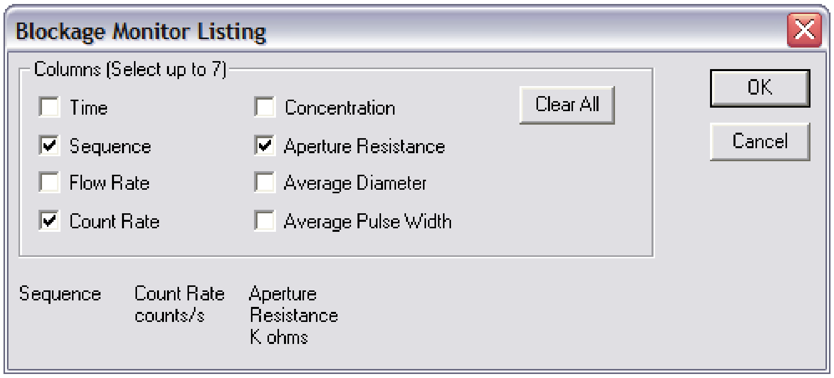
   c. Select From and enter specific points to limit the range without inserting cursors in a graph.

3. Click **OK**.

Blockage Monitors Column Format

If you select **View** > **Listing** then select **Blockage Monitors**, the Blockage Monitor Listing window opens when you select **Listing** > **Column Format**.

Figure 8.3 Blockage Monitor Listing



To Select the Data Columns that will Appear in the Blockage Monitor Listing

1 In the Blockage Monitor Listing window, select the data you want to display in the Columns pane.

2 Click **Clear All** to clear all selections.

3 Click **OK**.

Calculate Menu

For information on available menu items in the **Calculate** drop-down menu, see Viewing Statistics.

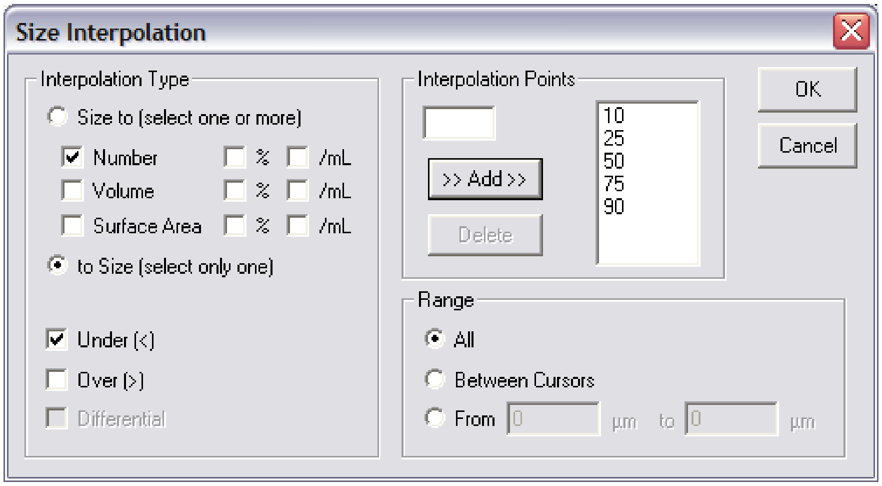
Viewing Interpolations

An interpolation allows you to divide a particle size distribution into any number of bins (channels) of any size. Interpolation is similar to viewing results by statistical percentiles, but offers greater flexibility.

The **Interpolate** menu option on the Run Menu bar is only available if you are viewing results by particle size (**View** > **Interpolate** and **Size** is the only possible combination).

1 Select **View** > **Interpolation** on the Run Menu bar, then select **Size.**

2 Select **Interpolation > Size Interpolation** on the Run Menu bar.



3 In the Size Interpolation window:

a. In the Interpolation Type pane, select Size to or to Size.

- Select Size to if you want to create interpolation points based on particle diameter size (that is, particle diameters define bin edges). You can select up to nine statistics to display based on particle size interpolation points
- Select to Size if you want to create interpolation points based on the statistic you select in the Interpolation Type pane (Number, Volume, Surface Area, %, and /mL). You can select only one statistical interpolation point to correlate to particle size.

b. Select one or more of these options: Number, Volume, and/or Surface Area, and select % and/or /mL for each selected interpolation type. If you are displaying statistics to Size, you can only select one option.

c. Select Under (<), Over (>), or (if you selected Size to), Differential.

Table 8.2 Size Interpolation Data

If you select...	The columns will display...
Under (<)	Statistical results below the interpolation point. The interpolation point becomes the upper edge of the bin.
Over (>)	Statistical results above the interpolation point. The interpolation point becomes the lower edge of the bin.
Differential	The number, volume, or surface area of particles between interpolation points.

d. In the Interpolation Points pane:

- To add an interpolation point, enter the new point in the field and click **Add**.
- To delete an interpolation point, select the point and click **Delete**.

e. In the Range pane:

- Select All to display results from the full range of the analysis.
- Select Between Cursors to display only results between your defined cursors. For information on setting and moving cursors, see Using Cursors in Graphs.
- Select From and enter specific points to limit the range without inserting cursors in a graph.

4 Click **OK** to view the interpolation.

Changing the Column Display

To wrap columns on the screen, select **Interpolation > Wrap Screen Listings** on the Run Menu bar.

To wrap columns in the printed report, select **Interpolation > Wrap Printer Listings** on the Run Menu bar.

When you select column wrapping, **Wrap Screen Listings** and **Wrap Printer Listings** remain checked until you clear them, or until you close the analysis file. These settings remain in effect when viewing both Listings and Interpolations.

To PrintReports with Column Wrapping

If you want to change the display of the open analysis file, but not report settings, select **Interpolation > Wrap Printer Listings** on the Run Menu bar. This setting will not change Printed Report options previously set in the Preferences window. To change printed report settings, see Preferences: Printed Report in CHAPTER 6, Setting View and Print Preferences.

Viewing Statistics

The **Calculate** drop-down menu on the Run Menu bar allows you to view statistics, regardless of whether the analysis window displays a graph, listing, or interpolation.

If you select **Pulses** or **Blockage Monitors** in the **View** drop-down menu, many of the options in the **Calculate** drop-down menu will not be available.

NOTE The statistics displayed in the pop-up window are based on preselected Preferences settings. To add or remove statistics, select **SOP** > **Edit Preferences** / **Statistics** on the Main Menu bar (see Preferences: Statistics in CHAPTER 6, Setting View and Print Preferences).

To View Statistics Between Cursors (Graph View Only)

If you are viewing a graph and have inserted two cursor points, **Calculate** > **Statistics** will display results only for the portion of the analysis run included between the cursors.

If you have not included cursors, or if you have placed only one cursor in the graph, **Calculate > Statistics** will display statistics based on the complete analysis run.

For information on placing cursors, see Using Cursors in Graphs.

To Save Statistics

To save statistics for the analysis you are viewing (for example, statistics between cursors):

1. Select **Calculate** > **Statistics** on the Run Menu bar.

2. In the Statistics window, click **Save.** The Multisizer 4e will include the statistics you save when you save the analysis run.

The statistics you save in the **Calculate** > **Statistics** window will appear when you select **RunFile** > **Get Info** on the Run Menu bar.

To View Statistics on the Screen (Graph View Only)

To view statistics together with a graph, select **Calculate** > **Show Statistics on Graph** on the Run Menu bar.

There are two ways to remove statistics from the graph view:

- Select **Calculate** > **Hide Statistics on Graph** on the Main Menu bar.
- Click the **X** in the upper right corner of the statistics box.

Comparing Statistics to Sample Specifications

If you created sample specifications (see Using Sample Specifications in CHAPTER 7, Analyzing a Sample), select **Calculate** > **Compare to Sample Specifications** to see analysis results compared to your pre-set parameters.

Comparing an Analysis to a Standard Run

If you have run an analysis as a standard for comparison with other analysis runs, you can use items on the Run Menu bar to compare the current analysis results to the standard.

To Compare with the Standard

1. Select **Calculate** > **Compare to Standard** on the Run Menu bar.

2. In the Open Standard Run window, navigate to where you saved the standard analysis file and click to select the file for comparison.

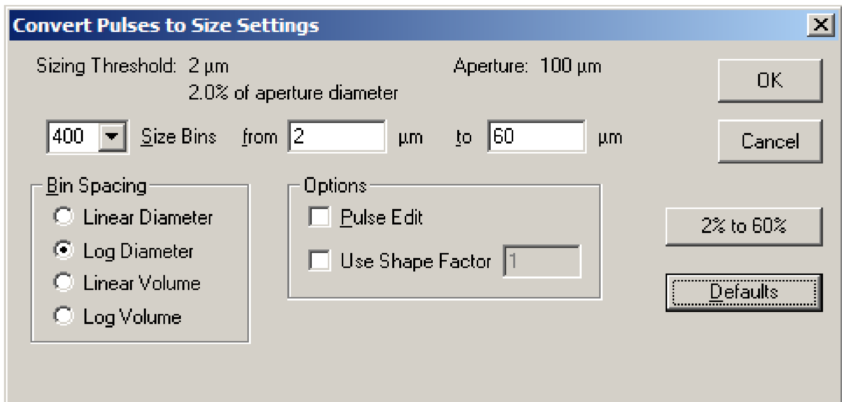
3. Click **Open.**

All display and statistical options will include comparisons to the standard file until you remove the standard file. To remove the standard file, select **Calculate** > **Remove Standard.**

Converting Pulses to Size

You can change the number of bins (channels) and select a range of particle sizes to display.

1 Select **Calculate > Convert Pulses to Size** on the Run Menu bar. The Convert Pulses to Size Settings window opens.



2 In the Convert Pulses to Size Settings window:

a. In the Size Bins field, enter the number of bins you want to use, or select a number of bins from the drop-down list.

b. In the from and to fields, enter the low and high limits for the range of particle diameter sizes you will be analyzing.

If you want to analyze the broadest possible distribution based on aperture size, click the **2% to 60%** button.

c. In the Bin Spacing pane, select how the Multisizer 4e will determine bin spacing:

- By an incremental count of particle diameters (Linear Diameter)
- By a geometric progression of particle diameter sizes (Log Diameter)
- By a linear progression of particle volume (Linear Volume)
- By a geometric progression of particle volume size (Log Volume)

d. In the Options pane, select one or more of the following, if desired:

- **Pulse Edit.** Select this option if you have a narrow distribution and want to ensure accurate analysis results by eliminating large pulses. Particles that do not pass through the center of the aperture tend to produce wider pulses. Pulse edit removes the widest pulses from statistical analysis and displayed results.
- **Use Shape Factor.** Select this option if you want to match your results to another particle sizing instrument, or if you need to adjust analysis results based on a known factor. Entering a shape factor of 1.03, for example, increases particle diameter or volume by 3%.

e. Click **OK**.

Converting Pulses to Size Between Cursors

To Convert Pulses to Particle Size Between Two Cursor Points on a Graph

1. Select bin size and other options in the Convert Pulses to Size Settings window (see Converting Pulses to Size).

2. In the graph view, place two cursors to delimit analysis results. For information on inserting cursors, see Using Cursors in Graphs.

3. Select **Calculate > Convert Pulses between Cursors** on the Run Menu bar. The graph view changes to display only results between the two cursor points.

   NOTE **Convert Pulses between Cursors** allows you to select a range of particle sizes for analysis based on cursor points. You can also enter specific diameter or volume limits in the Convert Pulses to Size Settings window. Limiting the particle size range using either cursors or text input allows you to increase the precision of your analysis by viewing the same number of size bins within a smaller distribution.

Converting Pulses to Size for the Widest Distributions

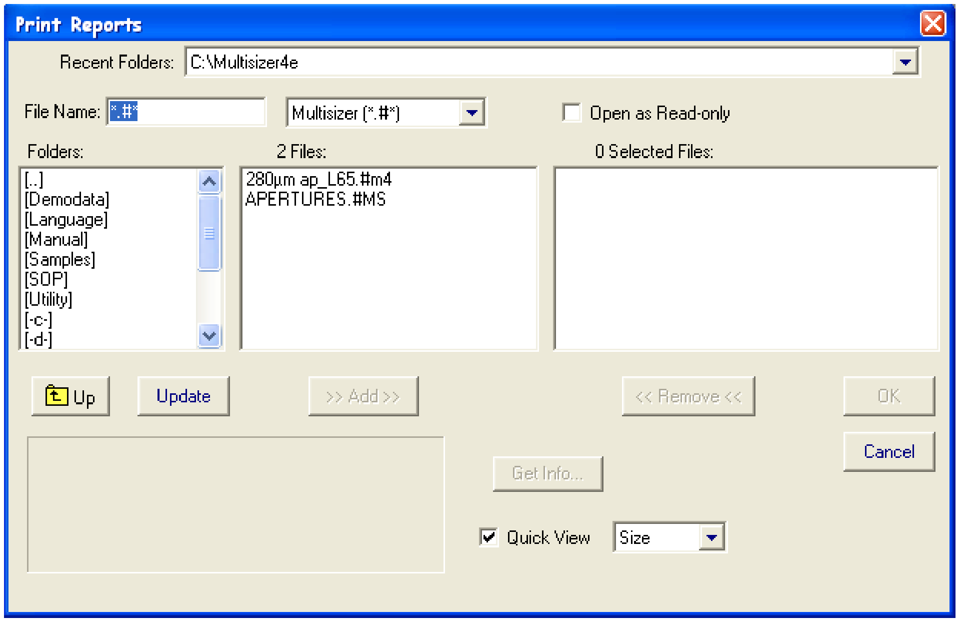
To automatically analyze the widest possible distribution based on aperture size, select **Calculate** > **Convert Pulses (2% to 60%)**.

To change the number of bins or other options associated with converting pulses, select **Calculate** > **Convert Pulses to Size** on the Run Menu bar. Enter new settings in the Convert Pulses to Size Settings window (see Converting Pulses to Size).

Printing Report(s) for One or More Analysis Files

If you want to print reports for multiple analysis runs or for a single analysis run that is not open in the Viewing Area:

1 Select **File** > **Print Reports** on the Main Menu bar. The Print Reports window opens.



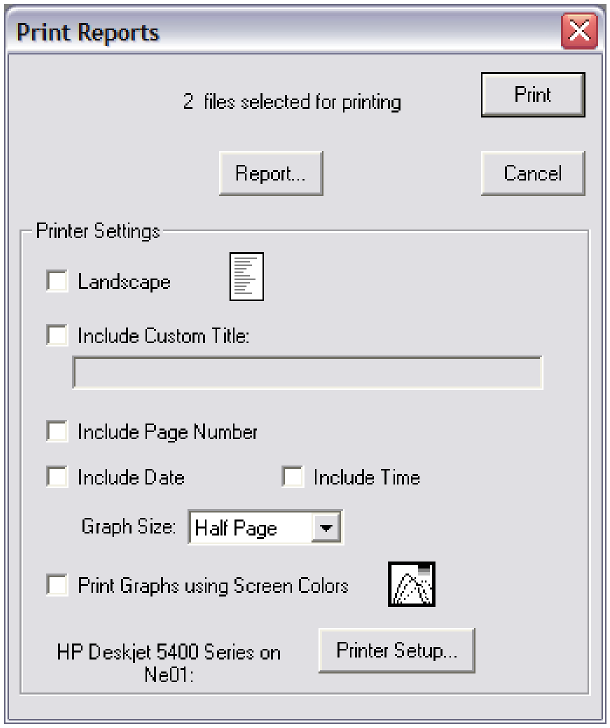
2 In the Print Reports window:

a. In the Folders box, double click the appropriate folder name. The folder contents appear in the Files box.

b. In the Files box, click to highlight the file(s) you want to print. To highlight multiple analysis files, hold the Ctrl key down while clicking each file name, or click one file and drag the files you want to include in the Selected Files box.

c. Click **Add**. The files appear in the Selected Files box.

d. Click **OK**. A second Print Reports window opens.

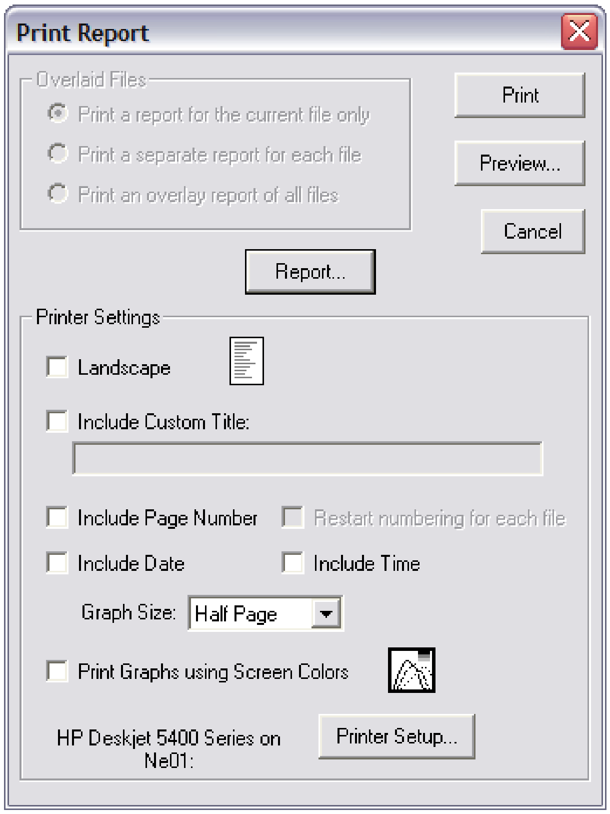


3 In the Print Reports window, click **Print**. For information on how to change printer settings or Preferences using the Print Report window, see Printing a Report of an Open Analysis File.

Printing a Report of an Open Analysis File

To print a report for an analysis file that is open in the Viewing Area:

1 Select **RunFile** > **Print Report** on the Run Menu bar. The Print Report window opens.



2 In the Print Report window:

Table 8.3 Open Analysis File Print Options

Select this option...	To...
Print a report for the current file only	Print a report for the open file in the Viewing Area only. This option is only available when multiple files are open.
Print a separate report for each file	Print a report for each open file in the Viewing Area.
Print an overlay report of all files	Print a report that includes statistics for all open files in the Viewing Area.
Landscape	Change the page orientation to Landscape (11 ´ 8.5). The page icon displays the current paper orientation (Portrait, or 8.5 ´ 11).
Include Custom Title	Enter a title for the report. Check this box only if you have not already included a title on the Preferences / Page Setup window (see Preferences: Page Setup in CHAPTER 6, Setting View and Print Preferences).
Include Page Number	Include page numbering in the report. Page numbers appear in the upper right corner of each page. Checking this box and other boxes in the Printer Settings pane will override settings you entered previously in the Preferences / Page Setup window (see Preferences: Page Setup in CHAPTER 6, Setting View and Print Preferences).
Restart numbering for each file	Include separate pagination for each file in an overlay file. This option is only available when you have opened an overlay, which consists of multiple files (see Using Overlay Files).
Include Date	Include the date you printed the report. The date will appear in the upper right corner of each page.
Include Time	Include the time you printed the report. The time will appear in the upper right corner of each page.
Graph Size	Use the drop-down list to select the size of graphs in the printed report. Selecting a new graph size will override the setting you entered previously in the Preferences / Page Setup window.
Print Graphs Using Screen Colors	Use the colors currently displayed in the on-screen graph.

- Click **Preview** to view the contents and layout of the report in a separate window before printing.
- Click **Report** to open the Edit Preferences window (see Creating a Preferences File in CHAPTER 6, Setting View and Print Preferences), where you can change print and view settings.
- Click **Printer Setup** to view printer settings.

3 Click **Print** to print the report.

Printing with Current Analysis Settings

You can print a graph or listing as it appears on the screen without changing Preferences settings.

To Print an Analysis as it is Displayed in the Viewing Area

1 Select **RunFIle > Print** on the Run Menu bar. The Print window opens.

2 In the Print window:

- **a.** Select the graphs, listings, and statistics you want to print. In the Printing Order box, your selections are listed in the order in which they will appear.
- **b.** In the Printer Settings pane, select print options. Click **Page Setup** to see more options, including margin and time/date settings.
- **c.** When you have made your selections, including graph size, color, and font, click **Print**.

Saving an Analysis File

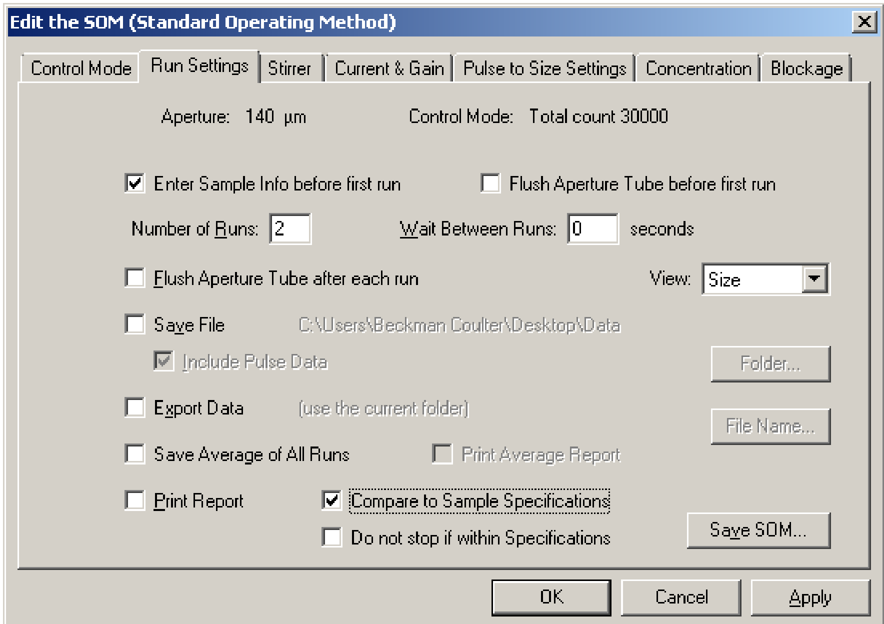
The Multisizer 4e can automatically save analysis data based on Standard Operating Method settings, or you can save sample data to a file when the analysis is complete.

Saving Analysis Data Automatically

To automatically save analysis data, select appropriate Run Settings in the current Standard Operating Method and enter sample information before the run.

1 Select **SOP > Edit the SOM** on the Main Menu bar. The Edit the SOM (Standard Operating Method) window opens.

2 In the Edit the SOM window, click the Run Settings tab.



3 On the Run Settings tab, select Enter sample info before first run. When this option is selected, the software opens the Enter Sample Info window when the operator starts an analysis.

4 Select Save File and, if desired, Include Pulse Data.

NOTE Selecting Include Pulse Data will increase the file size significantly.

5 Click **Folder.** The Select the Run Folder window opens.

6 In the Select the Run Folder window:

a. Select Use the current folder or Use a custom folder.

If you are using a custom folder, click **Browse** and navigate to the desired folder.

b. Click **OK** to return to the Edit the SOM window.

7 In the Edit the SOM window, click **File Name.** In the File Name Generation window.

8 Create the file name template by selecting components such as File ID, Date, and Time. For more information, see Using File Name Template Codes in CHAPTER 5, Selecting Analysis Settings: SOM and SOP, and click **OK.**

NOTE The file name generation template fields draw information from the Enter Sample Info for Next Run window (see Entering Sample Information in CHAPTER 7, Analyzing a Sample).

9 In the Edit the SOM / Run Settings window, click **OK**.

Saving Analysis Data Manually

If you have not used Standard Operating Method settings to save analysis data automatically (see Using a Standard Operating Method (SOM) in CHAPTER 5, Selecting Analysis Settings: SOM and SOP), or if you want to save analysis data using a new name, you can save the file after the analysis is complete.

To Save Analysis Data Manually When an Analysis is Open in the Viewing Area

1 Select **RunFile > Save** or **RunFile > Save As** on the Run Menu bar.

2 In the Save As window, navigate to the appropriate folder and enter a file name in the File Name field. If you want to write protect the analysis file, select Save as read-only in the middle of the window.

3 Click **Save**.

NOTE Analysis files are identified by the extension .#m4 or custom extensions created in the File Name Generation window (see Creating Automatically Generated Analysis File Names in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).

Opening a Saved Analysis File

1 Select **File > Open** on the Main Menu bar or click **Open** in the Main Toolbar.

2 In the Open window, navigate to the appropriate folder and select the file name.

3 Click **Open**.

Importing a File

You can import data from another program or transfer a size distribution from another instrument.

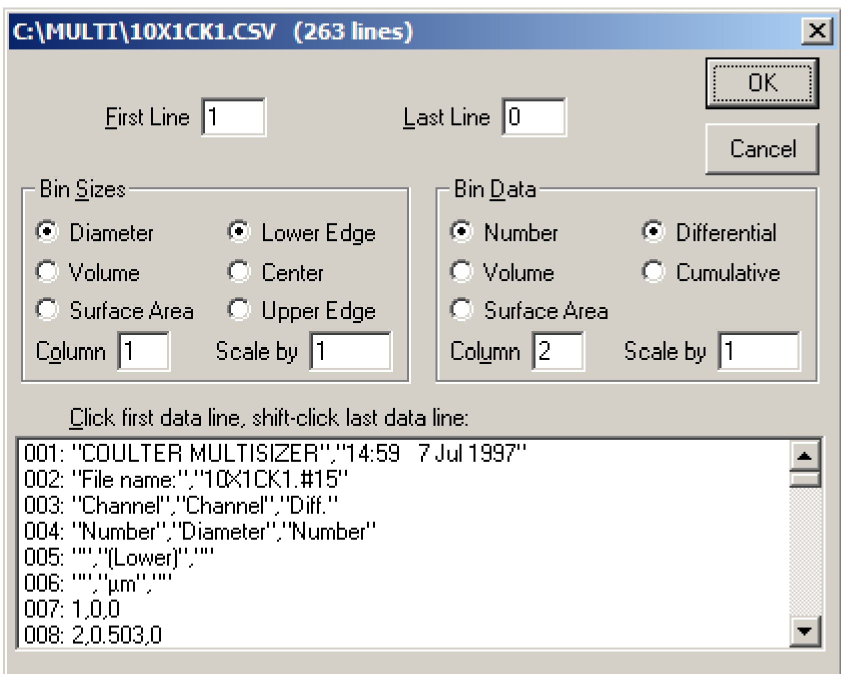
1 Select **File > File Tools > Import Size Data** on the Main Menu bar.

2 In the Import Size Data window:

a. Click the down arrow in the Files of Type text field to select the imported file type, or select All Files (*.*).

b. Navigate to the appropriate folder and select the file you want to import.

c. Click **Open**. An import file window displaying the path and name of the imported file opens.



3 In the import window, specify the type of size distribution to import.

4 Click and drag to select the data lines to import, or specify the lines in First Line and Last Line.

5 Click **OK**.

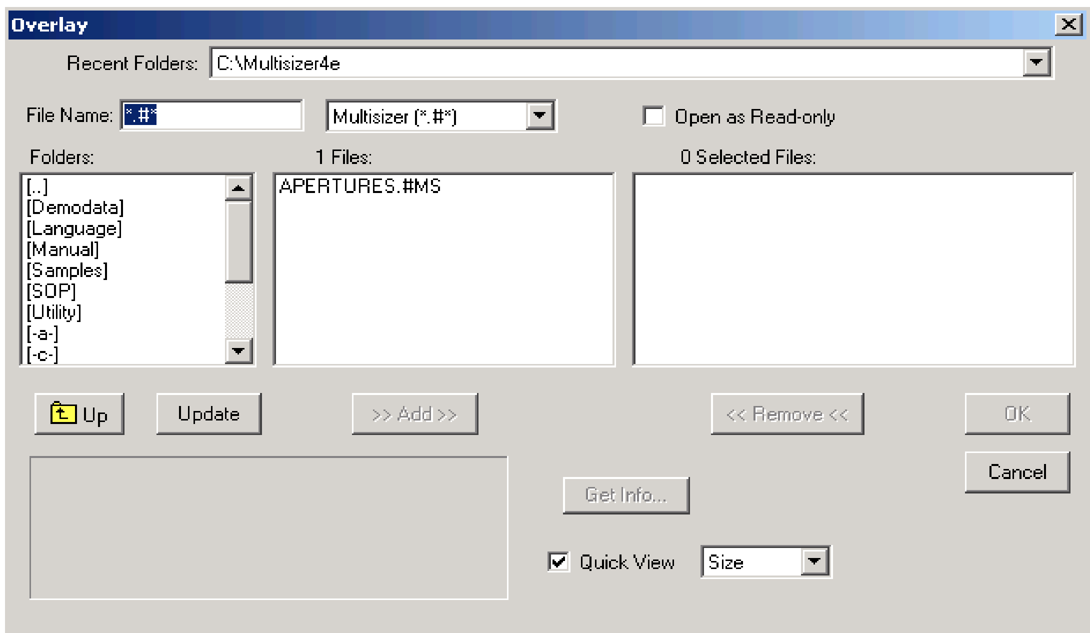
Working with Multiple Analysis Files

Using Overlay Files

Opening files for overlay allows you to view multiple analysis results on the same graph.

To Open Files for Overlay

1 Select **File > Overlay** on the Main Menu bar. The Overlay window opens.



2 In the Overlay window:

a. In the Folders box, double click the appropriate folder name. The folder contents appear in the Files box.

NOTE Selecting **Up** will allow you to move up in the directory.

b. In the Files box, click to highlight the file(s) you want to open for overlay. To highlight multiple analysis files, hold the Ctrl key down while clicking each file name, or click one file and drag over the files to be included.

c. Click **Add**. The files appear in the Selected Files box.

d. Click **OK**. The sample analysis data appear on one graph in the Viewing Area.

NOTE If the analyses you select have incompatible data or formatting, an incorrect file type error message will appear, and the files will open in individually tiled windows.

To Add a File to an Open Overlay

1 Select **RunFile** > **Open for Overlay** on the Run Menu bar. The Open for Overlay window opens.

2 In the Open for Overlay window, navigate to the file you want to add to the open overlay, and click **Open**.

3 The new analysis data appears on the graph.

Saving an Overlay File

An overlay file is saved as a list. The saved file does not contain the data from the original files, but points to the files in their current location. If you move or delete the original files, the overlay file will not open.

1 Select **RunFile** > **Save Overlay List** on the Run Menu bar.

2 In the Save a List of Overlaid Files window, navigate to the appropriate folder and enter a file name in the File Name field.

3 Click **Save**.

NOTE Overlay files are identified by the extension .ovr.

Opening a Saved Overlay File

1 Select **File** > **File Tools** > **Open Overlay List** on the Main Menu bar.

2 In the Open Overlay List window, navigate to the overlay list you want to open.

3 Click **Open**.

NOTE An overlay file does not contain the data from the original files, but points to the files in their current location. If you move or delete the original files, the overlay file will not open.

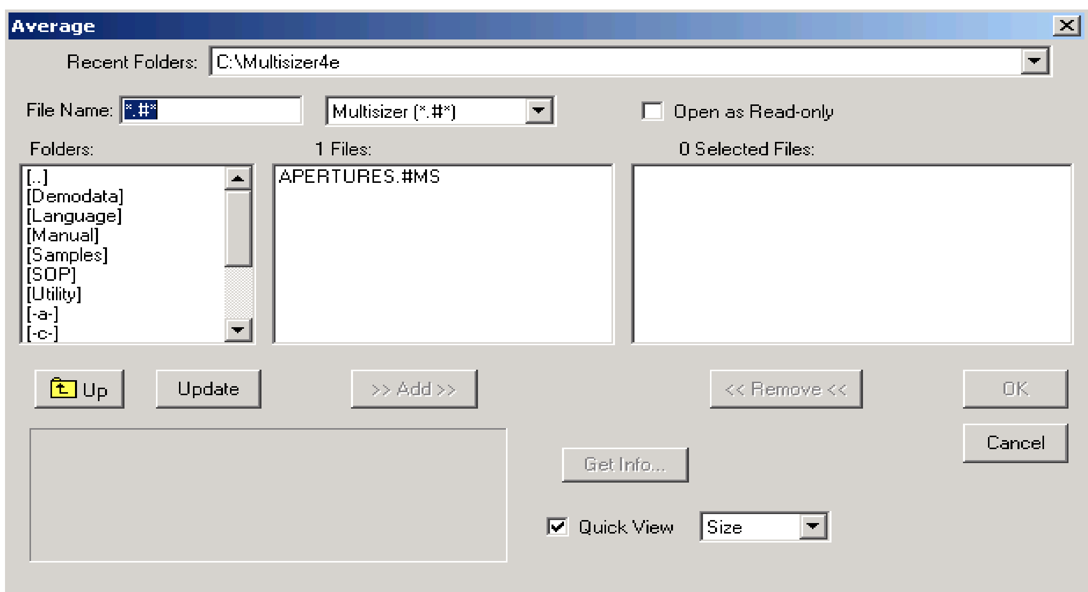
Averaging Analysis Files

The Multisizer 4e allows you to average the results of multiple analyses and view or print averaged statistics in a listing or graph.

Opening Files for Averaging

To View the Average of Two or More Analysis Files

1 Select **File** > **File Tools** > **Average** on the Main Menu bar. The Average window opens.



2 In the Average window:

a. In the Folders box, double click the appropriate folder name. The folder contents appear in the Files box.

NOTE Selecting **Up** will allow you to move up in the directory.

b. In the Files box, click to highlight the file(s) you want to average. To highlight multiple analysis files, hold the Ctrl key down while clicking each file name, or click one file and drag over the files to be included.

c. Click **Add**. The files appear in the Selected Files box.

d. Click **OK**.

Adding Data to an Average

1 Select **RunFile > Open and Add to Average** on the Run Menu bar. The Open window opens.

2 In the Open window, navigate to the file you want to add, and click **Open**.

3 The new average appears on the graph or listing.

Saving Averaged Files

You can save an average as a self-contained analysis file, or as a list of files (similar to an overlay file list, see Using Overlay Files). If you save an average as a list of files, the new file does not contain data from the original files, but points to the original files in their current location. If you move or delete the original files, the average file will not open.

To Save an Average as a Self-Contained Analysis File

1 Select **RunFile > Save** or **RunFile > Save As** on the Run Menu bar.

2 In the Save As window, navigate to the appropriate folder and enter a file name in the File Name field. If you want to write protect the averaged file, select Save as read-only in the middle of the window.

3 Click **Save**.

NOTE Averaged analysis files are identified by the extension .#av.

To Save an Average as a List

1 Select **RunFile > Save List of Averaged Files** on the Run Menu bar.

2 In the Save a List of Averaged Files window, navigate to the appropriate folder and enter a file name in the File Name field.

3 Click **Save**.

NOTE Average files saved as a list are identified by the extension .avg.

Opening a Saved List of Averaged Files

To Open a Self-Contained Average File (Extension .#av)

1. Select **File** > **Open** on the Main Menu bar, or click **Open** in the Main Toolbar.

2. In the Open window, navigate to the appropriate folder and select the file name.

3. Click **Open**.

To Open an Average File Saved as a List (Extension .avg)

1. Select **File** > **File Tools** > **Open a List of Averaged Files** on the Main Menu bar.

2. In the Open a List of Averaged Files window, navigate to the average list you want to open.

3. Click **Open**.

Creating a Size Trend Analysis

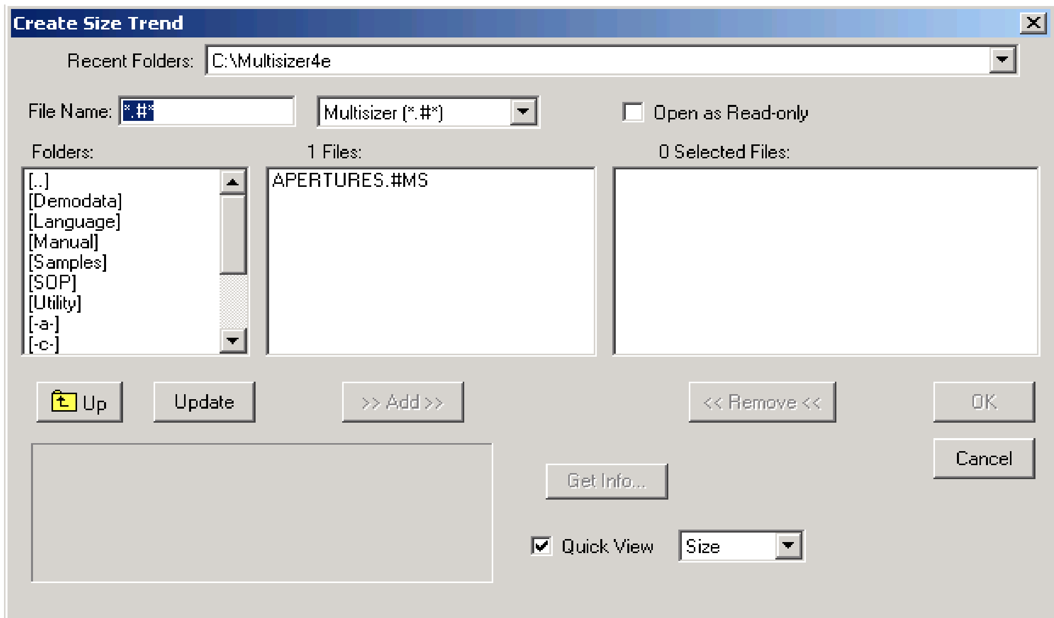
Creating a size trend analysis allows you to see variations in particle size over time or in relationship to other variables (such as pH). A Size Trend analysis combines data from multiple analyses into one graph or listing. You can update a Size Trend by adding additional analysis files at any time.

You can also use a size trend analysis to track changes in calibration (changes in the calibration constant, or Kd). For more information on Kd values, see Calibrating the Aperture in CHAPTER 4, Installing and Calibrating an Aperture Tube.

Opening Files for a Size Trend

To View Size Trend Data for Two or More Analysis Files

1 Select **File** > **File Tools** > **Create Size Trend** on the Main Menu bar. The Create Size Trend window opens.



2 In the Create Size Trend window:

a. In the Folders box, double click the appropriate folder name. The folder contents appear in the Files box.

NOTE Selecting **Up** will allow you to move up in the directory.

b. In the Files box, click to highlight the file(s) you want to include in the size trend analysis. To highlight multiple analysis files, hold the Ctrl key down while clicking each file name, or click one file and drag over the files to be included.

c. Click **Add**. The files appear in the Selected Files box.

d. Click **OK**.

Adding a File to a Size Trend

To Include Data from Another File to an Open Size Trend Analysis

1 Select **RunFile** > **Open and Add to Trend** on the Run Menu bar. The Open window opens.

2 In the Open window, navigate to the file you want to add, and click **Open**.

3 Analysis data from the new file is added to the size trend.

Saving a Size Trend

A size trend is saved as a self-contained analysis file.

1 Select **RunFile > Save** or **RunFile > Save As** on the Run Menu bar.

2 In the Save As window, navigate to the appropriate folder and enter a file name in the File Name field. If you want to write protect the trend file, select Save as read-only in the middle of the window.

3 Click **Save**.

NOTE Size trend files are identified by the extension .#tr.

Opening a Size Trend

1 Select **File > Open** on the Main Menu bar, or click **Open** in the Main Menu Toolbar.

2 In the Open window, navigate to the appropriate folder and select the file name.

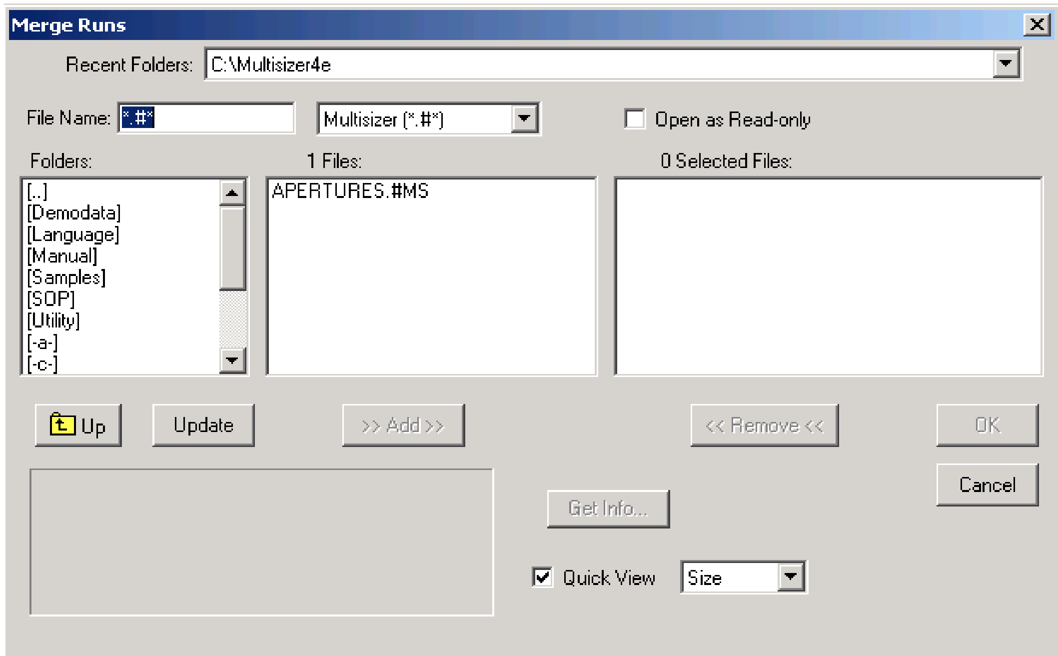
3 Click **Open**.

Merging Runs

Merging runs allows you to view combined pulse data and size distributions for multiple runs.

To Merge Two or More Analysis Files

1 Select **File > File Tools > Merge Runs** on the Main Menu bar. The Merge Runs window opens.



2 In the Merge Runs window:

a. In the Folders box, double click the appropriate folder name. The folder contents appear in the Files box.

NOTE Selecting **Up** will allow you to move up in the directory.

b. In the Files box, click to highlight the file(s) you want to merge. To highlight multiple analysis files, hold the Ctrl key down while clicking each file name, or click one file and drag over the files to be included.

c. Click **Add**. The files appear in the Selected Files box.

d. Click **OK**.

Saving a Merged Run

1 Select **RunFile > Save** or **RunFile > Save As** on the Run Menu bar.

2 In the Save As window, navigate to the appropriate folder and enter a file name in the File Name field. If you want to write protect the trend file, select Save as read-only in the middle of the window.

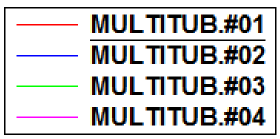
3 Click **Save**.

NOTE Merged files are saved as analysis files and are identified by the extension .#m3 or .#m4.

Moving Between Multiple Files (Cursor Menu and Shortcut)

There are two ways to move between overlay files (see Using Overlay Files) when a graph of the overlay is open in the Viewing Area.

- In the Main Toolbar, click **Cursor**. In the legend, the selected file (or cursor run) is underlined. As you click **Cursor**, the software underlines each run in sequence.



- Select **Cursor** on the Run Menu bar. The drop-down menu displays all analysis files in the overlay. Select a file name to display the cursors and statistics for the run.

To see statistics associated with the selected file (cursor run), select **RunFile > Get Info** on the Run Menu bar.

Filtration Efficiency

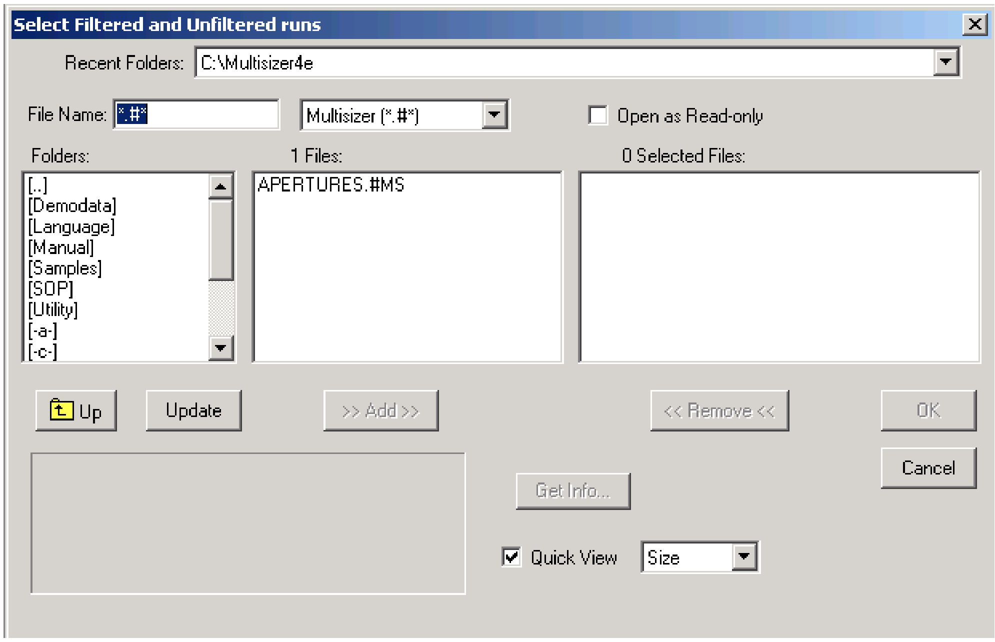
The Multisizer 4e allows you to measure filtration efficiency by comparing particle number and/or sizes in two runs, one before and one after filtering the sample.

For example, you can measure overall filtration efficiency. If the Analyzer counts 1000 particles in the first run, and, after applying the filter, counts 100 particles, then filtration efficiency is 90%.

For information on including filtration efficiency in printed reports, see Preferences: Printed Report in CHAPTER 6, Setting View and Print Preferences.

To View Filtration Efficiency

1 Select **File > File Tools > Filtration Efficiency** on the Main Menu bar. The Select Filtered and Unfiltered Runs window opens.



2 In the Select Filtered and Unfiltered Runs window:

a. In the Folders box, double click the appropriate folder name. The folder contents appear in the Files box.

b. In the Files box, click to highlight the two files you want to compare. To highlight both analysis files, hold the Ctrl key down while clicking each file name. Include the analyses in the order in which they were run (pre-filtration and post-filtration).

c. Click **Add**. The files appear in the Selected Files box.

d. Click **OK**.

Saving a Filtration Efficiency Analysis

1 Select **RunFile > Save** or **RunFile > SaveAs** on the Run Menu bar.

2 In the Save As window, navigate to the appropriate folder and enter a file name in the File Name field. If you want to write protect the file, select Save as read-only in the middle of the window.

3 Click **Save**.

NOTE Filtration efficiency files are saved as analysis files and are identified by the extensions .#m3 or .#m4.

Including Filtration Efficiency in a Printed Report

1. Select **SOP > Edit Preferences** on the Main Menu bar.

2. In the Edit Preferences / Printed Report window, select Filtration Efficiency. For more information on selecting filtration efficiency sizes, see Preferences: Printed Report in CHAPTER 6, Setting View and Print Preferences.

3. Click **OK**.

CHAPTER 9

Configuring Software and Security

Configuration Overview

The Configuration drop-down menu and certain configuration items in the Windows drop-down menu appear in the Multisizer 4e Main Menu bar only when you are running the software in No security mode. When running the software in a security mode, configuration menu items are available to Administrator and Supervisor accounts only. For information on Main Menu behavior in different security configurations, see Security Modes and Configuration.

Configuration Options

Depending on your security settings, configuration options will appear in the Main Menu in the Configuration, Supervisor, or Administrator drop-down menu. Multisizer 4e configuration options allow you to remove drop-down menu items from the Main Menu and Run Menu, change default file locations, and set the security level on menu items.

On the Windows drop-down menu, you can use the Status Panel Settings or Toolbars items to customize the Status Panel, the Main Toolbar, and the Instrument Toolbar (see Customizing the Status Panel and Customizing Toolbars).

Security Modes and Configuration

If you are running one of four security modes (see Customizing Security Settings), configuration options are available only if the user logs in as an Administrator or Supervisor. Menu items that appear in the Configuration drop-down menu when running the software with no security appear in the Administrator or Supervisor drop-down menu.

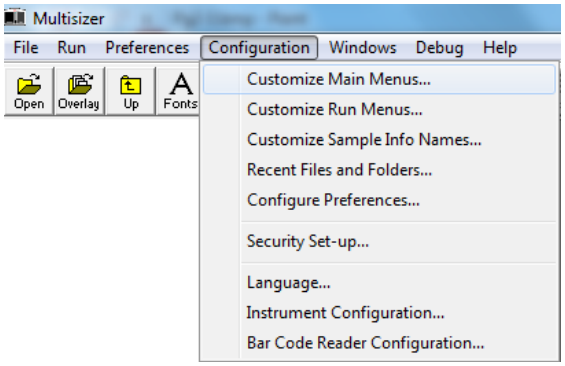
When a Reviewer, Operator, or Operator using advanced features opens the software and logs in (if necessary), no configuration options are available. The software displays a Security menu in the Main Menu bar.

Administrators can configure the security level to require different user types to log in with a username, username and password, or not at all. Depending on the security configuration the Administrator selects, items in the Security drop-down menu will change.

Possible Main Menu configurations for different user types and security levels are shown below.

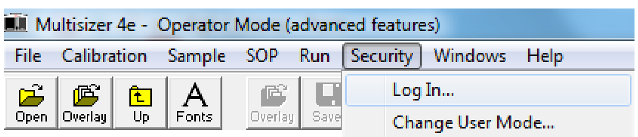
Main Menu: No security. Not connected to the Analyzer. Configuration options are available to all users.

Figure 9.1 Main Menu: No security



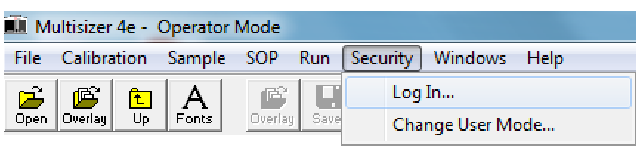
Main Menu: Medium security, Operator Mode. Username and password required. Configuration options are available to Administrator and Supervisor accounts only.

Figure 9.2 Main Menu: Medium Security



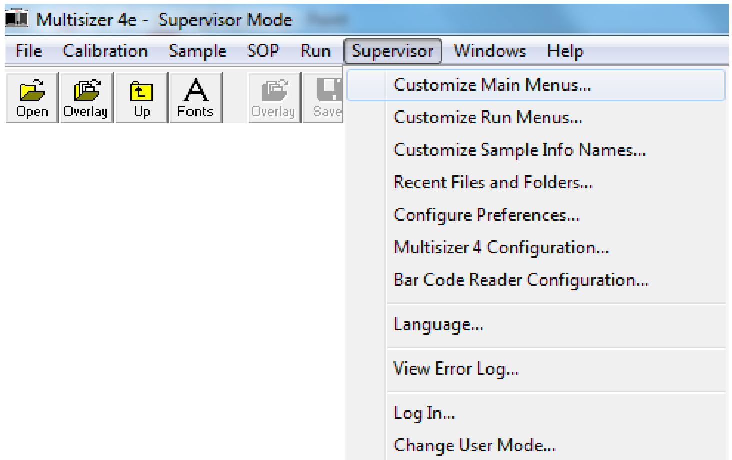
Main Menu: Low security, Operator Mode. Login required for Administrator and Supervisor accounts only. Configuration options are available to Administrator and Supervisor accounts only.

Figure 9.3 Main Menu: Low Security



Main Menu: Medium security, Supervisor Mode. Login with username and password required. Configuration options are available in the Supervisor drop-down menu.

Figure 9.4 Main Menu: Medium Security



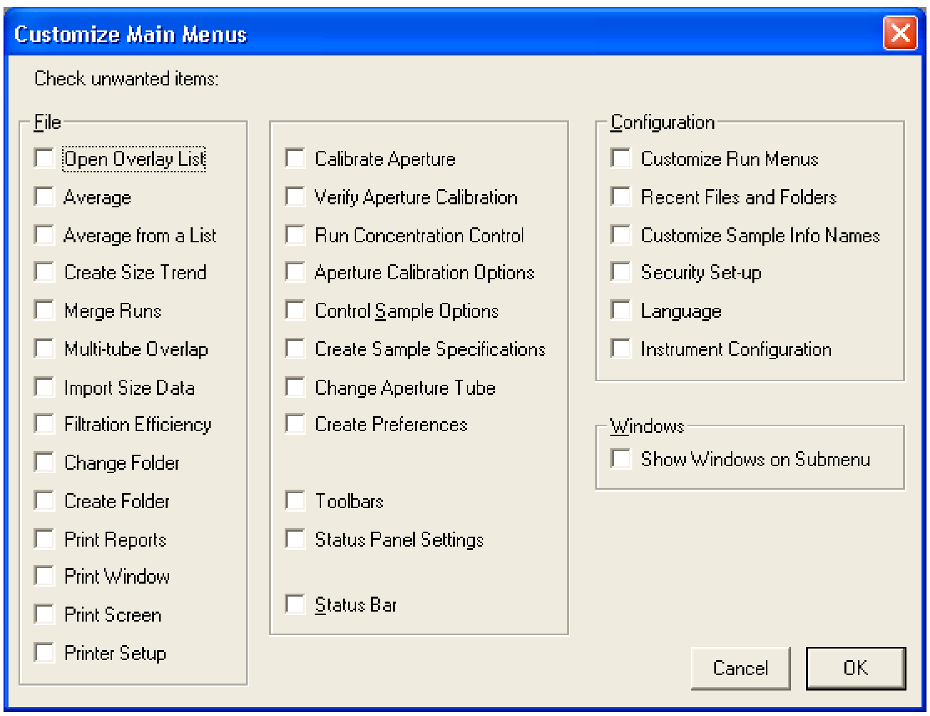
Customizing Main Menus

The Multisizer 4e software allows users with sufficient privileges (and all users when running no security mode) to hide menu items in the File, Calibration, Sample, SOP, Run, and Configuration drop-down menus on the Main Menu bar.

When you remove a menu item, the corresponding shortcut button, if one exists, remains on the Main Toolbar. For information on removing Main Toolbar buttons, see Customizing Toolbars.

NOTE When you hide a menu item, the Multisizer 4e software removes the item until you reinstate it using the Customize Main Menus window. Unlike menu items that are hidden, items that are temporarily unavailable during normal software function appear greyed out (for example, **Run** > **Last Run** is greyed out until you complete a first run).

Figure 9.5 Customize Main Menus



To Customize Main Menus

1 Select **Configuration** > **Customize Main Menus** on the Main Menu bar.

2 In the File pane, select the menu items you want to hide.

NOTE Some items in the File pane are located in **File** > **File Tools**. These items are Average, Average from a List, Create Size Trend, Merge Runs, Multi-tube Overlap, Import Size Data, and Filtration Efficiency.

3 In the center pane, select the menu items you want to remove from the Calibration, Sample, SOP, and Run drop-down menus.

- The following items in the center pane are located on the Calibration drop-down menu: Calibrate Aperture, Verify Aperture Calibration, Run Concentration Control, Aperture Calibration Options, and Control Sample Options.
- Create Sample Specifications is located on the Sample drop-down menu.
- Change Aperture Tube is located on the Run drop-down menu.
- Create Preferences is located on the SOP drop-down menu.
- Toolbars is located on the Windows drop-down menu. The Toolbars window allows you to customize button images and remove buttons on the Main Toolbar, remove the Main Toolbar, or remove the Instrument Toolbar.
- Status Panel Settings is located on the Windows drop-down menu. The Status Panel Settings window allows you to change the appearance and/or location of the Status Panel.

4 In the center pane, select Status Bar to remove the status text that appears in the lower left corner of the Multisizer 4e window. The Status Bar describes selected menu items or displays the default folder.

5 In the Configuration pane, select the menu items you want to remove from the Configuration menu.

6 In the Windows pane, select Show Windows on Submenu if you want to display the names of open analysis files in a submenu of the Windows drop-down menu. If this is not selected, open analysis files appear as items on the Windows drop-down menu.

Customizing Run Menus

You can hide items in the RunFile, Edit, View, and Calculate drop-down menus on the Run Menu bar.

When you remove a drop-down menu item, the corresponding shortcut button, if one exists, remains on the Main Toolbar. For information on removing Main Toolbar buttons, see Customizing Toolbars.

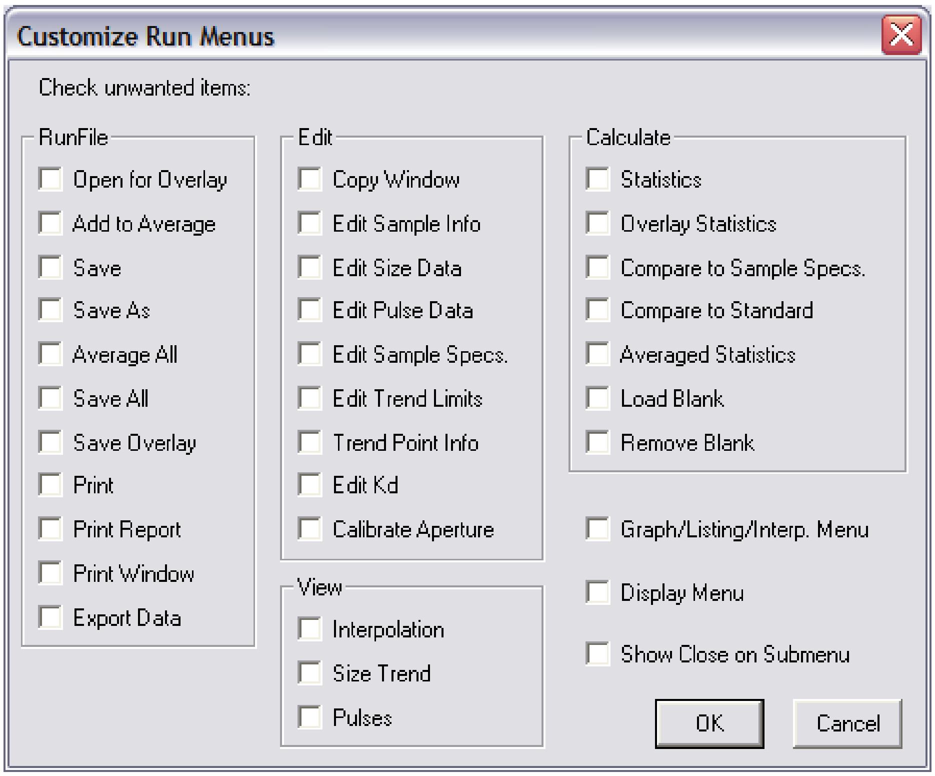
To Hide Drop-Down Menu Items

1 Select **Configuration > Customize Run Menus** on the Main Menu bar.

If you are running security mode, select **Customize Run Menus** in the Administrator or Supervisor drop-down menu.

2 In the Customize Run Menus window, select the menu items you want to hide and click **OK**. For more information on items listed in the Customize Run Menus window, see the table below. Changes take effect immediately. You do not need to restart the Multisizer 4e software.

NOTE When you hide a menu item, the Multisizer 4e software removes the item until you reinstate it using the Customize Main Menus window. Unlike hidden menu items, items that are temporarily unavailable during the normal functioning of the software appear greyed out (for example, **Run** > **RepeatLastRun** is greyed out until you complete a first run).



To Customize Run Menus

1 In the RunFile pane, select menu items in the RunFile drop-down menu that you want to hide.

2 In the Edit pane, select menu items in the Edit menu that you want to hide.

3 In the View pane, select menu items in the View menu that you want to hide.

4 In the Calculate pane, select menu items in the Calculate menu that you want to hide.

5 Select Graph/Listing/Interp. Menu to hide the drop-down menus that appear on the Run menu bar when you select Graph, Listing, or Interpolation in the View menu. These dynamically generated menus allow you to change view options for each type of analysis display. For more information on these menus, see Viewing Analysis Files in CHAPTER 8, Working with Analysis Files.

6 Select Display Menu to hide the Display drop-down menu that appears when you select Graph in the View menu. This dynamically generated menu appears only when you are viewing an analysis as a graph. For more information on the Display menu, see Viewing Analysis Files in CHAPTER 8, Working with Analysis Files.

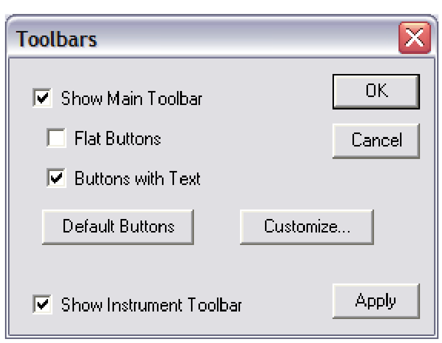
7 Select Show Close on Submenu to change the RunFile drop-down menu. If you are viewing multiple analysis files or overlay files (see Working with Multiple Analysis Files in CHAPTER 8, Working with Analysis Files), menu items appear at the bottom of the RunFile menu that allow you to close each of the open files. To move the Close items to a submenu, select Show Close on Submenu.

Customizing Toolbars

You can remove the Main Toolbar and/or the Instrument Toolbar. On the Main Toolbar, you can customize button images, add and remove buttons, and change button order. When you remove a button from the Main Toolbar, the corresponding drop-down menu item remains on the Main Menu or Run Menu. For information on removing Main Menu items, see Customizing Main Menus. For information on removing Run Menu items, see Customizing Run Menus.

To Customize the Toolbars

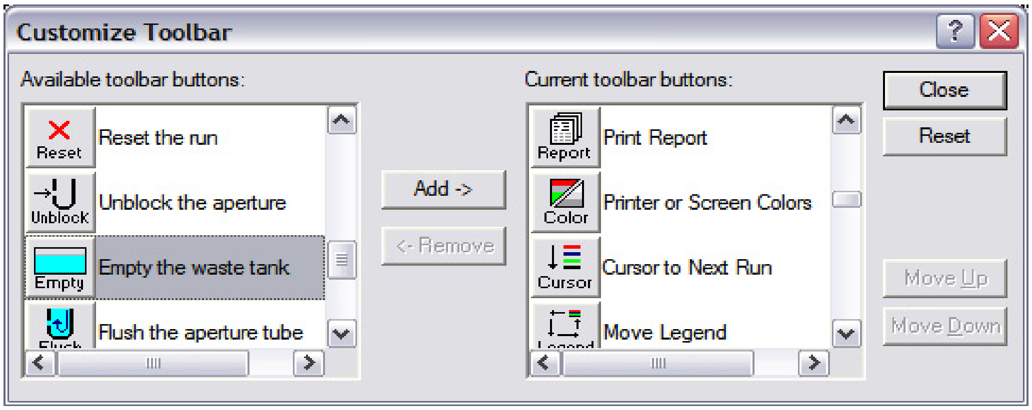
1 Select **Windows > Toolbars** on the Main Menu bar. The Toolbars window opens.



2 In the Toolbars window:

- To remove the Main Toolbar, clear the Show Main Toolbar checkbox.

- To display the buttons without three-dimensional edges, select Flat Buttons.
- To display buttons with text below the button image, select Buttons with Text. To display buttons with images only, clear this checkbox.
- To remove the Instrument Toolbar, clear the Show Instrument Toolbar checkbox.
- To return to the default Multisizer 4e software button configuration, click **Default Buttons**.
- To add or remove a Main Toolbar button, or to change the button order, click **Customize**. The Customize Toolbar window opens.



3 In the Customize Toolbar window:

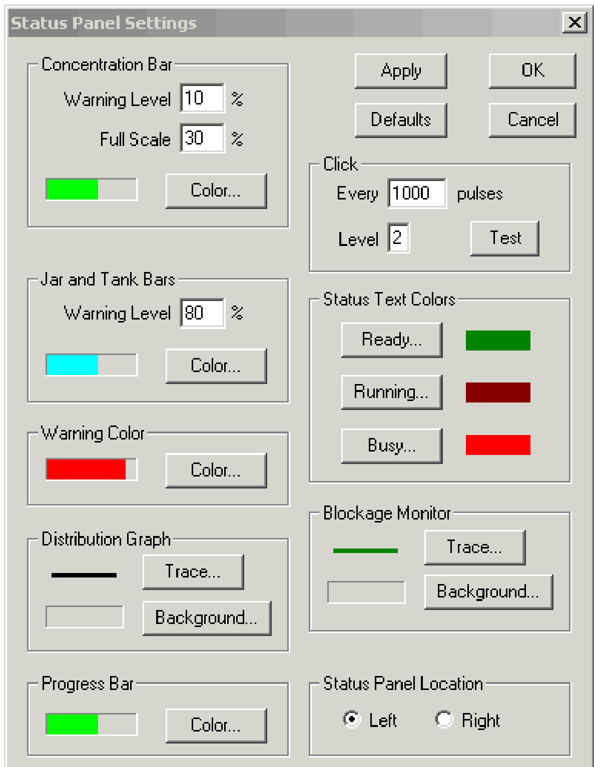
- To add a button to the toolbar, select a button in the Available Toolbar Buttons box and click **Add**.
- To remove a button from the toolbar, select a button in the Current Toolbar Buttons box and click **Remove**.
- To reposition the button on the toolbar, select a button in the Current Toolbar Buttons box and click **Move Up** or **Move Down**.
- Click **Close** to save your settings and return to the Toolbars window.

4 In the Toolbars window, click **OK** when you are finished. Changes take effect immediately. You do not need to restart the Multisizer 4e software.

Customizing the Status Panel

You can customize the colors that appear on the Status Panel, adjust warning levels, set the speed and volume of audible clicks, and change the location of the Status Panel.

1 Select **Windows** > **Status Panel Settings** on the Main Menu bar. The Status Panel Settings window opens.



NOTE **Windows** > **Status Panel Settings** is available only when the software is connected to the Analyzer.

2 In the Concentration Bar pane:

a. In the Warning Level field, enter the concentration or coincidence correction percentage that will trigger a warning (that is, change the scale color to red or the warning color you select). The default maximum concentration is 10%. For more information on suggested concentration levels, see Preparing the Analyzer for Calibration in CHAPTER 4, Installing and Calibrating an Aperture Tube.

b. In the Full Scale field, enter the total concentration percentage that the scale will display. Beckman Coulter recommends that you enter a value that is twice the Warning Level percentage. The default Full Scale percentage is 20%.

c. Click **Color** to change the scale color used to indicate that the concentration or coincidence correction level is within the acceptable range.

3 In the Jar and Tank Bars pane:

a. In the Warning Level box, enter the percent full that will trigger a warning (that is, change the Jar or Waste Tank display to red or the warning color you select). The default warning level is 80%.

b. Click **Color** to change the color used to display Jar and Waste Tank fluid that is below the warning level.

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4 In the Warning Color, Distribution Graph, Progress Bar, Status Text Colors, and Blockage Monitor panes, select the desired Status Bar display options such as lines and colors.

---

5 In the Click pane:

a. In the Every [X] pulses field, enter the number of electrical pulses that will occur between audible clicks on your computer.

b. In the Level field, enter a value between 0 and 5 to set the desired volume of the clicks.

c. Click **Test** to test the volume level.

---

6 In the Status Panel Location pane, select Left or Right to display the Status Panel on the left or right side of the Multisizer 4e software window.

---

7 Click **OK** when you are finished. Changes take effect immediately. You do not need to restart the Multisizer 4e software.

Customizing Sample Info Names

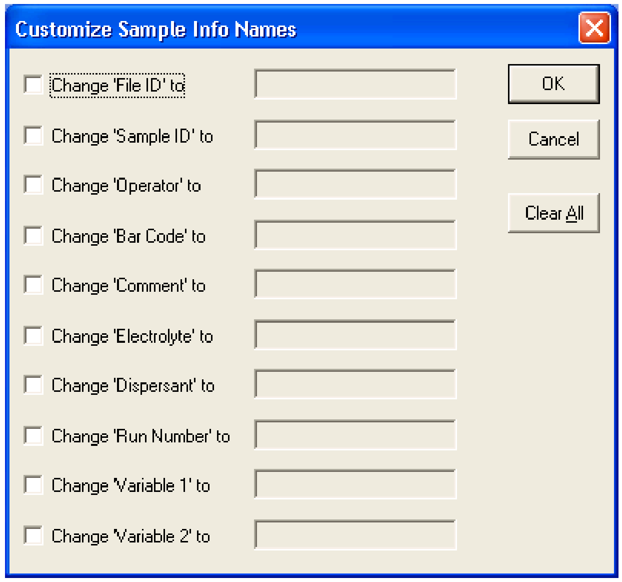
You can change the default terms used for field names such as File ID, Operator, and Variable 1. For example, you can change Operator to Technician, or specify a variable you will use in sample runs by changing Variable 1 to pH.

The new terms you configure in the Multisizer 4e software will appear in the Enter Sample Info window, the Edit the SOM (Standard Operating Method) window, throughout the software, and in printed reports.

---

1 In the Main Menu bar, select **Configuration > Customize Sample Info Names**. The Customize Sample Info Names window opens.

If you are running security mode, select **Customize Sample Info Names** in the Administrator or Supervisor drop-down menu.



2 In the Customize Sample Info Names window:

a. For each field name you want to change, select it and enter a new name in the corresponding text field.

b. Click **OK**. Changes take effect immediately. You do not need to restart the Multisizer 4e software.

NOTE If a Standard Operating Procedure (see Definitions: SOM, Preferences, and SOP in CHAPTER 5, Selecting Analysis Settings: SOM and SOP) is currently loaded, the Multisizer 4e software will not allow you to change field names. If you have loaded an SOP, exit the Customize Sample Info Names window, click **Remove SOP** in the Status Panel, and return to the Configuration menu.

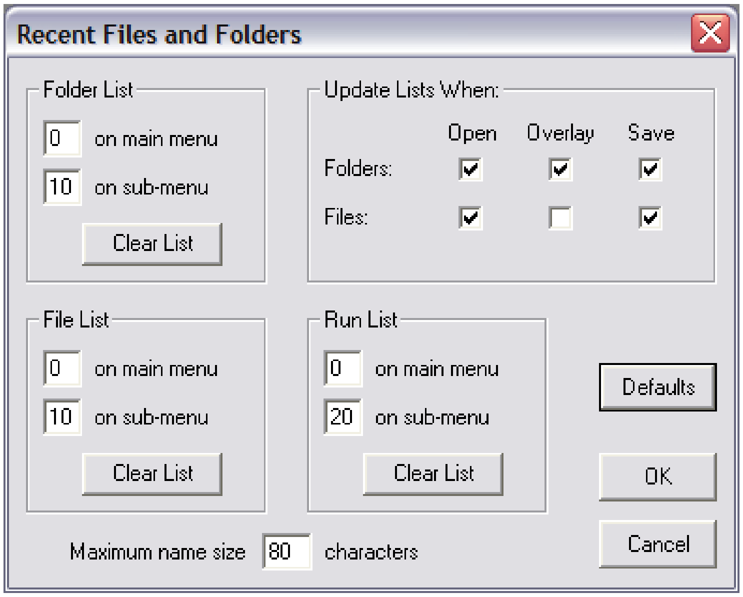
Customizing Recent File and Folder Lists

You can select how many recently used files and folders will appear in the File drop-down menu or in File submenus (Recent Folders and Recent Files).

You can configure the Multisizer 4e software to include a file or folder in the Recent Folders or Recent Files list when the file or folder is opened, used in an overlay (see Using Overlay Files in CHAPTER 8, Working with Analysis Files), and/or saved.

1 Select **Configuration** > **Recent Files and Folders** on the Main Menu bar. The Recent Files and Folders window opens.

If you are running security mode, select **Recent Files and Folders** in the Administrator or Supervisor drop-down menu.



2 Folder List pane options:

- In the on main menu field, enter the number of recently used folders you want to make available from the File drop-down menu. If you enter 0, no recently used folders will appear on the menu.
- In the on sub-menu field, enter the number of recently used folders you want to appear when the operator selects **File** > **Recent Folders**. If you enter 0, no submenu will appear.

3 File List pane options:

- In the on main menu field, enter the number of recently used files you want to make available from the File drop-down menu. If you enter 0, no recently used files will appear on the menu.
- In the on sub-menu field, enter the number of recently used files you want to appear when the operator selects **File** > **Recent Files**. If you enter 0, no submenu will appear.

4 Run List pane options:

- In the on main menu field, enter the number of recent runs you want to access from the File drop-down menu in the Main Menu bar. Recent runs will appear only if you select the option to automatically save the run in the **Edit the SOM** > **Run Settings** tab (see Using a Standard Operating Method (SOM) in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).

- In the on sub-menu field, enter the number of recent runs you want to appear when the operator opens **File > Recent Runs**. If you enter 0, no submenu will appear. Including recent runs in the File submenu can help operators locate analysis files saved automatically (see Using a Standard Operating Method (SOM) in CHAPTER 5, Selecting Analysis Settings: SOM and SOP). If you enter 0, no recent runs will appear.

5 In the Update Lists When pane, select options to define when the Multisizer 4e software will add a file or folder to the recently used list. By selecting a combination of checkboxes, you can choose to add a file to the list only when the file is saved, any time a file is opened, and/or when the operator adds a file to an overlay (see Using Overlay Files in CHAPTER 8, Working with Analysis Files).

6 In the Maximum name size field, enter the maximum number of characters that will display in the File drop-down list or the File submenus.

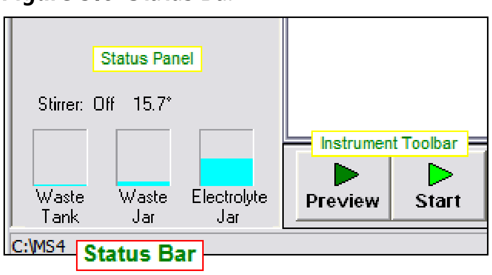
NOTE The number of characters you select for display in the Maximum name size field will not affect the maximum file name length you selected in the Standard Operating Method (see Using a Standard Operating Method (SOM) in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).

Selecting a New Folder from the Recent Folder List

Configuring the Multisizer 4e to display a list of recently used folders in the File drop-down menu or the Recent Folders submenu allows the operator to switch quickly between different folders.

The name of the folder that will open when you select **File** > **Open** or click **Open** in the Main Toolbar is displayed in the Status Bar. The Status Bar is located in the bottom left corner of the Multisizer 4e software window.

Figure 9.6 Status Bar



Adding a Folder to the Recent Folders List

You can populate the list of folders that will appear in the File menu or the **File > Recent Folders** submenu.

1 Select **File > Change Folder** on the Main Menu bar.

2 In the Change Folder window, select a folder and click **Add to List**.

3 Click **OK**.

Customizing the Save Preferences Default Setting

When exiting the Multisizer 4e software, you can determine whether the Multisizer 4e software will automatically save changes to default Preferences settings the operator makes.

To Automatically Save New Preferences Settings When Exiting the Multisizer 4e Software

1 Select **Configuration** > **Configure Preferences** on the Main Menu bar. The Preferences window opens.

If you are running security mode, select **Configure Preferences** in the Administrator or Supervisor drop-down menu.

If you...	Changes made by the operator...
Select Automatically save preferences	Will be saved to the default preferences file when the operator exits the Multisizer 4e software, overwriting previous settings.
Clear Automatically save preferences	Will not be saved when the operator exits the Multisizer 4e software. To save changes to preferences settings, the operator will need to save the settings to an alternate preferences file.

2 After you made your selection, click **OK**.

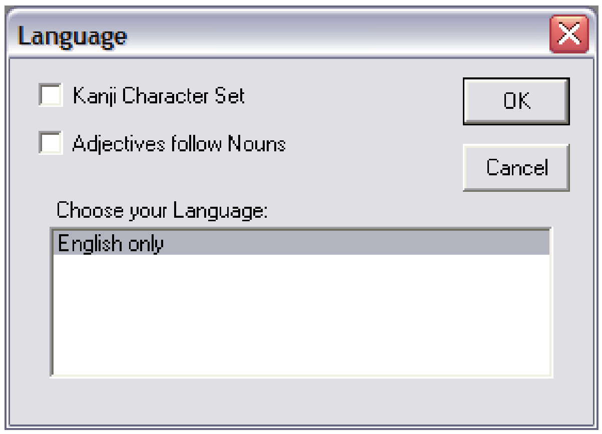
Choosing a Language

You can choose the language that will appear in menus, windows, and printed reports.

1 Select **Configuration** > **Language** on the Main Menu bar. The Language window opens.

If you are running security mode, select **Language** in the Administrator or Supervisor drop-down menu.

NOTE Select the Kanji Character Set if your system cannot display special characters such as the Greek mu (µ) correctly.



2 Select a language listed in Choose your language and click **OK**.

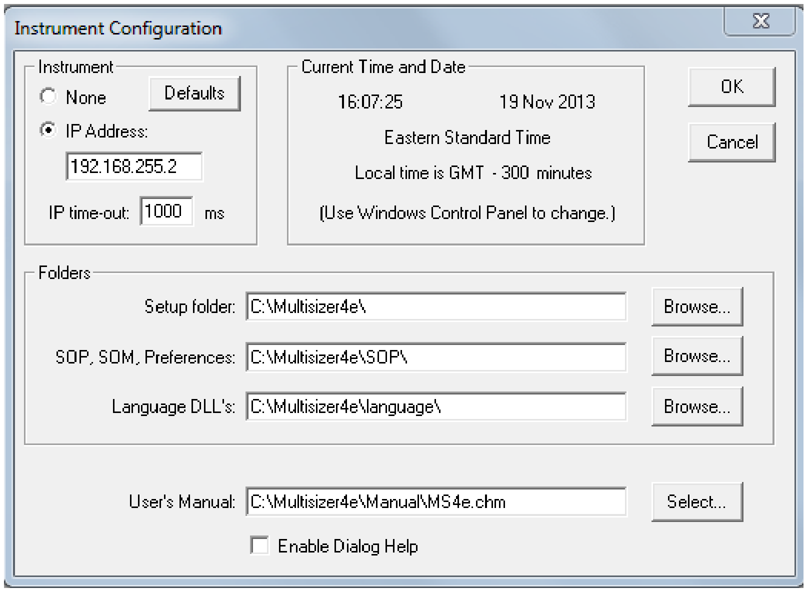
Multisizer 4e Configuration Options

The Instrument Configuration window allows you to set default folder locations for SOM, SOP, and Preferences files. You can also use this window to select new Help files (User Manual, Appendix, and Bibliography) as updates become available. If the Multisizer 4e software cannot find Preferences, Standard Operating Method, and other settings files on startup, one or more error messages will appear. If one of these error messages appears, use the Instrument Configuration window to reset default folder(s) for settings, preferences, and run files.

To Change Default File Locations

1 Select **Configuration** > **Instrument Configuration** on the Main Menu bar. The Multisizer 4e Configuration window opens.

If you are running security mode, select **Instrument Configuration** in the Administrator or Supervisor drop-down menu.



2 In the Folders pane, click **Browse** next to the Setup folder; SOP, SOM, Preferences; or Language DLL’s. The Browse for Folder window opens.

3 Navigate to and select the appropriate folder, and click **OK** to close the Browse window.

4 Click **Select** next to User’s Manual to select the location of the Multisizer 4e User Manual.

5 Select Enable Dialog Help to display a shortcut to context-sensitive online Help topics in Multisizer 4e software windows. Help topics are not available from all windows.

6 Click **OK** to save your settings.

Bar Code Reader Configuration Options

You can use the Bar Code Reader on the Analyzer Control Panel to scan:

- Beckman Coulter ISOTON II label
- Beckman Coulter Multisizer 4e Smart Technology Beakers
- Beckman Coulter Smart Technology Aperture Tubes

- Beckman Coulter Control material Assay sheets
- Customer sample packets and/or process sheets

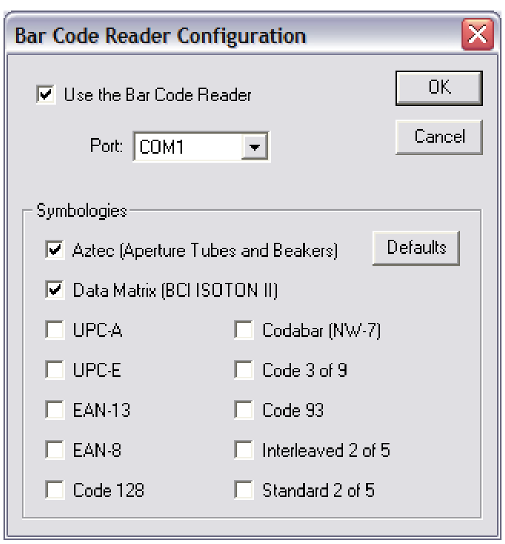
The Bar Code Reader (see Bar Code Reader in CHAPTER 2, Analyzer Overview) allows the operator to:

- Automatically load Beckman Coulter ISOTON II label information into the required software fields including lot number, expiration date, and the open date.
- Automatically load the Beckman Coulter ST beakers and allow automatic stirrer positioing. The Analyzer can then move the stirrer to the optimal position for that beaker to provide uniform particle distribution.
- Automatically load the Beckman Coulter ST Aperture Tube size, serial number, and Kd value after the tube has been calibrated.
- Automatically load the customer's packet labels or process sheet information into sample information.
- Automatically load the Beckman Coulter Material Assay sheet value, lot number, expiration date, and open date into the required fields for the Analyzer.

To Configure the Bar Code Reader

1 Select **Configuration > Bar Code Reader Configuration** on the Main Menu bar. The Bar Code Reader Configuration window opens.

If you are running security mode, select **Bar Code Reader Configuration** in the Administrator or Supervisor drop-down menu.



2 In the Bar Code Reader Configuration window, select Use the Bar Code Reader.

3 Select a port from the Port drop-down list. Because the software maps USB to COM ports, the correct port may be difficult to determine. If necessary, select COM ports sequentially until you locate the correct port.

4 In the Symbologies pane, select Aztec and Data Matrix. Select additional UPC symbologies only if you will be using the Bar Code Reader to scan sample packets or process sheets that you know use the specific UPC symbologies. Click **OK**.

Customizing Security Settings

You can run the Multisizer 4e with no security or one of four levels of password-protected security. With the exception of 21 CFR Part 11 security, the administrator determines the user privileges that will be associated with each security level (Low, Medium, or High).

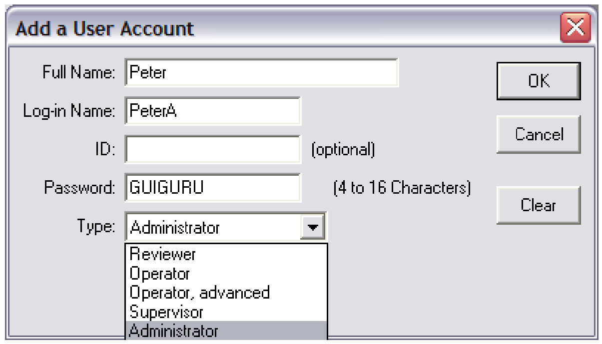
When selected, 21 CFR Part 11 security ensures that all settings meet the requirements of the Electronic Records and Electronic Signatures Rule (21 CFR Part 11). The FDA established the Rule to define requirements for submitting documentation in electronic form and the criteria for approved electronic signatures. For more information, see Regulatory Compliance in CHAPTER 11, Regulatory Compliance.

To Create an Administrator Account

1 In the Main Menu bar, select **Configuration > Security Set-up**. The Security Set-up window opens.

2 In the Security Set-up window, select a security level.

3 If you selected Low, Medium, High, or 21 CFR Part 11 security, click **Add a User**. The Add a User Account window opens.



4 In the Add a User Account window, in the Full Name, Log-in Name, and ID fields, enter user information.

5 In the Password field, enter a password between 4 and 16 characters in length.

6 Select Administrator from the Type drop-down list. The Administrator assigns privileges to each user type (Reviewer, Operator, Operator using advanced features, and Supervisor).

7 Click **OK**. A dialog box reminds you to record the Administrator’s log-in name and password.

8 Click **OK** to close the dialog box. A dialog box asks if you want to add a user account. Click **OK** to open the Add a User Account window or **Cancel** to close the window.

Logging In

Some security settings require all operators to log in with a username or a username and password. When log-in is required, the Log In window will appear when a user opens the software. If the user does not have log-in privileges, the software is locked. He or she must click **Exit** to close the software.

Figure 9.7 Log In

1 In the Log In window, enter your Log-in Name and Password, if required.

2 In the Log-in Mode pane, select a user account type. You can select the user type the Administrator assigned to you (for example, Operator using advanced features) or any user type with lesser privileges.

3 Click **OK**.

Logging Out

- If you are logged in as a Reviewer, Operator, or Operator using advanced features, select **Security** > **Log Out** from the Main Menu.
- If you are logged in as a Supervisor, select **Supervisor** > **Log Out** on the Main Menu bar.
- If you are logged in as an Administrator, select **Administrator** > **Log Out** on the Main Menu bar.

Changing Your Password

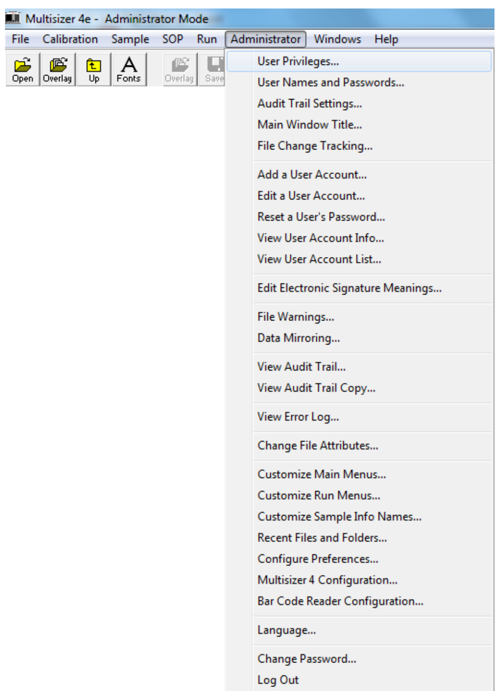
If you are logged in as a Reviewer, Operator, or Operator using advanced features, select **Security** > **Change Password** on the Main Menu bar.

Administrator Menu Options

When a user with Administrator privileges logs in to the Multisizer 4e software, the Main Menu bar displays the Administrator drop-down menu. The Administrator drop-down menu allows you to:

- Configure security requirements and privileges for each user type (see Customizing User Privileges)
- Add user accounts (see Customizing Security Settings)
- Create electronic signature requirements (see Maintaining Electronic Signatures in CHAPTER 11, Regulatory Compliance)
- View audit trails and error logs (see File History and Audit Trail in CHAPTER 11, Regulatory Compliance)
- Configure the software interface (see Configuration Overview)
- Change your password and log out.

Figure 9.8 Administrator Tab

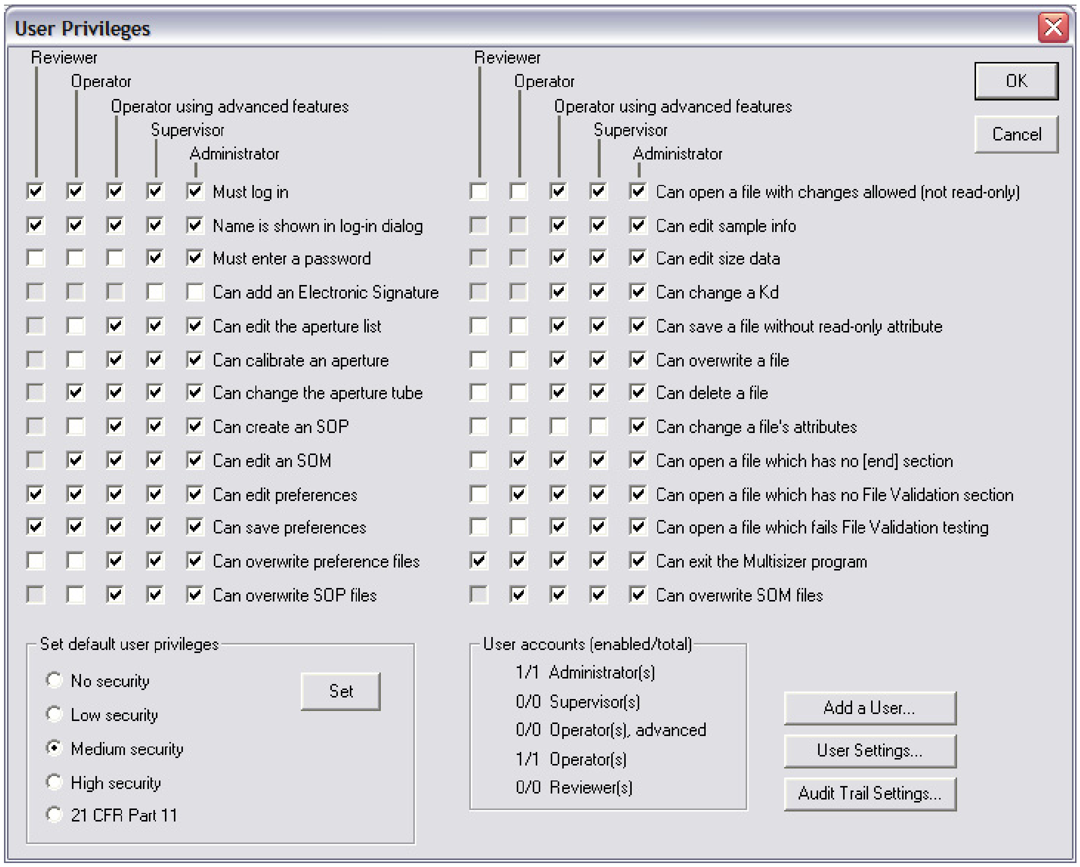


Customizing User Privileges

To Set Security Level and Customize User Privileges

1 Log in as Administrator.

2 Select **Administrator > User Privileges** on the Main Menu bar. The User Privileges window opens.



3 In the Set default user privileges pane, select a security level and click **Set**. The software returns the security requirement and privileges to the default configuration for the security level you select.

4 Select the security requirements and privileges for Reviewer, Operator, Operator using advanced features, Supervisor, and Administrator.

5 Click **OK**.

CHAPTER 10

Troubleshooting

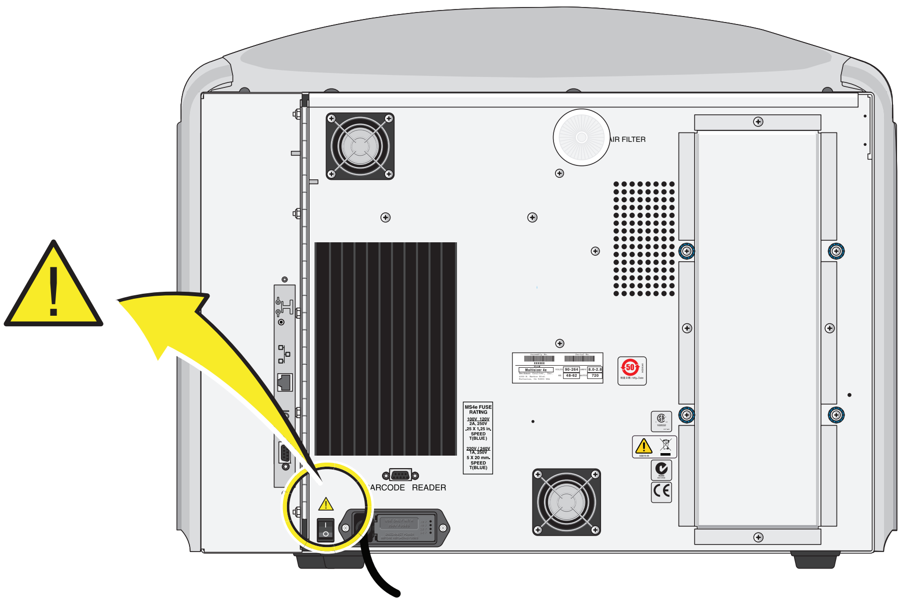
Hazard Labels and Locations

Carefully read the hazard warning labels on the instrument. The hazard labels are located on the instrument as indicated.

NOTE If a label is missing or unclear, contact your Beckman Coulter Representative.

Caution/Warning Label and Location

Figure 10.1 Caution/Warning Label on the Back of the Instrument

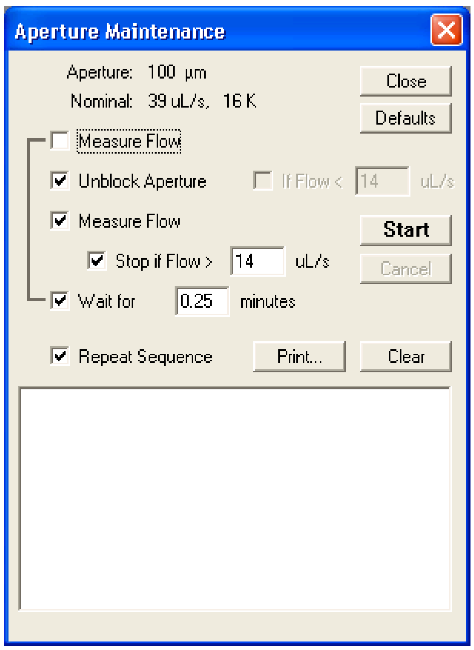


Resolving Common Problems

Aperture Blockage

The Aperture Maintenance function is the prefered method to be used to unblock apertures. The Aperture Maintenance function performs a continuous cycle of "unblock" functions until an acceptable flow rate for the aperture tube installed is attained. This is accomplished by performing an "unblock" function followed by Measure Flow Rate. This process continues until the result of the last Measure Flow Rate attains an acceptable flow rate for the aperture tube installed in the instrument.

Figure 10.2 Aperture Maintenance Window



To access the Aperture Maintenance function select **SERVICE** > **Aperture Maintenance** in the Main Menu. The Aperture Maintenance window will appear. See Figure 10.2.

The sequence can be adjusted for a "Wait Time" and an additional Measure Flow Rate.

The **Stop if Flow >** selection is adjustable to increase or decrease the target flow rate for the aperture tube installed. To return to the recommended values and/or setup sequence, select **Defaults**.

Once the Aperture Maintenenace is started, it will run unattended until the target flow rate is attained.

A log of actions and results are presented in the open field at the bottom of the Aperture Maintenance screen.

Normal Aperture Maintenance is performed with the aperture tube in an Accuvetter ST or a ST Beaker filled with filtered, high quality DI water and placed on the platform and raised to the up position. For this type of operation, the "Wait Time" can be increased to provide a "soak time" for the tube during a sequence.

NOTE The Aperture Maintenance function also works for aperture tubes larger than 280 µm. These larger tubes do not use Volume for Run Control. So, the metering pump cannot be used to attain flow rate values.

For larger tubes, Aperture Maintenance presents aperture resistance as a guide to the degree of a "blocked" aperture tube. These resulting resistance values are presented in the log area at the bottom of the Aperture Maintenenace screen. See Figure 10.2.

The Aperture Maintenance function has to be manually stopped when using aperture tubes larger than 280 µm because there is no **Measure Flow Rate** parameter to compare and enable a **Stop**.

NOTE The metering pump, Volume Run Control is not available for aperture tubes larger than 280 µm.

Monitoring the aperture tube resistance presented in the log area and comparing it to the "Nominal" value will give an indication of when to select **Cancel**.

NOTE The "Nominal" resistance value is listed at the top of the Aperture Maintenance screen under the installed aperature tube.

If the aperture maintenance cycle is unable to resolve high background counts, cleaning may be required. If this is the case, see Cleaning the Aperture Tube, Beakers, and Accuvette STs in CHAPTER 2, Analyzer Overview.

Reducing Noise

Noise often appears as an increased small particle count or varying particle counts that do not follow an expected size distribution.

Increasing or varying noise levels can be caused by:

- **Debris on the outside of the aperture tube and/or external electrode.** Clean up to a higher than the highest sample vial level will reach. See Cleaning the Aperture Tube, Beakers, and Accuvette STs in CHAPTER 2, Analyzer Overview for more information on cleaning the aperture tube and electrode.
- **Electrolyte solution.** If you are using a small aperture tube, increasing electrolyte salt concentration will allow the aperture current to be increased which will in turn increase the signal to noise ratio for particle sizes below 1 µm.
- **Low aperture current.** If you are using a small aperture tube or a concentrated electrolyte solution, the Multisizer 4e software will automatically select a low current/high gain setting.

- **High aperture current.** If the aperture current is set too high, the electrolyte solution in the aperture can heat or boil, causing noise. Reduce the current or increase the conductivity of the electrolyte solution.

⚠ CAUTION

Current set too high can saturate the amplifier, damage the aperture orifice, distort your results, or damage cells.

- **Atmospheric interference.** A dust-laden environment, platform vibration, or a loud environment can increase noise levels.
- **Electrical interference.** Any form of electrical interference can cause random noise patterns and pulses. Move the Analyzer to a different power source or a different location.
- **A damaged aperture.** If noise levels are increasing or varying (particularly on blank runs), the aperture may be damaged. Install a new aperture tube (see Installing an Aperture Tube in CHAPTER 4, Installing and Calibrating an Aperture Tube).

Reducing the noise level is particularly important if you are using an aperture tube smaller than 100 µm, or if you are measuring particles near the minimum threshold (2% of aperture diameter).

For instructions specific to 10 µm, 20 µm and 30 µm apertures, see Using Small Aperture Tubes (10 µm, 20 µm and 30 µm) in CHAPTER 4, Installing and Calibrating an Aperture Tube.

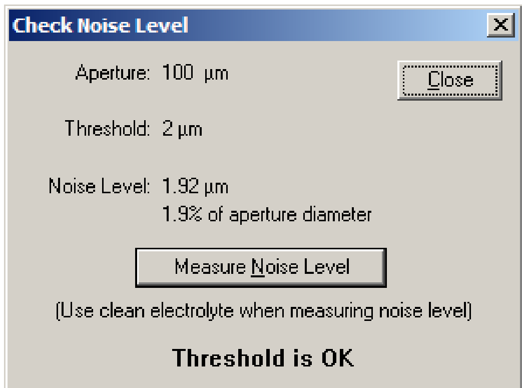
Measuring the Noise Level

When you install a new aperture tube using the Change Aperture Tube Wizard (see The Change Aperture Tube Wizard in CHAPTER 4, Installing and Calibrating an Aperture Tube), the Wizard prompts you to measure the noise level. After installing an aperture tube, you can measure the noise level at any time using the Troubleshooting menu.

To Measure the Noise Level

1 Place a beaker of clean electrolyte on the Analyzer platform.

2 Select **Run > Troubleshooting > Check the Noise Level** on the Main Menu bar. The Check Noise Level window opens.



3 In the Check Noise Level window, click **Measure Noise Level**.

4 When measurement is complete, the Check Noise Level window displays the noise level and threshold information.

NOTE If the Threshold is too high for the analysis run required, clean the outside of the aperture tube and the external electrode as well as the inside of the aperture tube. See Cleaning the Aperture Tube, Beakers, and Accuvette STs in CHAPTER 2, Analyzer Overview for more information on cleaning the aperture tube and electrode. After cleaning is performed, re-run the Measure Noise Level. Repeat as required.

If the threshold is set too low, increase the threshold in the Create SOM Wizard (see Using the Create SOM Wizard in CHAPTER 5, Selecting Analysis Settings: SOM and SOP) or the Edit the SOM window (see Editing SOM Settings in CHAPTER 5, Selecting Analysis Settings: SOM and SOP).

If noise levels remain high or variable after cleaning and troubleshooting, contact Beckman Coulter Field Service (see Requesting Service and Technical Support).

Purging the Metering Pump

Purging the metering pump removes air from a small chamber in the pump. Air can enter this chamber when the Analyzer is moved or shipped. When there is air in the metering pump, the Multisizer 4e software may display the following error message:

```
Flow Rate is too high, there may be air in the system. Try Flush Aperture Tube and Purge Metering Pump.
```

If this error message appears, select **Run > Troubleshooting > Purge Metering Pump** on the Main Menu bar.

The Status Panel displays progress.

If troubleshooting does not resolve the problem, contact Beckman Coulter Field Service (see Requesting Service and Technical Support).

Adjusting the Metering Pump

When you change the aperture size or adjust electrolyte viscosity, it is necessary to adjust the metering pump for the new tube or fluid. If you use the Change Aperture Tube Wizard (see The Change Aperture Tube Wizard in CHAPTER 4, Installing and Calibrating an Aperture Tube), this process occurs automatically when you click the Metering Pump button. When the metering pump requires a new adjustment, the Multisizer 4e software may display the following error message:

```
Adjust the Metering Pump
```

If this error message appears, select **Run > Troubleshooting > Adjust Metering Pump** on the Main Menu bar.

The Status Panel displays progress.

If adjusting the metering pump does not resolve the problem, contact Beckman Coulter Field Service (see Requesting Service and Technical Support).

Setting the Vacuum Regulator

The Multisizer 4e software resets the Waste Tank vacuum on start-up and at regular intervals. A Waste Tank vacuum malfunction can result from a mechanical flaw in the Analyzer. When the Waste Tank vacuum is malfunctioning, the Multisizer 4e software may display the following error message:

```
Check the Vacuum Regulator
```

If this error message appears, select **Run > Troubleshooting > Check Vacuum Regulator** on the Main Menu bar.

The Status Panel displays progress.

If resetting the vacuum regulator does not resolve the problem, contact Beckman Coulter Field Service (see Requesting Service and Technical Support).

Measuring Gain Stage Offsets

The Analyzer circuits that measure pulse height contain eight gain stages calibrated relative to each other. By default, calibration of these gain stages occurs automatically:

- At start-up and/or when a temperature change of 3 degrees C is measured in the Analyzer.
- Every 60 minutes after start-up or a after change in temperature of 3 degrees C.
- Every 500 minutes.

To ensure the precision of current and gain, it is possible to manually calibrate gain stage offsets at any time.

If gain stage offsets do not calibrate automatically and require re-calibration, select **Run > Troubleshooting > Measure Gain Stage Offsets**.

If calibrating gain stage offsets does not resolve the problem, contact Beckman Coulter Field Service (see Requesting Service and Technical Support).

Setting Custom Measurement Intervals

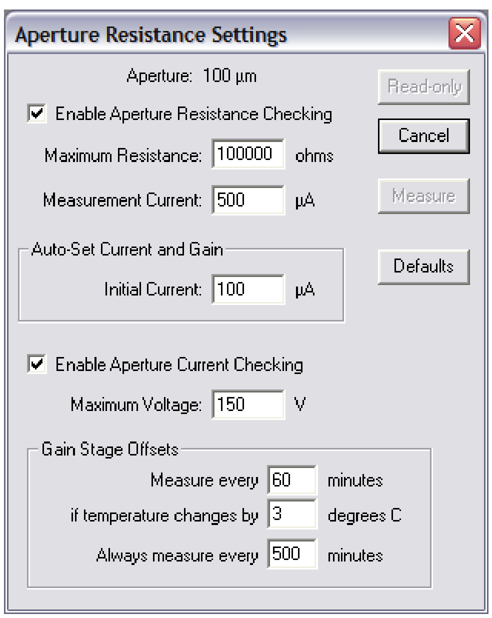
If you want to measure gain stage offsets more frequently, you can set custom intervals.

To Reset the Default Measurement Intervals

1 Select **Run > Troubleshooting > Enter Service Mode** on the Main Menu bar.

NOTE Entering Service Mode will allow you to change default settings in the Troubleshooting menu. If you do not enter Service Mode, the Gain Stage Offsets settings will appear as read-only values.

2 Select **Run > Troubleshooting > Aperture Resistance Settings** on the Main Menu bar. The Aperture Resistance Settings window opens.



3 In the Gain Stage Offsets pane, enter new values in the fields provided.

4 If you are not running Service Mode, a **Read-only** button appears. If you are running Service Mode, click **OK** to save your settings.

System Error: Unable to Measure Gain Stage Offsets

If the Analyzer is unable to measure gain stage offsets automatically, the Multisizer 4e software will display the following error message:

`Unable to measure gain stage offsets`

If this error message appears, power off the Analyzer, close the Multisizer 4e software, and restart the system. If restarting the system does not resolve the problem, contact Beckman Coulter Field Service.

Resolving Error Messages on Software Startup

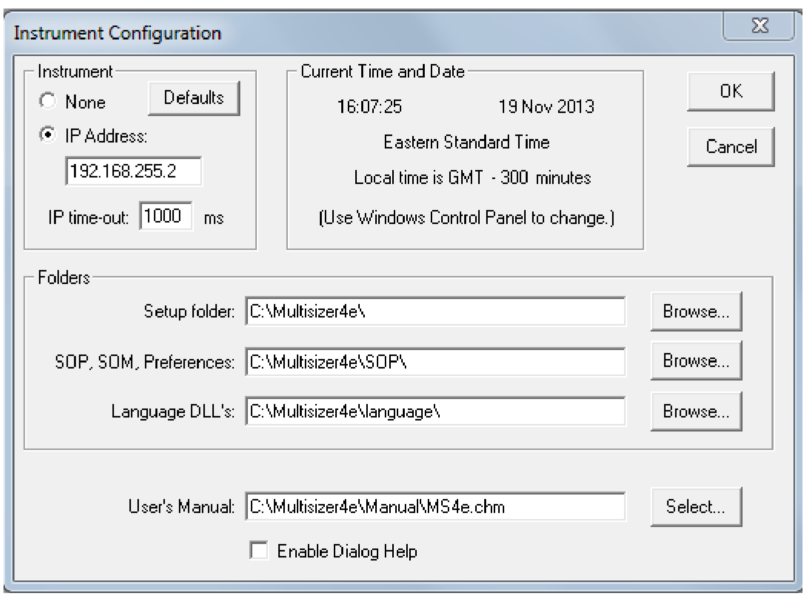
If the Multisizer 4e software cannot find Preferences, Standard Operating Method, and other files on startup, one or more error messages will appear. If one or more error messages appear on startup, use the Instrument Configuration window to reset default folder(s) for settings, preferences, and run files.

To Change Default File Locations

Select **Configuration > Instrument Configuration** on the Main Menu bar. The Multisizer 4e Configuration window opens.

If you are running security mode, select **Instrument Configuration** in the Administrator or Supervisor drop-down menu.

Figure 10.3 Instrument Configuration



For more information on using the Instrument Configuration window, see Multisizer 4e Configuration Options in CHAPTER 9, Configuring Software and Security.

Viewing Advanced Multisizer Settings

Waste Tank Settings

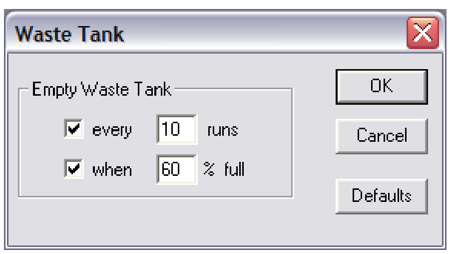
When the software is connected to the Analyzer, the Status Panel displays the Waste Tank fluid level. You can manually empty the Waste Tank into the Waste Jar at any time by clicking **Empty** in the Instrument Toolbar or selecting **Run > Empty Waste Tank** on the Main Menu bar.

You can select settings in the Troubleshooting menu that will automatically empty the Waste Tank based on fluid level or number of completed runs. If you do not change default Waste Tank settings, the Analyzer will empty the Waste Tank when the tank is 60% full.

NOTE You can change Waste Tank settings when you create or edit a Standard Operating Method (SOM). For information on setting automatic intervals for emptying the Waste Tank in the SOM, see Standard Operating Method: Control Mode in CHAPTER 5, Selecting Analysis Settings: SOM and SOP.

To Change Default Waste Tank Settings

1 Select **Run > Troubleshooting > Waste Tank** on the Main Menu bar. The Waste Tank window opens.



2 In the Waste Tank window:

- To empty the Waste Tank after a specified number of completed runs, select every [X] runs and enter the number of analysis runs.
- To empty the Waste Tank when the fluid reaches a specified level, select when [X] % full and enter a maximum fluid percentage.
- To accept system defaults, click **Defaults**.

3 Click **OK**.

Calibrating Waste Tank Sensors

If it appears that the Waste Tank is emptying too frequently or is not emptying at the correct fluid level, you may need to calibrate the Waste Tank sensors. This option is available in the Service Mode menu.

1 Select **Run** > **Troubleshooting** > **Enter Service Mode** on the Main Menu bar. The Service drop-down menu appears in the Main Menu bar.

2 Select **SERVICE** > **Calibrate Waste Tank**. The Status Panel displays CALIBRATING WASTE TANK.

⚠ CAUTION

Beckman Coulter does not recommend that you use the options available in Service Mode without the assistance of a trained Service Representative. To contact a representative, see Requesting Service and Technical Support.

Vacuum Settings

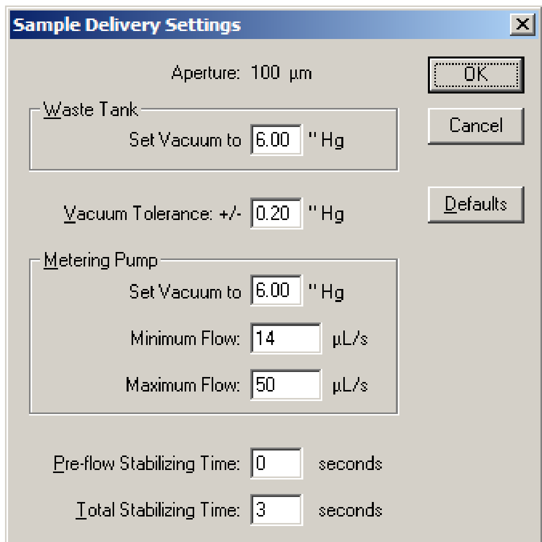
The Waste Tank and Metering Pump use vacuums to move fluid through the system. In both cases, the default vacuum is set to six inches of mercury (6.00" Hg) for most aperture sizes. If you are using time or count control mode and an aperture tube larger than 560 µm, you may need to change the Waste Tank vacuum setting.

⚠ CAUTION

It is not possible to change vacuum or Metering Pump flow settings unless you are running Service Mode. Beckman Coulter does not recommend that you use the options available in Service Mode without the assistance of a trained Service Representative. To contact a representative, see Requesting Service and Technical Support.

To View Vacuum Settings

1 Select **Run** > **Troubleshooting** > **Sample Delivery Settings** on the Main Menu bar. The Sample Delivery Settings window opens.



2 In the Waste Tank pane, the Set Vacuum to field displays the current Waste Tank vacuum setting. Set this value to 6.00" Hg unless you are using an aperture tube that is 560 µm or larger.

3 The Vacuum Tolerance: +/- field displays the vacuum deviation (in inches of mercury) that will trigger the error message Check the Vacuum Regulator.

4 In the Metering Pump pane, the Set Vacuum to field displays the current Metering Pump vacuum setting. This value must remain at the default, 6.00" Hg.

5 The Minimum Flow and Maximum Flow fields display the flow rates in mL/sec that will trigger error messages. For more information, see Flow Rate Error Messages.

6 The Pre-flow Stabilizing Time and Total Stabilizing Time fields display the time, in seconds, that the Analyzer will use for stabilization before beginning particle analysis.

7 The **Read-Only** button indicates that settings in this window are for viewing only. It is not possible to change vacuum and flow settings unless you are running Service Mode (see Working with Field Service).

8 Clicking **Defaults** displays system default settings. Although you can view the defaults, it is not possible to apply new settings unless you are running Service Mode (see Working with Field Service).

Flow Rate Error Messages

The default flow settings adjust for aperture tube size and are appropriate for most analyses. Under certain conditions—if you are using an unusually viscous electrolyte solution, for example, or if you have increased the vacuum for a large aperture tube—it may be necessary to adjust the flow rate to avoid unnecessary error messages.

When the flow rate is too low, the Multisizer 4e software will display one of the following error messages:

```
Flow Rate is too low, the aperture may be blocked
Flow Rate is too low. Unblock the Aperture?
Flow Rate and Pressure are too low, try unblocking the Aperture
```

When the flow rate is too high, the Multisizer 4e software will display the following error message:

```
Flow Rate is too high, there may be air in the system. Try Flush Aperture Tube and Purge Metering Pump.
```

If one of these error messages appears and the system is functioning properly (the aperture tube is not blocked and there is no air in the system), adjust the volumetric flow settings in the Sample Delivery Settings window (see Adjusting Volumetric Flow Settings).

Adjusting Volumetric Flow Settings

Adjust Metering Pump flow settings if the Multisizer 4e software displays the following error message while analyzing in volumetric mode:

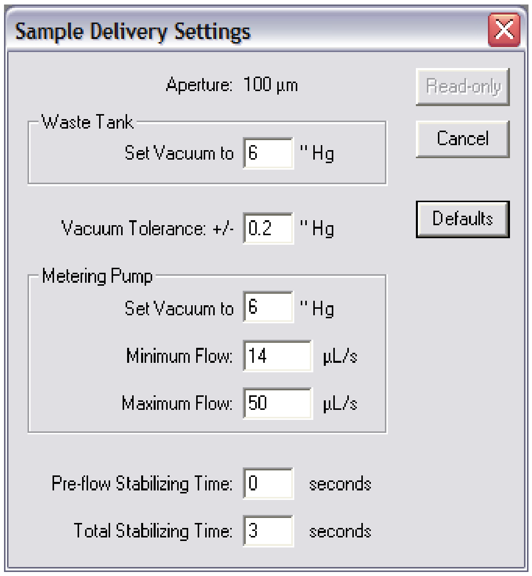
```
Adjust the Metering Pump
```

To Adjust Flow Settings

1 Select **Run > Troubleshooting > Enter Service Mode** on the Main Menu bar.

NOTE Entering Service Mode will allow you to change default settings in the Troubleshooting menu. If you do not enter Service Mode, settings in the Troubleshooting menu will appear as read-only values.

2 Select **Run > Troubleshooting > Sample Delivery Settings** on the Main Menu bar. The Sample Delivery Settings window opens.



3 In the Metering Pump pane, enter new values in the Minimum Flow and Maximum Flow fields, and click **OK**.

The Minimum Flow and Maximum Flow fields display the flow rates in mL/second that will trigger error messages. The default flow settings adjust for aperture tube size and are appropriate for most analyses. Under certain conditions—if you are using an electrolyte solution with a different viscosity, for example, or if you had to change the vacuum setting—it may be necessary to adjust the flow rate to avoid unnecessary error messages.

Fill, Flush, and Drain Settings

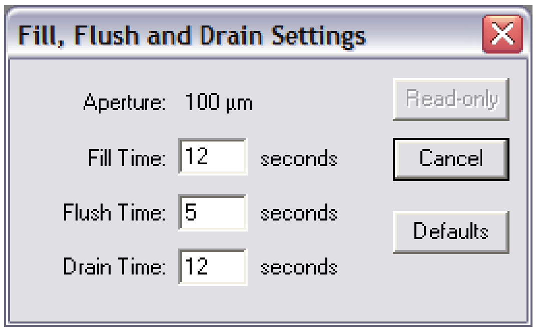
Viewing System Flush Time

You can select settings to automatically flush the system before and/or after each analysis run when you create a Standard Operating Method (see Standard Operating Method: Run Settings in CHAPTER 5, Selecting Analysis Settings: SOM and SOP). Flushing the system regularly is recommended in order to remove particles that settle at the bottom of the aperture tube.

Standard Operating Method settings do not allow you to adjust the length of time the system will use for a system flush. This option is available only in the Troubleshooting menu.

To View the Current Flush Time Settings

Select **Run > Troubleshooting > Fill, Flush and Drain Settings** on the Main Menu bar. The Fill, Flush and Drain Settings window opens and displays system Fill Time, Flush Time, and Drain Time in seconds.



CAUTION

It is not possible to change the system flush time setting unless you are running Service Mode. Beckman Coulter does not recommend that you use the options available in Service Mode without the assistance of a trained Service Representative (see Requesting Service and Technical Support).

To Adjust System Flush Time

1 Select **Run > Troubleshooting > Enter Service Mode** on the Main Menu bar.

NOTE Entering Service Mode will allow you to change default settings in the Troubleshooting menu. If you do not enter Service Mode, settings in the Troubleshooting menu will appear as read-only values.

2 Select **Run > Troubleshooting > Fill, Flush and Drain Settings** on the Main Menu bar.

3 In the Fill, Flush and Drain Settings window, enter the new system flush time in the Flush Time field and click **OK**.

NOTE A longer flush time removes more particles from the aperture tube, and can improve the accuracy of analysis results. On the other hand, increasing the flush time requires more electrolyte. You will need to fill the Electrolyte Jar more frequently.

Draining the System

CAUTION

Draining the system is recommended only when cleaning, shipping, or repackaging the Analyzer. As fluid evaporates within the system, remaining electrolyte deposits salt crystals that can obstruct fluid flow, increase sample analysis noise, and reduce instrument performance.

To Drain the System

IMPORTANT If the Electrolyte Jar is removed, place an empty beaker under the electrolyte pick-up tube to prevent spills.

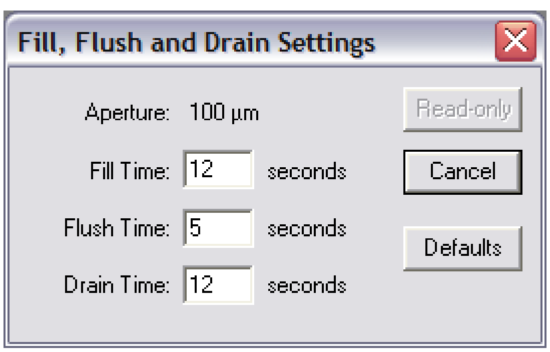
1. Select **Run** > **Drain System** on the Main Menu bar. A Beckman Coulter pop-up window directs you to remove or empty the Electrolyte Jar.

2. Remove or empty the Electrolyte Jar, then click **OK** in the pop-up window. The system drains.

To View System Drain Time

Select **Run** > **Troubleshooting** > **Fill, Flush and Drain Settings** on the Main Menu bar. The Fill, Flush and Drain Settings window opens and displays system Fill Time, Flush Time, and Drain Time in seconds.

Figure 10.4 Fill, Flush and Drain Settings



CAUTION

It is not possible to change the system drain time setting unless you are running Service Mode. Beckman Coulter does not recommend that you use the options available in Service Mode without the assistance of a trained Service Representative (see Requesting Service and Technical Support).

To Adjust System Drain Time

1 Select Run > Troubleshooting > Enter Service Mode on the Main Menu bar.

NOTE Entering Service Mode will allow you to change default settings in the Troubleshooting menu. If you do not enter Service Mode, settings in the Troubleshooting menu will appear as read-only values.

2 Select **Run > Troubleshooting > Fill, Flush and Drain Settings** on the Main Menu bar.

3 In the Fill, Flush and Drain Settings window, enter the new system drain time in the Drain Time field and click **OK**.

Aperture Resistance Settings

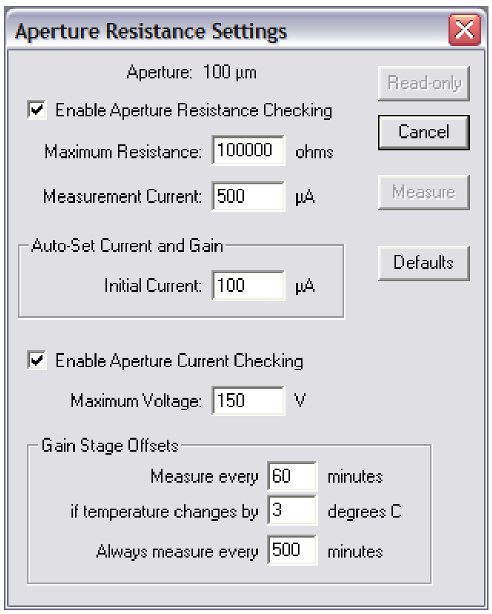
The Multisizer 4e software measures aperture resistance automatically at the start of each run. If resistance is too high, it is likely that air is being drawn through the aperture tube. The electrolyte level may be too low. Reset Multisizer 4e aperture resistance defaults only if you are using an electrolyte which causes unusually high current resistance.

⚠ CAUTION

It is not possible to change aperture resistance settings unless you are running Service Mode. Beckman Coulter does not recommend that you use the options available in Service Mode without the assistance of a trained Service Representative (see Requesting Service and Technical Support).

To View Aperture Resistance Settings

1 Select **Run** > **Troubleshooting** > **Aperture Resistance Settings** on the Main Menu bar. The Aperture Resistance Settings window opens.



2 If Enable Aperture Resistance Checking is selected, the Multisizer 4e will check aperture resistance before each run based on values entered in the Maximum Resistance and Measurement Current fields.

- The Maximum Resistance field displays the electrical resistance that will trigger the following error message:

  `Aperture Resistance High, check the Sample Level`

- The Measurement Current field displays the fixed current that the system will use to measure resistance.

⚠ CAUTION

Do not change the default Measurement Current setting. The measurement current is not the current setting that will be used during analysis runs.

- In the Auto-Set Current and Gain pane, the Initial Current field displays the initial current in µA.

- If Enable Aperture Current Checking is selected, the Multisizer 4e software verifies that the system can supply the voltages selected in the Measurement Current, Initial Current, and analysis current and gain fields. If the current exceeds the setting in the Maximum Voltage field, the following error message will display:

  Reduce Aperture Current or Preamp Gain

- The Gain Stage Offsets pane displays gain stage measurements. For information on these settings, see Measuring Gain Stage Offsets.
- Click **Defaults** to display system default settings. Although you can view the defaults, it is not possible to apply the settings unless you are running Service Mode (see Opening Service Mode).
- The **Read-only** button indicates that settings in this window are for viewing only. It is not possible to change aperture resistance settings unless you are running Service Mode (see Opening Service Mode).

To Change Aperture Resistance Settings

1. Select **Run** > **Troubleshooting** > **Enter Service Mode** on the Main Menu bar.

   NOTE Entering Service Mode will allow you to change default settings in the Troubleshooting menu. If you do not enter Service Mode, settings in the Troubleshooting menu will appear as read-only values.

2. Select **Run > Troubleshooting > Aperture Resistance Settings** on the Main Menu bar.

3. In the Aperture Resistance Settings window, modify the fields as necessary and click **OK**. For more information on fields and values in the Aperture Resistance Settings window, see Aperture Resistance Settings.

CAUTION

Beckman Coulter does not recommend that you change aperture resistance settings without the assistance of a trained Service Representative (see Requesting Service and Technical Support).

Calibrating the Electrolyte and Waste Jars

The fluid levels in the Electrolyte and Waste Jars are calculated by weight. It is usually necessary to calibrate the weight sensors only once, when you set up the Analyzer. If you switch to an electrolyte with significantly greater density, it is possible that you will need to recalibrate the Electrolyte and Waste Jar sensors.

1 Select **Run > Troubleshooting > Calibrate Electrolyte and Waste Jars** on the Main Menu bar. The Calibrate Electrolyte Jar and Waste Jar Level Sensors window opens.

NOTE When calibrating Electrolyte and Waste Jars, Beckman Coulter recommends that you calibrate the Electrolyte Jar “removed,” “empty,” then “full” to leave the jar in the full state. When calibrating the Waste Jar, calibrate the jar “removed,” “full,” then “empty”, to leave the jar in the empty state.

2 Remove the Electrolyte Jar from the Analyzer Compartment.

3 In the Calibrate Electrolyte Jar and Waste Jar window, select **Electrolyte Jar is Removed.**

4 Select **Calibrate**. When calibration is complete, Calibrated appears on the Electrolyte Jar is Removed.

5 Place the empty Electrolyte Jar in the Analyzer compartment (see Installing and Removing the Electrolyte and Waste Jars in CHAPTER 2, Analyzer Overview).

6 In the Calibrate Electrolyte Jar and Waste Jar window, select **Electrolyte Jar is Empty**.

7 Select **Calibrate**. When calibration is complete, Calibrated appears on the Electrolyte Jar is Empty line.

8 Remove the Electrolyte Jar, fill the jar, and replace the jar in the Analyzer compartment.

9 In the Calibrate Electrolyte Jar and Waste Jar window, select **Electrolyte Jar is Full**.

10 In the Electrolyte Jar pane, click **Calibrate.** When calibration is complete, Calibrated appears on the Electrolyte Jar is Full line.

11 Follow Steps 5 - 10 above to calibrate the Waste Jar (reversing the empty and full states), or click **Close** if you are finished.

To calibrate the Waste Jar:
Follow the steps to calibrate the Electrolyte Jar, above, reversing the empty and full states. Beckman Coulter recommends that you first calibrate the full Waste Jar, then empty the jar, to leave the Waste Jar in the empty state.

NOTE In the Calibrate Electrolyte Jar and Waste Jar window, Unknown is selected by default to prevent accidental calibration when the Electrolyte or Waste Jar is not in the intended state (empty or full).

Working with Field Service

Requesting Service and Technical Support

To Contact a Service Representative

1 Select **Help > Contact Beckman Coulter** on the Main Menu bar.

2 In the Contact Beckman Coulter window:

- Click **Service Support** to request help with instrument-related problems.
- Click **Applications Support** to request help with problems related to the Multisizer 4e software.
- Click **General Support** for all other questions.

Each button opens your default email editor and includes the email address of a Field Service Representative in your area.

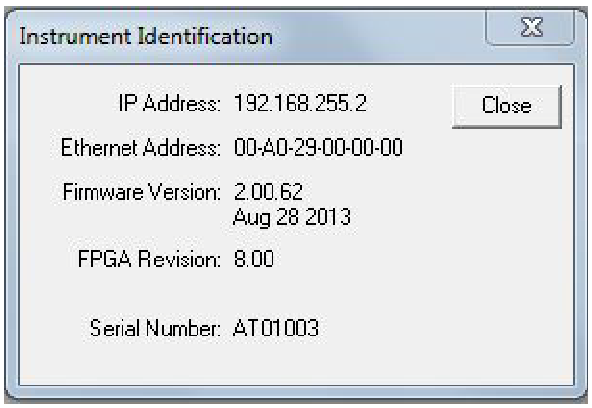
Identifying the Instrument

When you contact Beckman Coulter Field Service, the Service Representative will require software and firmware information in order to help you resolve the problem.

To Obtain Information About the Analyzer

1 Select **Run > Troubleshooting > Instrument Identification** on the Main Menu bar.

2 The Instrument Identification window opens.



- IP Address (Internet Protocol Address) is used to connect the Analyzer to the computer running the Multisizer 4e software. If your Analyzer is connected to a network, a Service Technician may have changed the IP address to identify the instrument on your network or intranet.
- Firmware Version is the version of the software running in the Analyzer.
- FPGA Revision is the version of the software controlling electronic circuits in the Analyzer.
- Serial Number uniquely identifies the instrument.

The Service Representative will also require information about the Multisizer 4e software version you are currently running.

To Locate the Current Version Number

1 Select Help > About on the Main Menu bar.

2 In the Beckman Coulter Multisizer 4e window, locate and record the software version number at the top of the window, and click **OK**.

Updating Firmware and FPGA Software

While the Multisizer 4e software runs on a PC connected to the instrument, firmware is the programming and software that runs within the Analyzer. If your firmware or FPGA software is out of date, the Multisizer 4e software will display one of the following error messages:

`Contact service for newer Multisizer 4e firmware. Instrument is version [X], should be at least version [X].`

`Contact service for newer Multisizer 4e FPGA version. FPGA is revision [X], should be at least revision [X].`

If one of these error messages appears, contact Beckman Coulter Field Service (see Requesting Service and Technical Support). A Service Representative will help you download a new version of the firmware or FPGA software to the Analyzer.

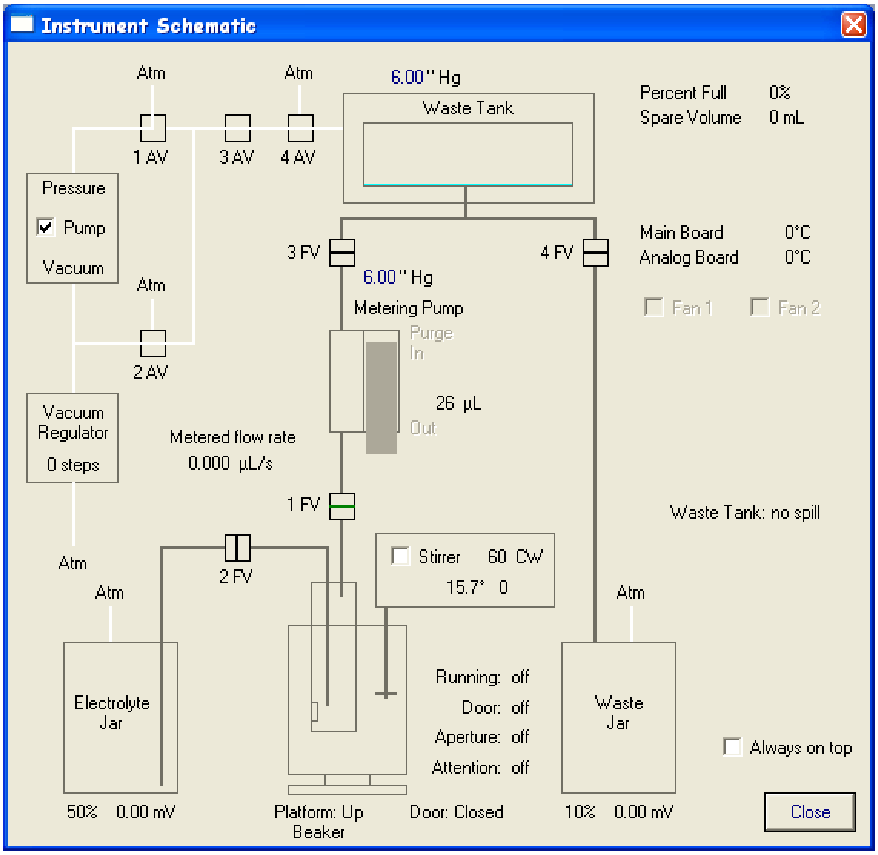
Viewing the Instrument Schematic

The Instrument Schematic Diagram allows you to view Analyzer processes as they occur (for example, Metering Pump actions, vacuum pressure, and valve function). When you call Beckman Coulter Field Service, a Service Representative will often ask you to open the Instrument Schematic to identify the source of the problem.

To View the Instrument Schematic Diagram

Select **Run** > **Troubleshooting** > **Instrument Schematic** on the Main Menu bar. The Instrument Schematic window opens.

Figure 10.5 Instrument Schematic



In the Instrument Schematic window, you can:

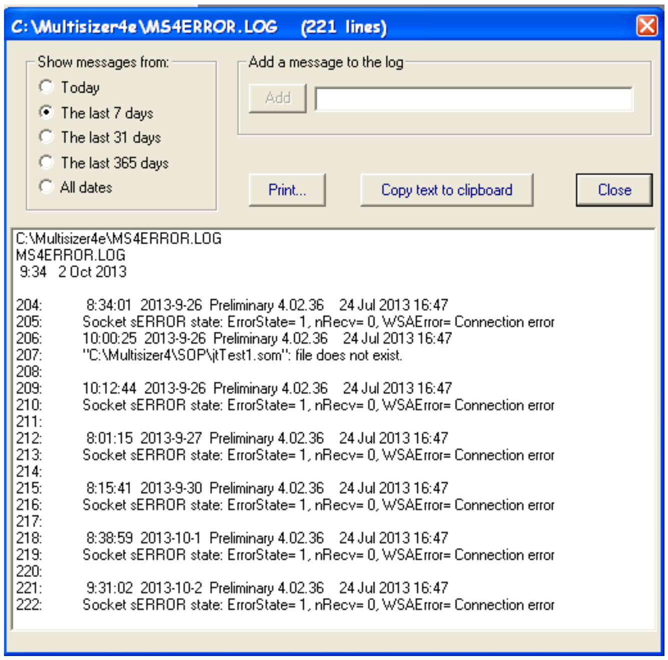
- Select Pump to turn the vacuum pump on and off. If you clear this section, Vacuum Pump Off appears in the Status Bar. Selecting the Pump will turn the vacuum pump back on.
- Select Stirrer to turn on the stirrer. You can also turn the stirrer off or on by selecting **Run** > **Stirrer Off** or **Run** > **Stirrer On** on the Main Menu bar.
- Select Always on top to keep the Instrument Schematic window visible during an analysis run. You may need to restart the Multisizer 4e software for this setting to take effect.

Viewing the Error Log

When you contact Beckman Coulter Field Service, the Service Representative may ask you to open the error log.

To Open the Error Log

1 Select **Run > Troubleshooting > View Error Log** on the Main Menu bar. The MS4ERROR.LOG window opens.



2 In the Show messages from pane, select the time period. You can add an error message manually, print error messages, or copy messages to the clipboard.

3 When you are finished, click **Close**.

Opening Service Mode

When you select **Troubleshooting > Open Service Mode** on the Main Menu bar, a new Service menu appears in the Main Menu bar. The Service menu includes advanced options for configuring and troubleshooting the Multisizer 4e software and Analyzer.

⚠ CAUTION

Beckman Coulter does not recommend that you use the options available in Service Mode without the assistance of a trained Service Representative (see **Requesting Service and Technical Support****).**

CHAPTER 11

Regulatory Compliance

Regulatory Compliance

Electronic Signatures (21 CFR Part 11)

The Electronic Records and Electronic Signatures Rule (21 CFR Part 11) was established by the FDA to define the requirements for submitting documentation in electronic form and the criteria for approved electronic signatures. Organizations that choose to use electronic records to meet record-keeping requirements must comply with 21 CFR Part 11, which is intended to improve an organization's quality control while preserving the FDA's charter to protect the public. Because analytical instrument systems such as the Multisizer 4e generate electronic records, these systems must comply with the Electronic Records Rule.

This section describes the relevant portions of the 21 CFR Part 11 regulations and their implementation using the Multisizer 4e control software. For more information, see Maintaining Electronic Signatures.

NOTE It is important to realize that implementation and compliance of the rule remains the responsibility of the organization or entity creating and signing the electronic records in question. Proper procedures and practices, such as GLP and cGMP, are as much part of overall compliance with these regulations as are the features of the Multisizer 4e control software.

Electronic Records

Electronic record refers to any combination of text, graphics, data, audio, pictorial, or other information representation in digital form that is created, modified, maintained, archived, retrieved or distributed by a computer system. This definition applies to any digital computer file submitted to the agency, or any information not submitted but which is necessary to be maintained. Public docket No 92S-0251 of the Federal Register (Vol.62, No 54) identifies the types of documents acceptable for submission in electronic form and where such submissions may be made.

FDA Requirements

The Multisizer 4e control software has been designed to allow users to comply to the electronic records and signatures rule. Any organization deciding to employ electronic signatures must declare to the FDA their intention to do so.

The FDA ruling includes these guidelines:

- *The agency emphasizes that these regulations do not require, but rather permit, the use of electronic records and signatures.* (General comments section)
- *The use of electronic records as well as their submissions to FDA is voluntary.* (Introduction to final ruling)
- If electronic submissions are made:

  Persons may use electronic records in lieu of paper records or electronic signatures in lieu of traditional signatures....provided that: (1) The requirements of this part are met; and (2) The document or parts of a document to be submitted have been identified in public docket No. 92S-0251. (Section 11.2 subpart A).

Section 11.3 Subpart A describes two classes of systems; *closed* and *open*.

- A ***closed system*** is one where *system access is controlled by persons who are responsible for the content of electronic records*. The people and organization responsible for creating and maintaining the information on the system are also responsible for operating and administering the system.
- An ***open system*** is one where system access is not controlled by persons who are responsible for the content of electronic records.

A typical Multisizer 4e system can be regarded as a closed system.

Security Controls

A Multisizer 4e installation needs to have a procedure designed to ensure proper operation, maintenance and administration for system security and data integrity. Persons who interact with the system, from administrators to users, must abide by these procedures. Therefore, the ultimate responsibility is with the organization generating electronic records and signatures. The Multisizer 4e software is a component, albeit a vital one, of the overall process.

The controls to be applied to a closed system are specified in Subpart B, Section 11.10.

Electronic Controls

The chief purpose of electronic controls is to *ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine*. Many of the controls described in Section 11.10 refer to written procedures (SOPs) required by the agency for the purpose of data storage and retrieval, access control, training, accountability, documentation, record keeping, and change control.

System Validation

Other controls are addressed either by the Multisizer 4e software itself, or in combination with end-user procedures. Most important is the *validation of systems to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records* [Section 11.10 Paragraph (a)]. The complete and overall validation of the system, as developed by the organization, ensures the integrity of the system and its data. The features of the Multisizer 4e software are designed to comply with the specifications of these regulations.

The Multisizer 4e software employs a system of usernames and passwords, consistent with the specifications of Subpart C, Section 11.300, *to ensure that only authorized individuals can use the system, electronically sign a record, access the operation or computer system input or output device, alter a record, or perform the operation at hand.*

Audit Trail

The Multisizer 4e software performs data input and operational checks that *determine, as appropriate, the validity of the source of data input or operational instruction, and enforce permitted sequencing of steps and events* (Subpart B, Section 11.10). These two features ensure that, as much as possible, valid data are entered into the system, and all required steps have been completed to perform the task at hand.

The purpose of all such data checking and validation is to allow the system *to generate accurate and complete copies of records in both human-readable and electronic form suitable for inspection, review, and copying by the agency* [Section 11.10, Paragraph (b)]. The Multisizer 4e software system is able to enforce strict procedures to record all changes that are made to data generated using the software [Section 11.10, Paragraph (e)]. See File History and Audit Trail.

File History and Audit Trail

File History or auditing can be enabled for any file created by the Multisizer 4e software. After auditing has been enabled for a file, it cannot be disabled, nor can it be bypassed. Under these conditions, all changes made to a file are automatically recorded. These changes consist of *computer-generated, time-stamped audit trails to independently record the data and time of operator entries and actions that create, modify, or delete electronic records.*

When a change to a file is detected, the Multisizer 4e software automatically records the identity of the user making the change, the date and timestamp of the change, the parameter that has changed, the old value and the new value. The user is also required to “re-sign” the record electronically and enter a reason for the change, from a pre-defined list or as free text.

The audit trail is stored as File History inside the file itself, so that *record changes shall not obscure previously recorded information*, and so the file is *suitable for inspection, review, and copying by the agency*. This ensures that a complete and continuous record of all changes to the file is maintained. The file protection and archiving capabilities of the Multisizer 4e software ensure compliance with the requirement that *audit trail documentation shall be retained for a period at least as long as that required for the subject electronic records* [Section 11.10, Paragraph (e)].

Accessing File History

1 Select **File > Open** on the Main Menu bar and navigate to the file you want to open.

2 Select **RunFile > Get File History** on the Run Menu bar. The File History window opens.

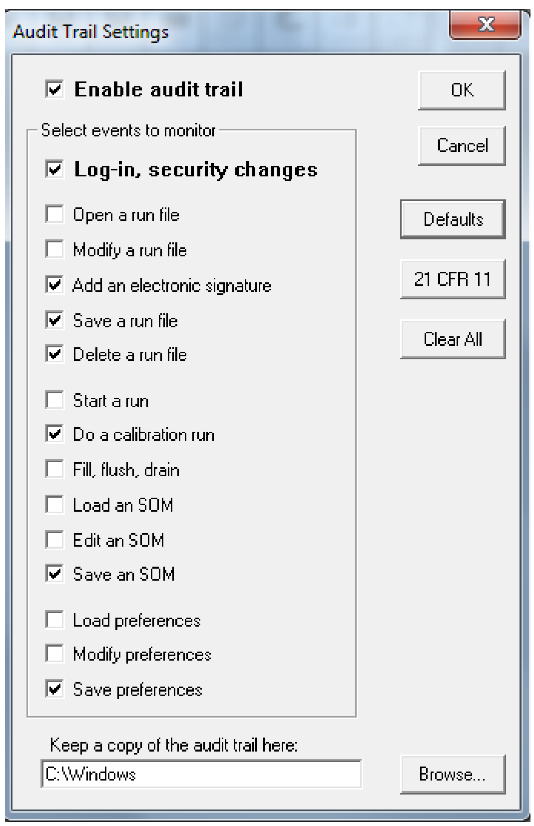
3 Click **Close** to exit the window.

Setting Audit Trail Options

In addition to the auditing associated with the electronic file itself, the Multisizer 4e software offers two additional levels of auditing. First, a general system audits records and stores information at a system level (for example, who logged in when, or when users were added to the system). Second, the Multisizer 4e software generates an error log that stores and records errors associated with the system (for example, missing files and bad communications).

To Select Audit Trail Settings

1 Select **Administrator** > **Audit Trail Settings** on the Main Menu bar. The Audit Trail Settings window opens.



2 In the Audit Trail Settings window, select Enable audit trail.

3 In the Select events to monitor pane, select the required options.

4 Click **Browse** to select the location of audit trail records.

5 Click **OK**.

NOTE The dialog has a button marked 21 CFR 11. When activated, this ensures all options are set so that the settings meet with the rules that apply to 21 CFR part 11.

Maintaining Electronic Signatures

*An **electronic signature** is a computer data compilation of any symbol or series of symbols executed, adopted, or authorized by an individual to be the legally binding equivalent of the individual's handwritten signature* (Subpart A, Section 11.3). *Each electronic signature shall be unique to one individual and shall not be reused by, or reassigned to, anyone else* [Subpart C, Section 11.100, paragraph (a)].

An electronic signature represents a user's computer identity, developed to ensure the distinct and unique identity of that user. Before any such electronic representation is applied, the organization must first verify the identity of that individual. The subsequent use of electronic signatures as the *legally binding equivalent of traditional handwritten signatures* must be certified to the agency in writing (Section 11.100).

Multisizer 4e software supports non-biometric signatures. ***Non-biometric signatures*** are computer generated and *employ at least two distinct identification components such as an identification code and password* (Section 11.200)

NOTE **Biometric signatures** are a method of verifying an individual's identity based on measurement of the individual's physical feature(s) or repeatable action(s) where those features and/or actions are both unique to that individual and measurable (Subpart A, Section 11.3). These methods generally require fingerprints or retinal scans, and require special scanning devices to read and interpret them.

Generating Electronic Signatures

The Multisizer 4e software employs User IDs and passwords to verify the identification of each user logging into the system. The regulation requires *maintaining the uniqueness of each combined identification code and password, such that no two individuals have the same combination of identification code and password* (Subpart C, Section 11.300). This section also requires that the *identification code and password issuances are periodically checked, recalled, or revised.*

The Multisizer 4e software supports both of these provisions.

To Add a User Account

1 Open the Multisizer 4e software and log in using an Administrator account.

2 Select **Administrator > Add a User Account** on the Main Menu bar.

3 In the Add a User Account window, enter the user's Full Name, Log-in Name, ID, and Password. Select a user type from the Type drop-down list.

4 Click **OK**.

The "identification code" or username of each Multisizer 4e user must be unique. Users must also supply a password to access the Multisizer 4e software. Passwords can be controlled to prohibit the use of duplicates and to force the selection of new passwords after a prescribed period of time.

To Authorize a User Type to Generate Electronic Signatures

1 Open the Multisizer 4e software and log in using an Administrator account.

2 Select **Administrator > User Privileges** on the Main Menu bar.

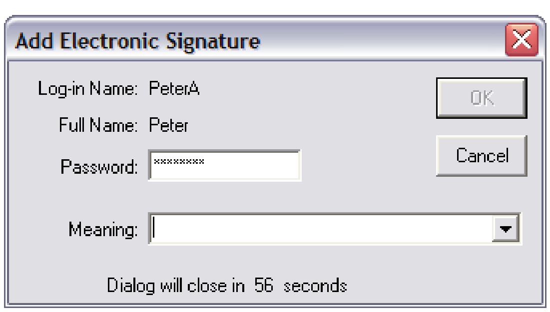
3 In the User Privileges window, select Can add an Electronic Signature. The checkboxes are aligned in five columns corresponding to user types.

4 Click **OK**.

To Generate an Electronic Signature:

1 Select **File > Open** on the Main Menu bar and navigate to the file you want to open.

2 Select **Edit > Add Electronic Signature** on the Run Menu bar. The Add Electronic Signature window opens.



3 In the Add Electronic Signature window, enter your password and select a meaning you previously recorded.

4 Click **OK**.

Applying Electronic Signatures

Subpart C, Section 11.200 stipulates several requirements for the control of electronic signatures. The regulations require that electronic signatures be used only by their genuine owners, and that signatures are administered and executed to ensure that attempted use of an individual's electronic signature by anyone other than the genuine owner *requires collaboration of two or more individuals*. Through the application of Multisizer 4e user and password configuration procedures, the system can be configured to ensure that use of these identifiers by someone other than the owner can be performed only with the owners permission.

Section 11.200 further specifies the use of electronic signature components during a period *when an individual executes a series of signings during a single, continuous period of controlled system access, and when an individual executes one or more signings not performed during a single, continuous period of controlled system access.* This section of the document represents the "heart" of electronic signature application. To comply with these provisions, the Multisizer 4e software uses the application of the username and password to authenticate the user making and saving the changes, in conjunction with file history and audit trails, *to independently record the date and time of operator entries and actions that create, modify, or delete electronic records.*

Additional Security Features

The Multisizer 4e software offers two important additional levels of security not required by the regulations. These levels make it easier to define and implement system policies:

- ***Data Mirroring.*** This feature, which can be accessed only in administrator mode, allows the administrator to securely store files in a separate location and in addition to locations specified by a user.
- ***File Attributes.*** This feature allows administrators to protect files in varying degrees using the file attribute options.

Starting the Multisizer 4e with Security Enabled

After the Administrator has configured security settings, the user is prompted to enter a user name and password every time he or she opens the Multisizer 4e software.

APPENDIX A

Aqueous Electrolytes

Aqueous Electrolytes for the Multisizer Instruments

WARNING

1. **Toxicity safety requirements and handling procedures of all reagents should be checked and adhered to (See Merck Index).**
2. **Care must be used in mixing some electrolytes. Violent reactions can occur.**

Electrolyte	Uses
ISOTON II (filtered solution based on 1% saline)	Water-insoluble materials
0.9-4% Sodium chloride (NaCl)	Water-insoluble materials, including drugs
5% Tri-sodium orthophosphate (Na_3P0_4)	Water-insoluble materials including negatively charged sample such as clays
2% Sodium acetate ($NaOOCCH_3$)	Metals such as zinc and others which react with chlorides

NOTE

1. Glycerol or sucrose (common sugar) may be added to most aqueous electrolytes to increase viscosity (and density) to assist suspension of large or dense particles (up to 70% by volume or 100% by weight respectively).
2. A preservative, for example, 5 mL/L formaldehyde solution (Formalin), or 0.1 wt% sodium azide, NaN_3, should be added to aqueous electrolytes before filtration to prevent micro-organism growth if the electrolyte is to be stored. The former can only be used in acid solutions. Formalin in alkaline solutions (for example, sodium phosphates) tends to precipitate out of solution, causing smaller particle counts on storage, and may interact with dispersant. The latter (azide) can only be used in alkaline and must not be used in acid solution.
3. Electrolytes must be filtered before use.

Before using electrolytes, check the stability of dispersion of the material. After dispersion and mixing for one minute using the stirrer on the beaker platform, take several counts at the coarse end (d max) and fine end (d min) of the size range. After the completion of the full analysis, or after 15 minutes further stirring, recount at the same levels. Comparison of the counts will indicate:

d max	d min	Symptom
Increasing	Increasing	Precipitation; Crystallization
Increasing	Decreasing	Flocculation
Decreasing	Decreasing	Dissolution
Decreasing	Increasing	Deflocculation
Decreasing	Steady	Settling of Large Particles
Steady	Steady	Stable Dispersion

Flocculation:	Try different electrolytes (for example, change to phosphate from chloride), different dispersants and pH.
Dissolution:	Presaturate electrolyte with sample before filtration. Use common-ion effect to depress solubility (for example, 5% sodium sulphate solution for barium sulphate; 2% calcium chloride for chalk or whiting).
Deflocculation:	Wait until stable. Use different electrolyte, dispersant, pH. Use ultrasonics (low power bath).
Settling:	Increase viscosity and/or density of electrolyte. Use round bottomed beaker perhaps with baffle.

With some materials it may be necessary to saturate the electrolyte before filtering to reduce the solubility of the particles being counted.

APPENDIX B

Non-Aqueous Electrolytes

Non-Aqueous Electrolytes for the Multisizer

Non-aqueous electrolytes are used if the particles to be counted are soluble in water, or if the particles are already dispersed in a liquid immiscible with water. This section is a general guide to non-aqueous electrolytes, their formulation and preparation.

WARNING

Most non-aqueous electrolytes are toxic, flammable, and may give off explosive vapors. The following four precautions should be observed:

1. **Install an extractor fan to remove the vapor.**
2. **Prohibit smoking and the siting of bunsen burners, commutator motors, and so forth in the vicinity.**
3. **When using flammable solvents, ensure that the aperture current is off before immersing the aperture tube in, or removing the aperture tube from, the beaker contents. (The automatic cut-off switch in the sampling stand door minimizes this risk).**
4. **Never mix electrolytes. When changing electrolytes, empty the waste bottle, and empty the aperture tube. Fill the electrolyte bottle with the new electrolyte solution. Set reset/count to reset and fill/close to fills to flush through the tubing.**

The mechanism of ion formation requires that the solvent be polar enough to cause a dissolved salt to dissociate into ions. A useful index of liquid polarity is the dielectric constant, which ranges from 1.0 for vacuum, 78 for water, to 110 and higher for such liquids as formamide which have very unbalanced molecular structures.

In addition to having a reasonably high dielectric constant (over 10), the liquid for a given analysis problem must be miscible with other liquids involved and capable of dissolving some salt which will then ionize in the liquid. It must not, of course, dissolve the particles to be analyzed. It should have moderate or low volatility, toxicity, flammability and cost.

Ion sources for non-aqueous electrolytes are less numerous than those for water. Resistivities achieved are seldom below the 100 to 200 ohm/cm range (1% NaCL in water is about 55 ohm/cm), which is adequate for most work using the Coulter principle, but may sometimes be a problem where lower resistivities are desired.

Solubility in various liquids may differ considerably among several useful types of materials, which include:

- Strong acids (HNO_3, HCl, and so forth);
- Ammonium (NH_4-)
- Substituted ammonium (for example, N $(CH_3)_4$-)
- Alkali (Li, Na, K) salts
- Thiocyanates (-SCN)
- Benzoates ($-OOCC_6H_5$)
- Perchlorates (-C104)
- Halides (-I, -Br, -C1, -F)
- Acetates ($-OOCCH_3$)

Where the material to be measured is water soluble (or otherwise affected), solubility considerations govern the choice of liquid, and chemical considerations governs the choice of the salt. It may be difficult to find suitable electrolytes for the previously listed salts. However, 4% NH_4SCN in isopropanol (40 g/L) forms a low-cost, useful electrolyte for many substances, and its resistivity may be lowered, if needed, by adding, for example, formamide. 4% LiCL in methanol is also extremely useful.

Where the problem is to measure contamination in non-polar liquids which are non-miscible with water (hydraulic fluids, lube oils, fuels, acetate 'dope', and so forth.) Miscibility is usually the governing factor in the choice of electrolyte. Frequently, in addition to the major polar liquid, a 'coupling' solvent is required, such as dichloroethane, trichlorethylene, tetrahydrofuran, and so forth. These often have rather low dielectric constants and their use should be minimal.

When the contamination levels are very low, it is desirable to incorporate at least 20-30% of the sample liquid to provide satisfactory count levels. This further raises the resistivity and may require careful investigation for an optimum electrolyte. Where high contamination levels exist, much less sample liquid is needed and this problem is greatly reduced. Good results and modest costs have been obtained with the following materials in formulations for various petroleum fractions (fuels, oils, and so forth) and other liquids:

1. Isopropanol with 30-40 g/L NH_4SCN - high dielectric – isobutanol might be used as a substitute.
2. Acetone with 1.5% KSCN - high dielectric - NH_4SCN decomposes rapidly in acetone, but KSCN does not (but is not as soluble); MEK may also be used, as acetone is rather volatile, evaporating rapidly from sample unless kept covered.
3. Dimethyl formamide - very high dielectric - dissolves and ionizes quite a few salts -expensive - more miscible with other liquids than formamide, which is otherwise preferable from cost and polarity view points.
4. Dichloroethane - medium dielectric, good solvent power.
5. Trichloroethylene - low dielectric, good solvent power.
6. Tetrahydrofuran - medium to low dielectric, good solvent power, but volatile.

(When using any of the electrolytes listed above or below, an aperture resistance <200 KOhm is desirable.)

Salt	Solvent
0.1% Sodium or potassium	Formic acid/water chloride (9:1)
6% Sulphuric acid	Acetone/water (8:2)
1% Magnesium perchlorate	Acetone/water (95:5)
2% Ammonium thiocyanate	Isopropyl alcohol
4-5% Ammonium thiocyanate	Dimethyl formamide
3% Ammonium thiocyanate	Formamide
2% Ammonium thiocyanate	Isobutyl alcohol
4% Ammonium thiocyanate	Acetone/methyl alcohol (9:1)
10% Ammonium thiocyanate	Acetone/ethyl alcohol/toluene (5:4:1)
2% Lithium chloride	Methyl alcohol
1% Lithium chloride	Acetone/methyl alcohol (2:1)
2% Lithium chloride	Benzene/n-butyl alcohol (1:1)
20% Lithium iodide	Isopropyl alcohol or isobutyl alcohol
Up to 40% Lithium iodide	Isobutyl alcohol
4% Zinc chloride	Acetone
10% Sodium iodide	Isopropyl alcohol/acetonitrile (85:15)
24% concentrated nitric acid	Chloroform/butyl alcohol (6:4)
1% Magnesium perchlorate	Methylene chloride/dimethyl sulphoxide (2:1)
2% Magnesium perchlorate	Methylene chloride/methyl alcohol (9:1)

NOTE

1. With some materials it may be necessary to saturate the electrolyte before filtering to reduce the solubility of the particles being counted.
2. Dimethyl sulphoxide increases the solubility of salts in many organic solvents.
3. Always melt lithium iodide before attempting to dissolve it in the solvent to eliminate crystallization water and therefore increase solubility.

Table of Electrolytes and Materials

Key to Electrolyte Solutions

NOTE

1. Unless otherwise stated, the electrolytes are dissolved in water.
2. Salts and solvents for non-aqueous use should be reasonably water-free.

a	10 g/L sodium chloride solution (33.3%(V/V) (often interchangeable with b)	aa	80 g/L ammonium thiocyanate in methanol in propane-1,2-diol)
b	Isoton II diluent (often interchangeable with a)	bb	60g/L lithium iodide in bis(2-hydroxyethyl)ether
c	4 g/L. sodium hydroxide solution	cc	50 g/L ammonium thiocyanate in (50 (V/V) methanol m cyclohexane)
d	0.1 mol/L hydrochloric acid solution	dd	3 g/L sodium chloride solution (+1 g/L cetrimide)
e	20-50 g/L trisodium orthophosphate dodecahydrale solution or 20 g/L to 50 g/sodium pyrophosphate decahydrate solution	ee	(20 g/L ammonium thiocyanate in 70% (V/V) propan-2-ol
f	200 g/L sodium chloride solution in propan-2-ol	ff	100 g/L concentrated hydrochloric acid in) in dichloroethane
g	10 g/L sodium nitrate solution	gg	40 g/L ammonium thiocyanate in butanone
h	40 g/L zinc chloride solution	hh	40 g/L ammonium thiocyanate in dimethylformamide
i	20 g/L to 50 g/L sodium sulphate solution in tetrahydrofuran	ii	50 g/L ammonium thiocyanate in (33.33% (V/V) of dimethylformamide and trichloroethylene)
j	10 g/L potassium silicate solution in acetone	jj	45 g/L lithium chloride in (90% (V/V) methanol
k	80 g/L ammonium thiocyanate in formic	kk	10 g/L potassium chloride in (90% (V/V) dimethylformamide acid in water
l	50 g/L lithium chloride in methanol	ll	7.5 g/L ammonium thiocyanate in (90% (V/V) butanone in trichloroethylene) (often interchangeable with m)
m	50 g/L ammonium thiocyanate or magnesium perchlorate in propan-2-ol	mm	40 g/L ammonium thiocyanate in 50% (V/V) propan-2-ol) + 40% (V/V) chloroethane
n	50 g/L ammonium thiocyanate or lithium chloride in acetone	nn	50% (V/V) (38 g/L lithium in propan-2-ol) in benzene
		oo	50% (V/V) (100 g/L ammonium thiocyanate in propan-2-ol) in dichloromethane

p	100 g/L to 400 g/L lithium iodide in butan-2-ol	pp	10 g/L hydrochloric acid in cyclohexanol
q	40 g/L to 50 g/L ammonium thiocyanate in butan-2-ol	qq	Up to 250 g/L lithium iodide in (50% (V/V) propan-2-ol in trichloromethane)
r	40 g/L to 50 g/L lithium iodide in acetone	rr	50 g/L ammonium thiocyanate in (33.33% (V/V) propan-2-ol, trichloroethane and tetrahydrofuran)
s	78 5 g/L sodium silicate solution	ss	50 g/L ammonium thiocyanate in (33.33% (V/V) of dimethylformamide, trichloroethane, and tetrahydrofuran)
t	10 g/L sodium carbonate solution	tt	40 g/L tetrabutyl ammonium perchlorate in (50% (V/V) propan-2-ol in dichloroethane)
u	60 g/L ammonium thiocyanate in dimethylformamide	uu	60 g/L ammonium thiocyanate in bis(2-methoxyethyl)ether
v	40 g/L ammonium thiocyanate (in ethanol) (V/V)	vv	40 g/L ammonium thiocyanate in (83% +5% (V/V) formamide butanone in light petroleum)
w	2.23 g/L sodium pyrophosphate decahydrate solution	ww	40 g/L potassium nitrate solution
x	100 g/L sodium chloride in (85% (V/V) propan-2-ol in acetonitrite)	xx	40 g/L tetrabutyl ammonium perchlorate in dimethylformamide
y	8 g/L sodium hydroxide solution	yy	65 g/L sodium acetate in (66.7% (V/V) ethanol in water)
z	7 g/L hydrochloric acid solution	zz	Karuhn's medium
yz	30 g/L lithium chloride in (50% (V/V) propan-2-ol in methanol)		

Materials and Recommended Electrolyte Solutions

NOTE Where more than one electrolyte solution is given, they are given in preferential order.

NOTE

+G - glycerol often added to help suspend large particles:

+S - sucrose often added to help suspend large particles;

satd - electrolyte solution should be presaturated with sample.

Material	Electrolyte Solutions	Comments
Acetylsalicylic acid	a satd, b satd	Aerosol OT
Acrylic emulsion or powder	a, b, e	See the chemical name.
Aldactone A	a, b	(Spironolactone)
Alumina	e	Any aluminas
	a, b	Coarse powders only
Aluminium	a	Alkaline electrolyte solutions react
Aluminium oxide	-	See alumina.

Material	Electrolyte Solutions	Comments
Aluminium silicate	e	(Andalusite)
Ammonium perchlorate	q satd, m	Common ion effect depresses solubility.
Ammonium phosphate	m	Up to 80 g/L ammonium thiocyanate may be used to increase the common ion effect.
Amphotericin (B)	a satd b satd	
Anionic bitumen emulsion	c	
Antimony	a	Daxad has been used as dispersant.
Asphalt emulsion	-	See anionic or cationic bitumen emulsion.
Attapulgite	e	
Avicel (microcrystalline cellulose)	a	
Azodicarbamide	m	Slightly water soluble
Ball clay	-	See clay.
Barium ferrite	r	See ferrites.
Barium sulphate	i	Common ion effect suppresses solubility.
Bark	a	
Barytes	e	
Bauxite	a	
Beef extract (dried)	m	
Bentonite	e	
Benzoic acid	b	
Benzyl procaine penicillin	a satd	
Beryllia	e, a, b	
Bitumen emulsion	-	See anionic bitumen or cationic bitumen emulsion.
Bone	e	
Boron carbide	a, b	Sulframine has been used as dispersant. Electrolyte solution has been presaturated with boric acid.
Brick dust	l	
Bronze	s, t	The recommended electrolyte concentration should not be exceeded because of chemical attack.
Cadmium sulphide	e	
Calamine	e	
Calcined magnesia	-	See magnesia.
Calcite	-	See calcium carbonate.
Calcium carbonate	l, m, a satd	
Calcium chromate	l, satd	
Calcium di-H phosphate	l	

Material	Electrolyte Solutions	Comments
Calcium oxide	l	
Calcium stearate	l, a, b, e	Difficult to wet; alcohols, or non-ionic dispersant with ultrasonics and spatulation, has been used.
Calcium tri-H phosphate	l	
Carbon (and carbon black)	e, a	Activated carbon will release gases on passage through the orifice to give spurious results. Disperse the powder in a little warm glycerol with spatula, add a little electrolyte solution, and boil for a few minutes. Cool and add remainder of electrolyte solution and count. Dispersion under vacuum has been used.
Carbonyl iron	a(+G), b(+G)	
Carbonyl nickel	a(+G), b(+G)	
Carborundum	-	See alumina and emery.
Casein	m	
Cationic bitumen emulsion	d	
Cellulose	n m	n is generally more suitable, but it cannot be used unless flow rate is reduced, with orifices above 280 µm, owing to whistling high frequency noise interference on rapid flow through the orifice and/or vapor bubble formation in the orifice.
Cement	l m	
Ceramic powders	e	
Cerium oxide	e	
Chalk	-	See calcium carbonate.
China clay (kaolin)	-	See clay.
Chloramphenicol	a satd	
Chocolate	m satd	Melt in beaker over water bath at 40ºC to 50ºC, spatulate with 50 g/L alcohol soluble fraction of Span 80 in cyclohexanol. Cool and add electrolyte solution. Presaturation with defatted solids (from petroleum ether) is essential for milk chocolate analysis, and sucrose may be used for dark chocolate.
Chromium powder	e	
Clay	e	Usually very fine, suitable for 30 µm or 50 µm/30 µm orifices. It is common to make up a dispersion and leave for one or two days before analysis. When the mass integration method is used, the density should be measured for the material suspended in the same electrolyte solution for a similar time; this is because of water, present in the lattice structure.
Coal	e a b m w	
Cocoa	l	

Material	Electrolyte Solutions	Comments
Coffee	l, m	For powder from coffee beans only.
Coke	a b e	
Copper	a b	No reaction during time for analysis.
DDT	a satd, e satd	
Diamond	e(+S), b(+S)	
Diatomite	-	See clay.
Dolomitic lime	l	
Dust (flue, coal, and so forth)	e a b	e is a better choice if metal particles are suspected in the sample, for example gas mains dust.
Electrolyte solutions	-	To measure particles in electrolyte solutions, for example, plating solutions, measure directly, or if particle size range is reduced because the orifice resistance is too low, dilute with filtered distilled water.
Emery	a(+G), b(+G)	Usually in very narrow size ranges, for example 75 µm to 85 µm.
Emulsions (including emulsions for rolling mills)	a b	Solids or oil in water can be lubrication and coolant measured. Two stage dilution of, for example, 1:100 each stage is preferable to reduce coagulation. Stability of emulsion over analytical period should be maintained. Measurement of water in oil is not possible at present.
Encapsulated particles	1, m	Dependent on encapsulant material.
Explosives: HMX,PETN,RDX	a satd	See also ammonium perchlorate and Guanidine perchlorate.
Felspar	a, b	
Ferrites	e	Addition of 500 g/L glycerol reduces reagglomeration rate. Heat treated ferrites are magnetic and cannot be dispersed to primary particles; the agglomerate size will be fairly stable and measurable. Preferably measure before heat treatment.
Fibres (paper pulp)	a	
Fibres (wool top)	m, x	
Fibres (in glycerine)	b	
Filters	-	See membrane filters.
Flint	a, e	
Flour	m, l	Usual range 10 µm to 125 µm; disperse
Fly ash	w	
Fullers earth	e	
Garnet	e	
Glass powder	a, b	Almost any electrolyte solution may be used.

Material	Electrolyte Solutions	Comments
Gold	a, b, e, g(+G)	
Graphite	e	Satisfactory for coarse graphites; very slow flocculation at 1 µm level.
	y	For fine graphite
Graphite in oils	-	See oil.
Griseofulvin	a satd	Add 0.1 g/L Goulac for dispersion.
Gypsum	m	
Guanidine perchlorate	q	
Herbicides	a b	Usually insoluble
Indomethacin	z satd	
Injection fluids (particles in parenteral fluids)	-	Usually need no added electrolyte solution. If no electrolyte is present, add filtered sodium chloride solution. Iron dextran apparently has never been analyzed successfully, possibly because of complex formation giving unstable counts.
Ink, ball point	aa	
Ink, silk screening	m, bb	
Ink, in toluene	cc	All particles are very fine; many particles are not measurable. Highly colored suspensions make observation of orifice difficult.
Ink, printing	zz	
Ion exchange resin	a, r	Saturate sample in electrolyte solution f to exhaust resin before analyzing.
Iridium	c(+S)	Extremely dense (22 g/cm^3): with 500 g/L added sugar, analyses can be performed up to 60-80 µm.
Iron	dd(+G), e	
Iron oxide	a(+G), b(+G)	
Kaolinite	-	See clay.
Kerosene (particles in)	ee	Add up to 20% (V/V) kerosene. Ketchup (catsup) a, b
Latex (rubber)	a, b, e	
Latex (synthetic)	a, b, e	
Lead	a(+S), f(+S)	Methanol has been used as a dispersant.
Lead (II or III) oxide	a(+G)	Up to 950 g/L glycerol has been used.
Lead, red	e	
Lignite dust	e	
Lime	a satd, 1, m	
Lycopodium powder	a, b	
Magnesia	m, l	Reactive in water when finely divided.

Material	Electrolyte Solutions	Comments
Magnesia, calcined	e(+G)	
Magnesium	e(+S)	Very slow reaction; will not affect accuracy of results in usual analytical time.
Magnesium hydroxide	p, e satd	
Membrane filters (cellulose nitrate) (particles captured by)	ff gg, hh, ii, jj	Dissolve the membrane in 30% (V/V) dimethylformamide in acetone before adding equal volume of electrolyte solution. The electrolyte solution and solvent mixture should be filtered and stored separately. Ideal for particles in oil: filter known volume of oil through membrane, then wash through with filtered petrol, carbon tetrachloride, trichloroethylene, and so forth and dry the filter before dissolving. Extra dimethylformamide increases membrane solubility.
Mica	e, a, b	
Powdered milk	m	For alcohol insoluble fraction.
	a	For water insoluble fraction. Disperse in a few drops of 900 g/L sodium hydroxide solution.
Molybdenum disulphide	y, e	
	m	More suitable for MoS_2 in oil.
Mud	a,. b, e	
Neomycin	l satd	
Nickel	a	See also Raney nickel.
Nylon, particles in	kk	Wet first with dispersant, then add electrolyte solution.
Ocean sediment	-	See sediment
Oil, cutting	a, b	
Oil, hydraulic and lubricating	ll	Will accept up to 50% (V/V) of added oil (DTD 585).
(oil specifications are for lubrication effectiveness,	mm	Electrolyte solution will accept 33% (V/V) to 50% (V/V) Skydrol.
Orange extract	a, b	
Paint (oil based)	gg, zz	
Paper pulp	a, b	
Peanut butter	m	
Penicillin	a satd	
Phenacetin	a satd	
Phenothiazine	a satd	
Phosphors	a, b, e	Size range approximately 1 µm to 40 µm.
Photographic emulsions	a, i	40 g/L potassium nitrate solution may also be used.
Pigments	e, a, b	
Plaster of Paris	l, k	Electrolyte solution should be anhydrous.

Material	Electrolyte Solutions	Comments
Plastics	a, b, e	
Plating solutions.	-	Analyzed with no added electrolyte or diluted with filtered distilled water.
Pollens	a, b	
Polyethylene	a, b	
Polypropylene	a, b	
Polystyrene	a, b	
Poly(styrene divinyl benzene)	a, b, e	Useful for calibration of orifices with most organic electrolytes.
Polytetrafluoroethylene	a	Almost any electrolyte solution is suitable. For finer powders use alcohols or ketones to ensure thorough wetting.
Polyvinyl acetate	a	
Polyvinyl chloride	a, b	
Polyvinyl propylodone	a	
Polyvinyl toluene	a, b	
Porcelain	e	
Potassium chloride	l	
Potassium sulphate	l	
Potato starch	a	
Quartz	e	
Raney nickel in xylene	l,m	Slow reaction with m.
River sediment	-	See sediment.
River water	a,h	
	e	If no calcium salts present
Rouge	a	
Rust in gasoline	e	Filter out and resuspend in electrolyte solution.
Rutile	e	
Sand	a,b,e	
Sediment	a,b,e	
Shale	e	
Silica	e	
Silica gel	l	
Silicates	e	
Silicon carbide	a,b,e	
Silicon nitride	m	
Silver halide	ww	

Material	Electrolyte Solutions	Comments
	a	For silver bromide
Silver oxide	a,g	
Slag (basic)	a,e	
Sodium (metal)	-	Dispersions are usually in oil or grease, and electrolyte solutions made from ammonium thiocyanate and alcohols or ketones are suitable (see oils). Coupling agents, such as l,l,2,trichloroethane, may be needed.
Sodium hydrogen carbonate	m	
Sodium chloride	m satd	
Sodium hydroxide	xx satd	Still not fully stable; use multi-channel models.
Soya flour	m,b	
Spironolactone	a,b	
Starch	a,e,l,m	Aqueous electrolytes suitable for hydrated form.
Stearates	a,l	Difficult to wet. Spatulate with alcohol and use ultrasonics.
Steel	a(+G), f(+G)	
Sugar	m satd	
Sulphadimidine	a satd	
Sulphur	a,e	Wet by spatulation with alcohol before diluting with electrolyte solution.
Superphosphate (of lime)	m	
Talc	e	
Tantalum	a(+G),e(+G),f(+G)	
Tin	a(+G), f(+G)	
Tin oxide	e	Disperse in 50 g/L Calgon solution.
Titanium dioxide	e	
Tomato juice	a	
Toner, xerographic	a	
Tungsten	a, e(+G/S), f(+G)	
Tungsten carbide	e(+G), a(+G)	Can cold-sinter within hours of milling to cause permanent agglomeration.
Uranium	c(+G), f(+G)	
Uranium dioxide	e, c	Can cold-sinter (sec tungsten carbide).
Viscose	c	Or dilute with filtered distilled water.
Water, contaminants in	a	Dilute as required into electrolyte solution.
Whiting	l, m,a satd	See calcium carbonate.
Wool fibre	m	

Material	Electrolyte Solutions	Comments
Yeast	a	
Yttrium iron garnet	e(+G)	Magnetic and needs frequent redispersing by ultrasonics.
Zeolite	-	See clay.
Zinc	yy	10 g/L Aerosol OT dispersant
Zinc cadmium sulphide	e	Range usually 1 µm to 15 µm.
Zinc oxide	l,a,e satd	Slightly soluble in water.
Zinc stearate	l	See calcium stearate.
	a, b, e	Wet with Sulframin AB.
	h,j	
Zinc sulphide	l	Not very suitable as the sulphide slowly oxides to sulphate.
Zirconium oxide	a	Wet with 40 g/L sodium pyrophosphate solution.

Standard Sieve Series

Table of Standard U.S. Mesh and Tyler Sieve Sizes

Sieve Size U.S. Mesh	Opening		Sieve Size Tyler
	Inches	Microns	
4.24	4.24	10 760 000	-
4.0	4.00	10 160 000	-
2.12	2.12	5 380 000	-
2.0	2.00	5 080 000	-
1 1/2	1.50	3 810 000	-
1 1/4	1.25	3 150 000	-
1.06	1.06	2 650 000	1.05
1.00	1.00	2 500 000	-
7/8	0.8750	2 240 000	0.883
3/4	0.7500	1 900 000	0.742
5/8	0.6250	1 6000 000	0.624
0.530	0.5300	1 320 000	0.525
1/2	0.5000	1 250 000	-
7/16	0.4375	1 120 000	0.441
3/8	0.3750	950 000	0.371
5/16	0.3125	800 000	2 1/2
0.265	0.2650	670 000	3.0
1/4	0.250	630 000 -	3.5
3.5	0.2230	5660	3.½
4	0.1870	4750	4
5	0.1570	4000	5
6	0.1320	3350	6
7	0.1110	2800	7

Sieve Size U.S. Mesh	Opening		Sieve Size Tyler
	Inches	Microns	
8	0.0937	2360	8
10	00787	2000	9
12	0.0661	1700	10
14	0.0555	1400	12
16	0.0469	1180	14
18	0.0394	1000	16
20	0.0331	850	20
25	0.0278	710	24
30	0.0234	300	28
35	0.0197	500	32
40	0.0165	425	35
45	0.0139	355	42
50	0.0117	300	48
60	0.0098	250	60
70	0.0083	212	65
80	0.0070	iso	50
100	0.0059	iso	100
120	0.0049	125	115
140	0.0041	106	iso
170	0.0035	90	170
200	0.0029	75	200
230	0.0025	63	250
270	0.0021	53	270
325	0.0017	45	325
400	0.0015	38	400
450	0.0012	32	-
500	0.0010	25	-
635	0.0008	20	-

Physical Properties of Organic Solvents

Table of Physical Properties of Organic Solvents

Name	Molecular Weight	Boiling Point (ºC)	Melting Point (ºC)	Density	Refrac. Index	Dielectric Constant	Flash Point (ºC)
Acetaldehyde	44.05	20.8°	-124°	0.780	1.3116	22	-38°
Acetic Anhydride	102.09	139°	- 73°	1.082	1.3904	20	53°
Acetone	58.08	56.5°	- 95°	0.792	1.3591	20.7	-18°
Acetonitrile	41.05	81.6°	- 46°	0.787	1.3341	38.8	56°
Acetophenone	120.15	202°	18.5°	1.033	1.5339	17.4	82°
Amyl Acetate(iso)	130.18	142°	-79°	0.876	1.4012	25	25°
n-Amyl Alcohol	88.15	138°	-79°	0.81	1.4099	13 9	33°
Aniline	93.13	184°	- 6°	1.022	1.5863	6.89	70°
Benzaldehyde	106.12	179°	-26°	1.046	1.5456	17.8	64°
Benzene	78.11	80.1°	5.5°	0.879	1.5011	2.28	-11
Benzoyl Chloride	140.57	197°	- 1 °	1.212	1.5537	23	72°
Benzyl Alcohol	108.14	205°	- 15°	1.043	1.5396	13.1	93°
Benzyl Chloride	126.59	179°	-43°	1.102	1.5396		67°
Bromobenzene	157.01	156°	-30.6°	1.495	1.5602	5.5	51°
Bromoform	252.75	150°	6.5°	2.890	1.5980	4.4	
n-Butyl Acetate	116.16	126°	-77°	0.882	1.3951	5.0	22°
n-Butyl Alcohol	74.12	118°	89°	0.810	1.3993	17.8	29°
sec-Butyl Alcohol (dl)	74.12	98°	-114.7°	0.808	1.3954	15.8	24°
tert-Butyl Alcohol	74.12	82.4°	24°	0.786	1.3847	11.5	11 °
Carbon Disulfide	76.14	46.5°	100°	1.263	1.6280	2.64	-30°
Carbon Tetrachloride	153.82	76.7°	23°	1.595	1.4607	2.24	
Chlorobenzene	112.56	132°	-45°	1.107	1.5248	5.71	28°
Chloroform	119.38	61.2°	-63.5°	1.489	1.4476	4.81	
2-Chloro-2-Methyl Propane	92.57	51°	-25°	0.847	1.3868		<0°

Name	Molecular Weight	Boiling Point (ºC)	Melting Point (ºC)	Density	Refrac. Index	Dielectric Constant	Flash Point (ºC)
Cyclohexane	84.16	80.7°	6.5°	0.778	1.4264	2.02	-20°
Cyclohexanol	100.16	161°	23°	0.962	1.4655	15	68°
Cyclohexanone	98.15	155.6°	-47°	0.948	1.4507	18.3	44°
Cyclohexene	84.16	81°	7°	0.811	1.4465	2.2	-20°
Dibutyl Phthalate	278.34	340°	-35°	1.046	1.4900	ca. 8	171°
o-Dichlorobenzene	147.01	180.5°	-17.6°	1.306	1.5515	99	66°
1,2-Dichloroethane	98.96	83.5°	- 35.4°	1.257	1.4443	9.3	13°
Dichloromethane	84.93	39.8°	- 77°	1.326	1.4244	9.09	
Diethanolamine	105.14	268.8°	28°	1.088	1.4753		172°
Diethylamine	73 14	55.5c	-50°	0.707	1.3864		-29°
1,2-Dimethoxyethane	90.12	83°	-58°	0.863	1.3813		-2°
N,N-Dimethylacetamide	87.14	166°	-20°	0.943	1.4373		66°
N,N-Dimethylformamide	73.10	153°	- 6°	0.945	1.4281		58°
Dimethyl Sulfate	126.13	188°	-32°	1.332	1.3874		83°
Dimethyl Sulfoxide	78.13	189°	18.5°	1.100	1.4795		95°
Dioxane	88.11	101.1°	11.8°	1.033	1.4175	2.2	12°
Ether	74.12	34.6°	- 123°	0.713	1.3529	4.3	- 45°
2-Ethoxyethanol	90.12	135°	-70°	0.929	1.4065		43°
Ethyl Acetate	88.11	77°	-83°	0.902	1.3719	6.0	-4.4°
Ethyl Acetoacetate	130.14	180.8°	-43°	1.026	1.4194	1.57	84°
Ethyl Alcohol	46.07	78°	-117°	0.789	1.3611	25	13°
44	108.98	38.4°	-119°	1.451	1.4241		-20°
Ethylenediamine	60.10	118°	8.5°	0.899	1.4499		33°
Ethylene Glycol	62.07	193°	-12.6°	1.113	1.4319	38.7	110°
Ethylene Glycol	76.09	124°	-85.1°	0.966	1.4028		39
Ethyl Iodide	155.97	72°	-108°	1.947	1.5168		
Formamide	45.04	210.5°	2.55°	1.133	1.4475	84	154°
Furfural	96.09	161.8°	-39°	1.160	1.5261	42	60°
Glycerol	92.10	290°	18°	1.25	1.4746		160°
n-Heptane	100.20	98.4°	- 90.7°	0.684	1.3876	1.9	- 4°
n-Hexane	86.18	68°	- 95°	0.660	1.3750	l .9	- 22°
Isopropyl Alcohol	60.10	82.5°	-88.8°	0.785	1.3772	l8	12°
Isopropyi Ether	102.18	68.5°	-85°	0.78	1.3680	3.85	-28°
Methyl Acetate	74.08	58°	- 99°	0.928	1.3614	7	- 10°
Methyl Alcohol	32.04	64.7°	- 98°	0.792	1.3292	33.6	11 °
Methyl Cyclohexane	98.19	100.4°	-126.4°	0.770	1.4253	2	-4°

Name	Molecular Weight	Boiling Point (ºC)	Melting Point (ºC)	Density	Refrac. Index	Dielectric Constant	Flash Point (ºC)
Methyl Ethyl Ketone	72.11	79.6°	-86°	0.805	1.3814	19	-6°
Methyl Iodide	141.95	42.5°	-66.5°	2.28	1.5293	7.0	
Methyl Isobutyl Ketone	100.16	118°	-80.4°	0.801	1.3962	18	23°
Nitrobenzene	123.11	210.8°	5.7°	1.204	1.5562	35	88
Nitromethane	61.04	100.8°	-28.5°	1.132	1.3817		35°
2-Octanol (dl)	130.22	178.5°	-38.6°	0.819	1.4203	10	71°
Octyl Alcohol	130.22	194°	-16.5°	0.827	1.4293	10	81°
n-Pentane	72.15	36.1°	-130°	0.631	1.3579	1.84	-49°
2,4-Pentanedione	100.10	140.5°	-23°	0.975	1.4512	25.7	34°
1,2-Propanediol	76.10	188.2°	-59°	1.036	1.4324	32	99°
n-Propyl Alcohol	60.10	97.1°	-127°	0.804	1.3850	20	15°
Propylene Oxide	58.08	34°	- 112°	0.830	—		- 35°
Pyridine	79.10	115°	- 42°	0.982	1.5102	12.5	20°
1,1,2,2-Tetrabromoethane	345.70	151°	- 1°	2.964	1.6380	7.0	—
1,1,2,2-Tetrachloroethane	167.86	146.5°	- 43°	1.586	1.4944	7	—
TetTachloroethylene	165.85	121 °	- 22°	1.623	1.5044		—
Tetrahydrofuran	72.10	66°	-108.5°	0.888	1.4070		-14°
Toluene	92.14	110.6°	- 95°	0.866	1.4961	2.4	4°
TributylPhosphate	266.32	289°	- 79o	0.973	1.4224		146°
Trichloroethylene	131.39	86.7°	- 73°	1.465	1.4784	3.4	—
Trietylamine	101.19	89.5°	-115°	0.723	1.4003	2.4	-9°
2,2,4-Trimethylpentane	114.23	99.3°	-107.5°	0.692	1.3916	1.94	-12°
Xylene	106.17	137°	- 25°	0.86	-	2.3	26°

Equivalent Names for Dispersants, Diluents and Some Materials

Solid Materials

Material	Equivalent Names
anhydrite	calcium sulfate.
alumina	aluminum oxide
alumina cement	cement, alumina
aluminosilicate (K,Na)	$(K,Na)_2OA1_2O_3$ $6SiO_2$
aluminosilicate (beryl)	3BeO $A1_2O_3$ $6SiO_2$
aluminosilicate hydrate	K2O3 $A1_2O_3$ $6SiO_2H_2O$
alundum	aluminum hydroxide
apatite	calcium phosphate fluoride
barite	barium sulfate
bartyes	barium sulfate
bauxite	aluminum oxide dihydrate = $A1_2O_32H_2O$
bentonite	aluminosilicate (Na ,K)
beryl	aluminosilicate (beryl)
calcium fluoride phosphate	$3Ca_3(PO_4)_2CaF_2$
calcium magnesium carbonate	$(Ca,Mg)CO_3$
calomel	mercurous chloride(Hg_2C1_2)
carborundum	silicon carbide
cerussite	lead carbonate
chalk	calcium carbonate
china clay	aluminosilicate (kaolin)
chrome yellow	lead chromate
clay	aluminosilicate (bentonite, kaolin)
coke	carbon
corundum	aluminum oxide
cryolite	sodium aluminum fluoride

Material	Equivalent Names
dolomite	calcium magnesium carbonate
feldspar	aluminosilicate (K,Na)
hematite	ferric oxide
fluorite	calcium fluoride
fluorospar	calcium fluoride
gypsum	calcium sulfate dihydrate ($CaSO_42H_2O$)
illmenite	ferrous titanate ($FeOTiO_2$)
kaolin	aluminosilicate
kieselguhr	diatomaceous earth
lead, red	lead oxide (Pb_3O_4)
lignite	carbon
lime	calcium hydroxide
limestone	calcium carbonate
litharge	lead monoxide
lithopone	barium sulfate/zinc sulfide
magnesite	magnesium carbonate
magnesium silicate hydrate	$3MgO4SiO_2H_2O$
mica	aluminosilicate hydrate
Perspex*	polymethylmethacrylate
plaster	calcium sulfate hemihydrate
pumice	aluminosilicate (Na, K)
pyrolusite	manganese dioxide
quartz	silicon dioxide
rutile	titanium dioxide
sand	silicon dioxide
sillimanite	$A1_2O_3SiO_2$
sucrose	$C_{ll}H_{22}O_{11}$
talc	magnesium silicate hydrate
thoria	thorium dioxide
titania	titanium dioxide
whiting	calcium carbonate
zircon	zirconium silicate

Liquid Materials

Material	Equivalent Names
butan-1-ol	n-butanol
butan-2-ol	isobutanol
butan-1-amine	n-butylamine
dodecane (mp -10°C bp 216°C)	mixture of alkanes (used for petroleum)
EneCOOH	an unsaturated fatty acid, for example, oleic acid = $HC_{l7}H_{32}COOH$, the principal component of olive oil
ethan-1,2-diol	ethylene glycol, glycol
heptane (bp = 98°C)	mixture of alkanes (used for kerosene)
hexadecane (mp 18°C)	mixture of alkanes (mp 0-20°C) (used for liquid paraffin) or mixture of alkanes (mp <20°C) (used for paraffin oil)
3-methylbutan-1-ol	isoamyl alcohol
octan-1-ol	octanol
propan-2-ol	isopropanol
propan-2-one	acetone
propan-1,2,3-triol	glycerol

Dispersants

Material	Equivalent Names
AlkCOOH	$HC_nH_{2n}COOH$ (n=11 to 17)
AlkPhPEOH	nonylphenyocta(ethox)anol Triton X-100 [Union Carbide] Lissapol NX [ICI] dispersant
AlkPEOH	alkyl polyoxyethylene Renex 648 [ICI] dispersant
EneCOO-	an unsaturated fatty acid salt for example, sodium linoleate = $HC_{17}H_{30}COO$- Na or sodium oleate = $HC_{16}H_{32}COO$- Na
KOrP	potassium orthophosphate (K_3PO_4)
2-methylheptane	iso-octane
NaAlkAryS	disodium methylene bis naphthalene sulfonate Dispersol T [ICI] dispersant
NaAlkPhS	sodium alkylbenzene sulfonate Teepol [Shell] dispersant

Material	Equivalent Names
NaDioctSS	sodium di-octyl sulfosuccinate Aerosol OT [American Cyanamid] dispersant
NaLigS	sodium lignosulfonate Daxad 23 [W. R. Grace] dispersant
NaPMP	sodium polymetaphosphate ($(NaPO_3)_3$) sodium hexametaphosphate Calgon dispersant Giltex dispersant Micromet dispersant
NaPyP	sodium pyrophosphate ($Na_4P_4O_{10}$)
PEOH	ethylene oxide condensate Nonidet LE [Shell] dispersant
PEOSorLau	polyoxyethylene sorbitan laurate Tween 20 [ICI] dispersant
SorLau	sorbitan laurate Span 20 [ICI] dispersant

Recommended Dispersants

Table of Recommended Dispersants

Material	Solvent	Dispersant
Aluminum	propan-2-ol	none
	hexadecane	lead naphthenate
Aluminosilicate hydrate (mica)	water	NaPyP
Aluminosilicate(bentonite)	water	NaPyP
Aluminosilicate (beryl)	water	sodium silicate
	water	NaPMP
Aluminosilicate (feldspar)	water	NaPyP
Aluminosilicate (kaolin)	water	ammonia
	water	hydrochloric acid
	water	KOrP
Aluminum oxide	2-methyl-heptane	SorLau
	butan-1-ol	none
	butan-1-amine	none
	water	hydrochloric acid (pH 3)
	water	AlkPhPEOH
	water	PEOH
	water	Perminal BX
	water	NaPyP
	water	sodium tartrate
	water	NaPMP
	water	AlkPhPEOH
Aluminum oxide (corundum)	water	NaPyP
Antimony trioxide	water	NaPyP
	water	NaPMP
Arsenates	water	NaPyP

Material	Solvent	Dispersant
Arsenic trioxide	hexadecane	AlkCOOH
	octan-1-ol	none
Ash	water	NaPyP
Barium carbonate	methanol	none
Barium sulfate (barytes)	water	NaAlkAryS
	water	NaPyP
	water	NaPMP
	water	KOrP
Barium titanate	water	NaAlkPhS
Boron (amorphous)	butan-1-ol	none
Boron carbide	water	NaPyP
Boron nitride	butan-1-ol	none
Boron oxide	water	NaPyP
Cadmium sulphide	ethan-1,2-diol	none
	water	NaPyP
Calcium carbonate	dodecane	none
	water	NaAlkAryS
	water	PEOH
	water	potassium citrate
	water	NaPMP
	water	NaPyP
	water	sodium silicate
	water	KOrP
	water	AlkPhPEOH
Calcium fluoride	water	NaPyP
Calcium hydroxide	ethanol	none
Calcium magnesium carbonate	water	NaPMP
Calcium oxide	cyclohexanone	none
	ethan-1,2-diol	none
Calcium phosphate monohydrate	hexane	none
	butan-2-ol	none
Calcium phosphate trihydrate	water	NaPyP
	water	NaPMP
	water	KOrP

Material	Solvent	Dispersant
Calcium sulfate dihydrate	ethan-1,2-diol	cobalt citrate
	methanol	none
	water	NaDioctSS
Calcium sulfate hemihydrate	water	potassium citrate
Carbon (activated charcoal)	butan-2-ol	none
Carbon (anthracite)	water	Perminal BX
	water	KOrP
Carbon (brown coal)	diethylphthalate	none
Carbon (charcoal)	water	EneCOONa
	water	NaPyP
Carbon (coal)	dodecane	none
	water	NaDioctSS
	water	EneCOONa
	water	PEOSorLar
	o-xylene	none
Carbon (coke)	water	Perminal BX
	water	EneCOONa
	water	EneCOONa
Carbon (diamond)	ethanol	none
	water	KOrP
Carbon (graphite)	ethanol	none
	butan-l-ol	none
	water	NaDioctSS
	water	EneCOONa
	water	tannic acid
	water	NaAlkPhS
Carbon (lignite)	diethylphthalate	none
Carbon black	water	NaDioctSS
	water	EneCOONa
	water	tannic acid
Cement (Portland)	ethanol	calcium chloride
	methanol	NaPyP
	2-methyl-heptane	SorLau
	heptane	EneCOOH
	butan-1-ol	none
Chromium oxide	water	NaPyP

Material	Solvent	Dispersant
Cobalt	diethylphthalate	none
	ethanol	none
	butan-l-ol	none
Copper	propan-2-one	none
	ethanol	none
	butan-1-ol	none
	water	NaPyP
Copper oxide	water	NaPMP
Copper oxychloride	water	NaPMP
Diatomaceous earth	water	none
	water	NaPyP
	water	NaPMP
	water	KOrP
Ferric oxide	ethanol	none
	water	NaPyP
Ferrous sulfide (pyrite)	water	NaPyP
Fly ash	water	none
	water	NaPyP
Food (cocoa)	propan-2-one	none
	diethylphthalate	none
	butan-2-ol	none
	butan-l-ol	none
Food (flour)	diethylphthalate	none
Glass, soda-lime	methanol	none
	butan-l-ol	none
	water	NaAlkAryS
	water	AlkPhPEOH
	water	none
	water	PEOH
	water	NaPyP
	water	KOrP
Iron	ethanol	none
	butan-1-ol	none
	water	AlkPhPEOH
	water	PEOH
	water	NaPMP

Material	Solvent	Dispersant
Lead carbonate	water	NaPMP
Lead chromate	water	NaPyP
Lead cyanamide	water	NaPyP
Lead monoxide	water	NaPyP
Lead oxide, red	ethan-1,2-diol	none
	water	NaPyP
Lead sulfate	water	KOrP
Magnesium carbonate	2-methyl-heptane	SorLau
	water	NaPyP
	water	NaPMP
Magnesium oxide	water	NaPMP
Magnesium silicate hydrate	water	NaPyP
Manganese dioxide	water	PEOH
	water	NaPyP
Manganese tetroxide	water	NaAlkAryS
	water	NaPyP
Molybdenum trioxide	water	PEOH
	water	NaPMP
Nickel oxide	water	AlkPhPEOH
	water	NaPMP
Phosphorus (red)	water	sodium silicate
Pigments (inorganic)	water	potassium citrate
Pigments (organic)	water	AlkPhPEOH
	water	PEOH
	water	potassium citrate
	water	NaPyP
Polymethylmethacrylate	water	KOrP
Polyvinyl acetate	water	NaPMP
Polyvinyl chloride	butan-2-ol	none
	water	EneCOONa
	water	KOrP
Selenium	water	NaPyP
Silicates	water	NaPyP
Silicon	ethanol	none
	water	Cetrimide
	water	NaAlkAryS

Material	Solvent	Dispersant
Silicon carbide	water	NaPyP
	water	NaPMP
	water	KOrP
Silicon dioxide (amorphous)	ethanol	KOrP
	water	AlkPhPEOH
	water	NaPyP
	water	NaPMP
Silicon dioxide (quartz)	water	NaPyP
	water	sodium silicate
	water	KOrP
Sillimanite	water	NaPyP
Silver halides	water	NaPMP
Sodium aluminum fluoride	ethan-1,2-diol	none
	water	NaPyP
	water:propan-1,2,3-triol	none
Starch	diethylphthalate	none
	butan-2-ol	none
Sucrose	diethylphthalate	none
	butan-2-ol	none
Sulphur	water	EneCOONa
Thorium oxide (thoria)	water	NaPMP
	water	NaPyP
	water	NaAlkPhS
Titanium	water	AlkPhPEOH
	water	NaPMP
Titanium dioxide (rutile)	water	Cetrimide
	water	AlkPhPEOH
	water	NaPyP
	water	NaPMP
Tungsten	propan-1,2,3-triol	NaPMP + PEOH
	aqueous sucrose	Renex 648
Tungsten carbide	ethan-1,2-diol	NaPyP
Tungsten oxide ($W0_3$)	butan-l-ol	none
	water	NaPMP
Tungstic acid (H_2W0_4)	ethanol	none
	butan-l-ol	none

Material	Solvent	Dispersant
Uranium dioxide ($U0_2$)	methanol	none
	butan-l-ol	none
	water	NaLigS
	water	NaPMP
Wood Pulp	water	sodium silicate
Zinc	ethanol	NaPMP
	butan-l-ol	none
	water	NaPMP
Zinc oxide	water	NaPyP
	water	NaPMP
	water	KOrP
Zirconium dioxide	water:propan-1,2,3-triol	NaPyP
	water	NaPyP
	water	NaPMP
Zirconium silicate (zircon)	water	NaPyP

APPENDIX H

Trade Names of Dispersing Agents

Table of Trade Names for Recommended Dispersants

Additive (Dispersing Agent)	Trade Name
alkyl phenol ethylene oxide condensate	Lissapol NX Nonidet P40
alkyl trimethyl ammonium bromide	Cetrimide
coconut oil amine adduct with 15 ethylene oxide groups (cationic)	Ethomeen C15 **(COULTER 3A)**
dioctylester of sodium sulphosuccinic acid	Aerosol OT
ethoxylated nonylphenol	Renex 648
9-10 ethoxyoctylphenol	Triton X-100 **(COULTER 1A)** Prawozell W-ON 100
monoester of the sulphonic succinic acid with ethoxylated coconut alcohol, Di-Na salt	Succipon K3
naphthalene stearosulphonic acid	Twitchell Base
naphthalene sulphonic acid condensate	Lomar PW
nonyl phenoxy polyethanol	Igepal CO-530
oleic acid-methyl-tauride-Na salt	Igepon T
petrol soap + trichlorethylene	Saponin K
polyisobutene succinamide	OLOA 1200
polyoxyethylene sorbitan monostearate	Tween 20 **(COULTER 1B)**
polyoxyethylene thioether	Sterox
secondary sodium alkyl sulphate	Teepol
sodium alkyl naphthalene sulphonate	Perminal BX
sodium alkyl sulphonate	Emulgator 30 Mersolat H
sodium ethylene dinaphthylsulphonate	Dispersol T
sodium hexametaphosphate	Calgon
sodium salt of condensed naphthaline sulphonic acid	Tamol SN

Additive (Dispersing Agent)	Trade Name
sodium salt of ethylene diamine tetra acetic acid	Chelaplex III Komplexon III Trilon B
sodium salt of the polyacrylic acid (sodium polyacrylate)	LW 300 Dispex 40
sodium salt of polymerized carboxylic acid	Daxad 30
sodium salt of polymerized substituted alkyl- benzolsulphonic acid	Daxad 23
sorbitan monolaurate	Span 20
wetting agent (for coal)	Neomerpin
alkylaryl sulfonate (anionic)	Nacconol 90F **(COULTER 2A)**
C_9-C_{11} linear primary alcohol ethoxylate	Neodol 91-6 **(COULTER 1C)**

APPENDIX I

Conductive Particles and Porous Particles

Conductive Particles

In theory, only some difference in conductivity is needed to generate a pulse but in practice, the discharge potential from conductor to ions (the reversible e.m.f.) creates a barrier that makes any conductor into a non-conductor provided the potential across the particle does not exceed the discharge potential.

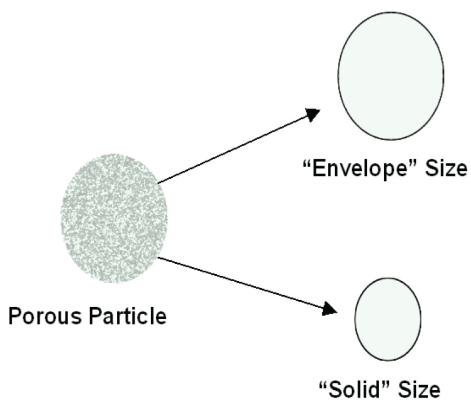
Particles of conductive materials can be accurately measured by the Electrical Sensing Zone method provided that the voltage applied across the aperture does not allow the barrier potential at the surface of the particles to be exceeded.

Particles of highly conductive materials which do not easily form surface layers such as copper, silver, and platinum can be sized correctly provided a very low voltage is used and the barrier increased by adding a 0.5% solution of Cetrimide.

In the case of cells enclosed by membranes, the linear response to cell volume changes at some “breakdown” value of applied voltage. This value has to be taken into account when setting conditions for the analysis. Blood cell analyzers are designed to operate below the blood cells’ breakdown potential.

Porous Particles

Particles with interconnected pores produce a size related to their solid volume not to their ''envelope'' volume. The electrolyte is able to fill the pores allowing the current to pass through the particles and therefore produce smaller pulses from the aperture. There are several techniques to solve this problem. They are based on the ability to fill the pores with an organic substance, usually a solvent, solid at room temperature.



APPENDIX J

Measurement Precision

How to Obtain Measurement Precision

In order to achieve maximum measurement precision, there are two potentially competing factors that must be considered. First, the total number of particles counted must be sufficiently high and second, the concentration of particles must be such that coincidence limitations are not exceeded. In other words, ideal counting conditions exist with large measurement volumes of low particle concentration. Generally these conditions are neither practical nor are they necessarily achievable. It is, however, important that any Particle Counter/Sizer should measure a large number of particles in order to achieve the greatest statistical confidence in the result.

Table J.1 presents precision statistics obtained from experimental data from 31 measurements of 10 µm latex particles.

Table J.1 Precision of Particle Counts at Various Count Levels

Average Count (n= 31)	SD (n-1)	CV %
106	9.46	8.92
1,076	30.8	2.86
9,755	85.8	0.88
10,005	85.0	0.85
10,500	99.4	0.95
49,444	221	0.45
98,559	234	0.24

Good statistics are of course indicated when there is little raggedness in consecutive channels in the DIFF mode, and when there is an absence of a regular rise in the data in the largest channels in the DIFF VOL mode, indicating that only one particle has been found in each channel Figure J.1, illustrates this effect of few particle counts; Figure J.2, shows the correct analysis after accumulating more data.

Figure J.1 Few Particle Counts

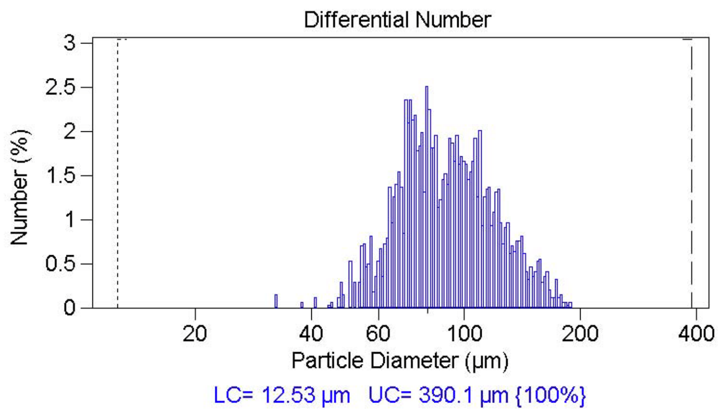
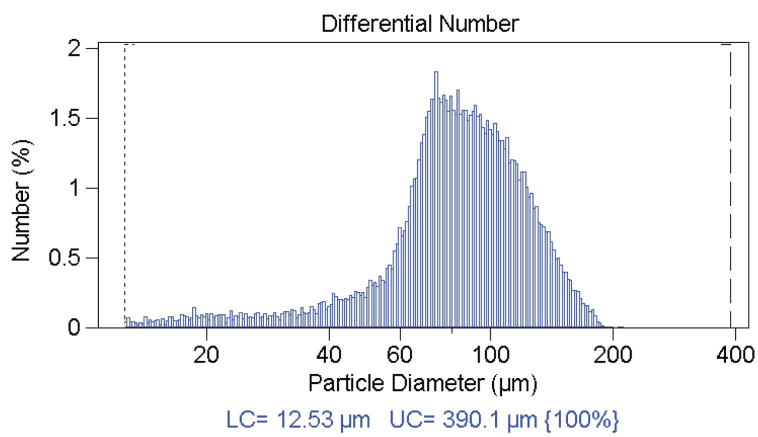


Figure J.2 Correct Analysis after Accumulating More Data



Abbreviations

The following list is a composite of the symbols, abbreviations, and acronyms either used in this manual or related to the information in it. When the same abbreviation (or reference designator) is used for more than one word (or type of component), all meanings relevant to this manual are included, separated by semicolons.

> — greater than

< — less than

% — percent

+ — plus

- — minus

± — plus or minus

°C — degrees Celsius

µ — micron

µA — microampere

µL — microliter

µm — micrometer (10^{-6})

A — ampere

Amp — amplifier

BCI — Beckman Coulter Incorporated

C — Celsius

CC — concentration control

CCW — counter-clockwise

CE — European Commission

cGMP — Current Good Manufacturing Practice

cm — centimeter

COM — communication port

CSA — Canadian Standards Association

CV — coefficient of variation

CV — coefficient of variation

CW — clockwise

DPP — Digital Pulse Processing

DSP — Digital Signal Processing

EMF — electromagnetic field

ESD — equivalent spherical diameter

ESZ — electrical sensing zone

FDA — Food and Drug Administration

ft — foot; feet

fL — femtoliter

GLP — Good Laboratory Practices

Hg — mercury

Hz — Hertz

IEC — International Electrotechnical Commission

in. — inches

IPA — Isopropyl Alcohol

IP Address — Internet Protocol Address

ISO — International Organization for Standardization

Kd — calibration constant

kg — kilogram

L — liter

lb — pound

LED — light emitting diode

m — meter

mL — milliliter

mm — millimeter

ms — millisecond

PC — personal computer

QC — quality control

RoHS — Restriction of Hazardous Substances Directive

SD — standard deviation; statistical deviation

sec — second

SOM — Standard Operating Method

SOP — Standard Operating Procedure

ST — Smart Technology

TD — Time delay

UL — Underwriter's Laboratory

USB — Universal Serial Bus

V — volts

VA — volt-ampere

VAC — volts alternating current

WEEE — Waste Electrical and Electronic Equipment Directive of the European Union

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Instructions for Use

PN B23937

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